

The Foreclosure of Summer

A novel of weather, debt, and citizenship across time

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The First Mild Winter

In Year 86, winter failed to close the east road.

By the second week of the usual storm season, the stone markers still showed above the grass. Meltwater ran clear beneath the footbridge instead of freezing into its culvert. Children from the canal quarter climbed the bare ridge after lessons and came home with mud to their knees, laughing as though the planet had made a clerical error in their favor.

The Vos household received the change first through the kitchen wall. Its vent stopped knocking at dusk. A brass fin that usually burned dust against its casing settled to the warmth of a sleeping animal. Ansel Vos placed his palm on it three times before he trusted the temperature.

“Keep your indoor coats nearby,” he told his daughters.

His wife, Pei, looked through the round pane above the sink. No black snow crossed the courtyard. The colony lamps shone without halos. “You said that yesterday.”

“I can say it again without using fuel.”

They ate at the table instead of crowding around the vent. Their youngest, Lysa, had turned six that autumn and could not remember a meal without the winter hiss behind it. She held her spoon near the brass fin, waiting for the customary tremor.

“Did they fix the sky?” she asked.

Ansel worked on radiator gates and understood enough of the Seasonal Continuity Lattice to distrust that verb. Engineers fixed couplings, drum teeth, canal seals. The sky accepted an adjustment and carried its consequences elsewhere.

“They moved some weather,” he said.

“To the mountains?”

“Farther.”

Pei gave him the look she used when work language entered the kitchen without wiping its boots. He broke bread and passed Lysa the soft center.

Later, after the girls slept, Ansel unfolded the public season notice. The sheet promised reduced canal draw, stable food storage, and six additional weeks of overland freight. Its final paragraph named the alteration a continuity improvement. Nothing on the front described where the cold had gone.

The reverse side held household heat credits. Their account had received forty-one degree-hours. Ansel knew the unit through labor: one degree held across one room for one hour, multiplied until bodies disappeared inside arithmetic. Forty-one would lower their winter assessment enough to replace the cracked inner pane before spring.

Pei read over his shoulder. “Can we trust it?”

“The warmth?”

“The credit.”

He rubbed his thumb across the printed seal. Ink remained dry. “They can audit heat.”

She waited.

“They can audit what happened here,” he amended.

At the Office of Seasonal Accounts, three streets east, a junior clerk entered the same transfer into a ledger bound with black thread. Public pages recorded the mild interval as recovered continuity. Technical pages kept older language beneath removable slips: cold account, storm burden, dissipation deferred.

The clerk had been instructed to copy quantities, not implications. She wrote the originating year, district gain, canal draw, and phase-drum setting. At the bottom of the entry, an obsolete field requested a settlement year of receipt.

Her supervisor leaned across the standing desk. “Leave that blank.”

“The drum gives a date.”

“The date identifies model depth.”

“It says Year 223.”

“Then write forecast horizon.”

The clerk fitted a paper strip over the obsolete field. She copied CONTINUITY MARGIN onto it, pressed down the paste, and sanded the wet edge. Beneath the strip, the old page retained two words and a number no public account holder would see.

COLD ACCOUNT: YEAR 223.

Outside, families walked after sunset because the air permitted it. The colony had known violent winters since landing. Doors opened only for work crews, bodies slept in shifts around shared heat, and frost sealed streets until canal steam cut a

path. That mild season offered another arrangement. People visited neighbors without calculating the cost of each threshold. The infirmary moved two patients home. Freight carts arrived with glass unbroken by cold shock.

No celebration required cruelty to feel sincere. Parents saw children gain weight. Repair crews slept through full nights. Pei Vos opened the kitchen vent and let outside air touch the bread dough, an extravagance she had never risked in winter.

Ansel stood in the courtyard with Lysa on his shoulders. Above them, a cloud comb combed the eastern bank into narrow silver ribs. Phase work pulsed below hearing. Each pulse coupled the present atmosphere to material laid down decades earlier, a path whose far end no one in the courtyard could see.

“Where did they move it?” Lysa asked again.

Ansel pointed beyond the ridge because children deserved an answer with a direction. “Down the long account.”

She accepted that. Adults around them raised cups. Someone played a reed pipe from an upper window. The music traveled farther through mild air than it had any right to travel in winter.

Before bed, Lysa drew the bare ridge on the household notice. She added five people beneath a wide sun, though winter suns on Orison stayed small and low. Pei folded the page around the heat credit and stored it above the kettle.

The Office ledger closed at midnight. Binding paste hardened over the renamed field. Along the black-threaded spine, the paper contracted by less than a hair.

No witness noticed.

In Year 223, a records porter carried that ledger from the founder stacks to the Office verification room.

It had outlived the clerk, her supervisor, Ansel, Pei, and the children on the ridge. Rebinding had replaced the black thread twice. Flood inspection left a blue thumbprint near the first mild-winter entry. Its removable strip remained in place, browned at the edges but still legible.

The porter set the volume beneath the legal observation slit. Canal Nine flashed below, hot enough to lift steam from a worker’s boots. Spring sunlight warmed the room. He signed the transfer docket and went to fetch the next box.

For several minutes, nothing occurred.

Then the old paste released.

The paper strip lifted at one corner. CONTINUITY MARGIN curled away from the page, exposing the founder words beneath. White grains formed inside the crease. They spread across the date field and gathered around Year 223 without wetting the paper.

A ledger thermometer on the wall registered no change. The canal kept its heat.

Clerks worked in the outer room, arguing over orchard retention figures and a late cloud-comb report.

Cold entered the bound volume through no visible opening.

It traveled along ruling ink, crossed a repair seam, and touched the household-credit page from Year 86. Lysa's child sun remained there in faded yellow pencil. Frost occupied its rays one by one.

The old phase setting woke beneath the account. Its pulse had crossed 137 years with the postponed weather, preserving every unpaid degree-hour while names vanished from the public face of the record. At the far side of that interval, ordinary rooms had received what the mild winter withheld. The ledger did not contain their lives. It held only the route by which present comfort had found them.

A hairline of ice reached the observation slit.

In the outer office, a morning bell sounded. Footsteps approached the verification room. The frost withdrew from the drawing but remained around the date, bright and exact.

When the door opened, cold breathed across Nera Voss's empty desk.

Act I

Chapter 1 — The Ledger Breathes

At nine minutes past the morning bell, Nera Voss pressed her seal into a season that had already happened.

The brass warmed beneath her thumb. On the forecast glass above it, three columns turned from amber to white: canal discharge, orchard retention, cloud-comb draw. She waited for the fourth. When it remained amber, she did not press harder. Instruments had no respect for insistence, and ledgers had even less.

Across the workroom, Edrin Sol folded yesterday's transfer schedules into their blue sleeves. He folded paper with the neat severity of a man refusing to make a private weather system out of his own breath.

"Canal Nine again?" he asked.

"Not yet."

"Amber means drift until it means otherwise."

"Amber means unverified."

His mouth moved by the width of an old disagreement. "Which is why ordinary people let notaries suffer over it."

Nera lifted her thumb. The seal retained a shallow impression of her whorl, crossed by the Office mark: two weather fronts meeting over an open hand. Everything else on the desk looked ordinary. The day schedules waited in their bone clips. Binding paste cooled in its dish. Through the legal observation slit, a rectangle of Canal Nine flashed white under the late-spring sun.

Then frost formed along the slit.

It began in the lower left corner, a fern no longer than her fingernail. Nera watched it branch across the glass. The workroom stood twenty-four degrees above freezing. Canal Nine carried discharge water hot enough to pink a worker's skin through gloves. No cold source ran through this wall.

Edrin stopped folding.

A second fern joined the first. Their meeting made a sound too slight for ice: paper taking breath.

Nera set the seal down. "Close the morning transfer."

"On an instrument anomaly?"

"Close this room."

He crossed to the public door without argument. That frightened her more than the frost. Edrin turned the lock, drew the privacy glass, then faced the desk with both hands empty and visible, the posture every notary learned for a disputed

record.

The fourth column changed.

Its heading should have read CONTINUITY MARGIN. Instead, letterforms surfaced beneath the amber field as if the forecast glass had been waiting under dirty water.

COLD ACCOUNT.

Nera knew the phrase. Every notary knew enough founder vocabulary to avoid using it in public. Before the Seasonal Charter standardized the ledgers, engineers had named transferred weather by what it cost. A mild winter created a cold account. A softened heat crest created a storm account. The modern Office used continuity margin because margin suggested room, and cold account suggested a creditor.

Below the heading, a number assembled one digit at a time.

-18,443,711 degree-hours.

Edrin exhaled through his nose. "That's not in the active lexicon."

"Neither is a negative transfer at this stage."

"Instrument drift can retrieve archived labels."

"Instrument drift doesn't frost legal glass."

"It may if Canal Nine has reversed a sink."

Nera looked through the whitening slit. A heatworker in silver canvas walked the canal gantry with a sounding rod over one shoulder. Steam lifted from the water around his boots. If a sink had reversed, alarms would have opened every street vent between the Office and Low Glass. She heard only the morning tram, the tick of Edrin's cooling sleeves, and the almost inaudible intake of the ledger.

They could test the local instruments without touching the source. Nera opened the maintenance drawer and took out the Office reference bulb, a plain glass oval containing a measured volume of brine. Every morning the schedule clerk placed it beside the observation slit and recorded the expansion line. Every evening a second clerk repeated the measurement. Redundant hands discouraged convenient weather.

The brine stood at the proper mark.

Edrin fetched the wall thermometer and held its copper foot against the frosted frame. The needle dropped from twenty-four degrees to eleven, then stopped. Two fingers away, the plaster remained warm.

"Localized," he said.

"Write the values."

"You don't need to order every sentence into the book."

"I need you not to remember them in a more comforting order."

He gave her a look that admitted the point while disliking its shape. The red book received the brine mark, the two temperatures, and the absence of a canal alarm.

Nera next isolated the ledger from the civic forecast feed. She released three mechanical catches and pulled the silver data tongue from its slot. A modern forecast glass without the feed should have gone blank except for its internal clock. The orchard, cloud, and canal columns vanished. The cold account remained.

Edrin circled the desk. "Internal contamination."

"Or a stored field."

"With a physical effect."

"Note both."

She laid a blank witness tile over the number. The tile copied whatever light reached its lower surface, a simple chemical reaction with no civic connection. When she lifted it, blue figures stained the enamel: negative eighteen million and change, complete to the final degree-hour. The frost had copied too, an opaque border that did not melt under her palm.

That evidence could survive a failed instrument. It could also survive an inquiry designed to erase one.

A knock sounded at the public door. The schedule clerk called, "Senior Voss? Column four has disappeared from the day board."

Nera glanced at Edrin. The local freeze had not yet been entered, but the public board had already lost a line.

"Instrument verification," she called back. "No action on the remaining schedule."

"How long?"

"Until the room gives you an admissible answer."

The clerk muttered something about orchards. Footsteps receded.

Edrin pointed to the empty place where the live columns had been. "The day board received a change before we issued one."

"Yes."

"Someone upstream may have isolated the line."

"Or downstream."

The last word entered the room and altered it. In Office usage, downstream meant the next machine, the next canal gate, the next ward receiving a transfer. It also meant time. Every seasonal schedule pushed consequence away from the hand that sealed it. Nera had used the word in a thousand findings without asking whether distance changed a claimant into an abstraction.

She reconnected the silver tongue. Three columns returned. The fourth

remained under its founder heading, blue-white and cold.

A public schedule note appeared beside it:

LINE WITHHELD PENDING OWNER CONFIRMATION.

No officer identification followed. The ledger had generated the note without her seal.

Edrin read it twice. "Owner of what?"

"The burden, perhaps."

"We don't assign ownership to weather."

"We assign liability."

"To institutions in the present."

Nera tapped the future credit attached to the negative figure. "This entry disagrees."

Outside, the tram bell passed west toward the orchards. She imagined redleaf buds under folded mesh, crews trusting the white columns to tell them when to open. A notary could hurt them by acting too quickly. A notary could also hurt someone unnamed by letting a false schedule proceed. Procedure offered no harmless position. It only required her to state where she put the risk.

"Record that the unsealed line has already altered the public board," she said. "Add possible remote custody action. Add that remaining transfers continue under ordinary authority."

Edrin wrote. "You are building the case before we know what the case concerns."

"I'm keeping it from becoming whatever the first steward says it concerned."

"That sounds almost political."

"It sounds sequential."

For once, he smiled. The expression vanished when the frost thickened around the source field.

A shallow line pulsed under the number.

CLASSIFICATION SOURCE AVAILABLE.

Edrin read it with her. "Don't."

"That's advice?"

"It's the procedure. An unscheduled source call can propagate corruption into the witness copy."

"Then make a hand witness."

His gaze sharpened. "Nera."

"You are deputy notary of this room. Witness the glass."

He took the red book from the emergency shelf, broke its wax thread, and wrote

the hour. His pen made no scratch. Red-book paper drank sound along with ink, a founder affectation retained because ceremony sometimes helped frightened officials keep their hands steady.

Nera touched the source line.

A file opened without the usual chain of custody. The first page carried a gray Office header from twenty-one years ago. The second carried a vent diagram she had seen once in a hospital corridor, while a clerk explained why the dead came home without their work coats.

Her mother's name appeared in the third line.

MARA VOSS, SECOND VENT ATTENDANT.

Nera read no farther for several breaths. Edrin moved half a step toward her, then stopped. Kindness from him always arrived as restraint.

She made herself continue.

Canal Six had ruptured during a spring sink change. Three attendants died inside the service collar. The inquiry found that the city schedule had diverted too much cooling capacity toward the Warm Ward's seed vaults, leaving the lower canal with a twelve-minute lag. Nera had been twenty. She had known the rupture, the lag, the seed vaults. She had built a career from the belief that the file contained the whole injury.

The final classification now stood in the center of the glass.

ACCEPTABLE CONTINUITY LOSS.

Not acceptable infrastructure loss, as the family copy said. Not compensable civic fatality, as the memorial register said. The hidden words carried a different grammar. A continuity loss belonged to the season system, not to a pipe. Someone had counted her mother against weather.

"Did you know this language remained in use?" she asked.

Edrin looked at the red book. "No."

"You hesitated before I opened it."

"I knew founder labels could damage admissibility. I did not know what it would retrieve."

"Read the footer."

He stepped close enough for his sleeve to brush the edge of the desk. Under the classification, smaller type identified the account credited by the loss:

WARM WARD WINTER RETENTION, SETTLEMENT YEAR 223.

Nera read it twice. Her mother had died in Year 202. No lawful schedule could credit a death forward by twenty-one years. The current winter retention had not yet been performed. It sat four seasons ahead, one of the white lines waiting under her seal.

Edrin whispered, "That's impossible."

"No." Nera heard the notary in her voice before she felt like one. "Impossible has no entry state. This has three."

She pointed: the casualty in Year 202, the credit in Year 223, the negative degree-hours gathering under a cold-account heading.

"Instrument drift," he said again, but without conviction.

"Write your objection."

"I haven't made one."

"Then write that you considered drift and did not assert it."

His eyes met hers. Twenty years of shared offices, promotion lists, contested commas, and funeral silences passed between them. He bent to the red book.

Nera moved the other schedules away from the glass. One false line could taint a day transfer. A tainted transfer could stop cloud combs over three districts and radiator flow through Low Glass. She had no authority to suspend the whole day without a second notary and an engineer's hazard declaration. She did have authority to freeze a single unauthenticated line for investigation.

The distinction felt insultingly small. It remained the distinction she possessed.

She opened the local action menu. CANCEL, RETURN, QUERY SOURCE, FREEZE LINE. Her finger hovered above the last choice.

"If you freeze it," Edrin said, "the schedule clerk will see the exception."

"They should."

"The clerk will send it upward. Steward Rusk will ask why a private workroom revived a pre-Charter term."

"He should."

"Before we know whether someone put your mother's file here to make you react."

That stopped her.

A ledger could not choose a wound, but a person could. Nera looked again at the source header. No custody chain. No retrieval stamp. A classification absent from the family copy. Her mother's name had arrived too exactly, at the first touch, from an account connected to Nera's ward and her present schedule.

"Add potential targeted contamination," she said.

Edrin wrote it.

"Add that deputy notary Sol advised against source access before retrieval."

His pen paused. "You're preserving my objection?"

"I'm preserving what occurred."

Something eased in his face, though not enough to call relief. "Then add that senior notary Voss ordered a hand witness before contact."

She did.

The cold spread beyond the observation slit. A white line touched the forecast glass, crossed the metal frame, and reached the ledger's lower corner. Binding paste stiffened in its dish. The brass seal cooled until condensation filmed its handle.

Down on Canal Nine, the heatworker with the sounding rod stopped walking. He looked up toward the Office windows.

Nera chose FREEZE LINE.

The ledger asked for scope.

She selected the negative degree-hours only. The current orchard retention remained live. Cloud-comb draw stayed white. Canal discharge held steady. Nothing beyond this room should change.

Her seal accepted the order, then rejected it.

TEMPORAL OWNER NOT PRESENT.

Edrin leaned so close his breath dimmed the glass. "Temporal owner."

"There is no such class in the Charter."

"Founder lexicon?"

"Not one I've seen."

The frost reached the first ledger page. Instead of wetting the paper, it sank between the fibers and drew a thin white border around the negative number.

Nera tried the freeze again under contested ownership. The Office allowed that status when water rights, crop shares, or emergency heat passed through an unresolved estate. It required a named claimant within ten working days. No claimant meant the line returned to the public schedule.

The seal hesitated.

CONTESTED OWNERSHIP REQUIRES COUNTERPARTY.

"Use the Office," Edrin said. "Seasonal Accounts as provisional counterparty."

"That makes us claimant and authenticator."

"For ten days."

"It makes us both for one second."

"The second may keep the record open."

He spoke from procedure, not panic. Nera had defeated him for senior appointment because he treated admissibility as a moral good. She had told the board that records served people, never the reverse. Now he offered the record a fiction so it could survive long enough to tell them whom it harmed.

Nera entered OFFICE OF SEASONAL ACCOUNTS, PROVISIONAL ONLY.

This time the seal warmed.

The cold-account line turned blue. A narrow bracket appeared around the number, the mark for a local freeze. On the street outside, no alarm sounded. The canal worker resumed walking after a glance at his wrist gauge.

Edrin closed the red book. "Ten days."

"Before automatic return."

"To a schedule that doesn't acknowledge it exists."

Nera pulled a clean sleeve from the drawer. Its paper cracked in the cold. "We go to Canal Nine before we notify Rusk."

"He'll call that concealment."

"He can call it source verification. If someone targeted my file, the trace begins at the instrument. If no one did, the trace still begins there."

"You're assuming this room remains isolated."

"I am refusing to assume anything."

Edrin glanced at the locked public door. Already, shadows moved behind the privacy glass. The schedule clerk had noticed the missing line or the frost on the outer pane. Offices perceived interruption faster than danger.

Nera copied the ledger state by hand. She wrote the cold-account figure, the source file, her mother's hidden classification, the Warm Ward credit, the temporal-owner refusal. Her pen tightened when she reached acceptable continuity loss. Twenty-one years ago, the phrase might have turned her away from this work. Now it accused the work she had chosen instead.

She remembered her mother's hands after canal shifts, reddened to the wrists despite insulated gloves. Mara Voss had kept a chipped heat receipt above the kitchen kettle and crossed out each service fee only after checking the public temperature log. When Nera complained that nobody else read the logs, her mother said nobody else would tell them whether they had received what they paid for.

At twenty, Nera had heard a lesson about thrift. At forty-one, the memory opened differently.

Edrin cleared his throat. "The clerk is about to use the override."

A brass key scraped beyond the door.

Nera sanded the red-book entry and handed him the witness copy. "Tell them we have an instrument hazard under seal."

"Do we?"

She touched the frost. It burned cold against her fingertip, clean and immediate.

"We have a record with physical effect. Hazard is the modest word."

He took the copy. Before he turned, the ledger made that paper-breath sound again.

A final field opened below the frozen line.

Not a claimant. Not an owner. A timestamp.

The year displayed first: 360.

Then the day, hour, and six-digit phase mark followed, all fixed one hundred thirty-seven years beyond the clock above their door.

Edrin did not reach for the lock.

Nera compared the figures to the workroom time. The minutes matched. The seconds advanced together, one clock in Year 223 and another impossibly resident in Year 360. Frost gathered around the future mark, but the numbers stayed clear.

The legal observation slit gave a small, sharp crack.

Neither pane broke. A hairline appeared between them, inside the sealed gap, where no tool or weather could reach. Cold breathed through it in measured pulses.

Six long. Six short. Six long again.

Sumi Quill had once shown Nera the emergency cadence used by phase crews who could not trust a voice channel. Attention, it meant. Not danger, not yet. Attention to the instrument. Attention to the other side.

Beyond the locked door, the clerk called Edrin's name.

Nera kept her eyes on the future clock. "Canal Nine," she said.

The cold pulse repeated.

Nera drew the future mark into the red book before wonder could make it imprecise. She recorded the six-long, six-short cadence, the matching seconds, and the absence of any registered transmitter in her own year. Beside Edrin's witness line she left a blank for the counterparty. An empty space carried no legal standing, yet refusing the space would decide the question before the unknown sender could answer it. The Office had taught her to distrust blank fields because later hands filled them carelessly. For the first time in her career, she guarded one because a later hand might already be reaching back.

This time, beneath the timestamp, the ledger left room for an answer.

Chapter 2 — A Hundred Thirty-Seven Years

Sumi Quill refused to open the phase room until Nera surrendered her brass seal.

“Magnetic load,” she said, holding out a three-fingered hand. Her left glove ended in two stitched flats where the last fingers should have continued. “You bring a notary seal near an exposed drum, it learns your office mark. Then every calibration clerk spends a month proving the machine has not developed civic ambition.”

Nera placed the seal in the wall safe. Edrin added his, though he had objected to visiting Canal Nine before reporting the Year 360 timestamp upward. The objection now occupied two pages in the red book under his own neat hand. It had not stopped him from carrying the witness copy inside his coat.

Sumi locked the safe and gave Nera a slate token in exchange. “If the building floods, this proves we owe you brass.”

“Does it flood often?” Edrin asked.

“Only when the canal forgets our confidence.”

She turned the wheel in the phase-room door. Warm air pushed outward, wet with mineral vapor and the bitter smell of scorched ceramic. Canal Nine passed beneath the floor grating. Its water struck baffles in repeating blows, each one felt through Nera’s shoes before sound reached her. Three black drums stood on stone cradles along the far wall. Each rotated slowly inside a cage of copper rails.

Nera had seen phase drums from public galleries. From there they looked seamless, almost simple. At arm’s length, the nearest drum carried a history of damage. Pale arcs crossed the black ceramic where calibration tools had scored it. Patches of fused glass marked old repairs. Numbers written in frost chalk climbed the cage posts, some recent and sharp, others preserved beneath clear lacquer.

Sumi watched Nera read them. “Your cold line began on drum three. Don’t touch the cage.”

“I know.”

“Notaries know prohibitions. Engineers know which prohibitions have funerals attached.”

Edrin unfolded the witness sleeve on a dry bench. “The ledger clock advanced in phase with ours for eighteen minutes. Then Senior Voss isolated the line from public observation.”

“Show me the copied mark.”

He slid the witness tile across. Blue numerals recorded the future timestamp and its six-digit phase field.

Sumi’s impatience left her face. She took a magnifier from her collar and compared the copied field with the chalk marks on drum three. Her thumb traveled

down the cage post without touching it.

“Who else saw this?”

“The schedule clerk saw an instrument-hazard notice,” Nera said. “No timestamp.”

“Steward’s office?”

“Not yet.”

“That answer shortens my life in several administrative directions.”

“It keeps your first observation from arriving after an instruction.”

Sumi looked at her for a moment, then nodded once. She pulled a hinged floor plate upward. Beneath it, a brass rail carried hundreds of narrow scars at regular intervals. Some had darkened with age. Others showed fresh metal where the survey stylus cut after each calibration.

“Phase material remembers position badly,” she said. “We make it remember honestly by wounding the rail.”

A small wheeled reader sat beside the opening. Sumi fitted it over the oldest visible scar, tightened two side clamps with her shortened hand, and turned a crank. Paper emerged from the reader in a thin curl. It printed drum orientation, lattice delay, source year, and predicted receiving horizon.

The first strip gave a delay of 137 years.

Sumi advanced to the next scar. Another strip emerged with the same delay. She repeated the process six times. The source years changed. Atmospheric load varied by thousands of degree-hours. Every receiving horizon remained fixed at 137 years beyond its source.

Edrin sorted the strips into order. “The public schedule says distributed horizon.”

“Public schedules say many things that keep schoolchildren from asking engineers for orbital mechanics,” Sumi said.

“Does the delay ever vary?” Nera asked.

“By eleven minutes during a drum shear. By less than a minute when the orbital pair crosses eclipse. Never by a year. Phase grain fixes the interval when we cast the ceramic.”

“Why that interval?”

Sumi crouched and pointed through the floor opening. A silvered rod ran beneath the scar rail into the stone cradle. “Year Three builders found one stable coupling. They could keep the colony from freezing or continue searching for a morally improved integer. People preferred surviving.”

Nera heard no contempt in it. Sumi spoke as she might describe a gate opened under pressure. The decision had occurred; material preserved its shape.

“Can the same drum receive from the far end?” Nera asked.

“No.”

Edrin glanced at her.

Sumi held up the magnifier. “That means I have never measured it, not that I have proved absence. We built for outward dissipation. Every sensor faces the transfer direction.”

“The timestamp entered against that direction.”

“Then either your ledger invented a phase field that matches this rail, or something loaded the coupling from Year 360.”

A canal baffle struck below them. The nearest drum turned through another handspan. White chalk lines crossed under a copper pickup and came away unchanged.

Nera took the oldest strip from Edrin. Its source year read 141, receiving horizon 278. “We need every anomaly tied to this drum.”

Sumi pointed toward a steel cabinet. “Maintenance only?”

“Maintenance, forecast drift, account exceptions, missing public columns, unexplained cold near legal instruments.”

“That search touches protected weather records.”

“I can authenticate a source inquiry.”

“Your seal waits in a box because the source inquiry can erase my calibrations.”

“Then you operate the search and preserve the method.”

Edrin closed his eyes briefly. “We have no case number.”

Nera wrote one on a blank hazard form. TEMPORAL CALIBRATION DISCREPANCY, CANAL NINE. She signed by hand and left the authority field open.

“That will not survive review,” he said.

“It only needs to survive until we learn what review concerns.”

Sumi read the heading. “You two sound different, but the paper suffers equally.”

She opened the cabinet.

Inside, brass spindles carried maintenance rolls dating back to the canal’s construction. Sumi chose drum-three records and threaded the first roll through a table reader. Nera expected mechanical complaints. The records offered them in abundance: chipped pickups, thermal drift, a shear that had taken Sumi’s fingers twelve years earlier, two floods, nine false alarms caused by mineral steam. Sumi marked each against the corresponding scar on the rail.

The cold anomalies hid among ordinary failures.

A frost trace on a disconnected forecast glass in Year 197. A ledger pane that cracked from inside its sealed gap in Year 204. Three minutes of negative dissipation during a cloud-comb shutdown in Year 216. Each report had received an adequate local explanation. Paste contamination. Trapped moisture. Reversed

canal sink.

Nera copied source dates onto separate tiles. Edrin read receiving horizons from the old technical rolls. Sumi checked physical scars.

Year 197 aligned with a disturbance registered in 334.

Year 204 pointed to 341.

Year 216 reached 353.

The present anomaly connected Year 223 to Year 360.

Nobody spoke for the next several entries. Canal blows filled the room. The figures kept their interval with a discipline no office policy could produce.

Sumi refused to let the interval stand on paper alone. She shut the cabinet and brought a wooden case from under the calibration bench. Inside lay twelve ceramic wedges, each cut from the same casting stock as drum three. Fine copper wire crossed their curved faces. Their labels named repair years rather than serial numbers.

"Witness pieces," she said. "We set one against the cage after every major cut. If the rail shifts later, these preserve the old relation."

Edrin inspected the case without touching it. "Independent of the maintenance rolls?"

"Independent until an engineer reads them. Nothing stays pure after a person forms an opinion."

She selected the Year 204 wedge and clamped it into a hand comparator. A narrow lamp projected its grain pattern onto white tile. Lines appeared like a river delta seen from too high above, branching around dark inclusions in the ceramic.

Sumi set the matching scar strip beside it. "Normal outgoing load leaves grain tails toward the east pickup. See how each inclusion draws a short line?"

Nera leaned close. The tails angled right. Near the wedge's outer edge, five bent left.

"What creates those?"

"A coupling pressed from the receiving side would do it. So would a hammer blow in precisely the wrong direction."

"Was there a hammer event?" Edrin asked.

"We inventory dropped tools because management prefers tools to unknown forces." Sumi opened a second roll. "No impact that month. No cage entry during the shift."

She checked the Year 197 wedge. Three inclusions carried western tails. Year 216 showed nine, clustered under the pickup that had registered negative dissipation. The present witness wedge, cut the previous morning before Nera opened the old source field, contained none.

Sumi took a fresh rubbing from drum three. Chalk touched paper through a

long-handled frame so no one approached the cage. When she drew the rubbing out, one western tail crossed the newest grain mark.

Edrin read the time from the wall clock and entered it in the red book. "Can drift reverse a grain tail?"

"Drift blurs. Pressure chooses a direction."

"Could someone have altered the witness pieces?"

"Someone could enter this room, replace ceramic cut from four different decades, reproduce our repair labels, falsify maintenance custody, and apply microscopic stress matching a phase event." Sumi returned the wedge to its padded slot. "That person would still need to know when your future clock appeared."

Nera looked at the drum. Its rotation seemed too slow to perform anything, yet each passage beneath the pickup moved weather no hand could lift. "Test the comparator with a known outgoing pulse."

Sumi's jaw tightened. "Any pulse enters the public schedule."

"A calibration load."

"Still public weather. Small does not mean private."

Nera checked the transfer board mounted near the door. Canal Nine had three approved calibration windows that morning. The next would release stored heat from an empty service tunnel into a deep sink, too little to alter a ward forecast but enough to mark direction.

"Use the scheduled ten-minute window," she said. "No additional load."

Sumi found the permit and held it up for Edrin. He compared its season seal with the copy logged in the red book. At the exact minute, she lowered a copper bridge across two contacts.

Drum three's tone changed below hearing. Nera felt it first in her back teeth. The comparator lamp trembled. On a blank witness wedge, one new grain tail extended east, clean and narrow.

"Expected direction," Sumi said.

She lifted the bridge when the permit ended. The room's public board confirmed that the service tunnel had discharged its authorized quantity. No unexplained deviation appeared.

Nera examined the fresh tail beside the western scars. Opposing forces had marked the same material through the same pickup. One belonged to a transfer their own year initiated. Whatever made the other marks had pressed through the fixed interval from its far side.

Edrin turned to an older page in the red book. "The Year 204 glass crack occurred during an approved calibration window."

Sumi checked the times. The outgoing pulse had begun eleven seconds after the

cold trace. "They overlap."

"Would our pulse conceal incoming pressure?" Nera asked.

"It could. Outgoing load keeps every sensor leaning the expected way." Sumi studied the westward marks. "We call deviations noise because the machine spends most of its life shouting in one direction."

She fetched a grease pencil and marked the four anomaly dates on the wall schedule. Each sat within or immediately before a substantial transfer. The future pressure had not appeared during quiet periods. It had pushed when the coupling carried enough load to make contact, then vanished inside the larger outward movement.

That finding changed what Nera could authorize. An idle drum test might prove nothing. A loaded test could impose weather beyond this room. Any attempt to hear more would also decide who absorbed the experimental burden.

"Add the calibration permit and witness grain," she told Edrin. "State that we introduced no unscheduled transfer."

He wrote the qualification before the finding. "And state that a reverse mark does not identify a sender."

"Yes."

Sumi glanced at Nera, perhaps surprised by the agreement. Nera had no interest in making evidence carry a person before it could support a direction.

She arranged the source tiles along the bench. "All of them sit exactly 137 years downstream."

Edrin pressed a fingertip to the Year 204 report. "Registered where? These later dates belong to forecast models."

"Models still have timestamps," Sumi said.

"They do not submit maintenance exceptions."

"These did."

"Under whose custody?"

Sumi enlarged the field. The old report showed an automatic origin code used by remote diagnostic instruments. No officer name appeared. The location field contained a belt coordinate, not a civic district.

Nera recognized the coordinate format from long-range demographic schedules. "Afterclime planning."

Edrin's expression tightened. "Forecast space."

"People occupy it in the planning models."

"Projected population occupies it. The Charter does not treat that as custody."

Sumi stopped the reader. "Ceramic does not ask the Charter before holding a phase."

Nera gathered the strips. "Take us to the forecast archive."

Canal Nine's demographic room occupied a dry chamber above the drum cages. Its walls held no public windows. Settlement maps appeared on horizontal glass under layers of crop demand, shelter capacity, migration probability, and seasonal mortality. Clerks used the room to test whether current transfers preserved future continuity in the abstract.

A faded label over the door read AFTERCLIME RESOURCE MODEL.

Nera had authenticated summaries from this model for years. She had never opened the household layer. Notaries verified sources and seals; demographers chose what their numbers represented.

Sumi entered the coordinate from the Year 360 timestamp. The map traveled east across the habitable belt, passed two current orchard districts, then settled on a future settlement marker in the wind terraces.

Edrin folded his arms. "A projected node."

The marker opened into streets.

They appeared as pale cuts beneath topographic shading, planned around heat corridors and buried reservoirs. A population field listed 18,420 residents. Beneath it ran school demand, meal distribution, recoverable body heat, and age bands in five-year increments.

Nera touched the youngest band. The model displayed 1,106 children under five. A second group attended lower instruction. Another required apprenticeship placements, reproductive care, elder support, and burial labor. The categories had no names, yet they required kitchens, roofs, clinic hours, and adults awake through storms.

"Why does a forecast node contain burial labor?" she asked.

"Continuity planning," Edrin said.

"That phrase does not answer the question."

Sumi opened the source list. "Demographic office uses ordinary settlement templates. Every future node gets inhabited before they run resource need."

"Inhabited," Nera repeated.

"Mathematically."

A shelter diagram showed six underground levels with temperature targets for each room. The lowest levels depended on exhaust heat from communal ovens. One failure branch moved children into storerooms for eleven winter weeks. Another converted a school corridor to an infirmary after cold lung cases exceeded available beds.

Nera thought of the public schedules she had sealed. Those schedules described the Afterclime as a buffer interval beyond actionable residence. Yet the same civic

system stocked its shelves, assigned its teachers, predicted its pregnancies, and counted how many bodies could sleep beside each heat wall.

Edrin tapped the model provenance. "These parameters derive from our present census. No individual exists here."

"No individual exists in next month's orchard forecast," Nera said. "We still refuse to schedule workers through lethal frost."

"Workers can object."

"The line tried."

Sumi enlarged the cold-load overlay.

A blue field descended over the settlement marker. It represented burdens sent from earlier source years, layered by date and intensity. The mild winter from Year 86 appeared as a narrow band. Later deferrals thickened it. By Year 360, the lines joined into a dark column that extended beyond the model's ordinary color range.

The household layer recalculated.

School heat closed first. Clinic reserve followed. Food stores rose in importance as orchard output fell. The model moved families downward through the shelter levels, compressing rooms until recoverable body heat met minimum survival targets.

No one had needed to invent a future city for the frost to harm. The Office already maintained one in greater domestic detail than it maintained Low Glass.

Nera opened the public assumptions attached to the same run. The future belt carried a status code: UNINHABITED DURING DISSIPATION RECEIPT.

She placed the two views side by side. On the left, no inhabitants. On the right, eighteen thousand people queued for ovens and burial shifts.

"Authenticate the contradiction," she said.

Edrin did not reach for his seal. "We left both seals below."

"Write a hand witness."

"For a model inconsistency?"

"For simultaneous official claims."

"The population has no legal identity."

"Then state that. State exactly what the Office uses it to calculate while denying it for receipt."

He looked from the empty-status code to the shelter rooms. His procedural resistance did not vanish. It altered direction.

"Show me the model version," he told Sumi.

She opened the archive header. It had been updated seventeen days earlier under an ordinary food-security cycle. No hidden access flag appeared. Any authorized demographer could have seen the households. Any seasonal notary

could have asked.

Edrin took out the red book. "Version, checksum, coordinate, population parameter, household layer."

Nera dictated them. He wrote without softening her nouns.

When they finished, Sumi returned to the source map and removed the cold-load overlay. Color came back into the future streets. A clinic reopened. School rooms resumed their listed use. The transformation looked like rescue, but nothing physical had changed. Only one set of assumptions had been hidden.

A small instrument beside the map clicked.

Sumi turned toward it.

The device resembled a compass mounted flat in a ceramic dish. Its needle usually leaned east, following outgoing phase load toward the fixed horizon. Frost chalk around its rim marked acceptable drift. The point now stood at center.

"What does that measure?" Nera asked.

"Backscatter." Sumi crossed the room in three quick steps. "Noise from imperfect coupling. It should favor the transfer direction."

The needle moved west.

Sumi opened a lower panel and disconnected the forecast glass. The map went dark. The needle continued across center, slow enough to count each etched division.

Edrin came beside Nera. "Could the demographic run have caused it?"

"No active load reaches this instrument from the model," Sumi said. "It reads drum three through a mechanical pickup."

She set her shortened hand against the ceramic dish to feel vibration. Her face changed before the needle stopped. Not fear. Recognition without a category.

From below came the regular impact of canal water against baffles. Between two blows, a finer pulse traveled up the table leg.

Six long. Six short. Six long.

The Year 360 cadence from Nera's ledger.

Sumi looked at the open red book, then at the dark settlement map. "It has found the diagnostic edge."

"Can you hold it?" Nera asked.

"Not without feeding immediate load into the drum."

"How much?"

"For a reading, perhaps one local front. For information, I don't know."

The needle pressed against its western stop. It had never been designed to point there. The fine brass tip bent, resisted, then climbed one fraction higher against the current.

Chapter 3 — The Return Channel

By Day 4, Sumi had built a receiver from instruments designed to deny reception.

She mounted the backscatter needle above drum three, replaced its bent tip, and ran a copper hair from the western stop into a retired forecast printer. The printer had once issued cloud-comb warnings in block capitals. It now sat on the phase-room bench with its casing open and six kinds of insulation packed around its feed gears.

Nothing in the arrangement resembled permission.

Nera stood at the transfer board reading the test order for a third time. Edrin held the red book against his chest. Two Canal Nine technicians waited by the manual gates, their gloves marked with white ceramic dust. Sumi had sent everyone else above the flood door and locked the public gallery.

“Read the affected area,” she told Nera.

Nera did not look up. “Service blocks nine through fourteen. Canal footway. East maintenance yard. No residences.”

“Temperature change.”

“Six degrees over forty minutes, with a possible eight-degree pocket at the yard wall.”

“Recovery.”

“Stored canal heat restores baseline within twenty-two minutes after closure.”

Sumi checked the wall gauges. “Forecast office gives us nineteen.”

“The order records twenty-two.”

“That difference concerns me less than a legal document claiming water moves faster than it does.”

Nera crossed out nineteen on the attached forecast slip, wrote twenty-two, and initialed the change. She had argued for a notarial test instead of an engineering improvisation. Sumi had argued that weather would not distinguish their professions while breaking equipment. Both had prevailed enough to remain dissatisfied.

Outside the phase room, late spring pressed warm air against Canal Nine. Workers had cleared the footway and clipped frost flags along the rail. A cargo tram waited beyond the eastern bridge because its brake lines would stiffen under the test front. None of the displaced tasks counted as laboratory cost. Sumi had added them by hand.

Edrin opened the red book. “Authority derives from an instrument-hazard inquiry and scheduled calibration allowance.”

“No extra future transfer,” Nera said. “The lattice accepts the immediate cold

here.”

“That phrasing implies we know where ordinary transfer goes.”

“We know the fixed interval.”

“We do not know who applied reverse pressure.”

Sumi tightened the printer feed with her three-fingered hand. “If you two need another hour to separate knowledge from grammar, the cold front will miss its cleared window.”

Nera faced the board. “Proceed under the written scope.”

Sumi signaled the gate technicians.

They turned their wheels together. Deep below, canal baffles shifted. Hot water diverted from the service loop into the main radiator channel instead of cushioning the incoming air. The room floor lost its familiar warmth. A gauge on the west wall dropped one division.

Drum three changed pitch.

Sumi felt the alteration along her jaw. Outgoing phase load usually pulled against the cages with a pressure like distant acceleration. This time, the copper rails eased. Cold stayed in their year. For forty minutes, six service blocks would take weather the schedule had expected to send away.

The backscatter needle moved west.

Its copper hair lifted. The forecast printer struck one letter against paper.

A.

The feed advanced twice, then printed a broken square.

Sumi held up her hand before anyone approached. “No touching the rig. The first movement belongs to the instrument state.”

Edrin wrote the time. Nera watched frost spread along the test-room observation slit. Unlike the ledger frost, this gathered from local air. Moisture clouded the inner pane, froze around the frame, and thickened as the service blocks cooled.

Another letter struck.

R.

Then three marks appeared too damaged to read. Sumi adjusted no control. Every adjustment could turn content into their own expectation.

The west needle climbed to the second division. The printer produced five more characters.

E A D ?

“Could be READY,” Edrin said.

“Could be BREAD,” Sumi replied. “Do not teach the noise a sentence.”

At the manual gates, one technician called out the east-yard temperature. Five

degrees down. The footway sensor reported four. Differences crossed the network because walls, water, and stone held yesterday's heat unevenly.

The printer struck again. Its roll showed a fourth damaged attempt:

> FIRST TRACE: ?RE?D

Nera moved closer without touching the bench. "It may be repeating."

"Repetition gives us error correction if we resist helping it." Sumi marked the raw sequence on a slate, including every blank and malformed symbol. "We need a second line."

"The local front has used twelve minutes," Edrin said.

"We authorized forty."

"A valid order does not make each minute necessary."

Sumi glanced at him. "That sounded almost like engineering."

The printer gears shivered. Letters arrived in groups separated by long pauses. Twice the paper tore at a feed hole. Sumi let the loose edge hang rather than risk contact. By minute seventeen, three overlapping sequences occupied the roll.

> SECOND TRACE: ARE?D? / THIRD TRACE: ?READ?

Nera read them in silence. "A question."

"Likely."

"Can it receive our response?"

"The ordinary load already travels outward. If we alter it in timed increments, the far-side pressure may distinguish the pattern."

Edrin shifted the red book under his arm. "That sends no language."

"Language begins as agreed differences." Sumi pointed to the six-long, six-short cadence copied from Nera's ledger. "Someone already found a difference we recognize."

The gate technician called the yard at seven degrees below baseline. Frost flags snapped in air that had started moving toward the canal.

Sumi had modeled three reply methods. The safest used binary load pulses, one small gate movement for a short mark and a larger movement for a long one. It could injure the baffle seals through repeated cycling. A slower method varied pause length while holding load constant. The far end might mistake those pauses for natural noise. The third routed a forecast-printer pattern through a phase pickup never intended to carry output. That risked feeding current into the witness circuit.

"We use gate timing," she said. "One phrase. READ."

Nera studied the test order. "Material increment?"

"No added transfer. We shape the retained front inside the authorized window."

"Service area variation?"

"Less than one degree if the technicians hold the main gate."

Edrin asked, "Can the pattern identify this facility?"

"Only if the receiver knows the phase scar. We send Canal Nine's machine cadence before the word."

Nera looked through the slit at the cleared footway. A frost flag had stiffened horizontal. "Send the facility cadence, then READ. Record every gate movement."

Sumi relayed the sequence. The technicians worked without seeing the letters their hands formed. Short movement, return to hold, long movement, hold again. Canal water struck different baffles as the gate changed. Each impact entered the room through stone.

For eight minutes after the final pulse, the printer produced nothing.

The local front reached its coldest point. Ice filmed the manual wheels. One technician flexed her fingers inside her gloves while the other checked the tram brake sensor beyond the cleared zone. Sumi watched every needle except the one that mattered, forcing herself to preserve the entire system rather than stare a reply into existence.

At minute twenty-nine, the copper hair rose.

The printer struck seven clean characters.

> READING

No one spoke. The word used ink intended for routine warnings, dark and slightly oily. It occupied less than the width of Nera's palm.

Sumi checked the western pickup, cage voltage, drum speed, and gate position. A false printer command might produce letters, but it could not press phase grain from the far side at the same moment. The newest witness strip showed three western tails.

Edrin wrote the word into the red book. "I need custody of the paper."

"You need to wait until the test ends," Sumi said.

"A loose roll has no admissibility."

"A hand near that feed becomes an electrical path."

Nera stepped between them without drama. "The roll remains in place. Edrin witnesses visually. Sumi preserves instrument state. I will authenticate after isolation."

"Authenticate what?" Edrin asked. "A machine output with unknown origin?"

"The time, material state, local load, drum response, and characters received."

"That does not authenticate a speaker."

"No. It prevents us from pretending uncertainty about one fact erases the others."

Sumi looked back at the service gauges. Thirty-two minutes. "If you want a

second phrase, decide now.”

Nera drew a blank return form from her case. “Can we transmit ordinary characters?”

“Not with today’s rig. We can send coded text if we reduce it first.”

“How much?”

Sumi had calculated the capacity from gate wear, pulse length, and remaining cold. “At this load, 180 characters per six hours. We have eight minutes, so perhaps four reliable characters after the facility cadence.”

Edrin gave a disbelieving laugh without humor. “You measured a six-hour rate from one word.”

“I measured the safe gate-cycle budget and the phase response. If we continue beyond it, someone can explain to the canal crews why a conversation ate their seals.”

Nera wrote four letters on the form.

> NAME

Sumi shook her head. “Too ambiguous. It could ask theirs or offer ours.”

> WHO?

“Cleaner,” Edrin said.

Nera crossed it out. “We do not know whether the other side means a person, instrument, office, or settlement.”

“Four characters do not support a hearing.”

“They support a question.”

Sumi watched the temperature clock. “Choose.”

Nera wrote:

> YOU?

Edrin objected at once. “Second person presumes a correspondent.”

“The printer already conjugated reading.”

“A machine can report its own action.”

Sumi placed the gate log in front of him. “Then ask the machine to object later.”

Nera initialed YOU? as interrogatory text, not authenticated identity. Edrin added his witness mark beside the limitation. Sumi gave the technicians the pulse sequence.

The last gate movement ended at minute thirty-nine.

“Close retained load,” she ordered.

Both wheels turned. Canal heat entered the service loop again. Drum three resumed its ordinary pull toward the fixed horizon. The backscatter needle fell but did not return east. It hovered just west of center.

The printer began to strike.

One character came through before the local front ended. Then another. The first line looked like a location code, but three symbols had no current alphabetic equivalent. A second line started while service-block temperature climbed.

> N? H?RE

"NO HERE?" Edrin said.

"Raw marks first," Nera told him.

Sumi copied exactly what the printer gave. The signal weakened as immediate canal heat restored the cleared blocks. A western grain tail appeared with each legible cluster.

> L?V?NG

"LIVING," Nera said.

This time Sumi did not caution her. The letter intervals supported it.

The printer stopped. The backscatter needle settled against center. Outside, workers lifted the first frost flags and found them frozen to their clips.

Sumi isolated the receiver in the order she had written before the test. Gate circuit. Copper hair. Printer feed. Mechanical pickup. Only after every conductor lay disconnected did she cut the paper roll with ceramic shears and pass it to Edrin.

He placed the roll between witness tiles. Nera pressed her recovered brass seal into the sleeve, authenticating the physical record without naming its origin. Sumi added the grain rubbings, temperature logs, gate sequence, affected work orders, and recovery time.

The service blocks warmed in twenty-three minutes, one beyond the order. Sumi recorded the overrun, then took Nera and Edrin outside before anyone could reduce the local front to a successful line on a form.

Cold had changed the cleared footway. Canal steam rolled across its rail in dense sheets and froze on the shaded uprights. The east maintenance yard held a shallow basin of white where meltwater had stood before the test. Workers moved through it in stiff steps, lifting tools that had not required winter handling when the morning began.

The tram mechanic knelt beside the waiting cargo car with a heat braid wrapped around its rear coupling. She had dismantled the brake housing after its seals contracted. Four crates of clinic glass remained strapped to the car, their delivery clock still running.

"Twenty-two minutes on the permit," she said when Sumi approached. "Thirty-one before I trusted the brake."

Sumi crouched beside her. "Air or metal?"

"Metal. The air came back close enough. This pin kept the cold."

The mechanic passed her a gauge strip. Its chemical bars showed a nine-degree

drop inside the housing, sharper than the yard sensor had recorded. Stone and standing air had moderated the official reading. Exposed brass had not.

Nera examined the clinic labels. "Any breakage?"

"Not yet. If the car misses the noon transfer, sterilization has to repeat at destination."

Edrin opened the red book against the canal wind. "Delay attributable to test front."

"Write seven workers held here," the mechanic said. "Three on the car, four waiting to unload. People cost hours even when the glass survives."

He wrote it.

At the yard wall, a split bucket had emptied sealant water over a drainage grate. The water had frozen before reaching the canal. A worker chipped it away with a wooden mallet so runoff would not flood the electrical trench when warmth came back. Beside him, early moss had blackened in a strip below the vent outlet.

"We can replace that," Nera said.

The worker looked up. "The moss?"

"The lost growth."

"Facilities will scrape it and bill phase operations for cleaning." He struck the ice again. "Growth does not have an expense code until it blocks something."

Sumi wrote the affected length herself. Eleven meters. The strip had no civic importance. It still marked a living surface that would not resume because three officials needed to know whether a word could travel backward.

One of her gate technicians waited at service block twelve, both hands around a cup. Talla Venn had worked the east wheel during the pulse sequence. Meltwater darkened her cuffs.

"Glove seam?" Sumi asked.

"Held. Wheel rim iced at the seventh pattern."

"Why didn't you call it?"

"You needed the last letter."

Sumi took the cup from her and checked the hands. Skin color returned evenly when Talla opened and closed her fingers. No numb patches. No stiffness beyond cold.

"I needed you to follow stop limits."

"The wheel stayed inside limit."

"Your hands belong to the limit."

Talla glanced past her toward Nera. "Did it answer?"

Sumi could not share unauthenticated text with a technician simply because the technician had supplied part of its cost. That rule felt less defensible on the

footway than it had beside the printer.

"Something read the pattern," she said. "That statement remains under witness review."

"Then put my name on the test record. Not just gate technician two."

Edrin heard. He turned a page and asked her to spell Talla Venn, though her badge sat in plain view. The deliberate question gave her control over the entry. Sumi recognized the courtesy because Nera had used it with machinery all morning: do not let the observer improve the source.

They walked the six blocks with a hand thermometer. Most walls recovered within prediction. A tool locker at block fourteen retained cold because crews had packed wet rope against its back panel. Two latch springs fractured when opened. A condensation trap overflowed after thaw, spreading mineral water across the lower steps. None of these damages threatened life or public heat. Together they turned four received letters into a maintenance shift, a delayed clinic delivery, and new entries on seven workers' time sheets.

At the final sensor, Sumi compared canal recovery with the model she had signed. Twenty-three minutes in open air. Thirty-four in exposed mechanisms. Unknown in roots and wall growth.

Nera read the figures over her shoulder. "The order needs a material-cost annex."

"Every return will need one," Sumi said. "If more immediate load improves the signal, characters have weather prices."

Edrin looked back toward the locked phase-room door. "We received more than four."

"Because pressure continued after closure. Either the far side stored enough phase strain to finish, or our six-hour estimate lacks half its variables."

"Can you still state 180 characters per six hours?"

"As an initial safe capacity, under this drum state and this amount of retained cold. Not as a promise."

Nera added the qualification to the test summary before applying her initials. The number remained useful only if it carried the conditions that gave it meaning.

They carried the evidence upstairs to the demographic room. No one suggested reopening the household model. The dark map already had a population in Nera's notebook and no recognized residents in the public code.

Edrin spread the return roll beneath a magnifying lamp. "We can reconstruct probable text, but probability cannot enter as quotation."

Nera aligned the repeated marks. "Separate transcription from reading. Preserve the damaged strings. Then state the highest-confidence interpretation."

"And if the interpretation changes?"

"Version it."

Sumi took the chair opposite them. Her left glove lay on the table, exposing a palm crossed by old white scars. She had lost two fingers because a phase drum kept moving after a safety gauge reported rest. A proper record of the gauge had mattered during the inquiry. It had not restored her hand.

"What will your Office do with an authenticated word?" she asked.

"Try to control its custody," Edrin said.

Nera did not deny it. "Which is why custody begins here."

The printer clicked in the phase room below.

All three heard it through the floor.

Sumi stood so quickly her chair struck the wall. The receiver had no electrical feed. She had disconnected the mechanical pickup. Yet another impact followed, then a rapid series.

They descended together.

The paper stub still threaded through the printer had advanced by half a page. Frost covered the severed copper hair. The backscatter needle pressed west without touching any connected circuit.

Letters filled the new paper in uneven capitals. Some printed twice. Others faded near invisibility. Sumi read the raw line from left to right and felt each western pressure through the bench.

> STO? CALLING ?S EM?TY

Nera gripped the rail outside the cage boundary. "Again."

A second line arrived with fewer breaks.

> STOP CALLING US EMPTY

The words exhausted the signal. Frost loosened from the copper hair and fell as dry grains onto the bench.

Edrin stared at the printer. "That responds to the model classification."

"Or to language sent long before today," Nera said.

The distinction mattered. Their test had not created an instant exchange across 137 years. It had opened a material path into one history already carrying their choices. Someone at the far end might have worked from records of this day, or from inherited knowledge of how the Office described them. Cause still traveled forward, even when a reply arrived against the load.

A final cluster struck below the demand.

The letters came slowly enough for the printer hammer to cool between them.

> NAME PERSON READING

Nera reached for the return form, then stopped before writing. A name would cross the line where instrument authentication ended and contact began.

Sumi watched the needle hold west, waiting on a person rather than a machine.

Chapter 4 — The Founder Sum

Edrin would not let Nera authenticate the return text until she proved the printer had no alphabet of its own.

On Day 6, he laid its retired service manuals across the verification desk and opened each to the command table. The printer could issue fifty-four warnings from mechanical input: cloud shear, canal breach, comb stall, frost hazard, evacuation. None included EMPTY. None could generate a request for the reader's name.

"That supports novelty," Nera said. "Not origin."

"It also supports admissibility."

"You spent yesterday arguing the opposite."

"I spent yesterday objecting to your certainty." Edrin fitted a custody strip across the return roll's sleeve. "Today I object to anyone else using uncertainty as disposal authority."

The legal observation slit still carried a white trace in its sealed gap. Canal Nine flashed beyond it under mild spring light. At the bottom of the glass, the cold-account column remained frozen under Nera's contested-ownership order. Four days had passed. Six remained before automatic release.

Sumi stood at the end of the desk with the printer hammer in a padded tray. She had removed it under witness and brought the entire striking assembly from the phase room. "Metal has no private vocabulary. Its marks came from pulse spacing."

Edrin drew a line beneath his conclusion. "Then we can authenticate received characters and reverse phase pressure. We cannot authenticate the sender, their legal capacity, or the meaning of a collective pronoun."

"US," Nera said.

"I can read capitals."

"You called it a collective pronoun."

"Grammar does not confer standing."

"No, but it records that the source claims plurality."

He hesitated, then wrote that qualification too.

A key turned in the outer door.

Nera's workroom had been locked under instrument-hazard rules. Only three people held an override: Nera, the chief records custodian, and First Steward Halden Rusk. Her own key lay on the desk. The custodian had sent a written refusal to enter.

Halden opened the door without an escort.

He wore an Assembly coat cut from gray vent cloth, warm enough for civic tunnels and plain enough to suggest he had not dressed for authority. The suggestion failed. Halden carried rooms with him. Clerks lowered their voices before he asked. Committees discovered shared positions after he named the danger of disagreement.

His gaze found the frost, the open manuals, and the printer hammer. It stopped on Edrin.

"You reported a contamination event," he said.

Edrin kept both hands on the red book. "I reported an uncontained evidentiary condition."

"When?" Nera asked.

"After the second return."

"You told me custody began here."

"It did. Custody also requires notice to the officer responsible for systemic risk."

Halden closed the door behind him. "We can discuss Deputy Sol's phrasing after we know whether the city has received an instrument warning or a political message."

Sumi's mouth tightened. "Weather does not become political by forming letters."

"We have rarely permitted weather to form them."

Nera indicated the witness chair. "If you intend to observe, sign the room entry."

Halden looked at the chair, perhaps measuring the distance between office courtesy and compulsory process. He signed.

Edrin gave him the evidence in sequence: frozen ledger line, future timestamp, fixed phase interval, demographic contradiction, controlled retained-load test, raw printer traces, material-cost annex. Halden interrupted only to ask for source fields or times. He did not challenge the 137-year delay. He asked twice whether the future settlement marker existed in any current legal registry.

"It exists in the Office's food and shelter model," Nera said.

"That was not my question."

"The distinction created the problem."

"The distinction may preserve lawful action while we investigate."

"By calling a settlement occupied for heat planning and empty for burden receipt?"

Halden turned to Edrin. "Has that contradiction been authenticated?"

"Under hand witness. The model remains live and can be reproduced."

"Then reproduce it under steward seal before anyone changes the version."

Nera had expected suppression. Preservation unsettled her more. Halden

understood which evidence would survive public conflict, and he had begun preparing it before he admitted conflict existed.

Sumi placed the printer hammer back in its tray. "The far-side pressure continues at low level. We have not opened another local front."

"Do not," Halden said.

"That requires your authority in writing. My scheduled calibration allowance resumes tomorrow."

"You will receive a written hold on experimental shaping, not ordinary maintenance."

"Specify whether passive receipt counts as experiment."

Halden looked at her scarred hand. "Passive receipt may continue if it imposes no new load and preserves raw output."

Sumi nodded. He had given her a boundary she could operate, which made refusal less simple than an order to close everything.

Nera opened the cold-account source record. Her mother's hidden classification filled the center of the forecast glass. ACCEPTABLE CONTINUITY LOSS. Beneath it remained the impossible Year 223 Warm Ward credit.

"This language entered from a founder lexicon," she said. "The source chain points to an inactive liability schedule. I want the complete schedule."

Halden's expression altered by less than Edrin's earlier smile. "Pre-Charter account language often lacks present legal meaning."

"Often?"

"Early survival records used provisional categories."

"Do you know this one?"

"I know that continuity losses included equipment, harvest capacity, migration windows, and deaths. The category did not distinguish them with modern care."

"Do you know where its liability schedule is held?"

"We should first preserve the current evidence."

Nera heard the institutional plural arrive. Halden used we when authority needed to appear shared and risk needed to lose an owner.

Edrin turned pages in the printer manual until he reached the sheet beneath it, a copied index they had found among the cold-account materials. "The inactive schedule has an annex code."

Halden's gaze moved to the page.

Edrin read aloud. "F-SUM. Custody class: sealed foundation. Access: two seasonal notaries, one phase engineer, steward notice."

"Not steward authorization?" Nera asked.

"Notice only."

Sumi leaned over the code. "Phase engineer means active lattice certification. They expected the annex to depend on material operation."

Halden held out his hand for the index. Edrin passed it over.

"You knew an annex existed," Nera said.

"I knew sealed foundation materials existed. There are eleven. Water allotment, seed custody, emergency burial, orbital salvage."

"And F-SUM?"

"I have seen references."

"Have you read it?"

"No."

The answer came without delay. Halden rarely lied outright. He chose categories that made truth perform less work.

"Then receive notice," Nera said. "We are opening it."

Halden did not move toward the door. "First we preserve the contradiction you intend to carry into that vault. If the model changes while we are below, the annex will appear to have supplied your interpretation."

He had found the procedural weakness before Edrin voiced it. Nera disliked the delay and accepted its necessity.

They returned to Canal Nine's demographic room under steward custody. Sumi brought the drum-three scar records in their wooden case. Edrin carried the sealed printer roll. Halden asked the records custodian to attend by public speaking tube, so no private aide gained access to the evidence.

The demographic model had changed since Day 2.

Its future settlement marker still occupied the wind terraces, but the household layer no longer opened from the ordinary food-security menu. A gray band covered the command.

MODEL DEPTH SUBJECT TO CAPACITY REVIEW.

Sumi read the update field. "Applied this morning. Demographic systems office."

"Routine?" Halden asked.

"No capacity review has ever hidden a local layer from phase operations."

Edrin checked the checksum Nera had copied two days earlier. The current value differed in four positions. "Our hand witness proves a prior version existed, not what it contained."

Nera looked at Halden. "Your notice traveled."

"No instruction left my office to alter this model."

"That sentence allows several kinds of instruction to have entered it."

"It also states what I know." He approached the blocked control. "Can the

previous version be restored without overwriting current state?"

Sumi opened the maintenance panel beneath the map table. Physical update spindles rotated behind a glass cover, each holding a narrow strip of punched alloy. "Demographic systems keeps three prior runs in local cache. They expected bad forecasts, not evidence disputes."

She issued two blank witness spindles to Edrin. He scratched the date and room code into their soft rims. Nera added her hand mark. Halden pressed his steward seal into a separate authorization tile granting read-only recovery.

Sumi cut power to the public display before opening the cache. The room went dark except for the archive lamp and the pale line under the door. She removed the newest spindle, set it in a padded cradle, then drew out the prior run with ceramic tongs.

"Version date?" Edrin asked.

"Seventeen days before the cold account appeared."

"Checksum?"

Sumi read each character while he compared it with Nera's notebook. All positions matched.

The old run went into a hand reader with no civic feed. Streets returned one punched coordinate at a time. Shelter levels followed. The population of 18,420 appeared after the third pass. Then came the household allocation: school heat, clinic reserve, communal ovens, burial labor.

Halden stood close enough to see every category. "Open the receipt status."

Sumi advanced the spindle.

UNINHABITED DURING DISSIPATION RECEIPT.

The two claims occupied separate paper strips now, not a changeable glass. Edrin placed them side by side beneath the red-book time field.

"Senior notary," he said, "state the conflict without inference."

Nera looked at the settlement strip. "The same model assigns resident needs to the coordinate and denies inhabitants when calculating transferred burden."

"Engineer?"

Sumi said, "Both outputs derive from one archived run. Neither requires a simulated person or manual addition."

"Steward?"

Halden took longer. "The Seasonal Office maintained incompatible operating assumptions within one official model. One assumption supported resource planning. The other supported dissipation accounting."

Edrin wrote all three statements. He then asked the custodian, still listening through the speaking tube, to repeat the archive time and room identity. Her remote witness could not prove what the strips displayed, but it fixed who had

occupied the room and when.

Nera examined the gray capacity-review band left on the current model. "Preserve that update too."

Halden's gaze shifted to her. "It may reflect an innocent system change."

"Then preservation will protect its innocence."

Something almost like approval crossed Edrin's face. He mirrored the update notice, its authoring office, and the altered checksum.

Sumi restored the current spindle without reopening its blocked household layer. The public system resumed with no sign that an earlier version had contradicted it. On the evidence table, the punched strips remained stubbornly physical.

Halden asked for the update origin. The display gave an internal service route through the Assembly capacity office, not an individual officer.

"We will investigate that route after the foundation access," he said.

"Why after?" Nera asked.

"Because whoever moved first already knows enough to remove a menu. If we announce which annex we seek, they may learn what we have not yet read."

The argument used concealment to protect discovery rather than authority. Nera could not reject it without pretending motive had no effect on procedure.

Edrin sealed the recovered model outputs in a blue sleeve. "The evidence now enters the vault with an authenticated contradiction and an attempted access change. If the annex supplies claimant language, no one can later claim the language caused us to imagine inhabitants."

"Agreed," Nera said.

Halden took his steward tile from the reader. "Now I have received notice."

The foundation vault occupied the oldest stone below the Office, behind a door that had no forecast glass. Canal heat reached it through bare pipes. Salt dried white around their joints. The records custodian met them at the stair with three keys and an expression of professional grief.

"Notary Voss and Notary Sol satisfy paired access," she said. "Engineer Quill's current drum certificate satisfies material standing. Steward Rusk has received notice. If any one party withdraws before the inner seal parts, access fails."

Halden stood outside the marked floor square. He could observe but not substitute.

The custodian issued linen oversleeves and mouth cloths. Nera tied hers, then washed her hands in distilled water. Edrin checked the wash seal. Sumi surrendered every magnetic tool and accepted a brass probe from the vault kit.

Three locks waited at different heights on the inner cabinet.

"The order matters," the custodian said. "Engineering establishes material relevance. Junior notary establishes intact custody. Senior notary accepts legal exposure."

Edrin looked at Nera. "The cabinet has opinions about rank."

"It predates our appointments."

"That has not prevented other records from holding grudges."

Sumi inserted the probe into the lowest lock. Instead of turning it, she pressed the flat tip against a ceramic contact. A tone moved through the cabinet. Dust lifted from one hinge.

"Phase grain remains active," she said. "Material relevance established."

Edrin unlocked the middle plate. A strip of founder wax appeared behind it, dark green and unbroken. He read the stamped custody mark, compared it with the vault register, and cut the wax with a glass knife.

"Intact custody established."

The upper lock accepted Nera's brass seal rather than a key. The metal warmed under her thumb.

A warning unfolded on a narrow paper tongue.

OPENING PARTY ACCEPTS DUTY TO IDENTIFY CLAIMANTS DESCRIBED HEREIN.

Edrin read over her shoulder. "That language creates an obligation merely by access."

"Or attempts to."

"Present law may not recognize it."

"The record still conditions opening."

Halden spoke from beyond the square. "You can stop. The current evidence remains preserved. We can seek Assembly counsel before adding an unknown duty."

Nera looked at the frost scar across her fingertip from the first ledger contact. The skin had reddened around it but not blistered. Her mother's file waited upstairs under a category designed to attach death to seasonal function. A message from Year 360 occupied a sealed sleeve and asked them to stop calling its senders empty.

"Counsel cannot advise on language counsel has not read," she said.

She pressed her seal fully home.

The cabinet released a breath of dry air. Its door opened three finger widths and stopped.

Sumi checked the probe tone again. Edrin examined the inner edge for broken custody threads. Nera drew the door wider only after both nodded.

Inside lay one ledger.

It had no title on the cover. A pattern of numbers covered the brown hide in five concentric rings. Each ring ended at the same figure: 137. A narrow bone clasp held the volume shut.

The custodian read the shelf card. "Colloquial designation: Founder Sum. Formal designation: Aggregate Deferred Dissipation and Downstream Claim Test."

Halden exhaled once.

Nera heard it. "You knew the designation."

"I knew the colloquial name."

"That answer has reached its limit."

He did not reply.

Edrin lifted the volume onto the reading cradle. Sumi placed her probe near the spine. The tone sharpened, responding to phase material embedded beneath the leather.

"Active coupling," she said. "The book was built to read lattice state."

Nera opened the bone clasp.

The first pages contained quantities from the colony's earliest winters: heat moved, storms delayed, crop days gained, shelter deaths avoided. Beside every present benefit, a second column calculated burden at the fixed receiving year. The writing did not call that year forecast space. It called it downstream settlement time.

Edrin turned one page with ivory tongs. "Here."

A clause crossed the full width in founder script, then repeated in later standardized type.

> All downstream persons materially burdened by deferred dissipation retain claim against the originating account, whether named, projected, unborn, unregistered, or beyond present civic reach.

Nera read it twice. The sentence did not grant those persons citizenship. It did something earlier and narrower. It refused to make distance cancel injury.

Below the clause, a formula assigned each transfer three values: present survival gained, future burden imposed, and claimant uncertainty. Uncertainty could delay distribution but could never reduce the burden to zero. A handwritten correction darkened the final term.

UNKNOWN IS NOT EMPTY.

Sumi touched the air above the page, careful not to make contact. "The return wording."

"Not the same words," Edrin said. "The same objection."

Nera followed the formula to a set of blank current fields. The ledger had never stopped receiving phase data. Tiny figures shifted at the page margins as drum three turned above them.

"How large?" Halden asked.

Sumi studied the calculation keys. "Not readable without loading the active transfer archive. This book has rates but not all Year 223 belt schedules."

"Can you connect them without altering the source?"

"With witness mirrors. It will take two days if the Office records cooperate."

Halden entered the marked square for the first time. The custodian moved to object, then saw that access had already completed. He stopped at the edge of the cradle.

"Do not connect the original," he said. "Build mirrored inputs. Every source carries two witnesses. No current schedule changes during calculation."

Nera looked at him. "And the claimant clause?"

"We preserve it under foundation custody."

"Preservation can become burial with better paste."

"It can also prevent an unaudited figure from emptying heat stores, closing orchards, and turning wards against one another before we understand distribution." His voice stayed quiet. "We have confirmed a claim test, not calculated the claim."

Edrin asked, "Who receives notice of this opening?"

"Only the statutory record custodians until the sum is complete."

"That excludes Assembly counsel," Nera said.

"For forty-eight hours."

"You have no authority to seal a foundation opening from its designated custodians."

"I am not sealing it. I am managing disclosure of an uncalculated systemic liability."

Sumi closed the tool case. "Plain words."

Halden's eyes remained on the moving margin figures. "No one outside this room, the vault custodian, and the required archive witnesses learns what we found. Passive return receipt continues. No further cold front. You calculate the Founder Sum under continuous custody."

Above them, a phase drum completed another fraction of its turn. The ledger margin accepted a new figure from the active lattice.

Nera set the first blank witness sheet beside it while Halden ordered silence around the calculation.

Chapter 5 — The Price of a Degree

The Founder Sum could not count a benefit until someone named what had been spared.

On Day 9, Nera sat beneath the foundation vault with twelve mirrored transfer rolls and three conflicting definitions of survival. The oldest treated any temperature above lethal shelter minimum as benefit. A later revision counted crop days and industrial continuity. Current public accounts counted only civic interruptions avoided, which made a warm bedroom disappear from the calculation unless a pipe had first failed.

Edrin worked opposite her at the long stone table. Sumi occupied the instrument end beside a bank of hand readers. Two archive clerks witnessed source changes from a screened alcove. Halden had not returned, but his managed-silence order hung by the door with its forty-eight-hour limit crossed out and replaced by PENDING CALCULATION.

Nera had refused to initial the alteration.

“Run the first belt roll under survival minimum,” she said.

Sumi fed punched alloy through the reader. Numbers traveled by mechanical linkage into a blank calculation slate. The foundation ledger remained in its cradle behind glass, isolated from every modern circuit. A brass arm copied each result onto witness paper without touching the original.

Edrin watched the first total settle. “Now current civic continuity.”

The same transfers produced a smaller figure.

“Difference?” Nera asked.

“Household and labor comfort outside declared interruptions,” Sumi said. “Also preventive crop control. If a schedule keeps frost from forming, public accounts record no frost event and therefore no avoided loss.”

“An absence can support a claim only after something breaks,” Nera said.

Edrin turned the source roll under the lamp. “That follows current tariff law.”

“We are not calculating tariffs.”

“No. We are deciding whether to assign liability through definitions abandoned before any living officer took office.”

Nera looked at the Founder Sum’s first column. Its writers had distinguished necessity from comfort without pretending comfort had no material value. A family able to sleep in separate rooms spent more heat. An orchard protected from a minor frost gained yield even if the trees would have survived. A ward with stable vents avoided labor, repairs, mold, and illness that never became emergencies.

“Use the founder categories,” she said. “Preserve present definitions as a comparison run.”

Edrin wrote an objection.

She waited until he finished. "Do you want to read it aloud?"

"I want it attached. Founder categories may overstate liability by treating ordinary utility as borrowed survival."

"Attach my response. Current categories understate it by erasing any benefit successful enough to prevent a claim."

The brass arm copied both statements beneath the run.

They worked from morning bell through the first canal shift. Every transfer required four checks: source schedule, phase scar, present benefit, downstream load. Repairs created gaps. Renamed districts obscured older accounts. Several early rolls measured cold in shelter-days rather than degree-hours, forcing Sumi to reconstruct wall area and population from seed ledgers.

Nothing permitted speed. A decimal misplaced in Year 48 could become a district's winter allocation in Year 223.

At midday, one archive clerk brought bread and mineral broth to the vault threshold. Food could not cross the custody line. Nera ate standing on the public side while the Founder Sum waited open behind glass.

The clerk looked at the stacked witness sleeves. "Are you pricing a canal rebuild?"

"Historical load review," Edrin said.

"How much history?"

"As much as the rolls contain."

The answer obeyed Halden's order and told the truth so narrowly that Nera lost her appetite. She returned the cup with half the broth untouched.

When work resumed, Sumi called up the district map. Twelve belt zones appeared on the horizontal glass, their boundaries changing as archived years passed. Transfers flowed across them in pale blue bands. Nera watched familiar streets form from older names. Low Glass began as Canal Labor Housing. Bellweather's orchards occupied three failed settlement grids before irrigation stabilized the western plain. The Warm Ward started as a seed-vault enclosure and expanded each decade around the heat security it already possessed.

"Allocation by source equipment gives the ward credit only after its local network opened," Sumi said. "Allocation by beneficiary address includes the seed-vault period."

"Which did the annex specify?"

"Originating account and material beneficiary."

Edrin traced the old boundary with a bone pointer. "The first residents did not choose placement. Seed staff slept near the vault because storms closed the road."

“Choice does not determine benefit,” Nera said.

“It may determine remedy.”

“That comes later.”

She heard how easily calculation could impersonate judgment. A total could identify load and still say nothing about who should take tomorrow’s cold. The Founder Sum created claimants. It did not demand symmetrical suffering.

They separated benefit from fault and ran the map again.

The Warm Ward brightened first.

Its layered vent glass reduced heat loss. Stored soil beneath civic gardens smoothed temperature swings. Radiator canals reached its buildings through protected tunnels rather than open gantries. Those advantages had required investment, maintenance, and public planning. They also allowed the ward to accept additional transferred warmth without local systems saturating. Each mild winter accumulated more benefit where mildness could be held.

Low Glass showed a different pattern. It received moderated weather but lost heat through older windows and open service corridors. Its residents supplied repair labor to the canals. Injury logs rose during every major transfer cycle. Under the founder categories, labor harm remained part of the originating account instead of vanishing into workplace compensation.

Edrin would not accept district color as proof. He took a jar of numbered archive beads from the custodian’s shelf, shook it, and drew one without looking. The number selected a winter transfer from Year 209.

“Random source test,” he said. “If the method attaches bodies only when we choose sympathetic records, it has no method.”

Sumi loaded the corresponding roll.

The transfer had softened a two-night cold crest over central Lume. Public summaries credited uninterrupted water service and seed protection. The detailed schedule divided benefit among 8,204 addresses. Nera chose no household. She asked the reader to display the first complete source chain.

A Warm Ward nursery appeared.

Twenty-six infants had slept in two connected rooms above a civic heat store. The transfer kept vent intake within the band required for their lungs. No child entered emergency care. The nursery used less stored fuel and released two attendants to other wards during the second night.

“That counts as survival under the founder rule,” Nera said.

“And avoided civic interruption under the current rule,” Edrin replied. “No dispute there.”

The next complete chain led to a Low Glass washhouse serving four tenements. Its pipes would not have frozen without the transfer, but warmer intake shortened

every wash cycle. Forty-three residents completed laundry before the next labor shift. The public account assigned no benefit because no failure had been forecast.

Edrin read the work log. "Convenience."

A witness clerk spoke from the screened alcove. "If the cycle had run long, six attendants would have missed canal shift."

Everyone turned toward the screen. Witnesses did not usually interpret source material.

The clerk continued more quietly. "I lived in that block. My father clipped these wash tokens above our stove."

Halden's silence order prevented Nera from explaining why the record mattered. It did not prevent a source witness from identifying her own history.

"State only what you can attest," Edrin said.

"Long cycles lost work priority. Two losses in a quarter cut a household's heat credit. That rule applied through Year 214."

"Did your household use this washhouse during the selected dates?"

"Yes."

Edrin opened a separate witness page. He took her name, childhood address, and the source of her knowledge. The archive clerk could not prove every attendance record, but her testimony established that the schedule's category called convenience could alter wages and household heat.

Sumi displayed the transfer's maintenance costs. Canal crews had changed two gate seals during the cold crest. One worker suffered a crushed foot when a stiffened hoist released late. Compensation records counted the injury under public works rather than seasonal operation.

"Present benefit and present cost in the same district," Nera said.

"Across districts," the clerk corrected from behind the screen. "The washhouse gained. The injured worker lived three canals south."

Nera amended her note. Aggregate maps could make a working district appear to receive and pay one balanced amount. Households did not share the balance merely because a boundary enclosed them.

They followed the selected load through the fixed interval.

The Year 346 receiving model contained no registered settlement at the exact phase coordinate, only a migration depot projected beside a buried reservoir. Its household layer placed 312 residents there during the two-night crest. The inherited burden raised shelter demand, reduced pump reserve, and moved eighteen people from outer rooms to a communal heat wall. No casualty appeared in that model branch. Sleep space contracted. Water pumping shifted to one daily window. Repair labor doubled for three days.

Edrin examined the output. "Modeled hardship, not recorded injury."

"Yes," Nera said. "And the nursery benefit also appears through a model reconstructed from schedules. We cannot treat modeling as reliable when it protects infants and imaginary when it compresses shelter rooms."

"Present nursery logs confirm the benefit."

"The return pressure confirms occupation at one later coordinate. We do not yet have household records for this depot."

Sumi rotated the screen toward them. "The load exists whether the demographic branch predicts one person or three hundred. Population changes who receives it, not whether we sent it."

The selected transfer contributed only a narrow line to the total. Yet it connected infants, laundry shifts, an injured canal worker, and an unnamed depot across one material operation. No single body represented the whole account. The ledger's scale came from repeating such uneven exchanges for more than two centuries.

Nera asked the clerk whether she wanted her testimony included in the sealed calculation.

"I want the washhouse rule included," the woman said. "My name adds nothing to its teeth."

"The rule enters as documentary evidence. Your testimony establishes use."

"Then include both, but do not call us beneficiaries without the worker's foot on the same page."

Edrin copied that limitation into the annex. The source test passed, though not in the clean form he had expected. Founder categories captured ordinary utility that current accounts erased. They also required distribution below district level if any later remedy meant to avoid charging the same households for costs they had already borne.

Nera found Mara Voss in a Year 202 source sleeve.

Her mother's death occupied one line among three vent attendants killed by the Canal Six rupture. The active public record assigned infrastructure loss and family compensation. The hidden schedule assigned continuity benefit to the destination ward and carried the injury as a cost of preserving that benefit.

Beside the names, a later clerk had written a question in brown ink: DOES A PRESENT WORKER BECOME DOWNSTREAM COST WHEN DEATH ENABLES FUTURE RETENTION?

No answer followed.

Edrin read the note over Nera's shoulder. "The clerk saw the problem."

"Then covered it."

"We don't know that this hand applied the public strip."

"No. We know someone asked before the Office stopped asking."

She copied the marginal question into the witness record. Her hand stayed steady until she reached her mother's surname.

Sumi gave her the time that steadiness required. She recalibrated a reader that did not need recalibration and argued with a gear until Nera had finished.

The next roll contained current-year winter retention schedules. Nera's own season seal appeared on four. She had authenticated source quantities, equipment readiness, forecast tolerances, and legal distribution. None of the forms displayed the receiving year. A field named REMOTE CONTINUITY absorbed that information in summaries.

"Expand destination burden," she said.

Sumi opened the hidden layer.

Year 360 filled the column.

The figure repeated beside household moderation, orchard frost avoidance, seed storage, public tunnels, and school heat. Every current schedule reached the same fixed horizon regardless of the benefit category. The public labels had multiplied. The material destination had not moved.

Edrin sat back. "Authenticate the expanded fields before we total."

Nera applied her seal to the first witness sleeve. Its brass felt warm in the vault air. She had pressed the same mark into the original schedules with no visible future date. Authentication had not made her informed. It had made the system's omission durable.

One by one, they sealed the fields.

The first full sum finished after evening bell.

Sumi did not announce it. She checked the reader's carry wheels, reversed the final roll, and ran the last century again. Edrin compared every ninth subtotal against the hand ledger. Nera watched the brass arm copy the result in groups of three digits.

4,702,118,640 degree-hours.

The annex rounded liability to the precision supported by early shelter conversions. On the formal sheet, Edrin wrote 4.7 billion degree-hours and drew a box around the assumption range.

No one in the vault spoke for several breaths.

The number did not describe a single winter. It combined cold, storm load, heat crests, failed crop windows, and household moderation across the audited belt. A degree-hour could mean one room held slightly warmer. Billions could mean clinics, orchards, factories, nurseries, seed vaults, corridors, and kitchens kept within their chosen bands by weather passed into another year.

"Comparison run," Nera said.

Sumi loaded the current civic definitions. The result fell by more than half. It excluded comfort without declared interruption, successful preventive transfers, household labor, and injuries compensated outside seasonal accounts.

Edrin stared at the smaller figure. "This one would survive present tariff review."

"Would it survive the annex?"

"No." He rubbed his eyes. "The annex itself may not survive present law."

"Still not the question."

"The question will become law the moment anyone sees that larger number."

Nera turned to the district distribution.

Colored bands divided the 4.7 billion degree-hours by material beneficiary. The Warm Ward held the largest share. Its total exceeded every orchard district and working canal zone, not because its residents had requested each transfer, but because ward infrastructure captured moderation efficiently and policy protected its temperature band before others.

Nera asked for a household overlay.

Edrin looked at her. "That level does not improve belt calculation."

"It tests whether the distribution follows actual use."

"It also gives you your own address."

"Yes."

Sumi opened the layer.

Buildings replaced district color. Nera found her residence along an inner vent arcade, three streets from the winter garden. Its account showed nineteen years of stabilized nights, reduced mold risk, protected water pressure, and ward-wide soil storage. Tovan had grown up sleeping without an indoor coat. After her husband died, Nera had walked the arcade at night because the public vent glass kept snow off the paving and warmth in the walls.

None of that comfort felt stolen in memory. It felt built, maintained, paid through rates, and shared with neighbors. The ledger did not dispute those facts. It added a source of payment no rate notice had named.

Sumi enlarged the winter garden.

Stored soil heat occupied a deep chamber beneath paths, citrus frames, and public seed beds. The garden served more than display. It supplied cuttings after crop disease, gave children exercise during storm closures, and kept older residents moving through winter. Its survival value remained real. Its moderation account also dominated the ward map because soil accepted warmth for months at a time.

"Present benefit has bodies too," Sumi said.

Nera thought of the east-yard mechanic heating a brake coupling, of Talla's hands around a cup, of her mother crossing a canal collar with cooling capacity

diverted toward protected stores. "Which is why the sum cannot become a punishment schedule."

Edrin pointed at the liability bands. "Halden will ask what can be repaid without breaking current shelter and food systems."

"He should."

"And if the answer requires continued deferral?"

"Then anyone receiving it needs standing to answer."

The words moved beyond Nera's authority as a notary. She could authenticate a claim clause. She could not create civic standing. Yet any calculation that assumed silence from the burdened side would repeat the defect that produced the number.

Sumi switched from beneficiary map to the Year 360 demographic layer. The model's household access remained blocked on the public version. Their preserved spindle opened it locally.

Settlement markers multiplied across the belt.

The first model view had shown one node in the wind terraces. The complete belt audit revealed dozens: buried heat towns, orchard shelters, canal settlements, migration depots. Some carried populations below a thousand. Others approached current district size. Ordinary demographic assumptions placed people in all of them because future resource planning required workers, children, meals, sickness, repair, and death.

Many settlement names appeared only as coordinate codes. The model had not yet chosen what their residents would call home. Nera could not mistake that uncertainty for emptiness after reading the household layers. Names changed. Heat demand did not wait for registration.

Sumi overlaid inherited load. Blue bands descended until several settlement colors darkened beyond display range. She reduced the scale twice. The wind-terrace node at the first return coordinate remained visible under the deepest column.

"The return pressure originates here," she said, touching the edge of the marker.

"Originates at the phase coordinate," Edrin corrected. "Not necessarily from this modeled settlement."

Sumi withdrew her hand. "Fair."

Nera wrote both statements into the calculation summary: exact coordinate; probable settlement association unverified.

A pipe clicked behind the wall. The passive return bell in the phase room sounded once through the speaking tube.

Sumi crossed to the tube. "Drum three?"

A technician's voice answered from above. "Western pressure. Printer active

without shaped load. Eleven characters clean, then a pause.”

“Do not touch the feed. Read raw.”

The technician spelled the first line. HOLDFAST.

A second printer burst sounded faintly through the tube.

“Next line,” Sumi said.

“K E S T. Full string appears separated by one space.”

Edrin opened the red book to a new witness page. “Time, drum state, outgoing schedule.”

The technician supplied them. Ordinary lattice load had created enough contact for the return. No local front had been accepted. No question had been sent since the test.

Nera looked at the model’s wind-terrace marker. Its public field still carried only a coordinate.

Sumi entered the received words into the local search.

The model did not recognize them. An archive of provisional terrain terms offered no match. The name had not come from Year 223 planning.

Nera wrote it beside the coordinate by hand: Holdfast Kest.

For the first time, the deepest burden column ended at a place that had named itself.

Chapter 6 — The Warm Ward

Tovan held the vent glass against the morning and found a flaw no wider than a hair.

He stood on the upper walk of the winter garden with one boot hooked under the rail, his apprentice sash tucked into his coat so it would not trail across the polishing table. Beneath him, three terraces descended under a roof of layered panes. Lemon moss climbed the retaining walls. Bean vines carried white flowers through air warm enough for shirtsleeves. Children followed a gardener between beds of black soil, each child holding a paper cup of thawed water.

Nera watched her son turn the pane by degrees. A pale line moved inside the glass, vanished, then returned when he angled it toward the roof vents.

“Stress feather,” he said. “Inner layer. It will open under a hard contraction.”

Master Olen, who had placed the defective pane among six sound ones, grunted approval. “Mark it without ruining the surface.”

Tovan touched frost chalk to the corner. He wrote a small V rather than circling the feather. Circles encouraged an installer to inspect the marked flaw and miss another one nearby. Nera had taught him that when he was eight, before he decided her profession consisted entirely of delaying other people's work.

Olen lifted the pane from Tovan's hands. “Your mother has come to arrest the roof.”

“She only arrests paper.” Tovan looked down at Nera. “Glass gets an appeal.”

“Paper does too,” she said. “It simply loses more often.”

His smile arrived quickly and left when he noticed the blue audit sleeve under her arm. On ordinary visits she brought lunch or a complaint about the abrasive dust on his cuffs. The sleeve carried the Office seal and a red corner stripe for disputed benefit.

Nera had delayed this inspection for three days. After the belt calculation, every path through the accounts led toward her own ward. The figures named radiator draw, cloud diversion, stored soil heat, emergency vents, seed preservation. They did not name the old women pruning lemon moss at the second terrace or the coughing child who had been brought under the roof for clean air. Numbers could reveal concealment while performing another concealment of their own.

Olen gave Tovan the remaining panes. “West rib. Check every edge.”

“Do you need me?” Tovan asked Nera.

“Not yet.”

He heard the qualification. He stacked the panes in a padded carrier and took the stairs slowly enough to keep them from touching.

The winter garden opened before the Seasonal Charter, though most of its

current roof dated from the last twenty years. Its public plaque described civic seed protection and respiratory refuge. Neither claim amounted to ornament. During the mold winter, the terraces had supplied clean greens to four infirmaries. The roof vents filtered black snow when street air clogged with ash spores. Nera had brought Tovan here through two childhood lung fevers and counted his breaths beside the citrus beds.

She crossed to the soil manifold at the lowest terrace. Copper pipes entered beneath a tiled bench, each fitted with a waxed inspection tag. The tags showed temperature, draw period, source canal, authorized use. Garden staff had arranged them in tidy fans that made excess look ceremonial.

A keeper named Veysa met her with both hands stained dark from soil. "Senior Voss. We received notice of a benefit audit, not a closure order."

"No closure order exists."

"People hear disputed benefit and imagine you carrying winter in that sleeve."

"I need the storage records from the last nine cycles, including emergency releases."

Veysa glanced toward the children. "Those records concern refuge operation."

"They concern public heat."

"So does a nursery."

Nera opened the sleeve on the tiled bench. "Then the record should show that use clearly."

The keeper studied her face, perhaps looking for reassurance that the audit would preserve every familiar distinction. Nera could not offer it. Veysa wiped her hands on her apron and unlocked a brass cabinet behind the manifold.

Warm air flowed from the open door.

Not room warmth. This air carried the mineral closeness of a radiator tunnel. Inside the cabinet, glass gauges stood in two rows above a buried cistern map. The garden stored heat in packed soil and brine vaults under every terrace. Nera had known about the cisterns. Her household levy helped maintain them. She had never understood their scale.

Veysa removed a record cylinder. "We charge the beds during surplus periods."

"Define surplus."

"When the lattice offers draw below the ward threshold."

"That describes price, not source."

"The source ledger sits with Seasonal Accounts."

Nera set her palm against the cabinet frame. "I have read it."

Veysa's expression tightened. "Then why ask me?"

"Because a transfer line records what entered this garden. It does not record

what the garden refused to release.”

For seven winters, the public schedule had designated the vaults as emergency civic capacity. Release required a ward council declaration. The council had declared three respiratory emergencies and one seed threat. Low Glass had requested heat during two canal ruptures in the same period. Both requests showed insufficient reserve certification.

Veysa drew another cylinder from the cabinet. “If we empty the soil stores during a rupture, we lose the beds. Not decoratively. Root death, fungal bloom, seed stock gone. Recovery takes years.”

“How much could you release without losing them?”

“That depends on outside temperature, roof integrity, occupancy, crop stage.”

“Someone calculated a figure before denying the requests.”

A pruning knife clicked shut behind them. One of the older gardeners had stopped working. The children still moved between beds, pouring their cups into shallow circles around seedlings.

Veysa lowered her voice. “Nine percent during the first rupture. Six during the second.”

“Requested amounts?”

“Seven and five.”

“So both requests fell within the survivable release.”

“Survivable for the plants, according to a model that assumed no second failure.”

Nera wrote the figures on a witness tile. “Who changed the certification?”

“The ward council accepted the conservancy recommendation.”

“Who wrote that?”

“I did.” Veysa put the cylinder down. “My youngest slept under this roof during the mold winter. I watched a nurse pull gray threads from his throat. You can call the calculation selfish. Do not call it imaginary.”

Nera looked across the terraces. Tovan had reached the west rib and knelt beside a vent frame with his measuring comb. Below him, a girl tipped her empty cup over a bean shoot and waited for one last drop.

“I won’t,” Nera said. “I will call it a choice that another district paid to preserve.”

Veysa’s chin rose. “Has your Office never made such a choice?”

“It has made too many without recording them as choices.”

The keeper resealed the cabinet. She did not refuse access again. Together they worked through the cylinders, matching stored heat against seasonal schedules. The apparent surpluses came from softened cold fronts. Each softening entered the public ledger as continuity management, a service without a visible

counterparty. Under the founder accounting Nera had opened, every charge pushed an equal burden forward.

At midday, the roof vents adjusted. Hinges turned in sequence along the central rib. The garden held its warmth, but a faint draft crossed the lower beds where old seals admitted outside air. Tovan and Olen moved above the leak, calling measurements to each other.

Nera found one cylinder from the winter of Tovan's second fever. The garden had drawn six hundred thousand degree-hours during nine nights of black snow. She remembered sleeping upright against the terrace wall because lying down made him cough. She remembered Veysa, younger and equally soil-stained, bringing broth without asking whether Nera could pay.

The ledger assigned most of that draw to a burden beyond Year 223.

Her thumb rested over the cylinder's etched date. Gratitude did not alter source. Debt did not cancel gratitude either. She copied the entry without correcting her memory into something easier to condemn.

One line on the release chart lacked a witnessed flow value. Nera asked Veysa to open the test loop.

"We do that at quarter change," the keeper said.

"Your denial relied on available release. I need a physical measure."

"Opening it cools the nursery bed."

"For how long?"

"Seven minutes if the lower gate seats properly."

Nera took out the red witness book. "Then we will learn whether it does."

Veysa sent a gardener to clear the nearest path. She unlocked a floor hatch beneath the tiled bench and exposed two hand wheels, one green with age and one polished along its rim. The green wheel controlled garden circulation. The bright one admitted stored heat to the civic trunk leading east under the ward. Nobody had touched it during ordinary maintenance because the ward council required an emergency token for public release.

A blank token hung on a chain inside the hatch.

"That cannot authorize flow," Nera said.

"It authorizes a test that reaches the trunk but not the district valve." Veysa threaded it into the wheel hub. "The council receives a pressure notice. They may send someone."

"Let them."

The keeper braced one knee against the tiles and turned. Resistance traveled through the pipe with a low iron complaint. Across the lowest terrace, leaves trembled although no vent had opened. A gauge needle fell four marks. The civic-side tube warmed under Nera's fingers.

At the upper walk, Tovan looked over the rail. Olen told him to keep measuring. He obeyed, but his next call came late.

Water began moving below the floor. Nera had crossed radiator canals all her life and knew their rough, continuous force. This flow sounded restrained, parceled through pipes designed to surrender no more than officials had already accepted. The test loop carried enough heat to warm a clinic hall. The vault map showed six more loops at equal capacity.

Veysa watched the nursery bed. Tiny condensation beads vanished from its glass cover. "One minute."

The gauge settled at the modeled release rate. Nera recorded inlet temperature, trunk pressure, soil loss. At two minutes, a brass annunciator rang inside the cabinet. Ward council inquiry, printed on its narrow tape.

No one answered it.

At four minutes, the bean flowers nearest the manifold unfolded as the circulation system drew harder from the upper terraces. Heat moved through the garden as a visible sequence. Leaves lifted near the stored-soil pipes. Farther away, they remained still. A refuge could preserve life while revealing exactly which beds had first claim.

"Your east trunk can take more," Nera said.

"The trunk can. The return gate may not seat if we drive it above twelve percent."

"Has anyone tested twelve?"

Veysa pointed to a shallow burn on the polished wheel. "During the ash fever. We sent heat to a wash station after its boiler split. Fourteen percent for nineteen minutes. The gate stuck. Two gardeners closed it with a chain bar."

"That release does not appear in the council summary."

"They called it internal conservancy support because the wash station served this roof."

Nera wrote the omitted event into the book. "Who worked the chain bar?"

Veysa named herself and the older gardener who had stopped pruning. The woman raised two scarred fingers from across the path.

At seven minutes, Veysa reversed the wheel. It moved halfway and caught. The soil gauge continued to fall.

She did not curse. "Chain bar."

Nera set down her pen. Together they took an iron lever from the hatch and fitted it through the spokes. The older gardener joined them without invitation. Three bodies pushed. Heated metal breathed through Veysa's gloves, leaving dark moisture along the rim. The wheel shifted one notch, then another. When the gate seated, the water below them struck the stop hard enough to shake soil from the

tile joints.

The nursery bed had lost less than one degree. Its smallest leaves lay flat under the cover, then slowly lifted as circulation recovered.

Nera entered the closure failure, the human labor required, and the measured capacity of the trunk. An honest audit did not prove release painless. It made the pain specific enough to distribute.

Tovan came down carrying the feathered pane. "Three bad seals on the west rib. Olen says the replacements can wait until evening draw."

"Can they wait through an unmoderated frost?"

He looked toward his master before answering. "Not a full one. The outer leaves would silver first. Then contraction would take the corner panes."

"How long to make replacements?"

"Glass is ready. Fitting takes a shift if the frames cooperate."

"Do frames cooperate?"

"Only in instructional diagrams."

Veysa almost smiled. Tovan set the pane on the bench, careful of Nera's records.

"You're auditing the stores," he said.

"Yes."

"Because of your cold paper?"

Nera had told him about the physical frost and the future timestamp. She had not told him the debt total or Holdfast Kest. Halden's managed silence order applied to institutional disclosure; it did not grant him custody over her household. Even so, she had chosen caution each night, placing food between herself and Tovan while finding reasons to discuss apprenticeship marks.

"Because the stores benefited from transfers under review," she said.

"Did we?"

The question included the garden, their apartment, his lungs, every mild winter he remembered.

"Yes."

He waited.

Nera closed the cabinet record. "The Office shifted weather away from us. It did not disappear."

"Where did it go?"

"Forward."

"How far?"

She met Veysa's gaze. The keeper stepped away to give them privacy without

pretending not to hear.

"One hundred thirty-seven years."

Tovan glanced up through the layered roof, as if distance in time might reveal itself as depth in glass. "People live then."

"Yes."

"You said the demographic pages could be forecasts."

"I said I had not authenticated them yet."

"And now?"

"Now I have authenticated a settlement record and enough weather response to know the burden has witnesses."

He touched the marked corner of the flawed pane. "Are they asking us to turn everything off?"

"We have not received a full demand."

"That means no."

"It means they have had less room to speak than you used asking the question."

A bell sounded at the garden entrance. Two ward attendants admitted First Steward Halden Rusk without ceremony, though everyone on the terraces recognized him. He wore a gray civic coat with rain beading on the shoulders. Outside weather had shifted while the roof kept its own morning.

Halden dismissed the attendants at the rail. "Senior Voss. Your office said you had opened a disputed-benefit audit here."

"My office reports quickly."

"It reports exceptions. We have trained it to." His attention settled on Tovan. "Apprentice Voss."

Tovan inclined his head. He did not leave.

Halden approached the soil manifold. "Keeper, may we have the lower terrace?"

Veysa called the children toward the upper beds. The gardeners followed, some reluctantly. Olen remained on the west rib and made a performance of checking seals.

Nera handed Halden the two denied-release entries. "These stores could have met both Low Glass requests within their own survival models."

"Models made before the mold reserve order."

"The order came four months after the first denial."

"Public institutions preserve capacity before need becomes visible."

"They also invent future emergencies to outrank current injuries."

Halden read the entries. "And sometimes the emergency arrives. Your son survived one here."

Tovan looked from him to Nera. She felt the old impulse to move between official language and her child. He had nearly reached adulthood under protection built partly from such impulses.

"Do not use him as a finding," she said.

"I am using a documented outcome. Concealment preserved lives, Nera. Not in theory. The early cold-account riots closed canals, emptied seed rooms, and killed people before any deferred burden reached its due year. The Charter gave citizens a schedule they could obey without reopening the founding panic every morning."

"It also removed the people who paid."

"It removed projections no court could examine."

"We can examine them now."

Halden's eyes shifted toward the roof. Rain traced faint paths over the outer pane. "A narrow contact does not make a stable polity. We have fragments, an authenticated settlement name, phase behavior, and a sum large enough to collapse every seasonal promise if spoken without a remedy."

"Silence does not supply a remedy."

"It supplies time."

"To the side already receiving it."

Tovan lifted the flawed pane. "This roof has a bad inner layer."

Halden paused.

"My master hid nothing," Tovan continued. "He put it among good panes to test me. If I install it, the roof still holds today. Maybe all winter. Concealing the flaw would preserve warmth."

"Tovan," Nera said, unsure whether she meant warning or gratitude.

He kept his attention on Halden. "We mark it because the person fitting it needs to know where failure goes."

Halden did not patronize him with praise. "Glass has fewer frightened neighbors than government."

"It can still cut them."

Rain struck harder above them. The roof vents closed one row at a time. Moisture gleamed on the bean leaves, supplied by no cloud the children could see.

Halden returned the witness tile to Nera. "Complete your audit. Do not release the figures outside the inquiry group. I will convene a risk session after the orchard records arrive."

"You have no authority over what Tovan knows in his own home."

"No. You retain that particular burden." He looked at the heat gauges. "I ask you to remember that warmth here has names as well."

After he left, the garden gradually resumed its ordinary sound. Cups touched

stone. Pruning blades opened. Olen called for Tovan to bring the replacement seals.

Tovan gathered the panes. "Was that permission to tell me less?"

"It was a request."

"Will you grant it?"

Nera slid the cylinders into their padded sleeve. "I will tell you what I can authenticate, and I will say when someone has ordered me not to tell the rest."

He considered this with the severity he used on vent edges. "That sounds inconvenient."

"It has begun to be."

They stayed through the evening draw. Tovan heated frame clamps while Olen removed the failed seals. Each opened joint released a thin thread of garden air into the rain. Nera helped Veysa chart the response of the soil stores. The lower gauges dropped by fractions as the roof admitted cold, then steadied when the new seals seated. For the first time, Nera watched the garden spend heat rather than merely possess it.

At dusk, she stood beneath the repaired west rib. The white flowers had closed. Children had gone home carrying damp paper cups. Under the tiles, the buried soil vaults retained enough summer warmth to outlast weeks of ordinary frost.

Tovan pressed his thumb against the new pane and checked for vibration. "Sound."

Nera's pocket witness tile chilled against her coat.

She drew it out. The tile carried no civic feed, only the narrow return trace Sumi had taught it to copy. Letters appeared in weak blue strokes, several broken at their verticals.

> KEST CALENDAR OFFICE. RECEIVED YOUR SETTLEMENT QUERY. I KEEP WEATHER AND SHELTER DAYS. MY NAME IS SIV, CALENDAR KEEPER.

Below the text, the instrument opened a signature field.

Tovan read over her shoulder. He did not ask whether people lived there now.

Rain cooled the outer roof. Stored heat rose from the soil and fogged the newly fitted glass between Nera's hand and the darkening sky.

Interlude I — The Children's Calendar

Orra warmed the blue chalk in her mouth before she marked the day.

The chalk tasted of salt and old coins. Siv had told her not to wet it because moisture weakened the stroke, but the calendar room held six degrees at breakfast and a dry piece scratched uselessly across the ceramic. Orra breathed on the white slip, drew a careful circle around its date, then made one blue line beneath it.

"Show me the intake sheet first," Siv said.

Orra held up the paper with both hands. Wind load: eleven. Inherited cold: none measured. Shelter draw: three body-units below allocation. Terrace lamps: unlit. The figures leaned in different directions, but each remained legible.

Siv checked the phase clock. "Again. Read the cold field aloud."

"Nothing from before."

"Read what it says."

"Zero carried load." Orra pressed her lips together. "That means the same thing."

"It means something we can defend."

Her daughter lowered the sheet. The blue line on the ceramic had already begun to pale at one end where her tongue had thinned the chalk.

Holdfast Kest kept its public calendar in three forms. The shelter office recorded legal days by phase clock. Terrace crews counted workdays from one survivable opening to the next. Children used fired slips because paper curled in the damp rooms and because a child could hold a day without needing access to the shelter ledger.

Most slips carried black marks for storm, white cuts for ration closure, or paired red points when inherited cold arrived on top of local weather. Blue belonged to a true day. Orra had marked four in eleven years.

Siv took the chalk from her and strengthened the failing edge. "Now it will keep."

"Does it count if we spent lamp heat on the intake needles?"

"The heat does not decide the weather's source."

"But we made the reading warmer."

"We made the instrument readable."

Orra considered this distinction, suspicious of any answer that let adults keep a machine warm while their children wore indoor gloves. She set the slip beside the other three blue-marked dates. The oldest had cracked through a corner during the western shelter move. Siv could still remember the sky from that day, hard gray above the terraces, snow falling only from clouds directly overhead.

A pipe struck twice behind the calendar wall.

Both of them stopped.

The first blow could come from contraction. A second at the same interval signaled a heat-account correction. Siv opened the wall cabinet and found the balance wheel turning toward deficit. Room Six had exceeded its dawn allocation. The needle moved slowly, as if delay might soften the number.

"Go ask which vent opened," she told Orra. "Do not accuse anyone of waste."

"What if they wasted it?"

"Then find the open vent."

Orra pulled her hood over her braids and went into the shelter passage.

Siv entered the correction on the room ledger. Kest measured recoverable body heat because bodies supplied part of its survival system. Six sleeping adults counted differently from four adults and three children. Fever changed a room's allocation. Pregnancy changed it. Death required immediate recalculation, which made every mourning household receive a clerk before it received enough quiet.

The process kept people alive. That fact did not make it kind.

Room Six housed two terrace workers, an aunt with damaged lungs, and five children during the daylight rotation. Their expected dawn contribution should have covered the wall loss. Siv checked the night register. One worker had taken an outer shift and not come back before the pressure doors closed. The shelter still counted his heat.

She adjusted occupancy downward. The account deepened.

Orra returned with a strip of felt in one hand. "The lower vent seal came loose. Aunt Dessi stuffed this in, but it froze stiff."

"Any pipe damage?"

"No. She wants to know whether the blue mark still counts."

"It counts."

"Even with a leak?"

"Local cold can hurt us without coming from an old account."

Her daughter placed the frozen felt on the desk. Meltwater darkened the wood beneath it. "That seems unfair."

Siv fetched a fresh seal from the repair bin. "Weather never signed a fairness rule."

"People signed the accounts."

The answer followed Siv down the passage.

Room Six had hung blankets from the ceiling to trap breath around the sleeping shelf. Aunt Dessi crouched by the low vent with a kitchen knife and a coil of wire. Frost covered the grille in a thick oval. The children watched from under one blanket, their feet tucked together.

Siv scraped the frame clean. Orra held the lamp near her hands. When the old seal came free, outside air entered with enough force to gutter the flame. Dessi pushed the new felt into place, then Siv tightened the wire until the hiss narrowed and stopped.

"Charge the repair to the room?" Dessi asked.

"To shelter fabric."

"The council will move it back next quarter."

"Then I will move it forward again."

Dessi looked toward the children. "Keep your blue day. They have so few."

Back in the calendar room, a third pipe signal sounded from beneath the floor. Not correction. Three evenly spaced taps belonged to the phase alcove.

Siv locked the heat ledger and took the alcove key from under her collar. Orra followed to the threshold but did not enter. Children could observe the receiving needle through a slit in the door. Only a calendar keeper could touch the contact glass because a wrong pressure cost six hours of shelter current.

The needle leaned backward.

For generations, phase equipment had brought weather forward as debt. Its reverse motion had begun nine days earlier, weak enough that the engineers blamed a cracked bearing. Siv had kept copies of every motion. Yesterday the contact glass produced a row of current-year numerals from one hundred thirty-seven years before her own.

Now pale letters occupied the lower strip.

> SETTLEMENT IDENTIFIED AS HOLDFAST KEST. STATE NAME AND RECORD FUNCTION FOR AUTHENTICATION.

Orra whispered from the door, "They called us a settlement."

Siv read the line twice. The sender had not used forecast zone, projected occupancy, or unallocated downstream terrain. Settlement granted no food and closed no cold account. Still, the word admitted walls, records, people.

A reply wheel stood beside the glass. Its counter allowed one hundred eighty characters before locking for six hours.

Siv drew a writing slate toward her. She first composed the message required by the shelter council: Kest population, nine-year death count, projected crop exhaustion, immediate demand for cessation. The figures filled the slate with an authority no greeting could match.

Then she looked through the door slit.

Orra had returned to the ceramic calendar. She arranged the four blue slips in a row, leaving gaps for years that had given her none. The newest slip still carried meltwater from her gloves. It showed a true day beside a room deficit, a failed vent, an absent worker. Accuracy did not require pretending the day had spared them. It required naming which harm belonged to which source.

Siv wiped the slate clean.

A casualty total sent without provenance could become a weapon or an excuse. People in the earlier year might reject it as coercion. Kest's council might prefer shock because shock could force attention, yet contact had arrived through a request for a name. Siv decided to make the first answer small enough to verify.

She wrote the settlement office, her record duty, and the fact that Kest had received the query. The counter left twenty-seven characters.

"Why not tell them how many?" Orra asked.

"Because I want them to answer the first truth before I give them the worst one."

"What if they only answer easy truths?"

"Then we will know what their name costs."

Siv placed her palm on the contact rim. Its stored warmth drained through her skin. In the room behind her, the balance wheel clicked as repaired Room Six began recovering. The reply wheel accepted each letter with a faint mechanical bite.

At the signature field, Siv wrote her name and office without a title the past might misunderstand.

She pressed SEND.

The blue mark on Orra's newest day held. Beneath it, phase current left Kest in a thread too narrow for a body, carrying Siv's name into the year that had entered their cold account.

Act II

Chapter 7 — Audit of the Orchards

Lev split the bud with his thumbnail and found the pear flower asleep inside.

Its folded petals carried no frost burn. The green tissue around them held enough moisture to shine, which meant the night schedule had done what Bellweather paid it to do. Orchard mesh delayed the cold descent until the soil coils could raise the root zone by two degrees. By noon, the redleaf pears would feel spring.

He dropped the opened bud into a tin cup. A healthy sample still counted as a sample destroyed.

Along Row Seventeen, Yara Fen tightened mesh clips while her twins followed on the service path, collecting fallen wire in a bucket. The children had agreed to stay behind the yellow stakes. They had also agreed, with equal sincerity, to stop asking when they could climb a pear tree.

“Tram at the west gate,” Yara called.

Lev did not look up. “Supply?”

“Blue Office carriage.”

His thumbnail slipped into the bud scale and drew blood.

Nera had sent notice two days earlier, a short institutional message that requested access to Bellweather's seasonal records and offered no reason beyond disputed transfer benefit. Lev had answered with the cooperative's standard consent. Refusal would give the Office power to seal their forecast feed. Consent at least kept him near the records while someone else interpreted them.

He sucked the cut, tasted soil, then wiped his thumb on his work coat.

The tram crossed the outer switch and stopped beside the packing shed. Nera stepped down carrying two blue sleeves and a field case. She wore city boots with narrow soles, wrong for the thawed paths. Once, when they lived together, she had kept orchard boots under his kitchen bench and complained that mud dried more honestly than binding paste. He wondered whether she had thrown them away or stored them with other objects that proved an earlier life without requiring its return.

Yara met her first. “You have until tomorrow's evening release. After that we lower the east mesh, whether your figures agree or not.”

Nera glanced at the buds. “Forecast requirement?”

“Redleaf opening. One bad hour takes the center flowers.”

“Show me the control ledger.”

No greeting, Lev thought. Then Nera looked at him.

"Your hand," she said.

"Occupational tragedy."

"You always cut toward yourself."

"The bud objects less."

Yara's gaze moved between them. "Good. We have established that the auditor knows where he bleeds. Packing shed."

Inside, tables filled the center aisle beneath hanging crop maps. Workers sorted dormant grafts at one end. At the other, steam rose from a kettle beside the temperature drums. Bellweather's records occupied flat cedar drawers because field paper swelled under orchard humidity. Each drawer covered one season and carried three seals: agronomist, crew, public forecaster.

Nera opened her field case. "I need nine years of transfer requests, actual mesh draw, crop loss declarations, emergency reallocations, and labor injury reports."

Lev leaned against the drawer wall. "You requested seasonal records."

"I have specified them."

"You have expanded them."

"Because the disputed benefit includes maintenance exposure."

Yara stopped pouring tea. "Whose maintenance exposure?"

"All districts connected to the audited transfers."

"That answer has office walls around it."

Nera took a blank witness tile from the case. "The Seasonal Accounts ledger concealed burdens attached to our weather moderation. I am tracing who received the benefit and who performed the dangerous work."

"Concealed from whom?" Lev asked.

"From the public schedule."

"Not what I asked."

Her fingers rested on the first drawer pull. "I can authenticate that the lattice moved weather burden exactly one hundred thirty-seven years forward. I can authenticate an inhabited settlement at that date. I can authenticate a large unpaid load across the belt."

"How large?"

"The sum remains under a managed inquiry order."

"You came to count our trees while keeping your total."

"I came to test which part belongs to your trees."

Lev pushed away from the wall. "No. You came because someone in Lume saw orchard draw in a column and decided food might look like comfort if arranged

correctly.”

Nera did not raise her voice. “Show me it does not.”

Yara set three cups on the table. “We can spend all morning proving each other cruel, or we can open the drawers before my children begin using grafting wax as breakfast spread.”

One twin had already removed the lid.

They began with current spring. Lev showed the schedule request for the next two nights: a four-hour delay on the northern cold front, root-zone retention at one point eight degrees, mesh closure before moonrise. Without that schedule, the outer rows could tolerate the frost. The center redleaf blocks could not. Their buds had entered the vulnerable stage six days early because the previous week arrived warmer than forecast.

“Did Bellweather request that warming?” Nera asked.

“No,” Lev said.

“Did you benefit?”

“We advanced pruning, reduced storage draw, and woke fungus early. Choose your column.”

Nera wrote all three.

Yara took them to the center block before the ground softened further. Her twins stayed with the packing crew, loudly disappointed. Orchard mesh stretched overhead in gray folded bands. Every fourth row held a soil coil under the drip line, fed by a branch from the west radiator canal. Frost chalk on each trunk showed the lowest safe bark temperature.

Lev cut four more buds. Nera held the sample cup while he opened them on the flat of his knife.

“This one has center damage,” he said.

She bent close. A brown point marked the innermost petal.

“From last night?”

“Probably the false dip before dawn. The public forecast averaged it out.”

“Can you prove timing?”

“Yara's crew logged bark temperature every quarter hour.”

Nera looked toward Yara, who had climbed a short ladder to inspect a mesh seam. “Not an automatic needle?”

“Needles fail under condensation,” Yara said. “Workers remain expensive but repairable.”

Lev waited for Nera to object to the phrasing. She only asked to see the log.

At Row Twenty, a soil-coil junction had blistered. The repair patch showed a dark oval where heatworker gloves had pressed against wet insulation. Nera

photographed the seal mark.

"Canal crew repaired that during the last frost?"

"Low Glass night team," Lev said. "Two people. We sent fruit credits."

"Names?"

"The service docket carries them."

Yara came down from the ladder. "One burned her wrist. The docket calls it a superficial contact event because she finished the shift."

Nera's face changed by almost nothing. Lev knew the signs. Her mouth stilled before anger entered her questions.

"Did Bellweather file an injury attachment?"

"We received no form," he said.

"Did you ask?"

"No."

The answer remained between them, neither softened nor enlarged. Lev had known about the burn. He had sent credits and a jar of pear salve through the crew chief. He had treated care as a private transaction because the canal service continued. Nera entered the worker's name from the docket, then gave the tile to Yara to witness.

They ate midday bread beneath the retracted mesh. Wind moved across the field with a metallic flutter overhead. Nera sat on an overturned coil crate, brushing mud from one boot with the other.

Lev passed her a cup of hot root broth. "You could have worn the old pair."

"I no longer know where they are."

"Storage bench by your apartment door."

"You noticed them there?"

"I put them there when I brought the last boxes."

Years had passed, yet the exchange carried the dangerous ease of a repeated household task. Yara took her food ten paces away, near enough to interrupt if they became foolish.

Nera drank. "The future settlement sent a name."

"A person's?"

"Yes. A calendar keeper."

"Did they ask you to stop?"

"No full demand has arrived."

"Authentication. Recognition of the settlement. A channel stable enough to state a demand."

Lev broke his bread in half. "You believe them."

"I believe the records I have tested."

"That careful distinction once made me want to throw plates."

"You threw one."

"It had cracked in the kiln."

"You selected it from a cupboard containing sound plates."

A brief warmth touched her face. It went no farther. "I have not authenticated every projected consequence. I will not tell you more than the evidence supports."

"But you expect a demand."

"Yes."

Wind lifted the edge of the crop map beside them. Lev pinned it with his cup.

"If their demand means immediate cessation," he said, "Bellweather loses this block. Not profit. Food. Redleaf keeps through the early winter when frost barley stores begin to thin. The city planned clinic rations around this yield."

"I know."

"No, you have read it. Knowing requires putting a knife through a healthy bud because the forecast changed after your crew slept."

Nera looked down at the opened samples. "Then show me."

He heard no challenge in it. That made refusal harder.

During the afternoon they ran two forecast cases through the orchard board. The first kept the current schedule. Ninety-two percent of center flowers survived under ordinary variance. The second removed all delayed cold without transition. Survival fell below one-third. Frost barley gained soil moisture, but not enough to replace the pear calories before winter.

Nera requested a third case.

"Staged reduction," she said. "Keep tonight's bud protection. Reduce post-flower draw. Accept local rain instead of stripping the cloud band."

Lev entered the conditions. The board clicked through stored weather patterns and produced wide error bars.

"That case lies," he said. "Not deliberately. We have no crop behavior for that combination."

"It shows uncertainty."

"It prints a center line. Officials will quote the center and forget the bars."

"Then do not give them one."

Yara leaned over the board. "Give them labor instead. How many night crews for hand covers if post-flower cold comes back?"

Lev calculated rows, cover weight, available workers. "Six crews for four nights. Ten if rain freezes the ties."

"We have four," Yara said.

"The city can send workers," Nera offered.

"City volunteers tear mesh and step on root crowns."

"Heatworkers know field systems."

Yara tapped the blistered junction. "Heatworkers already repair what we draw. Asking them to cover our buds adds another benefit to our side."

Nera entered that objection too.

By evening, the audit had produced no clean moral geometry. Bellweather fed districts that never saw an orchard. Its moderated weather protected calories and cooperative income. Canal workers accepted burns to carry that moderation. Future citizens received the displaced frost. Removing every transfer at once would make current hunger answer past concealment, while continuing unchanged would preserve a theft because repair looked dangerous.

Lev stood at the board as the scheduled evening release approached. "I reject your Office saying nobody lives there. I also reject a reckoning that proves its honesty by killing this crop."

"I have proposed neither."

"Your proposals begin after the count."

"Yes." Nera met his gaze. "I came before the demand because any answer needs facts from the people who will bear it."

"People here."

"People in both years."

The release bell rang from the canal branch. Heat entered the soil coils. Outside, workers began lowering mesh over the redleaf blocks, the gray bands spreading from row to row.

They carried the audit into the field after dark. Bellweather's night crew worked without broad lamps because light drew glass moths into the folded mesh. Each worker wore a hooded glow tile at the throat. From the packing shed, the rows resembled a ground constellation arranged for harvest rather than navigation.

Lev assigned Nera to the tension cart. "Walk inside the wheel track. If the line catches, let go. Saving your hand will cost fewer pears than holding it."

"I have operated a document winch."

"Paper has not yet dragged anyone through a hedge."

Yara took the front handle. Nera took the rear without replying. They moved toward the center block while Lev checked soil-coil temperatures behind them.

Cold arrived first along the drainage cuts. It gathered as a clear skin over standing water, then crept into the wheel ruts. The public forecast called for a mild descent. Lev's field needles showed a harder edge riding beneath it, the kind of narrow variation city instruments averaged into harmlessness.

At Row Thirty-One, the mesh stopped halfway down.

Yara raised one fist. Every nearby hood tile turned toward her. A draw cable had jumped its guide and wrapped around a pear limb. Continued tension would strip the branch or tear twenty meters of mesh.

"Release the west ratchet," Lev called.

Nera found the handle and pulled. Nothing moved.

"Up, then toward the trunk."

She changed the angle. The mechanism opened with a crack, releasing cable so quickly that the cart lurched. Nera let go as instructed. Its rear wheel dropped into a rut and flung thawed mud across her coat.

Yara climbed into the tree with a pruning hook. Lev held the trunk below her. The trapped limb carried hundreds of unopened flowers. Cutting it would free the cover in seconds and remove part of the year's yield. Untangling the cable preserved fruit while cold advanced through the row.

"How long?" Yara asked.

Lev checked the bark needle. "Eight minutes before center risk."

"Ten to free it clean."

A worker brought a hand cover. Nera spread it beneath the branch while Yara worked the cable loop back over each spur. Frost began silvering the grass outside the lowered sections.

Nobody discussed instant cessation now. They spent each minute against visible tissue. Bellweather already relied on a borrowed weather interval, but the bodies under the mesh belonged to the current night: Yara balanced above them, Lev braced in mud, Nera held rough cloth clear of the buds. An ethical account that ignored either fact would produce a clean sum and a false answer.

At six minutes, the soil coils warmed. Moisture rose around Lev's boots. At eight, the bark needle steadied one mark above damage. Yara freed the last loop and dropped from the tree.

"Draw," she said.

The crew pulled the mesh down by hand. Its edge reached the ground and sealed against the sand tubes. Pear leaves shivered beneath it, then stilled as trapped air warmed.

Nera touched the rescued branch through the hand cover. "Does this one now owe more than the others?"

"The record will assign draw by block," Lev said.

"Convenient."

"Accurate enough for cultivation."

"Not for distribution."

Yara coiled the loose cable. "Then improve the record without pretending each flower chose its pipe."

They finished the center block near midnight. The mesh changed the sound of the orchard, containing breath, footfall, and cable noise beneath a low fabric sky. Outside it, frost spread across the unused path. Inside, workers opened their coats.

Lev gave Nera a bark needle to witness. She compared its etched line with the automatic gauge, noted a three-tenths difference, and signed both values. Her hands had reddened around the knuckles. He almost offered his gloves. She had her own in the field case and had chosen not to use them because she needed the same contact the workers used. He recognized the gesture and distrusted its neatness, yet the mud on her coat had escaped intention.

At the packing shed, Yara entered the trapped-cable delay as labor, crop risk, and added coil draw. None of those fields existed on the old form, so she wrote across the blank reserved for celebratory yield notes.

Nera watched her press hard enough to mark the duplicate sheet. "Keep that method."

"I planned to."

Only after the field needles held steady did Nera close the current drawer. "Tomorrow I need the full emergency record."

"You have the declared events."

"I need undeclared shifts, local exchanges, and discretionary releases."

Lev frowned. "Bellweather cannot move a front. Only the Office can."

"Your cooperative can redirect within the west branch under crop-preservation authority."

"We use that for field balance."

"Then the records should show balance."

She had found a discrepancy. He could tell by the calmness with which she folded the witness tile. Nera became most formal when she intended to give someone time to choose candor.

The next morning, Lev unlocked the cellar archive before she arrived.

Old harvest drawers filled the underground room from floor to low stone ceiling. The air smelled of cedar, dry fungus, and the sharp oil used on record seals. He carried a lamp down the rows while Yara checked index cards behind him.

"Discretionary west-branch shifts," he said. "Last nine years."

"Drawer H through N."

"You knew?"

"I know where we hide choices by calling them cultivation."

Lev pulled H. Within it, each harvest record listed crop stage, yield, weather

deviation, worker hours. No column named displaced burden. The cooperative had inherited the form and never demanded one.

Nera arrived carrying mud on both boots this time. She had wrapped the narrow soles in sacking for grip. Lev noticed and said nothing.

They worked drawer by drawer. Most local exchanges stayed within Bellweather: one block surrendered coil heat so another could survive bloom; sunroot beds accepted rain while pear rows took a dry window. Nera traced each exchange against the city schedule without interpreting ordinary cultivation as fraud.

At noon, Yara found a frost mark filed under harvest celebration.

The card carried a silver ribbon from the largest redleaf yield in Bellweather's history. Under the ribbon, a handwritten note authorized an emergency west-branch diversion during a late cold descent. Lev recognized the former cooperative chair's seal and his own junior agronomist witness mark.

"I remember the night," he said. "The north cloud comb stalled. We had forty minutes before petal ice."

"Where did you send the frost?" Nera asked.

"Into an unused canal sink."

She laid the branch map beside the card. "Which sink?"

Lev followed the old routing marks. West branch to Canal Four, then east through a pressure gate labeled industrial reserve. The reserve connected beneath Low Glass.

"That gate should have absorbed it," he said.

Yara pulled the maintenance attachment. "Gate offline. The note says district air accepted residual load."

Lev read the line again. He remembered cheering when the mesh temperature rose. He remembered workers carrying lanterns between rows and the smell of pear flowers after ice retreated. He did not remember asking what accepted meant beyond the orchard boundary.

Nera placed a witness tile over the harvest card. Its chemical face copied the silver ribbon, Lev's younger seal, and the maintenance note beneath both.

"How much?" he asked.

"Enough for a local frost event. I need the district log to measure injury."

Yara removed the ribbon. Its pin had made two holes through the record. "My twins were born that winter," she said. "The cooperative sent us a celebration crate from this harvest."

Outside the cellar, the day's first pear mesh began to open. Sun entered through the stairwell in a widening bar.

Lev signed Nera's copy. Then he took the original harvest record upstairs, not to the archive drawer but to the public board where every Bellweather worker would

see which district had carried their celebrated frost.

Chapter 8 — The First Demand

The return counter opened at dawn with frost welded around its final digit.

Nera held a heated cloth against the brass rim while Sumi Quill worked a bone pick under the ice. The phase room below Canal Nine had grown colder with each completed exchange. No wall gauge registered the loss. Tools numbed fingers, breath stayed visible, and the black ceramic drums turned through their ordinary heat shimmer as if the room disagreed with its own instruments.

“Do not drip on the contact glass,” Sumi said.

“I know how water behaves.”

“You work above a canal. That gives you optimism, not skill.”

The ice released in one thin ring. Nera caught it in the cloth before it touched the glass. Across the return strip, weak letters waited from the night cycle.

> SIV, KEST CALENDAR. AUTHENTICATE YOUR OFFICE. GIVE ONE WEATHER ENTRY WE CAN TEST AGAINST OUR INHERITED LOAD.

A mechanical counter beside the strip showed 180. Each character Nera sent would lower it by one, including spaces and stops. At zero, the wheel locked for six hours while the phase drums rebuilt enough inversion pressure for another turn.

Edrin stood at the red witness book. He had inspected the room seals twice and found no reason to distrust them. Distrust remained in his posture anyway.

“She asks us to provide evidence,” he said.

“She should,” Nera answered.

“She gives no independent fact.”

“She gave a settlement name and office that match the demographic annex.”

“An annex inside our own system could supply both to a manufactured signal.”

Sumi pulled off one glove and flexed her shortened hand near a heat lamp. “Then ask something the system has not forecast.”

Nera opened the daily weather schedule. A southern rain front would cross Lume at the third afternoon bell. Its pressure signature appeared in public forecasts and future inheritance models. Siv could repeat it without proving access to her own year.

A small unallocated disturbance sat beyond the belt's eastern edge. The city forecast had measured it but not yet projected whether it would enter inhabited air. Nera copied its phase marks onto a slate.

“Too weak,” Sumi said. “By Year 360 it could be noise.”

“Then noise provides an honest result.”

Halden had authorized contact under managed inquiry, with every turn

witnessed and no public release. His order permitted authentication questions. It did not permit negotiation or any change to current schedules. Nera placed the limit beside the counter before composing her reply.

> NERA VOSS, SENIOR NOTARY, SEASONAL ACCOUNTS. TEST: EAST BELT DISTURBANCE PHASE 71-4C. REPORT ARRIVAL SHAPE, NOT OUR FORECAST.

The counter stopped at 56. She could add a question about Kest's legal records or ask why a calendar keeper controlled phase equipment. Every extra line risked folding several claims into one answer. Nera pressed SEND with the remaining space unused.

The return wheel engaged. Its teeth clicked beneath her palm. Cold moved through the contact rim and along the bones of her wrist. In the canal above, a pressure bell answered as the phase drums drew current.

Sumi recorded the material cost: one-tenth degree room loss, no change in public transfer, fourteen seconds of drum vibration beyond tolerance.

Edrin sanded the witness entry. "Six hours."

Nera looked at the blank strip. "We have records to test meanwhile."

They spent the morning building a custody chain for a person not recognized by the Charter. Kest's name appeared in a Year 208 demographic forecast, a Year 214 shelter-material allocation, and a hidden Year 219 burden table. Each source treated the settlement as future occupancy rather than a claimant. The records agreed on location and pressure-door count, but population estimates varied by thousands.

Nera refused to let the model fill the gaps. She opened a provisional counterpart file and wrote only what direct contact had supplied: settlement, calendar office, correspondent's chosen name, a functioning receiver in Year 360.

At noon, Halden entered the phase room alone. Condensation darkened the shoulders of his coat before he reached the lower rail.

"Have they stated an objective?" he asked.

"Authentication first."

"We should define the range of answers before a demand arrives."

"That would define their demand from this side."

"It would define our lawful capacity. We cannot promise cessation, compensation, or standing."

"We can promise not to misname them."

Halden glanced at the provisional file. "The Charter recognizes no temporal counterpart."

"The Charter also contains no return channel."

"A missing category does not create authority."

"No. It creates a missing category."

Sumi made a sound near the drum rack that might have covered a laugh. Edrin kept writing.

Halden placed a folded instruction beside Nera's slate. "Ask whether Kest can distinguish inherited load from native weather. Ask for a falsifiable event. Do not invite a remedy we lack power to grant."

"She has already requested the same test from us."

"Then perhaps she understands institutions."

The second contact window opened at the sixth hour. Frost crossed the counter from right to left, stopping before the first digit. Letters emerged in uneven groups.

> 71-4C ARRIVED YEAR 357 AS TWO PRESSURE CRESTS, 11 HOURS APART. SECOND CARRIED BLACK SNOW. KEST ARCHIVE HAS INTAKE NEEDLES AND ROOF DAMAGE LOG.

Nera checked the character wheel. The future reply had used nearly all its allowance.

"Year 357," Edrin said. "Three years before the sender's present."

"An archive claim," Halden replied. "Not current observation."

Sumi enlarged the phase trace. "Look at the second crest."

The city model still held the eastern disturbance as one diffuse system. Nothing in the active forecast predicted a split. Its phase composition allowed one, but only if the upper wind line turned north after crossing the unobserved ocean.

Nera wrote the claim in the red book without marking it confirmed.

Halden tapped his instruction. "Ask for the roof log."

"One hundred eighty characters cannot carry a document."

"Then an identifying checksum."

Sumi shook her head. "Their archive standards may not use ours."

Nera considered what Kest could send that Lume could test without trusting either era's terminology. A weather signature depended on measured intervals, pressure direction, particulate residue. Three quantities could fit in one turn.

She also needed to answer the question Siv had put first. Kest had asked the Office to authenticate itself. A title supplied no proof.

Nera unfastened her season seal and pressed it into the contact copy bed. The machine translated its physical ridges into a compact notarial pattern. Edrin witnessed the impression, then added his own deputy mark so Nera could not later claim sole custody.

> SEAL PATTERN FOLLOWS. SEND CREST INTERVAL, PRESSURE FALL, BLACK-SNOW MINERAL CODE. WE WILL TEST WHEN 71-4C ENTERS OUR FORECAST RANGE.

Twelve characters remained. Nera sent the message as written.

Halden read the frozen strip. "No question about their desired remedy."

"Not before evidence."

"You are delaying them."

"I am giving their evidence a form you cannot dismiss."

He studied her for a long moment. "That answer assumes dismissal requires technical cause."

"It requires cause in my record."

When Halden left, the phase room seemed to recover space without recovering warmth.

The third window would open near midnight. Nera went home for four hours, carrying a witness copy inside her coat. Tovan had left lentils under a warming lid and a note saying the garden's west rib held through rain. His apprentice sash hung from a chair, dusted white at the hem.

She ate standing at the counter. Every utensil in the apartment carried accumulated mildness: water that did not freeze in wall pipes, ceramic stored without thermal cracks, fresh greens from a roofed garden. None became shameful merely because its cost had surfaced. The point of an account did not lie in teaching a spoon to repent. It lay in deciding who could authorize the next charge.

Nera slept for ninety minutes with the witness copy under her hand.

At midnight, Canal Nine sounded louder than daylight traffic. Heatworkers moved along the upper gantry, their lamps passing over the phase-room ceiling through grated service gaps. Sumi had increased the drum cooling flow to protect the bearings. The change warmed the canal by a measurable fraction and left the chamber colder.

Edrin arrived with two sealed flasks of broth. "This does not imply friendship."

"Enter that reservation in the book."

He handed one to Sumi. "She would reject it as ceremonial language."

The contact strip cleared.

> INTERVAL 11H07M. PRESSURE FALL 19.6. MINERAL CODE FE-3/SI-8. SEAL RECEIVED. NOW RECEIVE OUR ACCOUNT: 31,604 DEAD IN NINE YEARS.

Nobody spoke while the drums completed their turn.

The figure occupied fewer characters than Nera's seal. She counted the digits although none required interpretation. Thirty-one thousand six hundred four. Nine years. No location breakdown, no definitions, no method for separating inherited weather from war, disease, native storm, or choices made in the future.

Her body understood the number before procedure allowed it. The broth flask felt too warm to hold. She set it beside the witness book.

Edrin recovered first. "Record exact punctuation."

"I have."

"Ask for casualty rolls," Halden's written instruction said from the desk.

Sumi stared at the black-snow code. "Ask what they want us to stop doing before we make them spend a month transmitting names."

Nera read the message again. Siv had placed the casualty figure after authentication. She had not demanded belief without a test. The weather details offered a future check, not immediate proof. Trust remained conditional in both directions.

The counter opened for Lume's turn.

"What can we lawfully say?" Edrin asked.

"That we received it."

"Anything else creates reliance."

"Silence creates reliance too. They may assume disbelief."

Sumi turned from the trace. "Give her one question, not five objections wearing a coat."

Nera drafted, erased, drafted again. Expressing sympathy would spend characters without altering Kest's risk. Accepting the figure as authenticated would exceed the evidence. Calling it alleged repeated the Office's habit of making people vanish behind classification.

> WE RECEIVED 31,604 / NINE YEARS. NOT YET VERIFIED, NOT DISMISSED. STATE FIRST ACTION YOU REQUIRE. PROVIDE CASUALTY METHOD IN LATER TURNS.

The counter retained 57. Nera pressed SEND.

Cold bit her palm harder than before. The return wheel slowed near the end, and Sumi reached beneath the console to assist its gear with a manual lever. Her three full fingers closed around the grip. The shortened two rested against polished metal without helping.

"Do not put that in the message," she said without looking up.

Nera understood. Injury could become theater as easily as a number could. She recorded Sumi's labor in the technical log and nowhere else.

The morning window brought no text.

Instead, the contact strip displayed a broken line of phase marks. Sumi compared them to the previous turn and found a missing interval.

"They lost current or chose not to answer," Edrin said.

"Can we distinguish?" Nera asked.

"Not until another turn."

Sumi did not wait beside the silent strip. She opened the lower service hatch and shone a lamp across the return governor. Its copper vanes should have rested after a completed window. One continued moving in short, irregular strokes.

"Residual pressure," Edrin said.

"Pressure has direction." Sumi lay on the grated floor and reached into the hatch. "This keeps pulling from their side."

Nera knelt beside her. Beneath the console, braided phase leads entered a ceramic block marked with hazard chalk. Frost coated one lead but not the others. Each weak stroke of the governor thickened that coating.

"Could a message be trapped?" Nera asked.

"Possibly. More likely their current fell during send."

"Can we recover partial characters?"

"If I raise local acceptance for ninety seconds."

Edrin looked toward the public transfer gauge. "Halden authorized contact, not schedule variation."

"Acceptance need not change a scheduled front," Sumi said. "We can take waste heat out of the phase bearings. The room loses another degree. Canal cooling gains the same load."

"Who carries that gain?" Nera asked.

"Night maintenance, if a bearing binds."

The answer stopped her hand before it reached the control. A small technical adjustment already had bodies attached. They could warm the room later. A worker could not postpone contact with a hot bearing.

Sumi pushed herself upright. "Then we ask the shift lead."

They climbed the iron stair to the lower gantry. Canal Nine ran beneath them, dark water carrying heat toward radiator branches. The night crew had stripped to inner shirts despite cold rain falling through the open grates. Two workers guided a ceramic scraper along the canal wall while a third watched pressure bulbs set behind wire cages.

The shift lead listened to Sumi's proposal without looking away from the bulbs. "Ninety seconds becomes how much at Bearing Three?"

"Two-tenths above current draw. Four if the governor catches."

"My crew can carry two. Four closes the west walkway."

Sumi offered a mechanical cutoff at three-tenths. The lead shook his head.

"Automatic cutoff reads the inlet. The danger sits at the collar." He pointed to a worker crouched where the drum housing entered the wall. "Put a hand witness there and let her call it."

Nera followed the worker to Bearing Three. The collar's outer shell glowed no color, yet heat reached her face from an arm's length away. A narrow sight tube showed oil circulating inside. Bubbles meant excess. A milky streak meant metal abrasion and an immediate stop.

The worker handed her a bell cord. "One pull for hold. Two for cut. If the tube clouds, do not wait to count bubbles."

"You trust an untrained witness?"

"I trust you to dislike being responsible."

Below, Sumi signaled readiness. Nera pulled once.

The phase room accepted its extra load. Bearing Three began with a low hum that traveled through the collar into Nera's boots. Oil moved faster in the sight tube. One bubble passed, then three together. The worker held a gloved hand near the shell without touching it.

At forty seconds, the sound sharpened.

Nera watched a pale thread enter the oil. Not milkiess, only reflected lamp light. It vanished when the worker shifted. She kept the bell silent.

At sixty-eight seconds, five bubbles crossed the tube. Heat pushed through her coat. The worker's glove darkened where sweat reached its cuff.

"Enough for them?" he called.

Sumi's voice came through the speaking pipe. "Governor still moving. Ten more."

The tenth second brought a gray fleck into the oil.

Nera pulled twice.

Current dropped. The bearing hum sank through two tones and stopped. On the gantry below, the canal scraper crew resumed movement only after the pressure bulbs steadied.

Sumi came up carrying a phase copy. "No characters. We recovered the send boundary. Their wheel engaged and failed after seven marks."

"So they attempted a reply," Edrin said.

"We can authenticate an attempt," Nera corrected.

The shift lead wrote the bearing temperature and gray fleck on Nera's copy. "Next attempt waits until we change the oil."

Nera signed the restriction. The return channel now carried a local operating limit witnessed by people outside Halden's inquiry group. Secrecy had gained another edge it could not fully seal.

The empty window cost Kest as much anticipation as it cost Lume. Nera spent the six hours in the Office while Edrin challenged the casualty file's admissibility. They listed possible definitions: direct cold exposure, crop-failure mortality, shelter accidents, migration losses. Each category required records. None justified treating the total as meaningless while they waited.

Halden ordered a private famine model. Nera objected to using an unverified figure as input.

"We model risk before proof," he said.

"Then label the output conditional."

"It will carry three uncertainty bands."

"Label the people conditional too, if that describes your position."

"We are trying to understand what immediate cessation would do here."

"They have not requested immediate cessation."

"Not yet."

Nera left his chamber with the model unsigned.

At the next opening, letters appeared before the frost reached the counter's edge.

> FIRST: FREEZE ALL NEW DEFERRALS. DO NOT ADD TO WHAT ALREADY COMES. EXISTING LOAD NEEDS NEGOTIATION. Stop calling us empty.

Capitalization weakened near the final sentence, as if current had fallen while Siv wrote. The words remained legible.

Edrin counted. "Within limit."

Sumi said, "And narrower than the Steward's model."

Nera copied the demand exactly. A freeze on new deferrals differed from immediate repayment. It left existing schedules, inherited infrastructure, and historical load unresolved. It also demanded authority Nera did not possess beyond a temporary emergency action.

She could spend the next turn explaining Charter limits. That would answer a moral claim with institutional architecture. She could promise to petition the Assembly, which might create false reliance. Before either response, the weather signature had to gain enough standing to survive Halden's risk chamber.

Nera built her message from three statements.

> DEMAND RECORDED: NO NEW DEFERRALS. I CANNOT GRANT PERMANENT FREEZE. I CAN AUTHENTICATE EVIDENCE AND PETITION. SEND ONE CURRENT KEST WEATHER MARK.

Sumi checked the counter. "Thirty-eight left."

"Send."

By the sixth window, Nera had been awake long enough to feel each phase-drum turn in her teeth. Rain struck the canal grates overhead. The southern front had arrived on schedule, but the eastern disturbance still lay beyond official range.

Siv's reply came in two parts. The first line named Kest's current roof pressure and native wind. The second gave a phase bearing for the old 71-4C disturbance, preserved from its arrival three years earlier. She included damage-log seal marks and one correction: black snow began after the second crest, not with it.

> CURRENT KEST: ROOF P 61.2, WIND NNW-8. ARCHIVE 71-4C BEARING 004.18, CREST GAP 11H07M. BLACK SNOW 6M AFTER. LOG SEALS K4-19/K4-20.

Edrin authenticated the internal consistency only. "It proves she maintains a detailed record. It does not prove the weather occurred."

"No," Nera said. "The storm must do that."

Sumi had stopped beside the forward forecast drum. Its return needle trembled where no scheduled input should reach it. She adjusted the magnification and laid Siv's bearing over the unallocated eastern trace.

The two lines matched.

Not broadly. The phase angles coincided to the second decimal. The official model still rendered one pressure crest because it lacked observations beyond the ocean. Siv's archive supplied a second bearing eleven hours behind the first, folded into the same disturbance like a hidden edge.

Sumi entered the bearing as a test condition. The forecast drum recalculated. One crest divided into two. A mineral uptake estimate appeared beneath the second, drawing iron and silica from a volcanic band the city had not yet mapped into the storm path. Six minutes after crest passage, the particulate field crossed the inhabited belt.

Black snow.

Nera checked the forecast issue clock. The public office would not ingest ocean-edge data until the next morning. No current forecast in Year 223 yet contained the split, the mineral code, or the six-minute delay.

Edrin put both hands on the witness book. "Seal the raw trace before the model writes over it."

Nera pressed her season seal into the phase copy. Sumi locked the eastern input drum. Outside, rain rang against Canal Nine, carrying no black mineral and no answer anyone could mistake for weather alone.

On the forecast glass, a storm that had not yet entered Lume's official future opened into the exact shape recorded by a calendar keeper one hundred thirty-seven years ahead.

Chapter 9 — Heatworkers

Beren Olt made Nera sign her own heat receipt before he let her cross the canal.

The receipt board hung at the Low Glass entrance to Canal Six, its metal face scarred by years of wet gloves. Beren set the household-size dial to one, selected visitor, and pulled the lever. A narrow paper tongue emerged with three figures: warmth received from district pipes, cooling labor charged to the canal crew, deferred burden attached to the current schedule.

The third field showed a blank.

“Your Office never gave the printer that number,” Beren said.

Nera held the paper over the board. “The ledger contains it.”

“Not where anyone paying can see.”

He wore an organizer's red cord over silver heat canvas. Mineral stains darkened both knees. At fifty-two, he moved with the care of someone who had stopped believing pain proved effort. Nera recognized him as the shift lead who had limited Sumi's bearing test. In daylight, the burn scar along his neck showed above his collar.

“You witnessed the return draw,” she said.

“I witnessed a notary stop when the oil showed metal. That bought you this walk.”

“What would refusal have bought?”

“Time to prepare for whichever moral discovery Lume plans to route through our pipes.”

Beren took the visitor receipt and clipped it to her coat where spray could reach it. His own receipt hung above his heart, folded until only the current draw remained visible.

Canal Six ran through a trench roofed in cheap vent glass. Condensation turned the district beyond it into moving color. Apartment bridges crossed above the water at each block. Laundry dried on warm rails. Clinic windows opened toward the canal despite the mineral smell because recovered heat cost less than wall coils.

Low Glass had benefited from seasonal moderation. No account could erase that fact. Pipes froze less often than they had in founder years. Children survived respiratory winters that once filled memorial walls. Workshops depended on steady canal temperatures. Yet every benefit arrived through equipment maintained close to boiling water, phase housings, and pressure gates that failed toward the nearest body.

Beren handed Nera a pair of grip-soled overshoes. “Your orchard sacking would come apart on the third ladder.”

"You saw that report?"

"Bellweather posted its frost diversion yesterday. Workers copy useful truths faster than offices."

They descended to the lower gantry. Canal water moved shoulder-high behind a slotted barrier, black under steam. Each radiator branch left the main flow through a ceramic throat. Workers tested the throats with sounding rods, listening for mineral buildup that could turn a controlled release into an explosion.

At Branch Eleven, a young worker struck the housing twice and frowned.

"Again," Beren said.

The second sounding carried a dull interval near the base.

"Scale pocket," she answered.

"Depth?"

"Two fingers. Maybe three."

"Never measure danger with fingers you intend to keep."

She corrected herself. "Four centimeters beyond safe wall."

Beren looked at Nera. "Write the first answer too."

The worker stiffened. "I corrected it."

"You did. The first answer tells us what training still asks you to risk."

Nera entered both statements on a witness tile. The instruction forms used by Canal Six still described manual depth checks for uncertain soundings. A later safety note recommended external gauges when available. The district had received two gauges for thirty-eight branches.

"Who allocates the rest?" she asked.

"Seasonal Infrastructure," Beren said. "Priority follows continuity value."

"Meaning flow volume."

"Meaning Warm Ward mains receive instruments before the branches that cool them."

The young worker opened a tool satchel. Two fingers on her right glove had been stitched flat at the ends to keep loose fabric from catching. The hand beneath retained all its fingers. The glove had belonged to someone else.

Nera watched Beren mark the branch for shutdown. "How much district heat does this remove?"

"Three apartment stacks and the dyehouse until evening."

"Can you reroute?"

"Through Branch Twelve. It leaks at the upper joint."

"Then both options carry injury."

Beren gave her a hard glance. "You expected a harmless side?"

"No. I am identifying which body each option approaches."

That answer did not satisfy him, but he led her onward.

They crossed beneath an apartment bridge where residents had clipped heat receipts to a public line. Some carried red arrears marks. Others showed credits for sheltering extra occupants during cold nights. A bakery receipt listed more recovered warmth than any home nearby because its ovens fed the district loop after dawn. Public accounting could reveal mutual support when the form allowed it.

A woman reached down from the bridge and handed Beren a cracked pressure bulb. "West riser again."

"Any cold rooms?"

"Top two. We moved the baby downstairs."

Beren passed the bulb to Nera. "District maintenance. Add it to your disputed benefit."

"I will add the repair and the warmth it protects."

"Do not make us innocent. That would be another way to make us simple."

"I have no innocent column."

"Your Office has always had one. It calls itself continuity."

At the canal collar, Beren stopped before a sealed gate marked with the date of Mara Voss's death.

Nera knew the numerals before she read them. The gate had been replaced after the rupture, its old iron removed except for a bent retaining bolt mounted on a plaque. Public text beneath it named three attendants and praised service continuity. Her mother's name appeared second.

"The private file says the Warm Ward seed vaults received cooling priority," Nera said.

Beren rested one hand on the rail. "The shift board said Canal Six could tolerate twelve minutes above collar limit. Mara said seven. Supervisor followed the board."

"You worked here."

"Second sounding crew. Other side of the gate."

"Did you testify?"

"I gave an instrument statement. They asked whether the alarm arrived before or after rupture. I said before. The final report said signal sequence uncertain."

Nera took the hidden classification copy from her sleeve. She had carried it since Chapter One's frost, first as evidence and then as an accusation she did not know where to place. Beren read acceptable continuity loss without changing his grip on the rail.

"We never saw that," he said.

"My family copy used infrastructure loss."

"Both versions kept the seed vault out of the memorial."

"Did the district receive any settlement beyond death compensation?"

"New gate. Safety week. Six months of household heat credit for the three families."

Nera looked along the gantry. The replacement collar had protected thousands of shifts since then. Its polished casing could support a story of institutional correction. The same story omitted who gained from the fatal limit and why the dangerous branch received upgrades only after people died.

Beren unlatched a narrow archive box beside the memorial. Inside lay handwritten injury cards that workers kept outside Office systems. Burns appeared as traced shapes. Crush injuries carried tool outlines. Respiratory exposures used black dots for each day a worker coughed metal dust.

Mara's card held a line drawing of the collar and three red marks where steam reached her through the ruptured suit.

Nera did not touch it.

"Why keep these here?"

"Because compensation files close. Bodies continue."

"May I copy them?"

"Ask the card holders. Ask families for the dead. No bulk seal."

She had arrived with authority to audit public transfers, not take custody of private injury. "I will ask individually."

Beren closed the box. This time his expression yielded a fraction. Not trust. Permission to continue earning it.

They ate beside the dyehouse while Branch Eleven cooled. Its workers had opened the high windows, releasing blue-tinted steam. District children collected condensed water from gutter taps and carried it home in stoppered jars.

Nera unfolded a burden map on the bench. "The future demand asks for a freeze on new deferrals. It does not demand immediate cessation of existing load."

"Does the steward know the difference?"

"He modeled the larger threat first."

"Of course." Beren bit into a canal bun. "A demand becomes easier to refuse when you enlarge it."

"I need a present distribution proposal before I petition."

"You need one the Warm Ward cannot call punishment."

"I need one that does not treat every current resident as equally responsible or equally protected."

Beren wiped salt from his thumb. "Then stop dividing by heads."

“Per-capita load remains the Assembly standard.”

“So did empty future.”

He drew four columns on the back of her map. The first listed historic degree-hours received. The second listed insulation and stored capacity. The third recorded maintenance exposure. The fourth listed essential outputs: food, clinics, water, shelter.

“Benefit first,” he said. “Capacity second. Exposure reduces what you assign, but it does not erase benefit. Essential output changes timing, not obligation.”

Nera examined the columns. “Maintenance exposure cannot become a permanent credit against all future burden.”

“No. Otherwise an institution could injure us until its debt vanished.”

“Warm Ward garden protects seeds and lungs.”

“Count the protection. Count its vaults too.”

“Bellweather protects food.”

“Count the crop. Count whose gloves keep the soil coils open.”

The formula resisted the simplicity of equal cold. It also resisted assigning all present obligation to affluent households. Low Glass had received stable heat, even while its workers carried disproportionate risk. Benefit created liability. Injury altered distribution. Capacity changed how immediate load harmed each place.

Nera entered sample figures from the audit. Warm Ward historic draw stood high. Its layered vent glass and soil stores gave it greater capacity. Bellweather draw varied by crop stage, with little storage and large food consequences. Low Glass received less moderation per resident while supplying most maintenance labor.

Beren watched her calculate. “Do not subtract burns from warmth. Put them on separate lines.”

“I have.”

“Your pencil moved toward subtraction.”

“It moved toward a ratio.”

“Ratios have buried more workers than gates.”

A bell sounded from Branch Eleven. The scale-removal crew had opened the ceramic throat.

Beren rose. “Come see what your ratio touches.”

At the branch, workers had isolated the flow between two pressure plates. The young worker with the inherited glove inserted a flexible scraper through the inspection port. Mineral scale came out in hot white curls. Each curl struck the catch tray and shattered into damp grit.

The branch temperature climbed despite isolation.

"Leak from Twelve," Beren said.

A worker at the upper joint called, "Seal cloth swelling."

Beren ordered the apartment feed reduced. A distant pipe knock traveled through the bridge supports. Above them, three stacks began losing heat.

Nera checked her receipt. The visitor paper had turned pink at its edge, indicating excess exposure on the gantry. Workers' receipts used the same dye, but their threshold sat higher under occupational rules.

"Why do their papers tolerate more?" she asked.

"Because changing the paper costs less than changing the pipe."

Beren took the young worker's receipt and held it beside Nera's. Her exposure field had not yet changed.

Nera wrote the difference into the public audit.

The swelling seal cloth split.

Hot water struck the barrier and burst upward in a fan. Beren pulled the young worker back by her harness. Nera reached for the emergency wheel, then stopped when two crew members shouted opposite directions.

"Clockwise!" Beren called.

She turned. The wheel fought her through half a rotation. Steam hid the branch. Someone rang the district reduction bell, three short strokes. Apartment demand fell. With less back pressure, the wheel moved again.

The spray narrowed and died.

Beren's sleeve had opened from wrist to elbow. Under it, his skin showed a red strip where water crossed the glove seam. He inspected the worker first.

"Face?"

"Clear."

"Hands?"

"Clear."

Only then did he let Nera pour cooling gel over his arm.

"Superficial contact event?" she asked.

"If you write that, I will put you in the canal."

"Partial-thickness scald, forearm, caused by Branch Twelve seal failure during reroute."

"Better."

The residents above had reduced their own draw quickly enough to lower pressure. Their rooms cooled so the repair crew could close a gate. The district did not divide neatly into workers who paid and households that benefited. People moved between those categories within a single alarm.

Beren refused the clinic until Branch Eleven carried safe flow. Nera could not order him as a patient. She could suspend the audit operation that had kept him near her map instead of his crew.

"Field inquiry closed for medical hazard," she said.

"You cannot close my shift."

"I can close my use of it. The burden meeting resumes after your arm receives a seal."

He swore once, with precision, and went to the clinic.

On Day Twenty-One, Nera returned to Low Glass with the first benefit-weighted table. Beren wore a loose white dressing under his canvas sleeve. Twelve heatworkers met them in a decommissioned bathhouse where wall pipes still clicked though no water entered the basins.

Nera pinned three maps above the empty plunge pool. One showed historical moderation. Another marked injury and maintenance hours. The third estimated immediate harm from retaining a small share of future transfers in present weather.

She did not propose a percentage. The evidence could support distribution before it supported scale.

A radiator fitter named Osric read the first map. "You count our household heat as benefit."

"Yes."

"We live beside the canal because rent farther uphill costs twice as much."

"Constraint does not make the warmth unreal."

"Then count the rent."

Nera added housing constraint to the capacity column.

From the bathhouse door, the woman who had reported the cracked west-riser bulb entered with a child asleep against her shoulder. She laid three household receipts on the old ticket desk. Each showed full district warmth delivered. Her room logs, written on the backs, recorded two nights when the top floor fell below shelter minimum.

"The printer counts heat entering the stack," she said. "It does not count whether heat reaches our rooms."

Nera compared the inlet mark with the handwritten temperatures. "Did you file a service dispute?"

"Twice. They closed both after the riser needle showed flow."

Beren pointed to the maps. "Benefit at district scale, failure at household scale."

"Then the unit needs to follow delivery," Nera said.

She divided Low Glass into pipe zones rather than treating the district as one

recipient. The bakery loop showed excess recovered warmth. Two upper residential zones showed partial delivery and high repair exposure. Their assigned obligation fell. The bakery's rose slightly because its ovens supplied capacity as well as consuming it.

The woman watched Nera make the change. "Do not reward the owner by charging the bakers."

"Worker food and wash heat come out before any commercial share."

"Put that on the sheet."

Nera wrote it into the formula, then asked the woman to witness the language. She signed with the hand not supporting her child. The alteration made the table longer and its politics harder to hide.

A clinic mechanic challenged Bellweather's essential-output credit. "Every orchard calls its crop necessary."

"Redleaf storage supplies early-winter clinics," Nera said. "The record supports partial protection during bloom, not full exemption afterward."

Another worker pointed at Warm Ward. "Their garden calls itself a clinic whenever anyone asks for heat."

"I separated respiratory refuge, seed capacity, and ornamental beds."

"Who decides which bean flower looks ornamental?"

"The garden keeper first. Public audit on appeal."

Beren sat with his bandaged arm on the table. "Still too clean."

"What have I missed?"

"Control. A ward that can refuse release has more capacity than a district whose receipts trigger automatic reduction."

Nera created a fifth column. Discretion over stored heat.

The table changed. Warm Ward's obligation increased because it held reserves and controlled their release. Bellweather retained timing protections tied to food. Low Glass gained credit for exposure and weak insulation but kept a smaller obligation for benefits actually received.

They tested the formula against one unit of immediate cold. Equal per-capita distribution lowered every district by the same nominal amount. In Low Glass, thin vent glass and leaky risers turned that figure into burst pipes and dangerous repair. In the Warm Ward, stored soil heat absorbed most of it before household temperature changed.

Nera adjusted for actual material burden. To impose equivalent risk, the protected ward would need to accept 2.4 times the per-capita load assigned to Low Glass.

Silence followed the calculation. Water ticked in an abandoned pipe behind the wall.

Osric asked, "Equivalent risk to whom?"

"Household baseline, adjusted for clinics and critical systems."

"Not workers on the canal."

"Maintenance exposure receives a separate reduction."

Beren shook his head. "Reduction implies someone can buy obligation with our scars. Put exposure into the order of operations. Protected districts take load first. Low Glass enters only after stores release and ward temperature reaches the agreed band."

"That creates a threshold, not a ratio."

"It creates both. Ratio tells you share. Order keeps their reserves from sleeping while our pipes crack."

Nera rewrote the procedure: stored capacity released before district reduction; higher-benefit wards accepted initial load; essential food protection received timed limits; worker exposure could not substitute for infrastructure repair.

On the third day, they carried the proposal to the canal assembly board. The paper covered six joined sheets. Workers gathered before shift change, wet gloves tucked under belts. Residents watched from the bridges above.

Nera read the assumptions aloud. She named every uncertainty and refused to call the table repayment. It distributed only new present burden that might arise from stopping further deferral. Historical accounts required negotiation with the people receiving them.

When she finished, Beren asked for objections.

A veteran drum cleaner raised one hand. "Why not equal cold? Easier to explain."

"Because equal temperature loss produces unequal injury," Nera said.

"That has always been true."

"Yes."

"And now the Office notices because future people complained."

Nera looked up at the apartment bridges, the receipt line, the replaced gate bearing her mother's date. "The Office had the injury records and chose not to join them to benefit. Contact removed one excuse. It did not create the evidence."

Beren stepped to the board. "Heatworkers accept that Low Glass benefited. We will not hide behind labor while someone else receives our deferred cold. We also refuse any plan that starts by making everyone suffer alike."

He took the equal per-capita sheet from beneath Nera's table. Twelve workers signed its margin, not in support but as witnesses to rejection. Beren drew a red line through the central assumption.

Above the canal, residents began clipping copies of their household heat receipts to the six-sheet proposal. By the time the shift bell rang, the equal-share

page had disappeared under evidence of who received warmth, who controlled it, and whose hands kept it moving.

Chapter 10 — The Steward's Offer

Halden Rusk began with the ration books because numbers behaved differently when someone had written hunger beside them.

The oldest lay open under the hooded lamp in his private records room. Settlement Year Eleven, north granary district, six weeks before the Seasonal Charter. Each household occupied one ruled line. Grain allotment stood first, then furnace salt, then bodies able to work. A clerk had added remarks in the margin without authorization. Child fever. Roof breach. Shared stove. Refused removal.

Halden moved a narrow reading weight down the page. Beneath it, the paper held still despite the civic tunnel fans. The Assembly above him had emptied after the evening bell, but heat continued through the floor pipes. First Stewards had always kept their records where the building's warmth could preserve paste and vellum. The same pipes fed two public shelters farther downhill. He knew their valve order. He had signed it every autumn for twenty-three years.

A second book waited at his elbow. That one recorded the Founding Crisis riots. The language had hardened by then. Persons became concentrations. A burned ration cart became distribution interference. Fourteen deaths at the south stair entered the ledger as a loss of queuing control.

Halden read the clerk's private remarks instead.

A woman had surrendered her meal counter to a neighbor whose milk had failed. Two brothers had slept against a granary wall to keep frost from opening a crack. A steward had ordered the crowd gate closed after seeing knives, then opened it again when he saw children pressed against the bars. He died before anyone entered his name in the formal report.

The famine model on the forecast glass did not contain him.

Halden touched the side control. The current city rose over the books in pale contour lines: Warm Ward stores, Low Glass radiator loss, Bellweather yields, the belt transfer due in nine days. He reduced the scheduled deferral by one quarter. Red spread through the orchard layer first. Three days later it reached the grain exchanges. By the sixth week, the model placed twelve thousand households below the winter reserve floor.

He restored the schedule and ran the public-disclosure branch.

The glass assumed a release of the cold-account figure, the Founder Sum language, and the messages from Holdfast Kest. It could not model shame, only behavior inferred from prior shortages. Ward purchases climbed. Furnace salt vanished from four markets. Two canal crews abandoned night shifts to protect their families. Bellweather closed its gates after a rumor that city officials would requisition seed grain. The projected deaths arrived as a range broad enough to absolve anyone who preferred disbelief.

Halden narrowed it by adding the ration records from Year Eleven.

The lower figure rose.

A knock sounded on the inner door. His clerk opened it without entering. "Senior Notary Voss."

"Send her down."

"She asked whether Deputy Sol may attend."

"No." Halden looked at the two chairs across from his table. "Tell her this concerns an offer, not a finding."

The clerk withdrew. Halden closed the riot book but left the ration ledger open. Concealment worked best when it looked effortless. Tonight he needed Nera to see the effort.

She entered carrying a hard document case against her ribs. Low Glass condensation had darkened the shoulders of her coat. She took in the books, the forecast branch, the two cups on the side tray. Her gaze stopped at the empty chair beside hers.

"You excluded my witness," she said.

"I excluded an audience."

"Edrin has already witnessed every material record."

"Then he can authenticate whatever survives this conversation."

Nera remained standing. "You said you had an offer."

Halden indicated the open ledger. "I have an explanation first."

"Explanations usually arrive after your orders."

"This one predates both of us."

She sat without removing her coat. The document case stayed closed on her lap.

Halden turned the ration ledger so she could read the margin. "Year Eleven. The winter before the Charter."

"I know the account."

"You know the authorized account. Read the left margin."

Her eyes moved down the clerk's notes. She did not touch the page. Notaries treated unsealed records as if fingerprints could alter tense.

"The north granary lost heat for nineteen days," she said. "The public history blames a failed sink."

"The sink functioned. The city had no reserve after rejecting the first broad seasonal transfer."

"That transfer had no public consent."

"No policy had public consent. Half the settlement believed the phase drums would poison births. The other half believed anyone opposing them wanted

children to starve. We had knives at ration gates and three wards threatening to divert canal flow by force."

"None of that makes a hidden claimant disappear."

"I have not said it does."

Nera looked up. Halden had expected anger and prepared his voice for it. Her attention proved harder to meet because it remained exact.

He brought the disclosure branch forward on the glass. "We release everything tomorrow. No preparation, no distribution plan, no explanation the public trusts. Here follows the first likely week."

She watched market lines turn red. "Likely according to whom?"

"Three civil models, two independent crop models, and the buying record from the vent-failure scare eight years ago."

"None includes testimony from Year 360."

"They cannot assign behavior to citizens the Charter does not recognize."

"That limitation belongs to your legal inputs, not to reality."

"It still shapes what happens here."

Halden expanded the Bellweather layer. The first redleaf loss appeared at thirty-six hours, followed by frost-barley failure under an abrupt retained load. "If fear closes the orchard tram, crews miss the mesh cycle. If the Warm Ward hoards furnace salt, Low Glass loses canal attendance. If workers walk off Canal Nine, your return channel fails along with present heating. Truth does not distribute its own consequences."

"Neither does concealment."

"No. Administration does, badly and under time."

He poured water into both cups. Nera did not reach for hers.

The forecast room carried a faint vibration from the Assembly's heat pumps. Halden had learned to read their changes during long sessions. A deeper note meant the north tunnel had opened. A brief shudder marked a ward requesting more flow. Tonight the machinery held an even tone, proof that no crowd stood above demanding a choice from him.

He said, "I knew the annex existed."

Nera's fingers tightened once on the case handle.

"I knew Founder Ell's committee sealed a competing liability sum after the ration riots," he continued. "I knew its language might support claims outside the Charter's present-ratepayer rule. I did not know the sum still operated in the lattice. I did not know a working return could cross the phase interval. Until Sumi Quill's authenticated test, I considered the far-side notation a founder contingency for settlements they might build later."

"They did build later. We built them by surviving."

"Yes."

The word had no defense inside it. Nera waited, perhaps for him to add one.

Halden folded his hands. "My predecessor showed me the sealed reference when I took office. He believed opening it would renew the founding dispute without changing any material condition. I accepted that judgment. I repeated it each year when the restricted index came for review."

"How many years?"

"Twenty-three."

"You signed a silence twenty-three times."

"I signed the conditions that kept seasonal authority intact."

Her expression altered by less than a frown. "That sentence makes the institution sound like a patient under your care."

"It has been one. You have seen what happens when a canal authority loses legitimacy. Crews stop believing hazard orders. Wards keep private stores. Growers falsify frost reports because they expect confiscation. Stability does not excuse the account. It determines whether we can pay any part of it without creating another."

Nera opened her case. Inside lay witness tiles, copied phase traces, and the narrow message strips allowed by the return channel. She placed one strip on the table.

The text read: WE ASK FIRST THAT YOU STOP ADDING WINTER WHILE YOU ARGUE WHETHER WE COUNT.

Halden had read the authenticated copy. Ink made it less manageable.

"They know we continue transfers," Nera said. "Every moderated front here reaches them as load. Your preparation period has weather."

"I know."

"Then tell me whose death range appears on the right side of your branch."

"The model cannot calculate Year 360 directly."

"You display our losses as red households. Their losses remain an absent field. That absence steers the decision before anyone speaks."

Halden moved the ration ledger between them. "This field carries names because the dead left books we can hold. I will not pretend equal precision where we have one narrow channel and an unverified population map."

"You have a verified settlement, a calendar keeper, storm signatures, and a casualty account."

"We have the beginning of evidence."

"We also have the continuation of harm."

He let the pumps sound through the pause. Refusing every point would insult

them both. Accepting each point without a plan would turn contrition into another form of vanity.

"The offer," he said, "creates time without closing the return."

Halden slid a prepared folder from beneath the riot record. Its cover bore no Assembly seal. A seal would have converted negotiation into directive.

Nera read the title aloud. "Framework for managed disclosure."

"Seven days. First, we brief canal crews, orchard cooperatives, ward councils, and public shelter stewards under accountable confidentiality. Second, we publish the existence of anomalous downstream load with the verified degree-hour range. Third, we establish an independent fund for future compensation, financed by a graduated charge on moderated-season benefits. Once distribution measures exist, we release the Founder Sum and the messages."

"Future compensation delivered how?"

"Material designs through the return. Crop strains, shelter methods, storm timing. Reduced transfers where our food models can bear them."

"Who decides what their losses cost?"

"A joint commission."

"With how many members from Year 360?"

"The channel cannot conduct a modern hearing."

"That does not answer me."

"One correspondent could advise the commission."

"Siv Ren could advise people deciding whether Siv Ren counts."

Halden kept his voice level. "A phased process can grow as evidence grows."

"It can also become permanent."

"Then place expiration clauses in it. You would hold notarial oversight. Edrin Sol would hold admissibility review. Quill would control technical statements. I am not asking you to trust the Steward's office alone."

Nera turned a page. "You reserve the right to delay each stage for continuity risk."

"The Assembly must retain emergency discretion."

"You have defined continuity risk to include public refusal."

"Public refusal can shut canals."

"It can also remove you."

"Yes." Halden leaned back. "I do not exclude that consequence."

The admission reached her. He saw it not as sympathy but as a recalculation.

He pointed to the model. "If the city believes seasonal moderation amounted to fraud, some will demand immediate cessation. Lev Marek's own records show what

that does to food. Others will deny every return and attack the Office for inventing claimants. Between them, stores empty. A rushed moral victory could foreclose summer for the people least able to buy private heat or food."

Nera closed the folder halfway. "You chose that phrase carefully."

"I chose it because crops do not care whether a catastrophe began with courage."

"Nor do they care whether it began with secrecy."

"No."

Halden rose and crossed to the wall cabinet. He opened it with his office key, then with the smaller private key he wore beneath his collar. Inside rested the Steward's copies of emergency ward plans. He removed three.

"These are not forecast branches," he said. "They are action orders I drafted after your first authenticated message. Low Glass receives priority furnace salt if markets run. Bellweather's seed reserve receives a guard chosen by the cooperative. Warm Ward private stores become requisitionable after a specified public threshold."

Nera took the top order. "Unsigned."

"If I sign now, clerks see the contingency and infer the disclosure."

"You ask me to accept hidden safeguards for harm caused by hidden policy."

"I am showing you the safeguards before I have protection for doing so."

She inspected the order line by line. "Warm Ward threshold sits too low. By the time public shelters reach this temperature, Low Glass apartments will have fallen farther."

"Amend it."

Nera looked at him over the page.

"I mean it," Halden said. "Write the threshold you can defend."

She took his pencil, calculated from the radiator-loss table, and reduced the trigger by four degrees. Her correction would move private stores into public control almost a day earlier. Halden initialed the margin.

For several minutes they worked without naming the larger dispute. Nera corrected the orchard guard's authority so Yara Fen's crew, not Assembly officers, controlled access. Halden added a ban on furnace-salt export after any partial release. She struck a provision allowing detention for unlicensed forecast distribution. He restored a narrower clause covering deliberate false alarms at canal gates. Their disagreement acquired verbs, thresholds, and responsible hands.

The third order concerned the public shelters. Halden had assigned every bed by ward capacity, a clean proportion that favored the buildings with stronger pipes. Nera turned the page sideways and recalculated from likely apartment loss

instead. Low Glass required more places despite having smaller certified shelters. The excess had to move through the civic tunnel into Assembly storage rooms.

"You intend to put families here?" Halden asked.

"I intend to keep them above freezing."

"This chamber has one sanitary line."

"Then your facilities clerk has seven days to add two."

Halden marked the work order in the margin. Opening the Assembly to canal families would anger members who treated public access as a security hazard. It would also put the consequences inside the building, where no forecast contour could soften them. He assigned the west committee room as a child sleeping area and struck its ceremonial furniture from the protection list. Nera read the changes, then initialed the shelter table as witness only.

When the orders could survive contact with frightened people, Halden set the disclosure framework on top.

"You could make this safer," he said.

"Safer for whom?"

"For enough people that the claim has a city left to answer it."

Nera read the final summary page. Halden knew the sentence that would stop her. He had argued with his clerk over it that morning and left it because the legal office would reject stronger language before any process began.

She found the passage with one finger.

"The Assembly acknowledges potential injury to forecast subjects occupying the modeled Afterclime interval." Her voice stayed quiet. "Forecast subjects."

"That wording passes current Charter review."

"They sent us a name."

"The summary opens a procedure in which the category can change."

"It opens by denying the fact the procedure exists to examine."

"If we begin with claimant, opponents challenge the entire framework before the first briefing."

"Then they should challenge it in public."

"They may win."

"They already win each day this phrase governs the account."

Halden returned to his chair. Fatigue pressed along the old break in his right wrist, an injury from a ration-gate crush before he joined the Assembly. The medical report called it minor crowd compression. His mother had called it proof that officials stood behind gates because they disliked being cold with everyone else.

"I can change subjects to affected populations," he said.

"Populations still makes their citizenship conditional."

"The Charter recognizes no citizenship across time."

"The annex does."

"The annex supplies a claimant clause, not a constitution."

"Then release it and let us discover what it supplies."

Above them, a heavy door closed in the civic tunnel. The pumps changed note as another route drew heat from the building. Halden imagined the city through pipes rather than districts: each valve opening because another narrowed, every protected room connected to a colder one.

He placed a signing seal beside Nera's hand. "Seven days."

She removed the amended emergency orders and returned the disclosure summary.

"I will authenticate these safeguards if you sign them," she said. "I will not sign a document that calls people forecast subjects after their message reached my desk."

"Without your name, the framework lacks notarial credibility."

"It should."

"Nera, an uncontrolled release may produce the very deaths you want the ledger to count."

"Then prepare openly for those risks. Do not purchase order with their legal absence."

She shut her document case. The return strip remained on the table between the ration book and the forecast glass.

Halden could have ordered the case seized. Restricted founder material had left authorized rooms under weaker pretexts. He could have suspended her seal pending inquiry, isolated Canal Nine, and let the working return collapse into a technical rumor. Every action had a lawful form. He had spent his career learning which forms a frightened city might survive.

Instead he signed the three emergency orders.

The first stroke committed Warm Ward stores. The second put Bellweather seed under cooperative control. The third protected Low Glass furnace salt. He sanded each signature and gave the pages back to her.

"Tomorrow," he said, "I will ask the Assembly for a closed risk session."

"I will bring the record."

"They may order the channel sealed."

"Then they will have to name what they are sealing."

Nera stood. She left his pencil beside the unsigned summary.

After she had gone, Halden reopened the Year Eleven ration ledger. The

anonymous clerk's remarks continued for two pages, then stopped halfway down a household line. No one had recorded whether the clerk died, fled, or received an order to write only quantities.

On the forecast glass, the disclosure branch kept running without its proposed notary. Store demand climbed. Canal attendance fell. Beyond the modeled city, the field for Year 360 remained blank.

Halden moved the return strip into that empty area and lowered the reading weight across its edge so the fans could not shift it. Then he drew the unsigned summary close. The phrase forecast subjects waited above Nera Voss's vacant seal box, exact, admissible, and no longer capable of passing for neutral.

Chapter 11 — A Cold Account

Nera held the season seal above the blank emergency line until the brass stopped trembling in her hand.

The Office clock showed six minutes before dawn on Day Thirty-Two. Canal Nine had not yet taken its morning load, but the pipe below the workroom already gave off enough heat to loosen the binding paste. The ledger lay open under three witness lamps. Its active schedule showed a broad blue transfer crossing the orchard belt at noon, stripping a predicted frost front from Bellweather and placing the cold account one hundred thirty-seven years forward.

Edrin stood opposite her with his own seal still in its case.

"You have not met the second-signature condition," he said.

"I have met the emergency threshold."

"Those are separate."

"I know what separate means."

Sumi Quill crouched at the disconnected forecast feed beside the wall. She had opened the silver housing and clamped a mechanical phase reader across its contacts. The missing fingers on her left hand rested clear of the teeth. "Argue more quietly. The drum trace is clean enough to hear you through it."

Nera lowered the seal without touching paper. The Charter allowed one senior notary to freeze an authenticated transfer for forty-eight hours when new evidence showed immediate material danger. It required a deputy's signature if the danger lay outside the recognized present. Her filing named Holdfast Kest and the winter line as materially burdened communities. Edrin had spent the last hour testing whether those names could enter a legal field designed for canals, wards, and crop cooperatives.

Outside the observation slit, a heatworker rang the preflow bell. Dawn had not reached the canal surface. Steam filled the narrow view like unbound paper.

Edrin said, "Read the scope again."

Nera did.

"Scheduled belt transfer F-223-32 is suspended at the primary phase entry for forty-eight hours from authentication. Local safety systems remain active. Present load shall be distributed by existing ward and agricultural hazard tables. No additional transfer may be attached to the frozen line. The purpose is preservation of evidence and prevention of continued injury to alleged downstream claimants pending Assembly review."

"Alleged," Sumi said without looking up.

"Required," Edrin replied.

"Convenient."

“Admissible.”

Nera watched the active schedule brighten as Canal Nine approached flow. “Test the return signature.”

Sumi tightened the clamp. The phase reader contained no civic software, only a coil, a smoked-glass needle, and a strip of frost-sensitive paper. If the ledger had fabricated the messages, it could not force the isolated coil to reproduce their phase scars. If an operator in their year had injected the signal, the delay pattern would show local harmonics.

The needle moved.

Six long marks crossed the paper. Six short followed. A narrow sawtooth appeared under each one, too fine for the eye until Sumi slid a magnifying plate over the strip.

“Remote phase wear,” she said. “Same spacing as the first storm signature.”

Edrin bent over the glass. “Could the current drum generate that wear after inversion?”

“Not unless it has been running for a century while we failed to notice.”

“Answer as an engineer.”

“I did.” Sumi shifted the magnifier. “The scar repeats every return cycle, but its edge decays toward us. Local injection decays away from us. Direction holds on all eight authenticated strips.”

Edrin took out a ruler and measured the marks himself. He copied the distances into the red book, then made Sumi reverse the strip and repeat the explanation. Nera understood his caution. A freeze imposed cold on people who had arranged roofs, shifts, and crops around the white schedule. If the Office used an invalid message to trigger it, every later claim would carry the defect.

The preflow bell rang twice.

A line of text surfaced in the disconnected ledger field. It arrived with gaps where the channel could not carry enough characters.

> KEST READY TO MEASURE LOAD. DO NOT EXTEND FOR OUR COMFORT IF YOUR PEOPLE LACK SHELTER. REPORT EXACT RETAINED FRONT.

Siv’s restraint unsettled Nera more than a demand would have. Holdfast Kest had recorded deaths across nine compound winters, yet its calendar keeper had spent part of the narrow channel warning them not to create indiscriminate suffering.

Edrin read the message. “The sender anticipates the freeze.”

“We told Siv the filing hour,” Nera said.

“You told the sender.”

“Siv Ren has supplied a name, role, settlement record, storm verification, and a child’s calendar sequence matching the phase.”

"A person under current law still requires contemporaneous identity."

"Current law also requires me to state the danger honestly."

He straightened. "Then do not confuse proof of origin with proof of legal status."

"I haven't."

"You wrote claimants."

"I wrote alleged downstream claimants, as you required."

Sumi snapped the reader housing closed. "If you two reduce a human being to a preposition, the noon line will open while you admire your grammar."

Edrin ignored the insult. He turned to the witness tiles arranged beside Nera's filing. The first held the Founder Sum clause. The second carried the cold-account figure and fixed offset. The third showed the storm signature Siv had sent before the storm existed in current forecasts. The fourth recorded Halden's signed emergency safeguards from the night before.

He lifted the fourth. "These orders alter present risk."

"They reduce it," Nera said. "Low Glass furnace salt cannot leave the city. Bellweather controls its own seed reserve. Public shelter thresholds now follow apartment heat loss."

"Warm Ward stores remain closed unless those thresholds trip."

"That order came from the Steward. My filing cannot change distribution."

"Yet."

Nera heard the warning and the possibility in the same word.

A second heatworker called from the gantry below. Canal valves answered with a sequence of iron impacts. The active transfer widened on the ledger. Frost over Bellweather would begin to lift within twenty minutes if they did nothing. By noon, the cold front would enter the phase system as a managed margin. In Year 360, Siv would receive whatever compound weather the system made from it.

Edrin placed his ruler down. "I want a live null test."

Sumi stared at him. "Now?"

"If I sign, the filing rests on a directional signal. I want proof the channel weakens when the schedule carries no accepted load and strengthens when we retain a measured fraction."

"That test itself changes the front."

"By how much?"

"A tenth of a degree across one canal sector for ninety seconds."

"Within ordinary balancing authority?"

"Yes."

"Do it."

Sumi looked to Nera.

"Run it," Nera said.

They moved to the observation slit. Sumi opened the manual balance panel and pulled the Canal Nine draw back by one mark. The pipe note under the floor dropped. On the return needle, Siv's carrier thinned until only two long pulses remained visible. Sumi held the balance for ninety seconds, then restored it. The carrier thickened.

"Now accept one mark locally," Edrin said.

"You asked for a null."

"I asked for direction."

Sumi set her jaw and advanced the local-load lever. Cold touched the slit almost at once. A clear rim formed along the inner glass, no wider than thread. The return needle swung hard enough to scratch the smoked strip.

Text appeared beneath Siv's prior message.

> CHANGE RECEIVED. KEST WEST THERMAL WALL RISES 0.03. SOURCE REDUCED ORISON-BELT FRONT. CONFIRM YOUR COST.

Nera looked through the slit. The heatworker below had pulled his collar higher. Ninety seconds and one balancing mark had reached his skin.

"Our cost begins at Canal Nine," she said. "Record the local temperature."

Sumi read the gauge. Edrin wrote it down, including the worker exposure estimate and the restored flow time. Then he turned the message strip over and checked its phase scar against the mechanical trace.

No one spoke while he compared them.

The dawn bell sounded across Lume. Light entered the canal steam and turned the observation slit white. On the ledger, the noon transfer advanced another minute toward automatic execution.

Edrin closed the red book. "The evidence meets emergency admissibility for a temporary act. It does not establish citizenship, standing beyond the interval, or final causation for the casualty count."

"Write every limitation," Nera said.

"I intend to."

"Will you co-sign?"

He looked at the filing, then at the frost thread left by Sumi's accepted mark. "I will co-sign the danger. I reserve judgment on the remedy."

"That is all this authority permits."

Edrin removed his seal.

They worked through the formal sequence. Nera read the transfer identifier. Sumi verified the phase entry against the physical drum. Edrin named each

unresolved objection. The schedule clerk, called in from the corridor, witnessed the current hour and the absence of any broader freeze. Nera pressed her brass into the filing first. Edrin placed his mark beside hers, offset by the narrow legal distance that meant concurrence without shared authorship.

The season seal heated under their hands.

For one breath, nothing changed.

Then the blue transfer line narrowed to black. A bracket closed around it. The ledger displayed the authentication hour and began counting down from 48:00:00.

Canal Nine sounded its hazard bell.

The cold front returned to the present.

It arrived first as pressure in Nera's teeth. Frost covered the lower observation pane, not the impossible fern from the return but ordinary ice carried on a real draft. Outside, the heatworker swore and secured his sounding rod before the gantry surface glazed. Sumi opened local radiator bypasses according to the hazard table. Steam rolled low across the canal.

The city board woke into amber warnings. Bellweather retention: suspended. Warm Ward draw: under recalculation. Low Glass canal loss: increased. Public shelter forecast: pending.

Edrin watched the warnings multiply. "We should notify the Steward before the boards reach the markets."

"He receives the same filing notice as every affected authority," Nera said.

"He will regard that as provocation."

"He signed the safeguards."

"He did not sign this."

"No one with the power to continue the transfer should receive private time to prevent its lawful suspension."

The schedule clerk had gone pale. "Senior Voss, Bellweather is calling. Their mesh forecast changed six degrees in three minutes."

"Put them through."

Lev's voice arrived over the wall horn under a wash of orchard wind. "Nera, tell me this comes from your hand."

"It does. One transfer, forty-eight hours. Your cooperative's hazard table now governs the retained frost."

"You froze it at dawn?"

"The evidence cleared at dawn."

"Redleaf crews are opening mesh for the old forecast. I need the front speed."

Sumi pulled the live belt contour across the glass. "Western edge reaches Bellweather in two hours eighteen. Temperature floor minus four at exposed rows,

minus two under intact mesh. False warm pocket on the southern slope may delay the first drop."

Lev repeated the numbers to someone beside him. Orders followed in the background: close west panels, wake Yara's crew, move frost chalk to the nursery rows.

"Nera," he said when the horn cleared, "this will take buds."

"I know."

"Do you know how many?"

"No. Send losses as they develop. Do not let the city model estimate your hands out of the record."

A pause carried wind and a metal latch striking shut.

"Good," Lev said. "Then write this too. We can take a frost with warning. We cannot plant under a government that changes the sky and calls the change neutral."

"I'll write it."

The line closed before either of them could find a private sentence inside the public one.

By the second hour, cold reached the western tram rails. Drivers fitted ice teeth to the wheels. Canal crews opened ward returns that had not moved in years. Nera authenticated each redistribution notice without leaving the workroom. Every notice showed where the retained front entered present life: a school dismissing children before sleet, a bathhouse extending public hours, a glassworks losing one furnace cycle to preserve apartment flow.

Beren Olt called from the Canal Six junction before the third hour. His voice carried through a field handset that knocked against metal with each step. "Your hazard table gives Warm Ward return water priority because their pipes hold pressure."

"That table predates the freeze," Nera said.

"Cold does not care when you printed it. My crews opened the Low Glass bypass and found half the insulation stripped for last year's civic repair. We can keep both routes alive for six hours, perhaps eight. After that, choose who gets the hot return."

Nera pulled the junction diagram onto the glass. The legal table indeed favored pressure integrity. Warm Ward's newer pipes could receive hot water with less immediate loss, so the automatic schedule protected them. Low Glass would then absorb colder discharge while its workers maintained the bypass.

"Can you reverse the sequence safely?" she asked.

"Not with the automatic valves. Manually, yes, if I put two people in the collar."

"What exposure?"

"Four minutes each at the current surface reading. Longer if the old wheel binds."

"Do not send them in yet."

"You have another source of heat?"

"No. I have an order requiring Warm Ward soil stores to support public thresholds."

Nera called Halden's duty clerk on a second line. The clerk tried to refer her to the ward council. She read the signed store order and named the Low Glass school already within one degree of opening as a shelter. The duty clerk asked for twenty minutes to verify. Nera gave him five and kept Beren on the field line while sleet rattled against his handset.

At four minutes, Warm Ward released one soil loop. Its heat entered the civic return below the garden and lifted Canal Six pressure enough for Beren to change the sequence without putting workers inside the collar. Low Glass apartments gained three hours before the next threshold.

"Write who opened it," Beren said.

"Steward order, notarial activation, ward operator Pell An."

"Good. Do not write that the system protected us. A person opened a valve after we forced the question."

Nera entered the operator's name in the impact annex. Beren's demand altered how she recorded every later intervention. Automatic response gave way to named hands, withheld permissions, and minutes of exposure avoided.

Warm Ward filed three objections. The first claimed damage to municipal gardens. The second warned of respiratory risk among older residents. The third challenged the use of messages from an unrecognized year. Nera admitted the health evidence and rejected the priority request. She forwarded the legal challenge to Edrin, who surprised her by marking it non-suspensive.

"You could have let their petition stop the clock," she said.

"The clock has started lawfully," he replied. "An objection cannot pretend it has not."

At hour six, Sumi opened the return window.

Bandwidth had increased with the cold held in Year 223. The channel still carried only fragments, but the letters arrived with fewer losses. Siv sent temperatures from three Kest walls, two terrace shelters, and a wind register outside the settlement. Each showed a small reduction in incoming load at the predicted phase.

> KEST CONFIRMS REDUCED DOWNSTREAM LOAD. WEST WALL DROP SLOWS. TWO SHELTER FIRES NOW WITHIN RECOVERY RANGE. NAME YOUR INJURED AND CROP LOSS.

Nera read the last sentence twice.

Siv had not thanked them. Gratitude would have made the transfer seem like a gift. The message asked for present names because repair that erased its own cost would teach the wrong lesson in either year.

She replied within the narrow allotment.

> ONE CANAL WORKER COLD EXPOSURE, STABLE. BELLWEATHER LOSS UNKNOWN. CITY SHELTERS OPENING. FREEZE EXPIRES EXACTLY AT AUTHENTICATED HOUR UNLESS ASSEMBLY EXTENDS.

Sumi counted the characters. "You have room for nine."

Nera added: NO PROMISE.

The phase drum took the message.

Daylight climbed over the city without warming it. From the Office roof, the cloud combs turned west toward the retained front. They could soften hail but not remove the cold without attaching it to another account. Sumi issued instructions that used pressure, timing, and existing heat only. Each limitation made the sky less obedient.

At midday, Nera crossed the civic tunnel to inspect the first public shelter opened under Halden's order. Assembly furniture stood stacked behind a rope. Facilities workers had added one sanitary line and marked the floor for another. Six Low Glass families occupied the west committee room, coats still on, heat receipts clipped to their bags as proof of address.

A ward clerk asked Nera whether the freeze would continue beyond the second dawn.

"I do not hold that authority," she said.

"Who does?"

"The Assembly."

"When will they decide?"

"They have not scheduled a hearing."

The clerk looked at the sleeping mats, then at the gold Assembly names above the door. "Weather has already scheduled one."

Nera carried that sentence back to the Office and entered it in the public-impact annex.

At hour twenty-nine, Bellweather reported the first split redleaf buds. At hour thirty-four, the Warm Ward garden lowered its soil temperature by two degrees to keep a Low Glass school above its shelter threshold. Halden's office issued no public statement. His emergency orders continued to function under their signatures.

The countdown reached 11:07:14 remaining when an Assembly courier arrived through the locked workroom door. He wore formal tunnel gray and carried a black

summons tube warm from the pneumatic line.

Edrin inspected the seal. "Closed risk authority."

"Of course," Sumi said.

Nera broke the tube. The summons named Halden, six Assembly delegates, a growers' representative, a heatworkers' representative, both notaries, and the chief phase technician. It ordered all supporting records into the heated tunnel chamber. Public attendance was barred under continuity-risk rules.

The hearing would begin at hour forty-one.

Nera set the summons beside the ledger. The black transfer line remained frozen. Beyond the observation slit, sleet struck Canal Nine and melted into the steam. On the countdown, seven hours stood between the Assembly's first question and the exact instant her authority ended.

Chapter 12 — The Winter Line

At hour twenty-six of the freeze, Holdfast Kest began passing voices it could not afford to keep waiting.

They arrived as text on Canal Nine's frost paper. Sumi fed the narrow strip through the return reader while Nera stood at the ledger with a pencil and a ruler. Each message carried Kest's phase mark first, then a second signature weakened by distance and relay. The first came from a terrace settlement called Silt Rung. The next named Deep Arable Four. By noon, seven places had entered the record.

> KEST RELAY. SILT RUNG REPORTS THREE COMPOUND FRONTS IN EIGHTEEN MONTHS. CURRENT POPULATION 6,904. REQUEST INCLUSION BY NAME.

Nera measured the scar beneath the letters. Kest's carrier formed the strong outer line. Within it, another rhythm pulsed at a slower interval.

"Could Siv invent the inner signature?" she asked.

Sumi lifted the paper toward a blue lamp. "Anyone can invent words. Not this phase wear. The relay crossed at least two active drums before Kest sent it backward."

"How far?"

"Distance does not translate cleanly through their damaged grid. More than one weather region. Less than the continental belt."

Nera entered the settlement as a source with independent phase evidence. The Office ledger had no field for a community outside its own year. She used the contested-counterparty annex and wrote the name in full.

The return reader clicked again.

> DEEP ARABLE FOUR. 11,202 LIVING. FORMER ROOT SEED RESERVE. NOW SHELTER CLUSTER. WE RECEIVE YOUR SOFT WINTERS AS LATE COLD.

Sumi swore softly. "A root reserve should never need habitation walls."

"You know the site?"

"I know the original plan. Deep Arable stations stored crop genomes below frost penetration. If people live inside one, the surface terraces failed or the shelters did."

Nera wrote both possibilities as unverified. She had learned how quickly a reasonable inference acquired the authority of a fact once it entered Office paper.

Through the phase-room windows, sleet ran down the black glass in diagonal lines. The cold retained in Year 223 had strengthened the return enough for Kest to send more than one short turn per opening. It also settled across Lume. Canal crews moved under white canvas capes. The tram above Low Glass passed with ice teeth grinding against the rail. Every additional character reached Nera through present discomfort that someone could point to and name.

Edrin had gone to prepare the closed-session record. Halden's summons waited in Nera's case. The hearing would begin in fifteen hours. Until then, the emergency clock continued without permission from anyone in the Assembly chamber.

Sumi adjusted the reader tension. "Next carrier has four nested scars."

Text came slowly, losing letters at each frost seam.

> KEST RELAY FROM NORTH CUT / PALE BASIN / HOLLOW MILE / SOUTH FURNACE. COMBINED VERIFIED LIVING 83,610. ALL CLAIM LOAD FROM SETTLEMENT YEARS 210-228.

Nera stopped at the date range. "Some of those accounts have not happened yet."

"In our year," Sumi said.

The active schedules on the side glass included transfers planned through Year 228. The public forecasts treated them as prudent reserves, commitments written in advance against harsh seasons not yet experienced. Along the winter line, those commitments had already arrived as weather.

Nera opened the future schedule index. "Can we match a relay signature to a planned front?"

"Not from the text alone."

"What would you need?"

"Phase onset, local pressure, duration, and the original drum family. Ask for one event, not their whole history."

Nera composed the request. The channel allowed more characters under the retained load, but not enough for politeness and precision together.

> TO SOUTH FURNACE: GIVE ONE INHERITED FRONT START HOUR, PRESSURE DROP, DURATION, DRUM SCAR. WE WILL TEST AGAINST UNEXECUTED SCHEDULES.

Sumi counted and sent it.

The reply could not arrive until Kest's next relay window. While they waited, Nera worked through the settlement names already received. Siv had included population methods. Some places counted only bodies with current shelter shares. Others counted travelers if they had spent twelve nights inside the thermal wall. One settlement counted a person after the community had accepted responsibility for that person's heat. None used the Charter's present-ratepayer language.

A shorter message entered between relays.

> SIV REN. YOU ASK HOW MANY STAND ON WINTER LINE. KEST HAS 61 VERIFIED SETTLEMENTS. OTHER RELAYS CLAIM 204. TOTAL RANGE 1.8 TO 2.1 MILLION.

Nera copied the number without rounding.

Two point one million people lived inside the broadest verified range. The quantity covered more than Holdfast Kest, more than the settlements capable of

reaching Kest, and perhaps more communities with broken drums that could not answer at all. A range preserved uncertainty. Its upper edge still exceeded Lume's population many times over.

Sumi read over her shoulder. "The Assembly will attack the method."

"They should test it."

"They will use the test as a reason not to say the number."

"Then we say both." Nera wrote beneath the entry: UPPER CLAIMED AND PARTLY VERIFIED WINTER-LINE POPULATION: 2.1 MILLION. DIRECTLY RELAYED SETTLEMENTS: 61. ADDITIONAL NAMED: 143. DUPLICATE REVIEW PENDING.

Nera sent Siv a request for the counting rules and waited through two carrier drops. Kest answered with a compressed table. The first column named each relay settlement. The second listed its last direct count. A third supplied the date and method. Some counts came from shelter heat rolls, others from meal allocation, burial subtraction, or crop-work rosters. Kest had excluded any number older than three local years from the verified subtotal.

One line carried no population at all: OPEN TERRACES EAST, CONTACT LOST AFTER DRUM FRACTURE.

"Why list them without a count?" Nera asked.

Sumi traced the failed relay path. "Because silence can mean broken equipment. It cannot prove an empty settlement."

The rule corrected an old Office habit. Current forecasts treated missing sensor data as zero occupancy unless another authority supplied a claim. That assumption kept blank regions from consuming model capacity. Along the winter line, the same convenience could erase a community precisely because its phase drum had failed under inherited weather.

Nera added a separate category: NAMED, CONTACT LOST, POPULATION UNKNOWN. Eleven settlements entered it.

Siv used the next window to challenge the upper range without being asked.

> DO NOT CALL 2.1 VERIFIED. UPPER COUNT INCLUDES NAMES REPEATED THROUGH TWO RELAYS AND TRAVELERS WHO MAY HAVE MOVED. SAY WHAT WE KNOW AND WHAT COLD PREVENTS US KNOWING.

Nera showed the line to Sumi. "Halden expects negotiation inflation."

"This cuts their own number."

"It preserves it as a range."

"Which an honest engineer does when the instrument shakes."

Nera composed a revised entry for Siv's review. It separated direct counts, recent relayed counts, duplicate exposure, and communities with failed contact. Kest answered with two corrections. Pale Basin's shelter roll had been updated six months later than the relay date. Hollow Mile counted a caravan also recorded at

South Furnace. The upper edge remained unchanged because other named communities had only minimum counts.

A final sentence followed the corrections.

> WE NEED YOU TO UNDERSTAND SCALE. WE DO NOT NEED A LARGE NUMBER MORE THAN WE NEED TRUE NAMES.

Nera placed that message beside Halden's forecast branch in her session file. The sentence did not ask the Assembly to accept an uncertain maximum as fact. It asked them to stop using uncertainty as permission to treat millions of possible lives as none.

Her pencil hovered over the category title. "Winter-line population" described geography shaped by cold, not a political union. "Claimant population" assumed the legal result before the hearing. Siv had rejected descendant. Nera wrote LIVING PERSONS REPRESENTED OR NAMED THROUGH THE KEST RELAY and kept the full range below.

The phase room door opened behind them. A young worker in a leather apron carried a narrow crate against his chest. Frost had silvered his hair at the temples.

"Tovan?" Nera said.

Her son shifted the crate to one arm. "Master Ilek sent replacement vent glass. Your requisition said legal observation panel, priority hazard."

"I expected a courier."

"So did I. The courier tram stopped at Low Glass."

Sumi took the crate before he dropped it. "You walked from Warm Ward carrying certified glass?"

"Only from the east junction. The case has suspension ribs."

"It has one cracked rib."

"That happened before I touched it."

Under other conditions Sumi might have smiled. She set the crate on the maintenance bench and began checking its seals.

Nera looked at Tovan's thin work gloves. "Where are your winter gloves?"

"At the shop. Nobody forecast sleet when I left home."

"The boards changed yesterday."

"The shop board called it a temporary notarial anomaly."

He used the phrase without accusation, which made it more pointed. Nera took his hands and turned them palm up. His fingers showed cold redness but no white patches. He pulled away after she had checked them.

"I am working," he said.

"So am I."

"That seems to cover everything lately."

The return reader clicked. Tovan glanced toward it.

Nera moved between him and the frost paper by habit, then disliked herself for the movement. "This room holds restricted evidence."

"I delivered restricted glass."

"You are not cleared for the channel."

"I know. Master Ilek made me recite the handling rule twice."

Sumi examined the crate's season seal. "He can wait by the warm pipe while I fit the pane. Sending him back into sleet without circulation would violate my hazard table."

Tovan sat on the low bench beside the radiator feed. Nera returned to the ledger, aware of every line that could arrive within his hearing.

Sumi removed the cracked observation panel. Cold entered through the open frame with enough force to lift the corner of Nera's notes. Tovan steadied them using a ceramic gauge weight. His eyes found the population line.

"Two point one million what?" he asked.

"People," Nera said.

"In Year 360?"

"Across settlements relaying through Kest. The count remains under verification."

He read the names below it. "These are all cities?"

"Some are shelter clusters. Some grew from agricultural stations. We do not know their legal forms."

"Do we know they are people?"

"Yes."

The certainty left her before she could moderate it. Sumi kept working at the frame.

Tovan touched none of the papers. "The shop says your Office froze weather for a message that might come from a forecast model."

"The shop is wrong."

"Halden's notice says disputed future source."

"Halden has not issued a public notice."

"The ward council copied his risk summary to registered workshops."

Nera understood then. Managed language had started moving without the managed-disclosure plan. Forecast subjects, disputed source, temporary anomaly. Each phrase reduced the claim before people saw the evidence.

She opened Halden's signed emergency order and showed Tovan the seal. "The Steward authorized safeguards because the signal has physical proof. Edrin

authenticated its direction. I froze one transfer under emergency authority. The Assembly decides tomorrow whether it continues.”

“Will it?”

“I do not know.”

“Does that mean the cold goes away if they say no?”

“It moves.”

“To them.”

“Yes.”

Tovan looked toward the reader. Frost paper emerged one narrow increment at a time.

The next Siv message began while Sumi pressed new glass into the frame.

> NERA VOSS. KEST ASKS WHY YOUR PUBLIC WORDS CALL US DESCENDANTS. DESCENT NAMES YOUR LINEAGE, NOT OUR CLAIM. WE ARE LIVING AT THE BURDEN END.

Tovan read it before Nera could remove the strip.

“Did you call them descendants?” he asked.

“I used the term in an early filing.”

“You always call me your present responsibility when people talk about future policy.”

Nera felt the accuracy of that. During ward meetings, she had rejected speakers who invoked unborn generations while refusing safer vent rules for apprentices standing in front of them. She had treated Tovan’s visible life as a correction to abstract descendants. Now the same word pushed Siv and millions of others away from the tense in which they breathed.

She wrote a reply.

> AGREED. I USED DESCENDANT TO MARK CAUSAL DUTY AND MADE YOUR CITIZENSHIP RELATIVE TO US. CORRECTION ENTERED: LIVING WINTER-LINE PERSONS.

Sumi paused before sending. “You have thirty-six characters left.”

Nera added: MY SON IS HERE. HE READS THIS.

Tovan looked at her. “Why tell them that?”

“Because Siv asked why our language fails. You are part of the answer.”

“I didn’t agree to become evidence.”

“No.” Nera lifted her pencil. “Do you want the sentence removed?”

The channel window remained open. Sumi waited with two fingers above the send contact.

Tovan rubbed warmth back into one knuckle. “Leave it. But do not call me your

reason for keeping the transfer.”

“I won’t.”

“You were going to.”

Nera did not deny it. She had pictured his workshop, Bellweather food, the Low Glass shelter, every present body that could suffer from abrupt cold. Tovan’s face had stood among them with a brightness she did not grant strangers. Love did not make the unequal scale dishonest by itself. Hiding the scale did.

Sumi sent the message.

The new pane settled into its seal with a low ceramic note. Tovan checked the pressure clips in sequence, tugging each one rather than trusting its color mark. His hands steadied when given material work. Nera recognized the habit and knew he had learned part of it from her.

He found one clip seated against a grain of old sealant and opened it again. A hurried fitter might have left the indicator green. Tovan scraped the channel clean, warmed the compound between his palms, and reset the clip until the glass carried pressure without a tremor. The work took four minutes. Nothing about it looked symbolic, which made Nera trust it more.

South Furnace answered through Kest before Siv replied. Its inherited front began at a phase hour three days beyond their current schedule. Pressure, duration, and drum family matched a transfer Lume had approved for Year 224 but not yet executed. The correspondence could not result from weather already sent.

Sumi laid current and future traces side by side. “They know the shape of a front we have only authorized.”

“Could our authorization leak forward through ordinary phase activity?” Nera asked.

“Not as weather. The load has not entered a drum.”

“Could a current operator have sent the schedule by another channel?”

“No other backward channel exists. Forward archives reach them with time, but that record would arrive one hundred thirty-seven years after execution. Their event sits in their lived past because our authorized transfer sits in our near future.”

Tovan had stopped fitting clips. “We already put storms on their calendar before they happen here?”

“The schedule commits them,” Nera said.

“Then your freeze stopped one receipt before the weather left the shop.”

“Something like that.”

He frowned. “Not something like that. Either it did or it didn’t.”

Nera looked at the matching traces. “It stopped this transfer before execution. South Furnace confirms other authorized transfers already shape their record.

Yes.”

Sumi certified the match on a witness tile. Nera added it to the closed-session case. The evidence changed the hearing. Halden could no longer describe future reports as interpretation of weather already observed. A settlement had supplied the exact body of a front that remained only an authorization in Year 223.

Siv’s reply came at the end of the window.

> TOVAN VOSS, IF YOU CONSENT TO NAME. YOU ARE NOT OUR ANCESTOR. WE ARE NOT YOUR CHILDREN. YOUR WARMTH AND OUR SHELTER SHARE AN ACCOUNT. ASK WHO SIGNED IT.

Tovan read his name on the frost paper. Nera watched him become still, not from fear but from the shock of an unknown adult addressing him without making him a symbol. Siv had asked consent after receiving the name, imperfectly and plainly.

“He thinks you signed my warmth,” Tovan said.

“Siv does not know who signed each schedule.”

“You did sign schedules.”

“Yes.”

“This one?”

Nera pulled the frozen transfer history. Her season seal appeared on the verification line from eleven months earlier. She had authenticated the canal capacity, orchard allocation, and modeled continuity margin. The future date had remained hidden.

“My seal verified its inputs,” she said.

“Did your seal make it happen?”

“It permitted the schedule to proceed after the Assembly authorized it.”

“So yes.”

“So I held one necessary part.”

Tovan’s jaw moved with the old resistance he showed when workshop masters hid responsibility inside team language. “If the glass I fit breaks, Master Ilek says the apprentice who signed the clip sheet cannot blame the furnace order.”

Nera examined her seal impression on the old line. “Master Ilek is right.”

The phase room warmed after the new pane closed, though sleet continued beyond it. Sumi packed the old cracked glass into the crate and marked it for evidence rather than disposal. Nothing that had touched the return channel could leave without a chain of custody.

Nera sent Siv one final message before the carrier fell.

> TOVAN CONSENTS TO NAME. I SIGNED VERIFICATION ON THE FROZEN TRANSFER BEFORE I KNEW ITS FUTURE DATE. I WILL ENTER THAT FACT AT

HEARING.

The reply window closed before an answer could come.

Tovan put on the spare canal gloves Sumi found in a locker. They hung loose at the fingertips. He carried the cracked pane case to the door, then stopped beside the ledger where the old schedule and the new population annex lay open together.

"Warm Ward says we paid for moderation through heat charges," he said.

"You did pay the city."

"But the city sent part of the cost forward."

"Yes."

He looked through the replacement glass at Canal Nine. A crew below spread frost chalk along the gantry, each mark bright against wet metal. The retained front had made the canal dangerous. It had also made the winter line easier to hear.

Tovan touched the heat receipt clipped inside his work apron, the small brass-edged card that proved his workshop fees were current.

"Does my warmth have a receipt?" he asked.

Chapter 13 — The Closed Session

The Assembly had hung the emergency clock where every witness would have to look past it to speak.

It occupied the heated tunnel chamber's far wall, red figures above a locked service door. 06:54:12. The remaining time counted down without sound. Beneath it, condensation gathered on old white tiles and ran toward the floor channel.

Nera placed the winter-line population annex on the witness table. Edrin set the admissibility book beside it. Sumi carried the return traces in a flat ceramic case strapped to her wrist. Across the central grate, Halden sat with six Assembly delegates under a row of brass heat fins. Their coats hung open. The tunnel had received enough flow to keep officials comfortable while sleet ticked against the street grilles overhead.

Lev Marek occupied the growers' chair with frost chalk on one sleeve. Beren Olt took the heatworkers' place and refused the padded seat, choosing a metal service stool beside the radiator map. Neither man greeted Nera. The clock stood between every private relation and the decision ahead.

Halden struck the table bell once. "This session proceeds under continuity-risk authority. No copy leaves the chamber unless released by vote. Senior Voss, state the act under review."

Nera remained standing. "Emergency suspension of transfer F-223-32, authenticated at first light on Day Thirty-Two. Scope limited to one scheduled belt transfer. The retained front remains in Year 223. My authority ends at the zero shown behind you."

A delegate from the Warm Ward glanced up at 06:51:40. "You filed without Assembly consultation."

"The Charter grants emergency authority because consultation may exceed the hazard interval."

"You had more than three weeks after first contact."

"I did not have admissible direction evidence until the live phase test."

Edrin opened the red book. "That statement matches the record. I co-signed after an isolated mechanical trace proved remote origin and directional coupling."

The delegate's gaze moved to him. "You previously challenged the source."

"I still challenge any claim beyond what the evidence supports. The temporary danger meets the required standard."

Halden said, "Enter Deputy Sol's distinction."

A clerk behind the privacy screen wrote it into the session ledger. The scratch of the pen carried through the chamber more clearly than the heat pumps.

Nera presented the frozen schedule first. She showed its legal identifier, original

authentication, and the forty-eight-hour bracket. Next came the Kest measurements after local load retention. Three wall temperatures, two shelter fires, one outdoor register. Each value changed at the phase time predicted by Sumi.

The agricultural delegate interrupted. "A change of three hundredths of a degree cannot establish material relief."

Sumi answered from her chair. "Not alone. The correlated carrier strength establishes direction. The wall response confirms reception. The transfer under review spans a much broader front than our ninety-second test."

"How much broader?"

"Enough to split redleaf buds and glaze two tram sectors here. Kest reports measurable shelter recovery. Exact future distribution remains unavailable at current bandwidth."

"So we impose known crop damage to obtain an unavailable result."

Lev leaned toward his microphone. "Bellweather lost buds because the Office changed a forecast we had treated as reliable. That harm belongs in the record. Do not use my orchard to call the other side empty."

The delegate turned. "You support extension?"

"I support an honest model. I have not seen one for extending this freeze beyond the next day."

"Then we agree."

"No. You want the old schedule restored because its losses arrive outside your hearing. I want staged retention with crop protection, labor input, and public notice. Those positions share no ground."

Nera watched Halden. He gave nothing away. The emergency orders he had signed sat in front of him, their thresholds already moving heat through the city. His disclosure summary remained absent from the session packet.

Beren tapped the radiator map. "Low Glass crews kept Canal Six open while the Warm Ward filed objections from rooms above twenty degrees. One soil loop moved only after Notary Voss invoked the Steward's order. If you extend, we need allocation authority now, not after a commission."

The Warm Ward delegate said, "Private and municipal stores cannot pass into general control on the testimony before us."

"They passed into your gardens on weaker evidence for years."

"Those stores preserve seed, respiratory health, and civic shelter capacity."

"Low Glass lungs count as respiratory health."

Halden rang the bell before the exchange could become procedural disorder. "Organizer Olt, state the condition you require."

"Benefit-weighted immediate load. Warm Ward takes 2.4 times Low Glass per

person until historical advantage receives full audit. Canal workers control exposure limits. Growers control crop triage. No official gets to spread cold equally and call equality fair."

The clerk's pen stopped, then resumed.

A central-district delegate addressed Nera. "Your filing contains no authority for that distribution."

"Correct. Only this body can change distribution."

"You created a hazard you lacked power to manage."

"I stopped a transfer after evidence showed the existing schedule moved hazard onto living people. The distribution defect already existed. It became visible here when the transfer stopped."

The clock read 05:38:19.

Halden requested the South Furnace match. Sumi opened the ceramic case and removed two transparent strips. One held the future settlement's reported pressure, duration, and drum family. The other held an authorized Year 224 transfer not yet executed.

"These shapes match within instrument tolerance," she said. "South Furnace recorded effects from a schedule that remains in our future."

"Could Holdfast Kest possess current authorization records through ordinary archival inheritance?" Halden asked.

"Eventually. Not before execution. Their account describes a lived front in their past. Our schedule has not entered the lattice."

Edrin added, "The match defeats accidental forecast coincidence as a complete explanation. It does not prove every relayed population count."

Nera placed the counting table before the delegates. "Kest does not claim that. Sixty-one settlements carry direct verification through the relay. Additional names and counts produce a range from 1.8 to 2.1 million, with duplicate review pending. Eleven named settlements have broken contact and no current count."

The outer-belt delegate lifted the page. "We cannot confer standing on a range."

"We can refrain from adding weather while we determine standing."

"By accepting severe weather here."

"By retaining one scheduled front for longer than two days, under a distribution plan this body has authority to write."

"Your son lives in the Warm Ward, does he not?"

The question entered with the false courtesy of relevance.

"He does."

"Your household receives moderated heat."

"Yes."

"Yet you ask growers and canal districts to absorb immediate loss from a policy whose benefits protected your child."

"I ask this Assembly to stop treating any beneficiary as proof that the account has no creditor. My own seal authenticated the frozen transfer before I knew its future date. Enter that too."

The delegate studied her for a private motive that might make the evidence easier to dismiss.

Halden intervened. "Household benefit belongs to distribution, not source authentication."

Beren gave him a hard look. "It belongs to both if only poor notaries can recognize theft."

Nera said, "My household's benefit should increase its present burden under a lawful formula. It does not alter the phase trace."

The clock passed five hours.

The legal officer asked to examine the freeze mechanism itself. A clerk carried the active ledger to the center grate without disconnecting its phase lead. The black bracket around F-223-32 pulsed with each drum cycle.

"Senior Voss, can you cancel this filing and issue another before expiry?" the officer asked.

"No. A notary may not renew emergency authority over the same hazard."

"Could Deputy Sol file a second suspension on independent grounds?"

Edrin answered. "Not lawfully if the material facts remain the same. Serial filings would convert temporary review into unauthorized policy."

The outer-belt delegate pointed to the South Furnace trace. "That evidence arrived after the first filing. New fact, new authority."

"It strengthens source verification," Nera said. "It does not create a separate transfer hazard. I will not use a technicality to hold weather after my seal expires."

Beren shifted on the service stool. "So the only person in this room willing to keep the line frozen refuses a loophole."

"I refuse hidden power. That includes power I want used."

"Easy to say when the load leaves your district after zero."

"It does not leave my district. Warm Ward benefits, Bellweather thaws, and Low Glass loses some retained cold. The burden reaches Year 360."

The legal officer requested a demonstration of expiry. Edrin placed a blank witness tile beside the countdown and called up the Charter sequence. At zero, the ledger would remove both notarial marks from the active bracket without deleting the filing. The record would persist. Control would revert to the Assembly-authorized schedule. No person had to press a key.

"Automatic restoration," the agricultural delegate said.

"Automatic execution of a prior political act," Nera replied.

Halden's attention sharpened. "Enter both characterizations."

The clerk did. On paper, they occupied adjacent lines. In the pipes, only one would operate.

Sumi leaned over the ledger. "Restoration also reduces return bandwidth. If you want further evidence after expiry, authorize Canal Nine to preserve inversion alignment. Otherwise the resuming front may knock the carrier out of phase."

"Can that alignment affect present weather?"

"Not while we merely hold the drum angle. It costs crew time and ceramic wear."

The delegates approved preservation by unanimous voice. It became the first decision of the session, narrow enough to pass because it did not alter who received the cold. Nera watched Sumi enter the order. Even refusal required maintenance if the Assembly wanted to hear what its refusal caused.

The Assembly recessed for twelve minutes while no one left the chamber. Privacy rules locked the doors until the vote. Clerks brought hot water and salt bread through the service hatch. Beren accepted water but not food. Lev broke his bread in half and carried one piece to Yara's crew representative waiting outside the chamber, only to meet the locked door. He returned and set the portion untouched beside his files.

Nera stood near the radiator map. The pipes behind it showed current flows in trembling amber. Warm Ward remained above target. Low Glass moved close to the school-shelter threshold again. Bellweather had no heat category, only crop-loss bands.

Halden joined her without a clerk. "You know the extension will fail."

"I know you have not recommended it."

"A recommendation without a distribution agreement produces a tie at best."

"Then recommend the agreement."

"Beren's ratio has one audit behind it and no full food model."

"The old allocation has no future-person term at all."

"I can count defects in both."

"The clock does not ask which policy deserves more study."

"No. It asks whether we replace an operating system with one assembled during a freeze."

Nera looked at the red digits. "You signed safeguards because you believed this cold might remain."

"I signed them because any lawful emergency requires preparation."

"You opened Assembly rooms to Low Glass."

"Six families now sleep under our committee names. I am aware."

"Does that change your recommendation?"

Halden's face held the fatigue of someone who had spent years making each concession small enough to preserve the power of granting it. "It changes what I will require after this session. It does not create votes that are not here."

The bell recalled them.

At 03:59:52, the legal officer presented three motions. The first would extend the transfer suspension for seven days under general hazard authority. The second would restore half the transfer and retain half for twenty-four hours. The third would allow the emergency act to expire while ordering a distribution and source review.

Sumi spoke before Halden recognized her. "The second motion cannot work as written. Half-transfer at the current drum angle will shear the carrier and create an unmeasured storm edge in both years."

The legal officer frowned. "Can it be corrected?"

"Not inside four hours. Give me a day and a phase crew."

"We do not have a day."

"You had years."

Halden struck the bell. "Technical objection sustained. Motion two withdrawn."

The removal narrowed the chamber. Extension or expiry. Present cold with no complete distribution rule, or resumed deferral with evidence of living recipients.

Lev presented Bellweather's loss sheet. Seven percent of exposed redleaf buds had split. The nursery rows survived under mesh. Frost barley showed no permanent damage yet. "Another seven days at this front would threaten the pear harvest unless cloud pressure changes," he said. "A day might be manageable. Seven without a staged forecast becomes hunger by administration."

The agricultural delegate seized on the phrase. "Then extension risks food."

"Everything here risks food," Lev said. "Deferral taught you to think one side had no mouths."

Beren submitted worker exposure. No serious injury had occurred. Two canal hands had cold numbness, and one glassworks crew lost wages after a furnace cycle shifted to apartment heat. He demanded immediate access to Warm Ward store levels before any further retained load.

The Warm Ward delegate refused disclosure under private utility protection.

"Then you want my crew to vote blind on how cold you can afford," Beren said.

Delegates began speaking over one another. Halden's bell failed to separate objection from answer. Behind him, the clock reached 02:41:08.

Nera did not raise her voice. She opened the Founder Sum witness tile and placed it beneath the central lamp. The clause became visible to the whole

chamber.

DOWNSTREAM PERSONS MATERIALLY BURDENED BY DEFERRED DISSIPATION
MAY CLAIM AGAINST THE SUM.

The room quieted enough for the pumps to return.

The legal officer said, "That annex lacks enacted procedure."

"It does not lack people," Nera replied.

"Senior Voss, your conclusion exceeds the text."

"The text names persons, burden, and claim. Procedure may remain disputed. Their categorical absence does not."

Halden looked at the tile longer than he had looked at any model. "Enter the original language."

The legal officer protested that the closed record could contaminate later annex review.

"Enter it," Halden repeated.

The clerk copied the clause.

At 01:53:30, the extension motion came to preliminary count. Two delegates supported seven days. Three opposed. One reserved pending Halden's recommendation. He requested a final risk statement from each witness.

Sumi said the return channel would weaken when the transfer resumed, but could survive if the phase drums remained aligned. She could not promise intelligible text under full load.

Beren refused continued retention without immediate store disclosure and worker control.

Lev supported one additional day, not seven, if Bellweather received honest front timing. No such motion remained before the chamber.

Edrin found the freeze lawful and the evidence sufficient for further investigation. He declined to call emergency authority a substitute for Assembly policy.

Nera stood last. "My filing expires whether you act or not. Expiry does not prove the danger ended. It proves the Charter gave one notary two days to make the danger visible. You have seen current crop damage, worker exposure, future shelter relief, independent settlement signatures, and a Founder clause naming burdened persons. If you refuse extension because distribution remains unjust, order a just distribution. If you refuse because evidence remains incomplete, stop adding load while you test it. Doing nothing selects the prior transfer."

The clock showed 01:19:04.

Halden asked for ten minutes to prepare his recommendation. He used nine.

"We cannot sustain seven days of retained load under current allocation without unacceptable food and ward risk," he said. "We also cannot restore confidence in

the former schedule by pretending its far-side effects remain hypothetical. I recommend expiry with immediate public distribution review, preservation of the return channel, and no approval of new multi-year schedules until the Founder annex receives a hearing.”

Nera heard the boundary. He would slow future borrowing but not stop the transfer already before them.

The final vote came at forty-seven minutes remaining. Two for extension. Four against.

The motion failed.

No one left. Privacy seals held until the emergency interval ended because every witness had to observe the transfer state. Lev sat with both hands around his untouched water. Beren copied the store-refusal exchange into his own notebook despite the ban on removal. Sumi reopened her trace case and attached a fresh frost strip to the carrier monitor.

At twenty minutes, Canal Nine crews called positions. Valves prepared to release the retained front into the phase entry. Bellweather reported mesh ready for the rapid thaw predicted after transfer. Warm Ward soil operators requested permission to close the loop they had opened to Low Glass. Nera denied closure until the black bracket vanished.

Halden moved from the delegates’ bench to the witness side. He stood beside the return monitor without speaking.

The red figures passed ten minutes.

Siv’s carrier appeared as a thin sawtooth. Text formed one letter at a time.

> KEST SEES PRESSURE RISING BEYOND YOUR SIDE. DID ASSEMBLY EXTEND. REPORT BEFORE LOAD ARRIVES.

Nera reached for the reply key. The privacy order prohibited external communication during session.

Halden placed his seal on the temporary release line. “Answer the factual question.”

She sent:

> EXTENSION REFUSED. AUTHORITY ENDS AT EXACT HOUR. PRESERVING CHANNEL. WE WILL CONTINUE CLAIM.

The message cleared with four minutes remaining.

At one minute, every active schedule in the chamber turned amber. Sumi watched the phase trace. Lev watched the orchard contour. Beren watched the radiator map. Nera kept her eyes on the emergency line she had sealed.

00:00:03.

00:00:02.

00:00:01.

At hour forty-eight, the ledger released her seal.

A gray notice crossed the filing: AUTHORITY EXPIRED.

The black bracket opened. Transfer F-223-32 turned blue and entered Canal Nine's phase feed.

Under the chamber floor, valves struck in sequence. Heat surged through one set of pipes while the tunnel air cooled along Nera's ankles. On the orchard contour, Bellweather's frost began lifting too quickly. The Warm Ward store request changed from pending to approved. The old distribution resumed because the Assembly had chosen no replacement.

Sumi's fresh trace darkened all at once. The sawtooth carrying Siv's reply compressed beneath a broad field of noise.

One fragment reached the frost paper.

> NERA WE

Nera held the frost paper flat after the letters failed, preserving the pressure of Siv's unfinished sentence even when no words remained to copy.

Static scored the rest of the strip black.

Interlude II — Weather Inheritance

In Settlement Year 140, the first receipt entered the Harrow household folded around a furnace-salt token.

Mira Harrow kept both in a blue cup above the stove. The token bought three nights of heat. The receipt explained that the household had benefited from a continuity margin created during the west-belt winter of Year 3. No sum appeared under debt. The lower half showed a civic credit and a future settlement date too distant for Mira's grandchildren to imagine.

Her son used the blank back to total root meal. Eight portions if the stove stayed lit. Ten if they ate before dusk and slept in one room. He left the official side unmarked because public paper could not be replaced without a fee.

When the third night ended, Mira surrendered the salt token. She kept the receipt.

The stove cooled. The margin continued.

By Year 176, Mira's grandson Pell farmed a strip below the cloud comb at West Tern. His crop book held weather in three columns: what arrived, what the civic board had promised, and what the plants endured.

One spring line recorded six days of hard rain under a clear-board forecast. The next named black snow that melted into oil-smelling furrows. Pell did not know which old moderation had produced either event. He wrote the Harrow receipt number beside both and drew a question mark through the civic credit.

An inspector later crossed out the marks. Household heat documents could not establish crop causation. The correction carried a proper seal.

Pell copied the receipt number again in the seed ledger, where the inspector had no authority. He chose frost barley from the communal store and returned half the expected yield. Beside the shortfall, he wrote no accusation. He listed the salt nights his grandmother had received and the ruined rows he could count.

The seed clerk accepted the figures because hunger made unofficial arithmetic useful.

In Year 211, West Tern abandoned its upper houses after three winters joined without a full thaw. The shelter committee converted the grain vault and issued twelve rules on a ceramic board.

Rule One assigned wall spaces by recoverable body heat. Rule Two placed infants between adults without lung illness. Rule Three forbade private stoves during sleeping hours. Rule Four required every incoming household to bring heat receipts, crop notes, or service records if they possessed them.

The papers did not buy admission. They told the committee which inherited loads might arrive together.

Pell's daughter Ena brought the blue cup. The furnace-salt token had disappeared decades earlier. Mira's receipt remained inside, rubbed white at the folds. Ena also carried the seed ledger page because Pell had cut it free before the farm office closed.

A shelter clerk copied both numbers onto wet clay. Civic credit became prior softening. Settlement date became expected burden interval. The clerk added West Tern's actual temperature loss beside the old household benefit.

No rule described guilt. The vault needed to know when its walls might fail.

During the next joined winter, the committee moved the Harrow family away from the north vent after their burden interval aligned with a pressure front. Two children avoided frost injury. A third resident took their former place and lost three toes when the seal split.

The shelter record kept both facts on the same tablet.

Year 248 brought a traveling repair crew from the southern drum stations. Its youngest fitter, Lio Harrow, carried the blue cup wrapped in wool. West Tern had broken apart after the grain vault flooded, and household records now traveled with whichever person could protect them.

Lio repaired phase housings he did not understand well enough to trust. At each station, he compared drum scars with the dates on his family receipt. The old line from Year 3 appeared as a shallow ceramic defect in four machines. Later transfers had widened it. Technicians called the accumulated mark ordinary wear.

He pressed frost chalk into the grooves and copied their spacing onto Pell's crop page.

At South Furnace, a supervisor told him to clean the chalk away before inspection. Lio did, then made a second copy on cloth using lampblack. Cloth held no certified standing. It survived bending better than paper.

His repair crew began trading such copies with shelter cooks. A cook could not reverse a phase drum, but she could store grain before a predicted burden interval. A roof crew could double-bind a vent. A family could decide which room to surrender first.

The inherited account became household weather practice long before any public office admitted a creditor.

Holdfast Kest received the Harrow bundle in Year 302. Lio's granddaughter Senn arrived with a caravan from the failed western vaults. She had never seen the blue cup used for drinking. Its glaze had cracked, so she packed the receipt, crop page, clay copy, and lampblack cloth inside a wall niche.

Kest assigned her family a heat share after nine nights. The calendar keeper recorded three kinds of obligation: heat contributed, heat received, and old load expected. Senn objected to the last category. She had not chosen Mira's stove nights. Her children had not planted Pell's barley.

The keeper agreed and left the obligation attached to the receipt rather than the family name.

That change mattered during the long cold of Year 317. Another household occupied the wall space predicted to take the Harrow interval. Kest moved both families, not because either deserved the weather, but because the receipt described a path through the lattice. The north room iced from the floor upward. No one slept there.

Afterward, shelter rules required inherited accounts to follow evidence instead of blood. Receipts could pass between households, settlements, or repair crews. Liability remained with the sending schedule. Risk traveled with the weather.

Senn's eldest child copied that rule onto a fresh ceramic slip. The old paper had begun to separate at Mira's root-meal sums.

In Year 349, a school group at Kest learned to read history from objects the public archive had once rejected.

They handled the blue cup with padded gloves. They counted Pell's ruined rows. They matched Lio's drum scars to Kest's wall repairs. Their teacher asked what the receipt proved.

One child said it proved Mira had stolen three warm nights. The teacher placed a furnace-salt ration in the child's palm and asked whether a family should have

refused heat without knowing who would receive the cold.

Another said it proved the city had lied. The teacher pointed to the honest old date hidden beneath the civic credit. Some engineer had recorded the destination before later offices changed the vocabulary.

Orra Ren, not yet born, would inherit a classroom shaped by that distinction: people could accept needed warmth without consenting to secrecy, and records could preserve a claim after institutions removed its name.

The Harrow receipt went back into the cup.

In Year 360, after the resumed transfer struck the winter line, Holdfast Kest opened its receipt room to shelter triage.

Siv Ren worked beside two heat clerks while noise buried the return channel. New cold arrived through the west thermal wall. They sorted household records by phase family, pressure onset, and the old sending schedules preserved in brittle print.

The Harrow bundle matched a front entering Kest's agricultural terraces. Mira's original margin formed only its earliest layer. Later entries had attached orchard softening, ward respiration reserves, canal protection, and civic gardens. Each benefit had entered under a separate public name. The phase scar joined them.

A clerk warmed the Year 3 receipt between ceramic plates until its hidden lower line darkened. Beneath the civic credit, a destination code emerged. Another field named the protected facility.

Siv copied both onto a new Kest heat receipt. The document did not charge the Harrow family. It assigned the incoming front to the originating schedule and listed every shelter drawing emergency fuel against it.

At the bottom, where Kest forms required a plain description of incurred debt, Siv wrote the words the old Office had omitted.

Warm Ward winter garden.

Act III

Chapter 14 — The Split Spring

Lev Marek found the first black bud at the eastern edge of Bellweather, where the orchard mesh had frozen halfway down its rails.

He held the branch beneath the inspection lamp. A redleaf bud should have resisted his thumb with the small, dry firmness of a seed. This one gave way. Brown water spread across his glove, sweet at first and then sour with ruptured tissue.

Behind him, Yara Fen called temperatures from the row gauges while two workers freed the mesh with wooden mallets. Metal tools could score the iced tracks. A scored track meant a torn curtain when the west wind returned, and Bellweather no longer had a forecast worth trusting about the west wind.

"Lower terrace reads minus four," Yara said. "Upper reads minus two. Canal return rose three degrees after midnight."

"Wrong direction."

"I know what numbers do."

Lev cut the bud lengthwise. Frost had entered from the stem after the air began warming. The resumed transfer had dragged cold across Bellweather at dusk, hard enough to skin the irrigation troughs. Near midnight, warm canal water arrived under the roots. Trees pulled moisture upward into tissue already made brittle by cold. Their own circulation had finished what the frost began.

He dropped the halves into a sample envelope and wrote EAST SEVEN, ROW NINETEEN. "How many rails held?"

"Forty-one closed. Six jammed above shoulder height. Three tore." Yara pointed through the rows. "The north crew used ladders until ice came off the crosswires. I sent them in when Jalen lost feeling in one foot."

"You should have called me."

"You were at the pumps."

"You still should have called."

"Then I could have listened to you tell me to send them in."

Her anger had acquired economy during the three days since the freeze ended. Bellweather had spent forty-eight hours preparing for cold that gathered honestly, one front moving over land at a speed they could measure. Then authority expired in Lume. The lattice resumed its deferred pull, leaving the orchard between incompatible instructions. Frost came down through open cloud. Root heat came west through the canals. By morning, spring had split at soil level.

Lev walked the row with his knife open. Every fifth tree, he cut one terminal bud

and one lower spur. He recorded black, bruised, or clear on a slate strapped to his forearm. Clear tissue did not promise fruit. A bud could survive cold and fail during bloom, especially after a false thaw woke it twice.

Near the service lane, two small figures waited behind the safety cord. Yara's twins wore identical orchard coats made from different years of patched canvas. One had tied both hoods under the chin; the other had lost a mitten and tucked that hand into an armpit.

"I told my sister to keep them at breakfast," Yara said.

"They look fed."

"They look curious. Curiosity walks under equipment."

The child with both mittens lifted a ceramic cup. "Mama, the kitchen says the pears are sick."

"The kitchen talks too much."

"Can we help?" asked the other.

Yara stepped over the cord and crouched between them. She checked the bare hand, rubbed each finger, then pulled her own spare liner from a pocket. "You can take this envelope to cold storage. You do not open it. You do not run. You tell Auntie Pel that Row Nineteen needs a tissue tray."

Both children reached for the envelope. Their argument continued down the service lane in fierce whispers.

Lev watched them pass stacks of folded frost cloth. Bellweather's labor books counted Yara as one skilled crew lead and the twins as two dependents. The categories never recorded who carried samples because every adult had both hands occupied, or who learned at seven to identify the smell of dead pear tissue.

A tram sounded beyond the western gate.

Yara straightened. "Office?"

"Likely."

"Did you invite them?"

"I sent every temperature record to Voss. Invitation would imply hospitality."

The tram crossed the cooperative spur at a cautious pace. Its forward glass held a pale scar where freezing rain had struck warm laminate. Nera stood alone on the rear platform with one hand around the rail. She wore an Office coat over field boots too clean for Bellweather.

Lev closed his knife. He had not seen her since the closed session, though her face had traveled out on every official notice. The Assembly had refused extension. Seasonal Accounts had resumed the transfer. Each notice used a different grammatical subject.

Nera stepped down carrying two instrument cases and no clerk. "The city board showed a four-degree recovery here."

“Come look at what recovered.”

He took one case before she could object and led her between Eastern Seven and Eight. Yara stayed close enough to hear. A notary visiting damaged acreage could become a finding before the mud dried, and workers had learned not to leave findings alone with officials.

At the first cut branch, Lev placed the lamp in Nera’s hand. “Press that bud.”

She did. Brown liquid marked the pad of her glove.

“Again, lower spur.”

The second bud opened green.

“What does that distribution tell you?” she asked.

“That your weather reached the branch from both ends.” Lev pointed toward the root line. “Frost closed the terminal tissue. Canal heat woke the lower wood. If the cold had held, we would have lost the exposed buds and kept the late spurs dormant. If warmth had held, we would be setting mesh for bloom. We received frost followed by a false thaw, with six hours between them and no usable warning for either.”

Nera looked along the row. “How much damage?”

“Ask after noon.”

“I need a provisional range.”

“You need a number that will sit still while someone reads it.”

Her gaze returned to him. “I need to know whether another interruption would make this worse.”

“There you are.”

Yara took the lamp from Nera and examined the lower cut herself. “Another abrupt interruption would kill more. Continuing the present schedule may kill more. Those statements can share a page.”

Nera removed her marked glove. “They will.”

“Public page?” Yara asked.

“Authenticated record.”

“That means no.”

Lev let the answer remain between them. He carried the cases toward the pollination shed, where the cooperative council had begun triage. Canvas maps covered the sorting tables. Blue pins marked intact mesh, red pins marked torn panels, white chalk marked rows where root temperature had reversed more than twice.

Three crew leaders moved aside for Nera. Nobody greeted her. Bellweather custom usually required hot pearleaf at any working table, even during disputes. Today an empty kettle stood near the stove.

Lev unrolled the latest bud count. "We can preserve frost barley on the south ground if we divert root water by tomorrow. Sunroot can wait. Redleaf needs stable air now. Not warm air. Stable."

Nera opened her narrow reference case. Inside lay a witness glass, sealing brass, and copies of the transfer schedule under gray thread. "The schedule predicts gradual warming for eleven days."

Yara laughed once. "Its last gradual warming arrived as ice."

"The resumed transfer produced phase recoil. The public schedule does not include the recoil term."

"Does your private one?" Lev asked.

Nera set down a page covered in hand corrections. "Sumi Quill derived a range before the hearing. Steward Rusk ruled it preliminary."

"So you had a range."

"I had an inadmissible calculation."

"We had workers on ladders."

The shed door opened behind them. Yara's children entered with Auntie Pel, a storage keeper whose white braid reached the waist of her work apron. Pel placed the sample tray beside Lev's map. Twenty-four envelopes lay in its slots. More than half carried black marks.

"Cold room has space for graft wood," Pel said. "Not for the full nursery stock."

Lev studied the tray. "Move all clean lower spurs from East Seven through Twelve. Tag maternal lines before cutting."

"We will lose the early crop."

"We may keep next year's wood."

Pel nodded and began assigning workers from the doorway. The choice crossed from slate to hands without a meeting. Saw kits came off the wall. Two crews left for the eastern rows.

Before the first saw touched wood, Yara stopped the crews at East Eight. She stripped bark from three thumb-thick shoots and laid them across a black tile. Green cambium shone on two. The third carried a gray crescent where warmed sap had met frozen tissue. She asked each worker to identify the injury without hearing the row number. Bellweather used blind tests after unstable weather because expectation could turn one damaged sample into a whole doomed field.

A young pruner called the crescent superficial. Pel disagreed and cut deeper. Gray continued toward the graft. They tagged that tree for removal, then opened a fourth shoot from the same root line. Its tissue remained clean.

"Not rows," Yara said. "Trees. Nobody clears a run because the first cut frightens them."

The instruction cost time they did not have. It also kept frightened hands from

converting uncertainty into acreage. Lev changed the work order: one tester ahead of each saw pair, every cut witnessed, disputed wood held until he or Yara could inspect it. Nera copied the protocol onto the back of Sumi's range sheet.

"Your allocation reports say damaged blocks get removed by percentage," she said.

"Our allocation reports assume weather has edges," Lev answered. "This injury follows circulation. It can skip three trees and take the fourth."

The crews went out again with black tiles tucked into their belts. Their pace slowed, but the first surviving tree remained standing between two tagged removals.

Nera watched them go. "You made that decision from twenty-four samples."

"I made it from twenty-four samples, soil probes, six torn meshes, and thirty years of watching trees dislike mixed instructions."

"The Assembly will ask for confidence limits."

"Tell them the trees declined to testify."

A corner of Yara's mouth moved, though the humor carried no ease.

Nera laid Sumi's handwritten range over the map. The recoil band crossed Bellweather, Low Glass, and three reservoir districts. It did not resemble the smooth public forecast. Its edges forked repeatedly, each line tagged with a different assumption about how much cold the lattice would send forward after the interruption.

Lev followed one branch west. "This model has us below freezing again in three days."

"One branch does."

"Probability?"

"Eighteen percent under current load."

"And if the Office freezes another line?"

"We cannot calculate that without knowing scope."

Yara leaned both hands on the table. "Every answer begins after you choose."

Nera did not defend herself. That made Lev angrier than another polished distinction might have. She had arrived with the damage already admitted in her posture, as though accuracy of regret could count beside split buds.

He tapped the gray-thread schedules. "Your Office had forty-eight hours of authority. Growers reorganized irrigation because you used it. Workers closed mesh under clear sky. Then your Assembly waited until the clock removed the decision from its hands. Now Bellweather absorbs the recoil while the Office says authority expired."

"I said it expired."

"You stood there and watched."

"I argued for extension."

"You argued inside a closed heated room. Outside, nobody knew whether to protect against the cold you kept or the warmth you might resume."

Nera's fingers rested beside the witness glass. "What would you have had me do at hour forty-eight?"

"Open the room."

"The records remained under session seal."

"Break it."

"And lose admissibility for the figures that could change the Charter?"

"Then choose the Charter. Choose the orchard. Choose the people in the year that answers you. But stop calling the interval neutral."

One of the twins had stayed inside. The mittened child stood near the stove, listening with the solemn attention children gave adults who forgot to lower their voices. Yara guided the child back toward the door without interrupting.

Lev lowered his own voice. "Every side absorbs uncertainty while your Office protects authority. The future takes the cold. We take the recoil. Low Glass takes the canal hazards. Your institution keeps its clean chain of custody."

Nera flinched at Low Glass, not at the accusation about her office. He remembered too late that she had grown up above those canals. Remembering did not make the claim false.

"The chain is no longer clean," she said. "My single-line order entered the public schedule. Its expiry did the same. The Office cannot describe either as ordinary continuity without falsifying the ledger."

"Can it publish that?"

"Yes."

"Will it?"

Her delay gave him the answer.

Outside, mallets struck the frozen mesh rails in uneven sequence. Each blow traveled through the shed frame. Meltwater had begun dripping from the eaves although the air still cut below freezing in shaded rows. Another split instruction.

Yara returned to the table. "We need a plan for the next three nights, not a confession."

Lev pointed to the recoil branches. "Can Quill model staged retention instead of a full interruption?"

"She has tried. Her runs depend on honest local burden maps."

"You have ours."

"We have the version Bellweather files for seasonal allocation."

Silence passed through the shed. Cooperative reports grouped labor housing, nursery glass, orchard rows, and storage into one heat district. The grouping protected Bellweather from urban rate formulas, but it hid where retained cold would land. Lev had defended it at six allocation hearings.

Yara spoke first. "Our map does not show the bunkhouse loop."

"No," Lev said.

"Or the kitchen store."

"No."

Nera opened a blank evidence sleeve. "Can you produce the working map?"

Lev looked at the council members. Pel had come back to the door. Two crew leaders held their arms folded. Nobody wanted Bellweather's internal heat routes exposed to a city that had repeatedly treated farm capacity as available margin.

"We can," he said. "Under reciprocal disclosure."

Nera's attention sharpened. "Define it."

"Bellweather opens its real burden map. Seasonal Accounts opens the transfer ledger that produced these recoil branches, including ward benefit and canal injury. Not a summary. Not managed disclosure."

"I do not control the sealed records."

"You control whether our evidence enters another closed session."

Yara pulled the white chalk toward her. On the canvas map she drew the unreported pipe from the root manifold to the crew bunkhouse. "We also need worker names attached to exposure hours. No aggregate capacity."

"Agreed," Lev said.

Pel added the nursery bypass. Another leader marked the kitchen store and the washhouse tank. Bellweather's official district dissolved under their hands into places where people slept, washed, ate, and tried to keep next year's trees alive.

Nera took no notes until they finished. Then she asked each person to state a name and role for the witness glass. She authenticated the map as cooperative testimony, not verified fact. The distinction sounded less like shelter when she said it in front of those who had supplied the risk.

By late afternoon, the sun lifted the orchard temperature above six degrees. Meltwater ran along frozen ground and pooled around the trunks. Lev ordered shallow cuts in the drainage berms. If water remained when night cold returned, it would sheath the feeder roots.

Nera worked beside the crews, carrying frost cloth from the store to East Nine. She did not remove her Office coat. Mud reached its hem. Nobody praised her for lifting things.

At the damaged row, Lev showed her how to wrap surviving lower spurs without trapping moisture. Her first knot compressed a bud.

"Looser," he said.

She retied it. "Like this?"

"Enough room for swelling."

"If it swells."

"If we work from that assumption."

They moved to the next tree. Years earlier, before Tovan's apprenticeship and before Lev left Lume, they had argued about whether care required optimism. Nera preferred conditions: if the graft held, if the storm missed, if procedure permitted. Lev had mistaken those clauses for reluctance. He understood now that she used them to expose where hope depended on someone else's burden. Understanding arrived after usefulness.

Near dusk, Sumi's recoil map proved right in one narrow respect. A warm wind crossed the western rows, carrying the smell of wet stone from the radiator canals. The temperature rose another two degrees in twelve minutes. Dormant bud scales loosened along East Three.

Yara swore and called for shade cloth. Workers reversed the morning's labor, lifting dark panels over branches they had just protected from frost. Her twins carried wooden clips in buckets, one child counting aloud while the other corrected every fifth number.

Lev stood beside Nera at the service lane. "This will happen again under any repayment plan."

"Yes."

"Staging reduces the discontinuity. It does not remove crop loss."

"No."

"Low Glass cannot take an equal share."

"Beren has made that clear."

"And future settlements require a way to dispute our maps."

Nera looked toward the tram line, where evening light had turned the rails the color of old brass. "That requires treating them as parties, not forecast inputs."

"Then put it in the model."

"Models do not confer standing."

"They expose what standing changes."

Lev removed the sample envelope from his pocket. Brown fluid had dried across the paper under his handwriting. "I will model staged repayment with Quill. Bellweather will supply ungrouped heat routes and crop thresholds."

Nera waited. She knew a condition approached.

"You open the public ledger," he said. "All of it. Founder terms, ward benefit, worker loss, future load. If your Office withholds the basis, I withdraw our maps and

say so in the Assembly.”

“I cannot promise publication alone.”

“You can promise which side you take when they refuse.”

The orchard mesh clicked as ice released from the last eastern rail. One side dropped cleanly. The other hung torn over blackening buds.

Nera held out her marked glove. Frost damage had stained the fingertips brown. “I will petition for the unredacted ledger and enter Bellweather’s disclosure as contingent on public access.”

Lev took her hand once, formally, before either of them could make the contact mean more than agreement.

Behind them, Yara called the night crews to the map. The revised canvas showed every hidden pipe in white chalk. At its eastern edge, beside rows no forecast could now promise to save, Lev wrote the first staged-load threshold.

Chapter 15 — The Forecast That Refused

Sumi Quill caught the summer order in the phase room's waste tray, folded beneath six pages of drum tolerances as if concealment could make it technical.

The order carried the Forecast Office's green border and Halden Rusk's countersign. ISSUE NORMAL SUMMER RANGE FOR DAYS FORTY-FOUR THROUGH NINETY. EXCLUDE UNVERIFIED RETURN-SIDE INPUTS. The second sentence had been printed in smaller type. Its ink remained just as dark.

She read it while Drum Four turned behind the safety glass. Each black ceramic segment passed the lamp, swallowed the light, then gave it back as a thin red line. The rhythm had altered since the forty-eight-hour interruption. No ear could detect the change. Sumi felt it through the handrail in her remaining fingers.

Pavel In, the morning modeler, looked up from the convergence desk. "That came through my queue."

"And moved itself into waste?"

"I printed it twice. I intended one copy for the furnace."

"Your courage has achieved duplication."

"It also achieved a run without their exclusion." He nodded toward the desk. "Bellweather sent an ungrouped map."

Sumi set the order beside the keyboard. Three model panes occupied the glass: public load, physical load, and phase displacement. A normal forecast reconciled them within twelve iterations. This run had reached four hundred and eighty-one. Each pane still displayed a different summer.

Public load showed warming on schedule, with three mild rain fronts and an ordinary reservoir decline. Physical load carried frost recoil west through the orchards, a heat crest over Low Glass, and two unstable cloud bands above Lume. Phase displacement contained narrow white scars where data from Year 360 entered the equations. Every attempt to flatten those scars moved error elsewhere.

"Did you isolate the return?" Sumi asked.

"First run."

"Not delete. Isolate."

Pavel pointed to a branch tree. "Here. Their storm signatures act like observations of our transfers after dissipation. If I remove the text content but keep authenticated phase timing, the fit still splits."

"Good."

"That did not sound like praise."

"It means the instrument noticed what the order asks us not to notice."

Sumi pulled the safety stool under the convergence desk and sat. The missing fingers on her left hand made ordinary keyboards tedious, so she used the phase crew's chord plate. Thumb, forefinger, and smallest finger combined into commands that Pavel still had to search for on printed menus. She opened the branch tree and began at the first disagreement.

Bellweather's working map gave root temperatures by occupied loop rather than allocation district. Low Glass had supplied canal injuries beside flow loss. The return channel contributed future storm pressure at the far side of each deferred load. Together they made the old forecast boundary impossible. The model could preserve a normal summer only by assigning heat to a place with no pipe, cold to a year with no claimant, or error to an instrument already authenticated by three independent scars.

At iteration six hundred, the run stopped.

A plain line appeared under the panes:

> SOLUTION REFUSED: CONSERVATION BOUNDARY INCOMPLETE.

Pavel leaned closer. "I have never seen that message."

"Because you use polite models."

"This came from the standard civic suite."

"Standard civic suites can count."

The run had not failed from insufficient computing time. It had refused to converge under mutually exclusive instructions. One constraint demanded that the summer remain normal. Another required physical conservation. A third admitted measured burden from the return channel. The solver could satisfy any two.

Sumi copied the run state to a black service wafer. She chose a wafer with a chipped corner, not because damage offered security, but because she could identify it by touch inside a pocket.

Pavel watched her hand. "Preservation requires supervisor authority."

"You called me."

"I called you because someone needed to tell the Forecast Office its order has no physical solution."

"That person sounds admirable."

"That person has rent."

She removed the wafer. "Then stay ordinary while I become difficult."

A warning bell rang at the outer phase lock. Two green notes, pause, one green note: administrative entry. Sumi disliked a code that made officials sound safe.

Deputy Director Vale came through wearing clean ear protection around his neck. He belonged to the Forecast Office rather than Canal Nine, and he walked the phase room as though every rail had already signed his expense forms. A clerk followed with a sealed model case.

Vale placed both palms on the convergence desk. "You received the issuance order."

"It arrived in waste."

"Technician Quill."

"Deputy Director."

His eyes moved across the stopped panes. "This run includes prohibited data."

"It includes authenticated timing."

"Return text lacks forecast standing."

"I removed the text."

"Return timing also lacks standing."

"Drum phase has standing. Your office has used it to certify every seasonal range since Year Eight."

Vale gestured to his clerk. The sealed case opened. Inside lay a green forecast wafer already labeled NORMAL SUMMER, DAYS 44-90. "We have completed a compliant run at the central office. Canal Nine needs to validate its phase baseline."

Sumi did not touch the case. "Show the residual."

"The summary falls within public tolerance."

"Show the residual by receiving year."

"We do not issue receiving-year fields."

"You calculate them."

"Internally."

"Then show me."

The clerk looked at Vale. His face remained smooth, but his thumb rubbed the case hinge. He knew what a withheld residual meant in an engineering room.

Vale shut the lid. "Your role concerns phase integrity, not policy boundaries."

"A forecast places physical load on equipment I maintain. That concerns integrity."

"Your certification confirms only that the drums can execute the schedule."

"The drums can execute many ways to injure people."

"This language does not help."

"It helps me remain accurate."

Behind the safety glass, Drum Four completed another rotation. Warning chalk along its lower bearing had flaked since morning. Sumi rose and crossed to the inspection rail. She pressed three fingers against it. The pulse arrived early on each seventh segment, exactly where the return channel strengthened under local cold.

"Run your wafer here," she said. "Full telemetry."

Vale hesitated. Refusal would make a clean procedural mark. Acceptance risked giving her the residual he had declined to show.

"We welcome validation," he said at last.

Pavel loaded the green wafer. Sumi opened every monitor channel, including maintenance harmonics usually suppressed from civic summaries. Vale objected to none of them. The clerk entered the central key.

The normal summer unfolded across the glass. Warmth advanced from the southeastern reservoirs. Cloud combs stripped two storm bands. Bellweather stabilized after Day Forty-Seven. Low Glass canals remained within labor tolerances. By the display's color, nobody suffered an unusual season.

Drum Four disagreed first.

Its bearing pulse rose above tolerance at the modeled edge of the first warm front. Drum Two followed. The return needle, mounted in a casing built for stillness, moved four divisions against the schedule. Then the model assigned the unbalanced cold to an unlabeled downstream field.

Sumi froze the display. "There."

Vale folded his arms. "A legacy field."

"Destination stamp?"

"Not relevant to summer issuance."

She expanded it. YEAR 360 appeared beneath the field, followed by the authenticated phase marks from Holdfast Kest and the relayed winter line.

The clerk stopped rubbing the hinge.

Pavel said, "Central's model saw it."

"Naturally," Sumi answered. "Numbers do not gain obedience from a green border."

Vale reached toward the control plate. Sumi covered the execution key with her left palm. The old injury made the gesture unmistakable. Two absent fingers lay over no controls; the remaining ones blocked enough.

"You will release the key," he said.

"When the run writes its conservation residual to local record."

"The summary already records tolerances."

"By receiving year."

"That field remains sealed."

"Then the key remains occupied."

For several breaths, the room contained only drum rotation and the canal pumps below. Pavel had stepped away from the desk. He had rent, Sumi remembered. He also had the good sense not to create a witness position he could not sustain.

Vale changed tactics. "The city needs a forecast. Growers cannot plan from a professional dispute. Reservoirs cannot release from a range that changes every time a message arrives from an unrecognized source."

"The source has three authenticated storm signatures."

"Not legal recognition."

"Cold does not ask legal recognition before entering a pipe."

"If we publish nonconvergence, public boards will show uncertainty across every district."

"That matches the instrument."

"Uncertainty can produce hoarding, work stoppage, and avoidable death."

"So can a false summer."

He looked older in phase light, though not kinder. "We are trying to hold enough continuity for the city to decide."

"You are issuing the decision as weather."

The clerk inhaled softly. Vale heard it too.

He gave Sumi a narrow choice. "Certify phase capability now, or I suspend Canal Nine from forecast service and transfer certification to Central."

Sumi removed her palm from the key. "Write the suspension."

His expression altered by less than a frown. He had expected argument to end at employment.

The clerk took out a notice slate. While Vale dictated, Sumi restarted the civic run and directed its full residual into the local witness spool. Paper emerged under the desk, warm from the printer, carrying row after row of phase error. Vale could stop the electronic record. He could not make the already printed figures reenter the machine.

At the bottom of the spool, the system calculated a checksum: forty-eight characters derived from every input, branch, exclusion, and result. Change one future timestamp or one Bellweather pipe, and the sequence changed with it.

Sumi tore off the strip. Vale held out his hand.

"That belongs to forecast custody."

"It belongs to the maintenance record until you sign transfer."

The clerk paused in his dictation.

Vale signed the suspension instead. "Central assumes service at the next bell."

Sumi folded the residual strip around the chipped black wafer. "Then Central can explain the bearing damage when its normal summer reaches the drums."

He left with the green case and the clerk. The phase lock sounded two red notes after them, indicating an incomplete administrative exit. Vale had forgotten to return his visitor token. Pavel picked it off the desk and set it in the waste tray.

“Now what?” he asked.

“Now we reproduce.”

They spent the next nine hours testing every respectable way the dissenting run might be wrong. Sumi changed sensor baselines within certified drift. Pavel replaced Bellweather’s working routes with filed district averages. They removed message content, casualty references, settlement names, and every human assertion from the future data. Only phase timing and storm pressure remained.

The first run stopped at three hundred iterations. A second produced no stable root. The third circled between two seasonal states until its safety counter ended the attempt. Run Four accepted a normal summer by assigning downstream dissipation an infinite capacity, a mathematical convenience with no physical place to hold it.

Before Run Five, Drum Four tripped its bearing-temperature latch. Sumi took Pavel below the model floor into the service gallery, where canal heat condensed on the ceramic housing and made every rung slick. The morning crew had already rolled back the guard. Inside, the phase bearing showed a pale contact stripe two handspans long.

Tesh, the senior drum fitter, held a frost-chalk gauge against the stripe. “Central’s range loads this segment at standard phase.”

“Standard against which destination?” Sumi asked.

He looked up. “The schedule doesn’t name one.”

“Then do not call it standard.”

Tesh passed her the gauge. Its chalk teeth recorded pressure as blue bands. Six bands lay evenly spaced. A seventh crowded the edge, showing where the bearing tried to execute present heat removal while authenticated downstream pressure pushed back through the same phase material.

Pavel crouched outside the guard. He had never entered a live drum gallery. “Would it shear?”

“Not today,” Tesh said.

Sumi inserted the gauge again at Central’s projected Day Fifty-Two load. The seventh band widened. “On the third warm crest, perhaps. Or it delaminates and gives us enough warning to congratulate ourselves.”

Her left hand remembered a different bearing before the rest of her did. A black segment had slipped its brake seven years earlier. She had reached for a release lever after the warning chalk failed. Two fingers remained inside the glove when Tesh pulled her clear. The inquiry praised rapid isolation and never asked why the drum carried an unlisted phase burden that morning.

“Photograph the stripe,” she told him. “Include the order number and the unlabeled destination field.”

“Maintenance images go to Central now.”

"Send them. A concealed mark does more work when everyone knows who concealed it."

They measured all sixteen segments. Four carried crowding at the same modeled dates. Tesh shifted no bearings and polished nothing. Instead, he scored witness notches on the removable guard, one notch per unstable segment, then signed the service slate with Sumi and Pavel. The physical record would remain even if Central cleared the digital alarm.

Back at the convergence desk, Pavel entered the measured contact pressure as an independent constraint. Run Five stopped sooner than the others.

"It believes the bearing," he said.

"The bearing has excellent professional standing."

They removed the phase-return observations entirely and kept only the service measurements. Nonconvergence persisted. The future message had revealed where to inspect, but ceramic under load supplied present evidence no order could classify as foreign.

At dusk, Sumi called Lev on the narrow civic line. Bellweather's image arrived in fragments, his face interrupted by bands of orchard gray.

"Your map has an unmetered loop between East Nine and the nursery," she said.

"Seasonal bypass."

"Capacity?"

"Variable."

"Everything varies. Give me limits."

He read pipe width, pump age, occupied hours, and minimum root temperature from a work slate. Yara's voice corrected his estimate of the bunkhouse draw from offscreen. Sumi entered both figures and marked the disagreement rather than averaging it away.

"Will that stabilize your run?" Lev asked.

"No."

"Then why ask?"

"Because failure should belong to the instructions, not missing pipes."

After the call, Pavel brought food from the canal canteen. The stew had formed a skin before either of them remembered it. They ate beside the convergence desk while Run Seven tested a staged load across Bellweather, Low Glass, and Warm Ward benefit bands. It found a physically possible range. Summer shortened. Orchard loss rose. Canal danger shifted toward districts with better insulation when the load followed historical benefit.

"That one converges," Pavel said.

"It does not meet the order."

"So the machine can solve the weather. It cannot solve the adjective normal."

Sumi saved Run Seven separately. "Machines have always struggled with ceremonial nouns."

Near midnight, she carried the witness spool upstairs to the legal observation room. Canal Nine showed through the slit as a black ribbon striped by service lamps. Edrin Sol waited beside the hand ledger, coat buttoned wrong at the throat. Nera had not come. That meant she either did not know about the meeting or knew enough to stay away from evidence offered outside a custody chain.

"You asked for a notary," Edrin said.

"I asked whether a checksum can receive a time witness without revealing its source record."

"It can. The resulting witness proves existence at an hour, not accuracy or provenance."

"Good."

"That answer usually disappoints people."

Sumi placed a blank tile on the table. She did not produce the wafer. "Record this sequence."

She read the forty-eight characters once. Edrin copied them by hand, repeated them, then impressed his deputy seal into the tile margin.

"What generated it?" he asked.

"A forecast run that refused to converge under a normal-summer constraint."

"Official?"

"Official inputs. Unofficial honesty."

"Did you remove the record from custody?"

"No. I preserved a maintenance copy before custody transferred."

His mouth tightened at the phrasing. "You know precisely where the disciplinary line lies."

"I maintain lines for a living."

Edrin slid the witnessed checksum toward her. "Why bring this to me?"

"Because Voss needs admissible proof that the dissent existed before Central replaces it. You can establish time without endorsing content."

"You are asking procedure to protect a leak."

"I am asking procedure to remember an event."

He looked through the slit at Canal Nine. "Central suspended your certification?"

"At the next bell."

"Then this may become evidence in your own inquiry."

"Everything useful eventually does."

Edrin kept one sealed copy and gave her another. He did not promise to show Nera. He did not need to. His decision had entered the brass while the wax remained warm.

On Day Forty-Three, Central sent its normal range to the public boards.

Sumi stood below the Office steps with Pavel and half the Canal Nine morning crew. Across Lume, forecast glass changed from uncertain amber to summer white. People stopped beneath tram shelters to read reservoir dates and orchard temperatures. Shopkeepers copied the warm range onto chalkboards. A vent-glass crew began removing storm braces from a roof.

For eleven seconds, the city possessed an ordinary summer.

Then Central's checksum reached Canal Nine for execution comparison. The drums tested it against the preserved local run. Every phase error Sumi had printed returned through the safety layer. Drum Four rejected its load. Drum Two entered conservation hold. The public boards lost one white column, then another.

No alarm followed. Forecast failure had no civic siren because its designers expected forecasts to fail privately.

The glass above the Office door went black. Along the canal, tram boards darkened in sequence. Farther uphill, the Warm Ward's weather ribbon vanished from beneath its heated arcade.

Pavel watched the darkness travel. "Did you cause that?"

Sumi felt Drum Four's refusal through the pavement, a small regular pulse under both boots. "I kept it from being called agreement."

Across Lume, people looked up from their copied summer dates. The official forecast remained dark, while the city's actual air warmed against every promise printed that morning.

Chapter 16 — No Neutral Season

Tovan had covered the kitchen window with rejected vent glass, three imperfect panes leaned edge to edge against the frame so no one outside could read the blue ceramic slips spread across the table.

Nera recognized his apprenticeship marks in the corners. One pane carried a trapped bubble. Another had cooled around a crooked reinforcing wire. The third showed a white stress bloom that would fail civic inspection. He had brought home the pieces he could not put into anyone's wall and turned them into privacy.

Beyond the window, Lume's forecast boards remained dark for a second day.

Tovan sat with his back to her, reading a transcript under the kettle lamp. He had arranged the copied slips into rows by date. Blue chalk marks appeared on only four. The rest carried temperature, shelter occupancy, ration cuts, or the name of a storm inherited from an earlier summer.

"You opened my evidence case," Nera said.

"I opened the kitchen drawer."

"The drawer held sealed return transcripts."

"It also held spoons."

She set down her coat. "Those records have not passed provenance review."

"You left the key in the flour cup."

"That does not grant access."

"No. It describes your confidence in the lock."

His voice had changed during the past month. Seventeen still showed in the impatience, but his questions had begun to acquire the narrow edge of a tool ground for actual work. Nera had taught him to ask who signed a statement. She had not expected the lesson to reach her own kitchen so directly.

He lifted one copied slip. "What does this mean?"

The line came from Orra's calendar, transmitted in fragments by Siv and reconstructed over four return windows. A child's hand had pressed a blue mineral bar beside Day Twenty-Eight. Beneath it, the notarial transcript read TRUE DAY: SNOW FROM OUR OWN SKY. NO INHERITANCE MARK DETECTED.

"It records local weather without deferred load," Nera said.

"I understand the words."

"What are you asking?"

"Why a child has to mark which weather belongs to her year."

Nera pulled out the opposite chair. "Because most days in Kest carry additional cold or storm burden from ours."

"From this year?"

"From many years. Some from ours."

"How many of these marks would she have without the lattice?"

"We cannot calculate that from the transcript."

Tovan pushed the rows toward her. "Then calculate something else. How much heat did our winter garden keep while her shelter cut meals?"

"The garden's allocation appears in the sealed ledger."

"I have been inside that garden since I could walk. You took me there last month and said the soil stores were civic resilience."

"They are."

"They also have a receipt in another year."

Nera looked at the pane nearest him. The trapped bubble distorted one blue mark into a small pale moon. "Yes."

His anger sharpened because she had not resisted. "How long have you known?"

"About the specific garden entry, since Day Twelve."

"And you let me go back to work beneath Ward glass without telling me."

"The transcript remained restricted. We did not know whether public contact would survive. I needed to protect you from being pulled into an inquiry you could not affect."

"You protected me into ignorance."

"I protected your apprenticeship."

"My shop fabricates the panes over those soil stores."

"You did not choose the transfer schedule."

"I still fit the glass."

Nera understood the argument too well. Her Office authenticated cold accounts without setting Assembly policy. Heatworkers maintained canals without choosing which ward received moderation. Growers opened mesh under forecasts they did not issue. Each role offered a clean boundary until all boundaries joined around the person carrying the cost.

"You are not liable for work performed before you knew," she said.

Tovan looked down at the slips. "That sounds like an Office finding."

"It sounds like your mother refusing to hand you guilt that belongs to institutions."

"And if I keep fitting it now?"

The kettle clicked as cooling metal contracted. Nera wanted to tell him to finish the apprenticeship, keep his head down, let her fight where procedure gave her

standing. That answer preserved his path by making another person's path irrelevant for a little longer.

"You can ask the shop to assign you to repair glass in Low Glass or the canal works," she said.

"They will call it political refusal."

"Yes."

"Will you sign the reassignment petition?"

"If the stated reason remains your own."

Tovan sat back. "You mean you won't use me."

"I mean you cannot serve as evidence that I am a good person."

Some of the anger left his mouth, though none left his eyes. He turned another slip. This one recorded three children sleeping in one shelter niche because body heat had become part of the room budget. Orra had corrected the adult tally in the margin: FOUR, COUNT ME.

"Does she know I read this?" he asked.

"No."

"Does Siv?"

"No."

"Then you used her too."

Nera folded her hands. "I intended to show the transcript only to the claimant review."

"There is no claimant review."

"Not yet."

Tovan tapped the row of slips. "You keep putting people behind the procedure that might recognize them. What recognizes them before that?"

Nothing in the Charter answered. Nera could have said the Founder Sum, but a sealed annex with no enacted clause conferred no present protection. She could have said the return channel, though contact did not create rights by itself. Accuracy required an answer more painful for its smallness.

"We do," she said. "Or we do not."

A knock came at the service door. Beren Olt entered without waiting, bringing canal damp into the kitchen and a rolled burden map under one arm. Lev followed with an orchard tube and mud dried white along his boots. They had agreed to meet at the Office. The black public boards had driven a crowd into Canal Nine, so Nera moved the session home without telling Tovan.

He looked from the men to her. "You were going to send me out."

"I was going to ask."

“Ask now.”

Beren placed his roll beside the ceramic slips. “If this concerns public load, the apprentice already belongs in the room. Warm Ward glass does not fit itself.”

Lev gave Tovan a brief nod. “Stay if you can bear hearing adults turn households into colored bands.”

“I have been reading a child’s weather calendar,” Tovan said. “I can bear your map.”

Nera removed the rejected panes from the window. Privacy no longer justified blocking the view. Across the canal quarter, the nearest forecast board held a black rectangle over the tram stop. People gathered beneath it every time cloud moved across the sun, looking up for an answer the city had ordered into darkness.

Beren unrolled Low Glass first. He had redrawn every radiator route by historical benefit and maintenance injury. Rust-colored lines followed wards receiving long-term moderation. Black marks named shears, burns, vent ruptures, and service deaths. Nera found her mother’s canal among them before Beren pointed to it.

“Load stays where benefit accumulated,” he said. “Warm Ward accepts the highest retained share. Civic center follows. Low Glass receives less because it already paid through labor and poor insulation.”

Lev opened Bellweather’s canvas beside it. White chalk divided orchards by crop threshold, bunkhouse occupancy, nursery survival, and food value. “If you follow benefit alone, you lose early pears, frost barley, and two sunroot plantings in the same sequence. By autumn, everyone buys hunger at an equal price.”

“Not equal,” Beren said. “Low Glass pays first when food prices rise.”

“Which supports preserving staple acreage.”

“Your map protects agricultural plantings before canal households.”

“It protects food before luxury crop. Redleaf takes the first cut.”

“Bellweather benefited from moderated springs.”

“Less than the ward that ordered pears under glass in winter.”

Tovan traced the edge where Beren’s rust band met Lev’s white threshold. “What happens here?”

Both men answered.

“Retained cold,” Beren said.

“Pump instability,” Lev said.

They stopped.

Tovan pointed to the label: EAST WORKER HOUSING. “Your maps assign the same place by different rules.”

Beren examined the canvas. “Those households draw from Canal Seven.”

"Seasonally," Lev said. "During bloom they draw root return from our lower manifold."

"Then your workers receive orchard benefit."

"They also sleep there."

Nera placed a blank sheet between the maps. "We need an order of questions. First, which locations cannot survive a given load. Second, who benefited historically. Third, which productive systems protect people beyond their owners. Fourth, who has already paid through injury or exposure."

Beren shook his head. "Survival limits can absorb every privileged preference. Warm Ward will classify ornamental soil as food resilience."

"Historical benefit can ignore dependencies," Lev replied. "A canal pump serving a rich district may also keep a hospital warm."

"Then separate the branches."

"That requires the unredacted ledger."

Their disagreement did not cancel either map. It exposed the harm each one could conceal alone. Nera began a composite index, refusing averages. A district could receive a survival threshold, benefit share, dependency note, and prior-loss credit without compressing them into one neutral score. The method produced no simple ranking. It did make every proposed load show its reasons.

They worked through midday. Tovan cut rejected vent glass into small squares and placed them over disputed locations. Clear glass marked shared facts. Bubbled glass marked uncertain occupancy. Stress-bloomed pieces marked places where both maps predicted failure for different reasons.

At the Warm Ward winter garden, Beren placed a rust weight near the maximum.

Lev added a food dependency note. "Its seed vault supplies graft lines after regional loss."

"The soil promenade does not preserve grafts," Beren said.

"No. Separate the vault from the garden."

Tovan used his cutting wheel to score a line between them. The glass split neatly. He moved the seed-vault half toward protected load and left the garden half under Beren's high-benefit marker.

Nera wrote the distinction into the index. Her own household appeared two rows below. Warm Ward residential, good insulation, moderate historical draw. Under Beren's formula, she and Tovan accepted more immediate cold than a Low Glass family. Under Lev's dependency map, their building lost little productive capacity.

"Put us in the upper band," Tovan said.

Beren studied him. "You understand what that means for interior glass?"

"I repair it."

"You may repair it while rooms are occupied and cold."

"I know."

Nera's pen hovered. "You do not decide for the building."

"No. I decide whether our unit appears as unwilling."

"There is no consent field in this model."

"Make one."

Lev rubbed orchard soil from a map edge. "Consent cannot replace allocation. A ward with warm houses will volunteer for someone else's pipe limits."

"Agreed," Tovan said. "But leaving it out lets the Office say every present person opposed the cost."

Nera added a testimony column. It carried no controlling authority. It did prevent later summaries from turning silence into refusal.

By late afternoon, the kitchen table held a model too complicated for any public board. That complexity no longer looked like failure. The old schedule achieved smoothness by omitting time, labor, and destination. Their composite held its rough edges because actual households occupied them.

Beren rolled his map halfway, then stopped. "This still gives no authority to the winter line. We can calculate every present burden and decide among ourselves how much of their cold they keep."

"They need standing before allocation," Nera said.

Lev looked at the blue slips. "Can the Charter recognize people outside the current year?"

"Not under enacted text."

"Founder clause?"

"Sealed and disputed."

"Then what can you file?"

"A petition to establish a class."

Edrin's witnessed checksum lay inside Nera's evidence case. It proved that the city's own forecast could not close while future measurements remained excluded. Siv's authenticated messages proved contact. Orra's calendar supplied a lived record, though using it required consent Nera did not yet have. The burden maps proved present allocation could not claim neutrality.

She drew a fresh legal sheet from the case and wrote the heading by hand:

PETITION FOR RECOGNITION OF TEMPORALLY DISPLACED LIVING PERSONS.

Beren read over her shoulder. "Do not call them displaced. They have not moved."

Nera struck the word.

Lev offered, "Downstream."

"That describes our accounting relation," she said. "Not their citizenship."

"Future?" Tovan asked.

"The Assembly uses future subjects for forecast populations."

"Use their year."

Nera wrote: PETITION FOR RECOGNITION OF LIVING PERSONS IN SETTLEMENT YEAR 360 AS CLAIMANTS UNDER DEFERRED SEASONAL LOAD.

The sentence fit badly across the page. Its length pleased her. Short titles had hidden more than they clarified.

She listed the requested findings. First, the return channel carried authenticated communication from living citizens in the same causal history. Second, present transfers imposed measurable burden at the fixed phase destination. Third, exclusion from current-year ratepayer status did not erase material injury. Fourth, no new deferral should proceed without representation from those receiving it.

Before anyone signed, Nera carried the draft through the civic tunnel to the Office registry. The passage lamps had been turned low under an emergency conservation notice, though the Assembly chamber above remained fully heated. Every third wall niche held citizens waiting with handwritten forecast disputes. A reservoir keeper wanted authority to release water without a summer range. Two school stewards needed a temperature basis for closing glass classrooms. None of them could file because the public board offered no certified condition to contest.

Edrin met her at the registry rail. Sumi's checksum tile hung from his evidence cord under a folded cloth.

"You cannot submit this as a ratepayer petition," he said after reading the heading.

"They are not current ratepayers. That defines the defect."

"It also defines the filing category."

"Use extraordinary claimant review."

"That requires a person asserting injury and a present action capable of remedy."

Nera placed Siv's authenticated phase marks beside the petition. "Here lies the assertion. New deferrals provide the action."

"The marks prove a source and hour. They do not prove that Siv Ren authorized you to petition."

"Initial contact demanded a freeze on new deferrals."

"A demand does not appoint counsel."

People behind Nera shifted along the wall. Their own blocked papers made an audience she would not have chosen, yet secrecy had already protected too much authority.

"What can proceed without claimed representation?" she asked.

Edrin turned to the Charter index. "A present beneficiary may petition to determine whether a scheduled benefit lacks a lawful counterparty. The Office used that mechanism for contested water estates before the Charter revision."

"That treats Year 360 citizens as absent owners."

"Only at filing. The petition can ask the Assembly to recognize them directly."

"It makes our standing primary."

"It makes your door available."

Nera disliked the shape because it began with present discomfort rather than future harm. She also knew a closed door improved no one's dignity. "How many beneficiaries?"

"One can file. A class prevents the Assembly from dismissing the matter as private conscience."

The reservoir keeper had begun listening openly. "If a season schedule benefits every current ratepayer, every current ratepayer has standing."

Edrin glanced toward him. "Notarial advice does not extend beyond the rail."

"It did not cross," the keeper said.

A school steward stepped forward with her blank forecast dispute. "My classrooms receive deferred moderation. Put the schools in the class, provided the filing does not commit children to an unbounded cold share."

"That reservation can accompany a signature," Nera said.

Within minutes, the waiting niche changed purpose. Citizens turned over their unusable forecast forms and wrote names, occupations, benefit received, and limits they refused to surrender. The reservoir keeper named pump safety. The school steward named minimum classroom heat. A tram dispatcher described power stored under mild-winter assumptions. None offered blank consent. Each admitted a benefit while insisting that repair retain material boundaries.

Edrin watched the list grow. "This remains unverified testimony."

"Time-witness it."

"You have not completed the burden index."

"Witness existence, not accuracy or provenance."

Her use of his checksum language almost produced a smile. He set the public clock beside the pages and sealed their lower corners. "You have until tomorrow's final registry bell to attach maps and founding authority. After that, extraordinary petitions move to the next session."

Nera returned home with twenty-three beneficiary statements and less than a day. Beren read every reservation. Lev cross-referenced addresses against the dependency map. Tovan cut smaller glass markers until the table carried schools, pumps, trams, homes, and the winter garden as distinct places rather than one

civic appetite.

During the night, the four of them tested each requested finding against a remedy. Recognition without notice could let the Assembly keep borrowing during review. A moratorium without distribution rules could push immediate cold toward Low Glass by default. Benefit weighting without crop protection could replace seasonal debt with food debt. The petition therefore requested standing first, provisional notice on every new transfer second, and a public burden proceeding before any load change.

Morning light reached the table before they finished. Orra's blue slips lay above the signature class, separate from it. Nera would not use the child's private record as proof without Siv's consent. She cited only authenticated storm pressure and the direct demand already received. Restraint weakened the emotional case and strengthened the petition's right to exist.

Lev signed as agronomist and cooperative officer. Beren signed as heatworker organizer, then added a reservation that future standing could not justify equal present suffering. Nera preserved the reservation in full.

Tovan reached for the pen.

"You cannot sign as a legal representative of Orra," Nera said.

"I am not trying to."

"You cannot consent for future citizens."

"I understand."

"Then state your capacity."

He looked around the kitchen: rejected panes, dark forecast beyond clear window, blue slips copied from a shelter 137 years ahead. "Present beneficiary."

The phrase made no innocence claim. It also refused inherited guilt as an excuse for inaction.

Nera wrote it beneath his name. Tovan Voss, vent-glass apprentice, present beneficiary of deferred seasonal load. He signed with the blunt workshop stylus he carried on a cord, pressing hard enough to leave an impression through all three petition copies.

Outside, the forecast board flickered. A single white square appeared and vanished, too brief to offer weather. Nobody beneath it moved away.

Nera sanded the signatures, folded the burden index behind them, and pressed her season seal into the petition. When she lifted the brass, Tovan's stylus mark remained visible through the page under hers.

Chapter 17 — The Holdfast Voice

Siv Ren began the recount with a name that had no shelter.

Maret Kol appeared on the Kest winter roll as dead during Storm Year Four, age fifty-two, ration class kilnworker. The location field named Outer Holdfast Bel. Kest had no Bel shelter book in its archive. Only a copied burial mark and a heat claim had crossed the ridge before the signal towers failed.

Siv laid the slip beside Orra's calendar in the schoolroom. Blue true-day marks occupied one shelf. Death rolls occupied three tables and part of the floor.

"Count the mark or reject it," said Warden Tahl.

"On what provenance?"

"A burial heat claim required two witnesses."

"Bel changed its witness code after the east roof collapse."

"Then tag the uncertainty."

"The first figure had no uncertainty band."

Tahl's jaw tightened. He had not slept at home for six nights. Nobody working the recount had. Kest's council had ordered all outer-holdfast slips opened after Nera's claimant question reached them, and the buried numbers had spread across the school like a lesson designed for adults who preferred not to learn.

Siv turned the copied mark over. A child had practiced sums on the reverse before someone used the ceramic for burial data. Seven groups of four. One crossed out. Under Kest shelter rules, reused school slips entered the archive only when no clean clay remained.

"Accept provisionally," she said. "Code the witness change. Link the heat claim."

Tahl pressed gray chalk into Maret Kol's row. "That takes us to two thousand four hundred and eight outside the first return."

"Not all verified."

"All dead."

"We do not know that from every record."

"You knew enough to leave them out."

Siv looked toward the schoolroom door. Orra sat in the corridor copying ration fractions with six other children because the archive had taken their tables. The children knew not to enter while gray chalk remained in use. They also knew every name adults lowered their voices around.

"We had three classes when Nera asked," Siv said. "Kest verified, winter-line relayed, outer unverified. The council authorized the first two."

"You wrote the return."

"Yes."

"You wrote one number without naming the omitted class."

"Yes."

Tahl set down the chalk. "Say the reason in the room."

Siv wanted to tell him there had been 180 characters, that trust had not yet existed, that sending an uncertain total would have offered the past an easy objection to every certain death. All of that remained true. None described the whole choice.

"We needed the number to hold," she said. "The council believed a narrower verified count gave us leverage. I agreed."

"Leverage for what?"

"A freeze on new deferrals. Recognition. Shelter timing."

"Did we get them?"

"We changed one transfer for forty-eight hours."

"And then their weather came back through our roofs."

A gust struck the snow packing above the school. Dust sifted from a ceiling seam. Children in the corridor covered their copy slips without being told.

Tahl returned to the rolls. "Then send the corrected count before they make law from the first one."

Siv took up the next slip. "We send when the provenance table can travel with it."

"At current bandwidth, that takes days."

"Yes."

"Delay looks like another omission."

"A number without its basis repeats the first mistake faster."

By the end of the shift, they had rebuilt the count from four sources. Kest's signed shelter books supplied names and dates. Winter-line relays supplied district totals with phase stamps. The outer holdfasts contributed burial heat claims, roof-collapse lists, and ration books whose missing pages required upper and lower bounds. Duplicate names appeared where families moved during storms. Two hundred and eleven people had death marks in one settlement and later meal entries in another. Siv removed them from the confirmed total. Forty-six names appeared under changed household names; she held them for witness review.

The confirmed number settled at 34,112.

It did not include every probable death. It included more than the number Siv had sent.

Orra entered after the gray chalk had been covered. She carried two cups of boiled root water and placed one at Siv's elbow.

"Does the past child know about my calendar?" she asked.

"Tovan is not a child to himself."

"That means yes."

"He read a transcript."

"Did you tell him I count myself in shelter?"

"I sent the slip because it corrected the occupancy record."

Orra considered this. "Good."

"You wanted him to know?"

"I wanted the room to have four."

Siv looked across the tables. Every dispute in the recount sought the same thing. A room, a ration line, a burial claim, a count that did not discard a person because the record crossed damaged ground.

"Nera asked to use the calendar in a petition," Siv said. "I have not answered."

"Will it stop weather?"

"Not by itself."

"Will it make them count us before they send it?"

"That is what she says."

Orra touched the shelf where her original slips stood. "Then she can use the blue ones. Not the day I wrote about Mama."

Siv had not known Orra understood which private line she had withheld from the transcript. "I did not send that one."

"I know." Orra pushed the cup closer to her. "You leave things out when they might make someone stop listening."

The child went back to the corridor before Siv could decide whether the words accused or merely observed.

At the signal room, the phase needle gave them one open window before rest cycle. Siv composed the correction in blocks, then erased every version that began with explanation. Explanation first would resemble persuasion. The number needed to arrive before the defense.

She wrote:

> CORRECTION. CONFIRMED WINTER-LINE DEATHS: 34,112. PRIOR TOTAL EXCLUDED OUTER FIGURES. KEST COUNCIL WITHHELD UNVERIFIED ROLLS AND SOUGHT NEGOTIATION LEVERAGE. I AGREED. DOCUMENT PROVENANCE FOLLOWS.

The message exceeded one window. Signal keeper Hesh divided it at sentence boundaries and added sequence marks. "You can soften 'leverage.'"

"No."

"You can say 'preserved bargaining position.'"

"That uses more characters to hide the same decision."

Hesh loaded the first block. "Once this crosses, they may close the channel."

"Then send the number first."

The phase needle moved. Frost formed on the ceramic contact plate, though the room's air remained above shelter minimum. The first block entered the fixed delay.

Siv watched 34,112 leave her year.

The council called her upstairs before the second block cleared the transmitter. Kest's meeting room occupied the warmest level because its builders had once believed governance required fingers loose enough to write. Tonight the wall pipes stood cold. Warden Tahl had diverted their heat to the school archive, a practical rebuke nobody mentioned.

Five councilors sat around the stone table. Between them lay the transmission order Siv had signed and a proposed restriction on further records.

Councilor Venn pushed the restriction toward her. "You admitted negotiation strategy without authorization."

"I admitted my own agreement with it."

"You named the council."

"The council authorized the count."

"We authorized a verified figure."

"You authorized omission of a record class."

Venn's scarred thumb pinned the paper. "Outer data lacked verification. If the past publishes raw claims and finds duplicates, every death becomes suspect."

"If we withhold the rolls, every correction becomes persuasion."

Another councilor, Mera Silt, opened a ration ledger. "The carrier cost matters. One hour at full return pressure strips enough heat from the upper shelter to close its nursery room. We cannot transmit names for moral effect."

"Not for effect," Siv said. "For reconstruction."

"How many hours?"

"Hesh estimates sixty-eight for legends and high-level rolls. Personal fields require hundreds more at ordinary compression."

Mera's finger remained on the nursery allocation. "Then no."

The refusal held material reason. Kest could not prove its dead by endangering its living without reproducing the logic it accused the past of using. Siv turned the restriction over and wrote three conditions on its blank side: transmit record legends first; send rolls in source order rather than selected examples; suspend whenever shelter temperature reached nursery minimum.

Venn read the conditions. "You still assume transmission."

"I assume the past will ask why the next number deserves trust."

"They may not ask. They may use this to dismiss us."

"Then our answer remains available to our own children."

"That is calendar language, not negotiation."

"Negotiation gave us the first omission."

The room divided without a formal vote. Two councilors backed full restriction. Tahl supported controlled release. Mera wanted a heat budget before any carrier opened. Venn wanted to send only the provenance table, not personal fields. No one proposed pretending the revision had not happened.

Orra appeared at the stair with a sealed cup for Siv. Children did not enter council rooms during winter allocation, but she held her ground outside the threshold.

"My shelter teacher says you need the calendar consent," she said.

Venn turned. "This session remains closed."

"She is the record owner," Siv replied.

Orra lifted a small packet tied with blue thread. "You can send true days. Not the line about Mama. Not the drawing on the back."

Mera accepted the packet and checked the knots. "Do you understand the past may keep copies?"

"They already keep weather."

Nobody at the table found a harmless answer.

Tahl moved for controlled release under the three conditions, with a fourth: personal child fields restricted unless the record owner or guardian consented. Mera added a hard nursery-temperature cutoff. Venn recorded his objection and then signed the source-order rule, preventing Siv from choosing only clean records.

The decision did not restore trust. It transferred control from a persuasive total to an auditable process.

Siv returned to the signal room carrying the council sheet. Hesh had finished the three correction blocks and begun calculating packet sizes. When he saw the signatures, he unlocked the raw-roll encoder.

"Legends first?" he asked.

"Then Kest, relayed winter line, outer source order. No curated names."

"Child fields?"

"Restricted flag. Orra released blue calendar slips except one private line."

Hesh threaded a copper heat lead from the transmitter to the upper shelter exchanger. The return field required a gradient, and Kest had little warmth left to spend. At ordinary power, the packets would take months. At continuous carrier, the exchanger would cool occupied rooms while the phase plate threw part of their

stored heat into a signal 137 years backward.

Siv wrote the cutoff temperatures beside the control. "We stop at nursery minimum even if a packet remains incomplete."

"An incomplete group will need full retransmission."

"Then it does."

Hesh rested his hand on the carrier lever. "If they answer with silence?"

"We send enough for them to know how to rebuild the count. After that, we wait."

The correction blocks had cost characters. The rolls would cost rooms. Siv pressed her calendar-keeper seal into the release field, accepting responsibility for both.

Nera received the correction at Canal Nine with Edrin on her right and the claimant petition still damp under its registry seal.

The first block carried only the revised figure. Sumi had rigged the return glass to print each character as it arrived rather than wait for a complete message. Numbers appeared one at a time in blue-white light. When the final digit settled, Edrin drew a line through the original total on his comparison sheet. He did not replace it.

"Difference?" Sumi asked.

"Two thousand five hundred and eight," Nera said.

The count matched no branch in the early demographic model. Nera checked the phase signature. Holdfast Kest. She checked the sequence mark. One of three.

"Do not notify the registry yet," she said.

Edrin looked at her. "The petition cites the prior figure in an attached evidence schedule."

"Then hold the schedule."

"Registry closes at the next bell."

"We file a correction or withdraw the quantity."

"A late correction invites dismissal."

"An uncorrected number deserves it."

Sumi turned from the return glass. "You do not know whether this one carries better support."

"Which supplies another reason not to publish."

The second block began. PRIOR TOTAL OMITTED OUTER FIGURES. KEST COUNCIL WITHHELD UNVERIFIED ROLLS.

Nera read it twice. The room seemed colder, though the wall thermometer held

steady. She remembered every time Siv had corrected her language about descendants, every demand for living names rather than forecast subjects. During those exchanges, Siv had retained names of her own.

Edrin finished copying. "This attacks provenance for all prior casualty claims."

"It attacks one count," Sumi said.

"The Assembly will make it larger."

"That does not make their inference sound."

"It makes it available."

Nera touched the petition seal. Twenty-three present beneficiaries had signed around a claim that future citizens could speak for their own injury. The revised count did not erase that claim. Concealment could let Halden recast contact as advocacy rather than testimony.

The third block arrived: AND SOUGHT NEGOTIATION LEVERAGE. I AGREED. PROVENANCE FOLLOWS.

Sumi swore under her breath.

Edrin asked, "Do we have a duty to disclose immediately?"

"We have a duty to preserve immediately," Nera said. "Disclosure follows authentication."

"You argued against managed disclosure."

"I argued against an institution choosing a comfortable summary and calling it safety."

"How does this differ?"

"Because I will publish the correction with its support or publish that support failed. I will not release a number I have not audited merely because it injures the other side."

Edrin folded his comparison sheet. "The registry may treat delay as concealment."

"Then witness the reason and hour."

He opened the red book.

Nera dictated without reducing the breach. A future correspondent supplied a revised casualty total. The sender admitted that prior figures excluded outer-holdfast records for mixed reasons: incomplete verification and negotiation leverage. Seasonal Accounts suspended publication pending provenance review. No finding rejected the underlying claims of material burden or living source.

Sumi listened with both arms folded. "You are angry."

"Yes."

"That belongs in the record?"

"No."

"It belongs somewhere."

Nera looked at the return glass. "It can belong in my next message."

The standard window allowed no answer for another three hours. They used the interval to test what they could. Edrin compared sequence marks against prior contact. Sumi measured phase noise and confirmed that all blocks came through Kest's established source. Nera examined the language. Siv had separated confirmed from probable, named the omitted region, and promised provenance. Those choices favored audit. They did not excuse concealment.

A runner arrived from the Office registry asking why the petition schedule had been withheld. Nera wrote a temporary amendment removing the casualty quantity while preserving storm signatures, settlement population, and direct testimony. The claimant case could proceed without a death total. It would lose force in the Assembly. Force obtained through an unaudited number had caused enough harm already.

At the next window, she sent:

> CORRECTION RECEIVED. I WILL NOT PUBLISH 34,112 UNTIL PROVENANCE AUDIT. SEND RECORD CLASSES, DUPLICATE RULES, WITNESS CHANGES. YOUR LEVERAGE ENDANGERS CLAIMANT FILING. WHY SHOULD I TRUST THE NEXT TOTAL?

Sumi counted the characters and cut two spaces to fit. "You could remove the last question."

"No."

"It asks for an answer no checksum can prove."

"It asks Siv to decide how to answer."

The return did not come at the next six-hour interval. Nor the one after.

During the silence, rumors reached the Office. Someone had seen the changed schedule at registry. A canal sheet claimed Seasonal Accounts had discovered invented future casualties. Warm Ward notices demanded restoration of the normal forecast. In Low Glass, Beren issued a shorter statement: A disputed count does not warm a pipe.

Nera refused three requests for the revised figure. She gave Halden's clerk the red-book entry and nothing more. The choice cost her the easy appearance of transparency. It also kept the claimant petition from depending on a number that might shift again.

On Day Fifty-One, frost returned to the observation slit outside the normal window. Sumi came at a run from the phase floor.

"They are holding a continuous carrier," she said. "Low amplitude. Raw packet structure."

"That exceeds their ordinary window," Edrin answered.

"It exceeds ours too. The receiver is drawing against the canal sink just to remain open."

Nera placed both hands on the rail. "Can we accept it without changing scheduled load?"

"For a few minutes. After that the safety layer closes."

A header crawled across the glass:

> SHELTER ROLL TRANSMISSION. UNEDITED SOURCE ORDER. KEST / WINTER LINE / OUTER. RECORD LEGENDS FIRST. PERSONAL FIELDS INCLUDED. RESTRICT CHILD DATA.

Nera felt the anger remain. Trust had not returned merely because Siv sent more material. Yet the message offered exposure rather than another assertion.

"Copy to sealed local storage," she said. "No demographic model. No public mirror."

Sumi opened the witness spool. Edrin broke a new red-book thread.

Data arrived in compressed groups: shelter codes, date spans, witness keys, migration marks, burial heat claims. Each group carried parity characters and a source hash small enough for the return channel. The raw rolls would require hundreds of windows. Siv had begun with the legends needed to read them, sacrificing the rhetorical advantage of leading with names.

Cold spread along the phase casing. On the future side, carrier pressure fluctuated as if someone fed warmth into the coupling by hand. A line appeared between packets.

> TRUST NEXT TOTAL ONLY AS FAR AS YOU CAN REBUILD IT. I WILL PAY TRANSMISSION COST. ORRA CONSENTS TO BLUE CALENDAR SLIPS EXCEPT PRIVATE LINE.

Edrin read the sentence aloud. "Pay how?"

Sumi touched the casing and pulled her fingers back. "Their transmitter is shedding shelter heat into phase loss. They cannot hold this rate safely."

"Tell them to stop," Nera said.

"We have no outbound window while their carrier occupies the channel."

Nera watched source legends accumulate. Each one exposed where Kest had drawn a boundary: verified, relayed, copied, inferred, contested. The categories did not clean the first omission. They made its reconstruction possible.

The safety layer began counting down from one hundred eighty seconds. Sumi reduced the receiver aperture, buying another minute at the cost of lost packet clarity. Edrin called back every parity failure. Nera marked which blocks would need retransmission.

At twelve seconds, a final group crossed. Then the carrier collapsed. Warm canal water struck the phase sink below them with a sound like stones poured into a

metal basin. Frost fractured across the observation slit and fell onto the desk in thin white plates.

The last complete field held no argument, only the first page of names from the outer holdfasts, preserved in their original shelter order.

Chapter 18 — The Short Summer

Edrin authenticated the dead by trying to make them disappear.

He began with migration duplicates, the easiest defect for an Assembly counsel to demonstrate in public. On the left side of the workroom table he placed Kest shelter rolls. On the right he placed outer records received through the continuous carrier. Between them lay a blank exclusion book.

"Name?" Nera asked.

"Maret Kol."

"Conflict?"

"Death mark at Bel, later ration issue at Kest under same name."

Sumi looked over from the phase console. "Then your dead kilnworker walked eighty kilometers through compound winter."

"Or two people shared a name." Edrin turned both packet prints toward the witness lamp. "Different birth code. Different household mark. Do not exclude."

Nera drew a line through the proposed duplicate and recorded the reason. Their task did not require belief in the final count. It required them to reproduce every inclusion from rules disclosed by Siv, then attack those rules with the same suspicion they applied to current-year evidence.

The return channel had delivered six source legends and a fraction of the rolls. Child fields remained restricted. Names from unsupported burial lists arrived as coded hashes until a second witness class could confirm them. Kest had not sent a clean archive. It had sent an archive that revealed where cleaning would alter fact.

Edrin selected the next conflict. "Sera En, death mark at Ridge Shelter, meal issue eight days later at Hallow."

"Migration time?" Nera asked.

"Possible under clear ridge conditions. Impossible under the storm recorded by Kest."

"Current record for that storm?"

Sumi opened the phase ledger. "Deferred load from Year Two Hundred Twenty-One. Pressure crest reached their date inside a six-hour band."

"Then hold both entries pending weather verification."

Edrin used gray thread for unresolved records. Accepted names received blue. Exclusions received white and never vanished; the book preserved each rejected claim beside its reason. By noon, the table held fewer certainties than either side wanted and more support than the initial total had carried.

Nera read Siv's record legend again. Kest's witness codes changed after roof collapses, migrations, and shelter mergers. A code that looked inconsistent across

nine years often described a settlement forced to move its signing table. Current Office rules treated stable custody as evidence of reliability. Future custody survived by changing shape.

"We need their rule for emergency witness succession," she said.

Sumi checked the outbound clock. "Next window in forty-three minutes. You get 180 characters unless we widen."

"Not yet."

The phrase had become a discipline. Widening meant keeping more immediate heat and weather burden in their own year. Sumi had derived several small test loads, but no public authority covered a material schedule change. A maintenance allowance did permit her to accept heat produced by the receiver itself. Until now, she had vented that heat into the deferred sink.

Edrin tied off the first exclusion-book section. "This authenticates process, not total."

"Yes," Nera said.

"Halden will ask whether we confirm the revised number."

"We confirm only what we can rebuild."

"And the petition?"

"Needs living source and material burden. Death strengthens injury. It does not create personhood."

Edrin's pen paused. "Put that in the audit cover."

Nera wrote it beneath the source inventory.

The outbound window opened. She compressed their first questions into procedural nouns: WITNESS SUCCESSION AFTER ROOF LOSS? MIGRATION DUPLICATE RULE WHEN STORM BLOCKS ROUTE? SEND TEMPERATURE LIMITS FOR CONTINUOUS CARRIER. Edrin added their exclusion count and hash. Sumi transmitted.

A response would take hours at ordinary width. During the wait, they carried the disputed storm dates to Bellweather.

Lev had opened the oldest redleaf block to the wood.

Forty trees stood with narrow cores drilled from their trunks. The cores lay in labeled channels under the pollination shed lamp, each one a striped record of growth, cold arrest, fungus, and recovery. Yara moved along the table with a lens fixed over one eye.

"You asked for crop signatures," Lev said. "These cover fifty-three years. Earlier blocks were replaced after blight."

Nera examined a core from East Three. Pale spring wood narrowed around a

dark seam. "What makes that a storm signature rather than drought?"

"Cell shape." Yara slid the core under her lens. "Cold rupture collapses the outer ring while leaving the inner vessels round. Drought narrows both. Fungus discolors across the grain."

Edrin opened the current transfer ledger to a deferred storm in Year Two Hundred Twenty-One. Its pressure and duration matched the future Ridge Shelter date, but a ledger match alone would only compare two institutional records.

Lev arranged six cores by row elevation. "A phase-shifted storm leaves a sequence. Upper rows rupture first. Canal-warmed lower rows show delayed damage. The interval gives wind direction and root heat."

"Can you distinguish six hours?" Nera asked.

"Not from one tree. From a slope, perhaps."

Yara marked the first ruptured cell on each core. Elevation increased across the table. The injury moved downward in a pattern consistent with a northern pressure front lasting between four and eight hours. Sumi's phase record placed six hours of peak load into Year 360.

Edrin wrote the result as a range. "Supports storm obstruction. Does not prove Sera En remained at Ridge."

"It proves the route faced the weather in their ration record," Nera said.

Lev brought out a second set from the false thaw. Fresh damage showed the same two-direction sequence he had demonstrated after the closed session. "Send this signature to Kest. Ask whether their shelter instruments see the far-side pressure from our split spring."

Sumi shook her head. "Cell descriptions will consume three windows."

"Then send the pattern code."

"They do not know Bellweather's code."

Yara removed her lens. "Teach it."

They built a compact legend from six marks: elevation, cell collapse, root delay, front duration, recovery. It required one full outbound window. The actual pattern required another. Sumi objected to every redundant character. Edrin insisted on a version mark so future replies could not be interpreted under later definitions.

By dusk, the audit had become reciprocal in more than name. Kest exposed shelter custody and exclusions. Bellweather exposed how crops carried weather after public forecasts smoothed it away. Canal Nine supplied phase timing. Each record class tested the others without claiming authority over them.

Lev leaned against the sorting table. His eyes had not left the damaged cores. "If they confirm the split-spring signature, does that make your claimant case stronger?"

"Yes."

"Does it make the orchard safer?"

"No."

"Good."

Nera looked at him. "Why good?"

"Because people in Lume are starting to speak as if recognition changes weather. I need the petition to tell the truth about its limits."

"It asks for standing and notice, not relief from all load."

"Then publish that beside the Founder language."

Yara set a blackened fresh core among the older samples. "Publish the map too. If the city sees only totals, it will choose a district to blame."

Edrin recorded Bellweather's consent with conditions: worker housing separated from crop loops, names restricted, nursery routes public, injury records unredacted where families had already authorized Office use.

On the tram back to Lume, Nera watched people cluster beneath dark boards at every stop. Handwritten forecasts had begun replacing official glass. One shop promised thirty warm days. A canal sheet warned of immediate snow. Neither cited instruments. Uncertainty had not emptied the public field. It had made room for whoever wrote fastest.

Sumi widened the return on Day Fifty-Five under a maintenance authorization no one had designed for politics.

The authorization permitted Canal Nine to retain receiver heat for diagnostic stability. In ordinary operation, the phase sink moved that heat downstream with the rest of the schedule. Sumi closed the sink gate by one measured notch. Warm water remained in the present canal.

Temperature rose along the lower gallery. Workers removed their outer gloves. Aboveground, vapor thickened over Canal Nine and drifted toward Low Glass. The load remained too small to change a district forecast, but it made the cost of communication visible on gauges everyone knew how to read.

"Receiver aperture?" Nera asked.

"Up fourteen percent."

"Safety margin?"

"Enough for one extended response. If the bearing stripe widens, I reopen the sink."

"Who bears the retained heat?" Beren asked from the gallery rail.

Nera had asked him to witness distribution. He carried a pocket thermometer and distrust enough for three institutions.

"Canal workers first," Sumi said. "Low Glass air second. Warm Ward receives

nothing from this local rise.”

“Then close it.”

“We need the return to audit the rolls.”

“You need Low Glass to accept immediate heat so Warm Ward can receive a legal answer.”

Sumi’s remaining fingers rested on the gate wheel. “Correct.”

Beren looked at Nera. “Your reciprocal audit begins with the same people paying the instrument cost.”

Nera opened Beren’s burden map on the console. A maintenance test could proceed without Assembly distribution, but authority did not make its burden neutral. “How do we shift the receiver heat?”

“Open the civic bypass to the Ward storage loop.”

“That loop remains sealed under residential priority.”

“Then obtain a co-signature or stop.”

Sumi glanced at the outbound clock. Six minutes remained before Kest’s expected response.

Edrin took the maintenance authorization. “A diagnostic load can use any connected sink if the primary route exceeds worker tolerance. Beren’s thermometer supplies the tolerance record. Nera can authenticate. I can witness.”

“Ward control may reject the route,” Sumi said.

“Let it reject in writing.”

She opened the civic bypass. The first request came back denied: NO RESIDENTIAL NEED. Nera appended the canal temperature and burden-map citation. Edrin sealed the diagnostic necessity. The second request hesitated, then granted a narrow draw into the Warm Ward soil stores.

Heat left the lower gallery. Beren’s thermometer stopped rising.

“Now widen it,” he said.

Sumi closed the phase sink another notch. She accepted the immediate heat load into the current-year system, but the benefit map redirected part of it toward infrastructure that had received the greatest historical moderation. Receiver aperture climbed to twenty-two percent.

The return glass filled with packet headers.

Kest sent its emergency witness rule, migration table, and nursery cutoff. Then came a response to Bellweather’s pattern legend. Their shelter pressure gauges had recorded the split-spring signature at the far side of the phase interval: upper terraces first, lower heat lag, six-hour crest. Lev’s trees and Kest’s walls described one transfer from opposite ends.

Edrin verified source hashes against the raw-roll legends. Sumi checked phase

timing. Nera compared every interpretive code to the version Bellweather had sent. At the end, all three signed one sheet titled RECIPROCAL AUDIT, ROUND ONE.

The audit did not ask either era to trust a total offered alone. It required each to disclose method, uncertainty, and material record. A claim survived only as far as the other side could reproduce its path.

Sumi reopened the sink before the bearing stripe reached limit. The extended window closed. Canal heat settled below worker tolerance, while the Warm Ward storage gauge showed a small, undeniable rise accepted from the communication cost.

“Enough to rerun summer?” Nera asked.

“Enough to stop pretending the return has no forecast weight.”

The honest summer run converged on Day Fifty-Eight.

Pavel projected it onto the public calibration wall with Central’s sealed model beside it. The two ranges shared reservoir levels and cloud-comb capacity. They diverged wherever Central assigned downstream load to an empty field.

Sumi entered the authenticated shelter pressures, Bellweather crop signature, and burden-map routes. The warm season contracted from both ends. Late frost risk extended forward. The first autumn storm advanced under retained receiver heat and known phase recoil.

Pavel read the result twice. “Summer shortens by nineteen days.”

No one in the room corrected him. The range held across three solvers and every physical constraint they had been able to verify.

Halden arrived before they issued it. He came without Vale or an Assembly clerk, carrying the restraint of someone who knew witnesses already filled the room.

“A nineteen-day revision without public preparation will cause hoarding,” he said.

“A normal range has already caused planting and store release under false assumptions,” Nera answered.

“We can announce staged uncertainty.”

“The model has converged.”

“Models change.”

“Then publish its inputs and revision rule.”

Halden looked at the reciprocal-audit sheet. “You authenticated future shelter records after admitting manipulation by their custodian.”

“We authenticated a method and the records that survived it. The exclusion book remains attached.”

"You are moving from inquiry to governance without Assembly approval."

"No. We are returning a forecast to the public."

He could have blocked issuance through executive seal. Instead, he asked for six hours to prepare reservoir and food notices. Nera granted two, the time required for the boards to synchronize. Halden signed the delay under protest.

At the second bell, weather glass across Lume lit amber. The new range appeared with uncertainty branches visible. Beneath them, one line stated the loss plainly: EXPECTED MODERATED SUMMER REDUCED BY NINETEEN DAYS FROM PRIOR PUBLIC RANGE.

People read the figure before they read the method.

A crowd formed at the Office doors. Winter-garden families demanded to know whether stored soil heat would be seized. Market porters shouted about food prices. Someone struck the forecast glass with a heat receipt rolled around a stone. The outer pane starred but held.

Nera went into the vestibule before the second stone could follow. Edrin unlocked only the inner door, leaving reinforced glass between her and the street. Voices struck the pane without separating into words. She raised the forecast run where people could see its amber seal.

A woman in a seed-vault coat pushed to the front. "Does nineteen days mean nineteen days of winter added to this year?"

"No," Nera called through the speaking grille. "It means the moderated warm range ends nineteen days earlier than the range issued under excluded data. Individual fronts remain uncertain."

"Then why print one number?"

"Because the season length changed by that amount across the verified branches."

A market porter lifted an empty produce crate. "Bellweather already cut pears. What else?"

"Redleaf takes the first reduction. Staple planning enters tomorrow's food notice."

"Tomorrow does not fill this." He shook the crate.

The school steward who had signed Nera's beneficiary class stood near the tram rail. "My classrooms need closure temperatures, not a season argument."

Nera found the branch sheet behind the forecast run and pressed it to the glass. "The school thresholds appear here by district. Copy them now. They remain provisional until reservoir review."

People nearest the door began reading aloud to those behind. The crowd changed shape. Some anger moved toward specific dates and temperatures. Other anger sharpened when people found that their district carried no clean answer.

A young man in a vent-shop apron reached the front with white glass dust in his hair. Nera knew him from Tovan's shop but not by name. "Ward council ordered us to brace garden panes through the revised autumn edge. Are the soil stores staying local?"

"The audit load entered them yesterday. Future allocation requires a public burden proceeding."

"So they can hold heat while Low Glass runs the receiver?"

"Not without the transaction appearing in the open map."

"That does not answer whether they can."

"No. The Assembly still holds distribution authority."

The admission produced another surge of voices. Someone called the audit another experiment paid by workers. Someone else demanded immediate closure of the lattice. Beren's route markers had given Low Glass a language for burden; the dark boards had given every rumor equal space. Now one honest forecast could not supply the governance that years of smooth forecasts had displaced.

Tovan appeared behind the apprentice and put both palms against the glass. He had left work without removing his forearm guards.

"Tell them what the ledger contains," he said.

Nera used the speaking grille. "Ward benefit, canal injuries, crop transfers, future shelter load, and the rules used to omit each from public schedules."

"Names?" asked the seed-vault woman.

"Worker names where families authorized release. Future child records remain restricted. Institutional signatories remain visible."

A man at the back shouted, "Why not now?"

"Because sixteen volumes cannot pass through this door without becoming whatever sentence the loudest person selects."

The porter raised his crate again. "Then open the hall."

Assembly Hall had copying tables, public witness rails, and enough doors to prevent a crush. It also placed Halden's authority in the room. Nera looked through the starred pane at faces reflected over the amber forecast. Delay carried risk. So did release into a street where a thrown receipt had already become a stone.

"I will register a full public reading," she said. "The notice will show hour, place, redactions, and every attached method."

"Before the next forecast changes?" the school steward asked.

"Before the next scheduled run."

Tovan stepped back from the glass. The apprentice beside him lowered a second stone Nera had not noticed in his hand. He unwrapped the heat receipt and smoothed it against his apron instead.

In Low Glass, canal crews refused to clear Beren's route markers. At Bellweather, Lev sent a notice that redleaf acreage would be cut before staples. Those explanations moved more slowly than anger.

Nera stood inside the damaged door with the unredacted seasonal ledger on a cart. Founder columns, ward benefit, canal losses, future load, and her mother's classification filled sixteen bound volumes. Edrin had attached Sumi's dissenting checksum and the first reciprocal-audit report. No summary could carry all of it without choosing what the public saw last.

Halden stopped beside the cart. "If you release this during unrest, people will read injury as instruction."

"If I withhold it, they will invent an instruction from nineteen days."

"You could wait until the Assembly sets terms."

"The Assembly had terms when it closed the session."

Outside, another blow struck the glass. This one left a white ring around the first crack.

Nera opened the publication registry. She entered Day Sixty-One, first bell, Assembly Hall, full public copy. No redactions beyond child privacy and living-worker addresses. Edrin witnessed the schedule. Sumi attached the forecast run. Lev and Beren's conditions entered as separate notices rather than a unified endorsement.

When Nera pressed her seal, the registry printed a notice across every restored board. The crowd outside stopped hitting the glass long enough to read where the ledger would open.

Chapter 19 — The Public Ledger

Nera carried the unredacted ledger into Assembly Hall with both hands under its boards.

The book had gained no mass during the reciprocal audit, yet its brass corners pressed through her gloves. Edrin walked at her left with the witness folio chained to his wrist. Sumi kept pace on the right, favoring the hand that still had all its fingers. They entered through the heated civic tunnel because the square above had filled before dawn.

Voices reached them through stone. Growers had come from the western tram line in waxed field coats. Low Glass crews occupied the lower steps with radiator keys hanging from their belts. Warm Ward residents waited under pale vent-glass awnings brought against the shortened summer's first hard rain. Nobody had received the full account, but everyone had lost nineteen days from the forecast.

At the hall door, an usher held out a custody slate. "Steward's office requests the reading copy."

"This is the reading copy," Nera said.

"The First Steward intends to present a verified summary before public inspection."

"No summary carries my season seal."

Edrin lifted the chained folio. "Deputy witness concurs."

The usher looked toward Halden's clerk, who stood beyond the bronze doors with two ward marshals. Sumi placed her damaged hand on the ledger's top board. Her shortened fingers made the gesture look like a measurement.

"Move us or open the door," she said. "Either choice will enter the live record."

The doors opened.

Assembly Hall had been built to keep argument warm. Radiator pipes crossed beneath the stepped floor, and every bench carried a brass foot rail supplied from the civic tunnel. Today the heat felt accusatory. Damp coats steamed. Orchard mud loosened from boots and marked the stone aisles. Above the central dais, the public forecast glass showed only the current temperature and a blank field where summer confidence had once stood.

Halden Rusk waited beside the reading table. He wore no steward's sash. That omission might have suggested humility if six clerks had not arranged themselves behind him with sealed packets.

"Nera," he said quietly, while the room found their faces. "You agreed to authenticated publication."

"I brought the authenticated account."

"You agreed that provenance would precede accusation."

"It does. Pages one through eighty-six establish provenance."

His gaze rested on the book. "The city has already begun hoarding food."

"The city began hoarding when your normal forecast went dark."

"We can still separate public hazard from legal claim."

"Not without altering the annex."

A bell struck once. Assembly members took their curved benches. Beren Olt sat in the heatworkers' petition section with both palms over the head of his radiator key. Lev had sent Bellweather's crop signature but stayed with the orchards. Tovan stood at the rear rail among vent-glass apprentices, his hair wet from the square. Nera caught his eye. He did not smile or nod. He held up Orra's ceramic-calendar transcript so she could see its blue edge.

Halden noticed. "Your son should not have to carry this."

"He chose what to carry."

"Children often do, after adults narrow the choices."

"He is seventeen."

"So were half the vent crews during the Founding Crisis." Halden's voice stayed low. "Read the summary first. Give the wards one hour to absorb the operational facts."

Nera opened the ledger on the table. Cold had warped the pages around the Founder Sum, leaving a shallow basin in the paper. "An hour gave us centuries of euphemism."

The second bell ended their private exchange.

Assembly Speaker Amel called the session under emergency disclosure rules. She named the reciprocal audit, the future casualty revision, the nineteen-day forecast loss, and the petitions waiting under seal. Her voice crossed the hall through old speaking tubes. No amplification lattice touched proceedings about the lattice itself.

"Senior Notary Voss," Amel said, "you will authenticate the object before you characterize its contents."

Nera set her mother's brass seal beside the Office seal. The older tool bore no authority. Its chipped handle had once authenticated heat receipts over a kitchen kettle. She had brought it because private records had exposed what public language concealed.

She gave the ledger's retrieval date, custody path, phase signature, witness chain, and material tests. Edrin confirmed each point from the folio. Sumi described how accepted current load widened the return channel enough to receive raw shelter rolls. She did not dramatize her missing fingers or the phase room's danger. Her technical nouns held the hall more firmly than an appeal would have done.

A Warm Ward member rose. "Can the future records have been prepared in response to our questions?"

"Yes," Sumi said. "A person prepared them."

"Can they be independently verified?"

"Storm signatures, crop failures, shelter allocations, phase timing, and burial heat reductions all cross-check against outputs generated here before contact. The records also contain errors. People made those."

Murmurs passed up the benches.

Nera read the first hidden schedule. A winter softened for the original landing hospital had pushed cold 137 years forward. The entry named no recipient because Year 140 lay outside the Charter's living public. The next page financed seed vault protection. Another protected a migrant camp after a cloud-comb failure. No clean division separated mercy from theft. Founders had saved lives, then assigned the material consequence to lives they refused to count.

She kept reading.

By page seventeen, the hall had stopped murmuring at every figure. Numbers accumulated too quickly for shock. Nera changed method. She asked the public glass to display one line from each ward, paired with its downstream weather signature. Warm Ward winter gardens faced Kest shelter rationing. Bellweather fruit retention faced black snow over the northern terraces. Low Glass factory continuity faced a future radiator canal freezing solid while workers remained inside its service collar.

Beren stood without recognition from the Speaker. "Show benefit share."

A clerk objected, but Nera already had the map prepared. "Per resident or per thermal holding?"

"Both. Let them see why equal cold would lie again."

Two maps opened side by side. The first divided current benefit by population and made the wards look nearly alike. The second traced stored soil heat, protected transit, industrial continuity, orchard yield, maintenance injury, and emergency access. Warm Ward darkened first. Several civic institutions followed. Low Glass received less benefit while its workers carried the highest bodily risk.

A woman in a pale insulation coat pushed through the Warm Ward petition row and gave her name to the floor clerk: Nial Vosper, superintendent of communal kitchens. She did not wait for recognition.

"My buildings show as protected holdings," she said. "That heat keeps four thousand residents fed when household vents close. Are you calling a communal stove private comfort?"

Nera enlarged the kitchen district. "No. The audit separates subsistence draw from reserve draw. Your ovens take from both."

"Because the ward council connected the reserve manifold after the grain riots.

We cannot uncouple it in a day."

"Can you meter it?" Sumi asked.

Nial looked up at the phase technician. "By building, not by oven."

"Then your kitchens can show what they consume while the soil vault shows what it keeps. That gives us an operational distinction."

A man three rows below Nial held up a scarred forearm. "Our canal already gives you a distinction."

The clerk recorded him as Ove Cas, night valve tender, South Low Glass. Ove had spent twenty-six years keeping a return manifold clear of mineral scale. His block received moderated heat for nine fewer weeks each year than the civic kitchens, yet injury costs from both districts entered the same public utility ledger.

"If her oven needs reserve water," Ove said, "someone turns a key beneath my sleeping room. Every ward plan starts at the warm end and reaches us as pressure."

Nial faced him across the aisle. "Do your children eat from our kitchens during a canal closure?"

"They do when the tram runs."

"It did not run during last winter's glass strike."

Ove's raised hand lowered. "No."

"Then stop drawing the map as if every protected pipe serves only the people above it."

Beren watched without intervening. Nera understood why. The exchange complicated his claim without weakening it. Infrastructure crossed ward lines; benefit did not become equal merely because emergency use occasionally did.

She switched the display from residence to service destination. Fine lines joined the communal kitchens to six Low Glass shelters and two tram depots. Other reserve lines ended inside garden vaults, private galleries, and a heated walking court. The room reacted differently to each terminus. A kitchen line drew guarded approval. The walking court brought laughter without humor.

"This ledger can distinguish them," Nera said. "Prior schedules chose not to."

Nial studied the new map. "If the court closes, its glass will crack under the first retained front."

Tovan called from the rear rail, "Then drain it before the front. Apprentices can set salvage cuts in the lower panes."

Several heads turned toward him. He opened Orra's ceramic transcript like a work order rather than a banner.

Ove pointed at the garden reserve. "How much can that manifold release without fouling our canal?"

Sumi answered with a range, then marked two uncertainties in red. "We need

your scale readings. Not the ward maintenance average. Your night log.”

“I have thirteen years.”

“Bring three coldest cycles and every rupture week.”

Nial added, “I can separate kitchen meal hours from vault retention if someone gives me two meter crews.”

No vote had authorized release. No distribution rule had yet survived debate. Still, clerks opened a working folio beneath the maps. Ove crossed the aisle to write his canal designation beside Nial's buildings. Their cooperation did not reconcile the wards. It produced the first pair of measurements the city could use without pretending their burdens matched.

Someone shouted, “You want us frozen for their gardens.”

Beren turned toward the voice. “I want their stores opened before our pipes split.”

Speaker Amel struck the rail for order.

Nera displayed the Founder Sum without commentary. Founder-era script filled the public glass in narrow black lines. The clause appeared near the lower margin, after formulas for survivorship and before the riots had forced the annex under seal.

She read it once in the founders' exact legal grammar: “Downstream persons materially burdened by deferred dissipation retain standing as claimants to the account, notwithstanding temporal nonpresence at execution.”

A strange stillness followed. Not silence, exactly. Wet fabric shifted. Radiator pipes knocked beneath the floor. A child in the public gallery asked which words mattered, and an adult whispered all of them.

Halden requested the floor.

He stood at the central speaking mark rather than the steward's bench. “The clause has never been enacted. It conflicts with the Charter's requirement that claimants present themselves through a living ratepayer register. Publication should not misstate archival language as current law.”

Nera answered from the reading table. “The petition asks whether current law can exclude the people whose burden gives the utility value.”

“We agree the question requires adjudication.”

“You sealed the document that permits the question.”

“My predecessors sealed it. We preserved that classification.”

“You renewed annex custody four times.”

“To prevent unauthorized destruction.”

“Also unauthorized reading.”

Halden accepted the distinction with a slight incline of his head. “We believed a

premature claim could cause ward conflict before any practical remedy existed.”

A grower called down, “Does one exist now?”

“No complete remedy,” Halden said.

His answer cost him more than denial. Nera saw it in the clerks behind him. Two stopped holding their packets at matching height. One lowered her eyes.

Nera turned pages until her mother's file lay under the glass reader. The vent diagram appeared above them, enlarged beyond any private scale. Canal Six, service collar, twelve-minute cooling lag. The public classification named infrastructure loss. The benefit ledger had preserved different words.

Her mouth dried. She had rehearsed this line with Edrin in the Office and again in the tunnel. Rehearsal had not included Tovan listening from beneath Orra's calendar.

“Mara Voss,” she said, “died with two other attendants during a vent rupture in Year 202. Her family received a finding of compensable civic fatality. The hidden schedule assigned her death to acceptable continuity loss and credited Warm Ward winter retention in Year 223.”

The diagram marked the three bodies as small red squares.

“Read the dependency,” Edrin said. His prompt sounded procedural. She knew it as kindness.

Nera read how the later ward schedule depended on cooling capacity withheld twenty-one years earlier. Her mother's death had not purchased the entire garden season. No ledger could make suffering exchangeable so neatly. Yet officials had counted Mara's life inside the continuity cost and denied her family the grammar of that count.

Tovan lowered the ceramic transcript. His face had gone colorless.

A Warm Ward councilor asked, “Were current residents told that retention depended on this casualty?”

“No,” Nera said.

“Then why present us as authors of it?”

“I am presenting you as beneficiaries with choices now.”

That answer brought anger from both sides. Low Glass residents demanded why ignorance should excuse comfort. Ward residents demanded why inherited infrastructure should create personal guilt. A Bellweather delegate named children whose food depended on stable nights. Someone in the high gallery began reading the revised death total aloud, one digit at a time, until marshals asked her to stop.

Halden raised both hands. “We must not confuse responsibility with blame.”

Beren replied before the Speaker could recognize him. “Then stop protecting stores as if blame owns the valves.”

The hall broke into overlapping arguments.

Nera closed the ledger, not to conceal it but to protect the binding from the hands crowding toward the table. Edrin moved beside her. Sumi climbed onto the phase technician's step and pulled the public forecast blank. In its place she opened the live return pane.

Noise crossed it in pale horizontal bands. The shortened summer had raised the current load enough to keep a narrow channel active. Characters formed, failed, and formed again. Siv had been told when publication would begin. She could not hear the hall. She would receive its outgoing account 137 years after the room generated it, inside the same history that had carried her answer back.

Sumi pointed to the witness field. "The phase mark authenticates current receipt."

The first line stabilized.

> SIV REN. CALENDAR KEEPER. HOLDFast KEST. PRESENT UNDER YEAR 360 SHELTER REGISTER.

A clerk corrected the capitalization artifact on the display copy but left the raw line untouched. Siv's second transmission arrived in a burst that made the radiator pipes tremble.

> YOU ASK IF WE ARE CLAIMANTS OR WEATHER. STOP CALLING US EMPTY. WE HAVE NAMES FOR EVERY ROOM WE LOST.

No speech could have quieted the hall so quickly.

Nera read the phrase aloud for the official record. "Stop calling us empty."

Speaker Amel leaned toward her clerk. "Enter the received statement as testimony subject to phase authentication."

Halden crossed the floor. "A return cannot testify under current procedure."

"Neither could a claimant excluded by date," Amel said. "The petition contests that exclusion. I will not decide admissibility by enforcing the contested rule before hearing."

For the first time since Nera had known him, Halden looked past procedure and found no smaller rule available. He turned to the Assembly members, measuring whether authority remained where he had left it.

The public doors opened behind Tovan.

Rain drove into the vestibule. The square had overflowed onto the canal bridges, and relay readers had carried the publication line by line to those outside. A chant began near the lower steps, roughened by distance. Nera expected the death count or a demand to open the ward stores.

The crowd repeated Siv's words.

"Stop calling us empty."

Marshals moved to close the doors. Halden told them not to.

He returned to the central mark. "We cannot repair 4.7 billion degree-hours

through one public emotion. We cannot expose this city to a compound winter without shelter calculations, food protection, or a distribution schedule. Full disclosure does not repeal material limits.”

“No one has proposed total repayment today,” Nera said.

Sumi brought up the retained-load model. “One scheduled transfer. Thirty-one percent held here for one day. Instrumented at every canal, ward store, orchard mesh, and phase drum. Automatic abort remains active for lethal thresholds.”

A Low Glass representative shouted, “Our pipes hit threshold first.”

Beren faced the dais. “Then load cannot remain equal. Open protected stores by benefit before temperature reaches us.”

Warm Ward members protested. Lev's crop map arrived through a side clerk from Bellweather, showing barley, sunroot, and redleaf tolerances under honest forecast ranges. Nera watched the argument change shape. An hour earlier, speakers had contested whether future citizens existed in law. Now they argued over valves, crop priorities, insulation classes, and who would carry how much cold.

That change did not make them just. It made refusal harder to hide inside vocabulary.

Speaker Amel called for a preliminary vote: whether to authorize preparation for one controlled retained-load day, with distribution terms due before execution. The board lit above the benches.

Halden could have delayed it by invoking civic panic. His packets likely contained the findings required. He looked toward the open doors, where rain blew across the threshold and the future testimony kept its place on the glass.

“We withdraw the request for managed release,” he said.

Three clerks behind him lowered their packets together.

Votes entered by ward. Bellweather approved with crop conditions. Low Glass approved only if protected stores opened first. Warm Ward divided, then crossed the threshold by four seats when a vent-glass guild member changed her mark. The final board held long enough for every relay reader to copy it.

AUTHORIZED: CONTROLLED RETAINED-LOAD DAY, DISTRIBUTION PLAN REQUIRED.

Tovan reached the table after the Speaker adjourned. People pressed around Nera with petitions, accusations, proposed valve schedules, and names for the claimant register. He slipped Orra's transcript beside Mara's brass seal.

“You read her file,” he said.

“Yes.”

“All of it?”

“All that belongs to the account.”

He touched the old seal but did not lift it. "Then give me a store inventory. The apprentices know which ward panes can take a pressure change."

Nera looked toward Halden. He stood beneath the voting board while Beren and Sumi argued over emergency thresholds in front of him. After a moment, he removed the civic-tunnel key from his sash and set it on the reading table for the record.

Outside, the chant had ended. Rain struck stone, orchard coats, vent glass, and the open bronze doors. Clerks brought fresh folios to receive the city's first preparations. Nera opened the ledger to a clean witness page. This time, no one asked her to leave the counterparty blank.

Chapter 20 — Lev's Harvest

Lev cut the first redleaf branch before sunrise on Day 62.

The pruning hook passed through green wood with a wet click. Twelve pears, each no larger than his thumb joint, dropped into the grass. They carried the dull rose color that usually deepened through summer. This year had no ordinary summer left to offer them.

He bent to collect the fruit. Waste still offended his hands, even when calculation required it. Bellweather's honest forecast hung from the nearest mesh post in a wax sleeve. Three temperature bands crossed the page instead of one confident line. The warmest spared most fruit but emptied the city stores. The middle band killed the western redleaf rows. The coldest endangered frost barley in the lower field and turned sunroot crowns to fiber before they could set sugar.

Above the orchard, retractable mesh waited in black folds. Each panel could hold radiated soil warmth over one crop block. Bellweather had enough intact mesh for sixty-one percent of its planted ground. Before publication, Lev could have requested an extra cloud-comb draw and pushed risk into a column nobody inhabited. Now every degree had an address.

Yara Fen came down the row with a chalkboard under one arm. She had tied her hair under a cut grain sack against the rain. "You started without the crew."

"I started with my own block."

"Your name sits on every block."

"The pears here belong to my winter share."

"Not after they enter the cooperative bins." Yara nudged one fallen pear with her boot. "You want to prove you can make the cut before you tell everyone else."

"I wanted the hook tested."

"On twelve pieces of fruit?"

Lev wiped sap from the blade. "It cuts."

"So do you."

Workers gathered at the row end, too far away to hear the exchange but close enough to see its posture. Lev disliked authority most when he caught himself performing it.

"Show me your board," he said.

Yara set it against the mesh post. She had redrawn his crop plan by labor hours rather than thermal value. Redleaf thinning required forty people for six days if they salvaged usable fruit. Barley trenching needed twelve people and both canal crawlers. Sunroot insulation demanded dry straw they had not yet purchased. His plan protected the highest calorie yield. Hers exposed who would remain in wet rows while temperatures fell.

"You assigned the east crew to pears through Day 75," she said. "They sleep in field housing with no floor canal. If we keep them on salvage, their rooms lose the same heat we claim to save in fruit."

"The housing loop has priority feed."

"On paper. The south valve stuck twice last winter."

"I signed its repair."

"You signed the request."

That distinction landed cleanly. Lev had read the maintenance closeout because he read every cooperative file. He had not crawled under the housing with a lamp to watch the valve cycle.

Yara pointed to the lowest row on her board. "My twins eat barley meal through winter. They also sleep over that valve. Stop speaking as if crop and household loads occupy separate maps."

Lev looked across Bellweather. Mist held between the redleaf rows. The barley field beyond them had gone silver at the tips, not from frost but from seed hair catching weak light. Sunroot leaves lay flat and broad over the black beds. Three forms of food, three tolerances, one crew already short from tram closures.

"Bring the east workers in," he said. "We rebuild the plan at breakfast."

Yara did not pick up the chalkboard. "With them speaking?"

"With them speaking."

"Not approving a plan you already finished?"

"No."

She studied him for the easy concession he sometimes gave when he intended to preserve the decision elsewhere. "Then leave the hook here."

Lev set it in the grass.

At breakfast, twenty-three workers crowded the grading shed around tables made for fruit crates. The honest forecast covered one wall. Nobody asked for the old forecast back. They asked which error band could kill a child, which could spoil grain in storage, and whether the controlled retained-load day would be canceled if Bellweather crossed a crop threshold before Low Glass crossed a human one.

Lev answered only what the plan guaranteed. "Human thresholds trigger automatic abort. Crop thresholds do not."

"Then why instrument the fruit?" asked Kem, the irrigation keeper.

"So the next transfer does not use invented damage."

"That helps next time."

"Yes."

"What helps this time?"

Lev moved three wooden crop markers on the table. "We sacrifice the main

redleaf harvest. We open its mesh during the retained front, let those rows radiate freely, and use their soil loop to protect the frost barley. Sunroot gets straw, trench walls, and the spare south panels. We preserve seed stock from every redleaf strain, but not market fruit."

The room held the proposal without accepting it.

Redleaf had paid Bellweather's equipment debt for nine seasons. Barley fed more people but brought a controlled price. Sunroot could survive cold if its crowns stayed dry, though harvesting frozen ground broke tools and wrists. A rational plan still needed workers willing to enact it.

Yara stood near the door. "Name what disappears."

"Seventy-eight percent of saleable pears under the middle band. Possibly all under the lower."

"Wages."

Lev looked at the cooperative sheet. "Eight weeks of harvest shifts."

"Debt payments."

"Deferred unless the Assembly refuses crop-loss credit."

"Field housing rent?"

He hesitated.

Yara tapped the table once. "There. Your command assumption."

A few workers shifted. Nobody smiled.

"The cooperative owns the housing," Lev said.

"Which means you counted rent as income while removing wages as expense."

"I did not remove wages."

"You moved them into an emergency petition. We cannot buy meal with a petition."

Lev took the chalk from her board. "Housing rent suspends from Day 65 through winter inventory. Meal shares continue at current work credit even if pear shifts disappear."

"Can the cooperative carry that?"

"Not without selling two mesh crawlers."

Kem looked toward the machinery bay. "Sell the north crawler. Its wheel seal leaks."

"It leaks because you patch it with packing cloth," another worker said.

"I patch it because the parts order went to cloud combs."

Argument spread through the shed, practical and unceremonious. By midday they had a plan no single person had authored. Pear crews shifted to barley trenches. Housing repair preceded crop work. Sunroot straw came from a brewery

contract Yara knew had stalled. The cooperative would keep one crawler and strip the other for seals before sale. Lev retained final responsibility, but his original schedule survived only in the crop priorities.

He retrieved the hook after lunch. This time forty hands cut beside him.

By Day 71, Bellweather smelled of green fruit and opened soil.

Rows that should have carried shade stood exposed. Workers hauled immature pears to fermentation pits because even failed fruit could feed yeast for winter grain. The redleaf leaves remained on the trees to return what energy they could to roots. Lev walked each row with a strain register, marking two protected branches per tree where viable buds might harden for next year.

Nera arrived on the afternoon tram with a case of sealed temperature wafers. She stepped down wearing city boots unsuited to orchard mud and did not complain when the ground took them past the ankle.

"You cut them," she said.

Lev handed her one of the last sound pears. "Not all. Enough to stop pretending."

She weighed it in her palm. "The Assembly wants a crop-loss ceiling."

"Crops do not observe ceilings."

"They need a number for the abort schedule."

"If you abort when redleaf fails, nothing happens. It has already failed."

"I know."

"Do they?"

Nera looked toward the open rows. "They will when your inventory enters the public ledger tonight."

He heard exhaustion in the precision of her answer. During their years together, she had grown more exact as fatigue deepened. He had once mistaken that for distance. Sometimes it had been distance. Sometimes exact words had kept her upright.

They walked to the sunroot beds. Workers had laid straw between broad leaves, then fixed it with narrow mesh so the next wind would not uncover the crowns. Nera knelt to inspect a sealed wafer beside the irrigation lip.

"The city offers four more sensors," she said.

"We need twenty."

"You have twelve."

"Two drift in rain. One reads the canal, not the bed. One belongs to Yara's housing loop now."

Nera rubbed mud from the wafer's seal. "Sumi can calibrate six before Day 85 if

the phase room holds.”

“Then we still need two.”

“Warm Ward has garden instruments.”

“Warm Ward has guards.”

“Halden opened the inventory.”

“An inventory does not open a gate.”

She stood. “Come with me tomorrow.”

“To the ward?”

“To claim them under the benefit schedule.”

Lev looked back at the pear rows. “I leave for a day, Yara runs the orchard.”

“That appears to frighten you less than it did last week.”

“She already runs half of it. Last week I was frightened she'd make me admit it.”

Nera's mouth nearly formed a smile. “And now?”

“Now it has an entry state.”

The phrase belonged to her, from an older argument about failed pumps. She recognized it and looked away toward the barley.

Their former intimacy lived in such recognitions. It did not oblige either of them to resume what had ended. Lev had wanted her to leave the Office when Tovan was young. Nera had heard a demand to abandon the only authority she could use. She had wanted him to trust verified risk over instinct. He had heard contempt for knowledge carried in hands and crops. Neither misunderstanding had been complete.

“I cannot offer you a safe test,” she said.

“I would distrust one if you did.”

“I cannot promise the city will keep crop relief after people feel the cold.”

“Then seal what they promise before it starts.”

“I am trying.”

Lev turned the pear in her hand so the stem pointed down. “Trying no longer counts as a private virtue. The ledger took that shelter from all of us.”

Her fingers closed around the fruit. “Will you lock the mesh?”

“When the workers approve the final stations.”

“I asked whether you will.”

“Yes.”

“Knowing a bad front could take the barley too?”

“Yes.”

Nera nodded once. She placed her free hand against his forearm, where sap had

dried in dark streaks. The touch carried neither apology nor claim. It acknowledged the answer and the body required to carry it.

At the tram stop, he gave her six sensor requests and kept the last pear. She left him the sealed wafers and a copy of the controlled-day conditions bearing her notarial mark.

The frost arrived eleven days early.

At two hours before dawn on Day 83, warning chalk along the west mesh posts turned blue. No civic transfer had begun. This cold came from the shortened natural season, sharpened by the load Orison no longer deferred under routine assumptions.

Lev woke to Yara striking the iron pipe outside field housing. Three beats for temperature, two for mesh failure. He pulled on boots without socks, dragged his coat over his nightshirt, and ran toward the lower sunroot beds.

Wind had caught the spare south panels before their locks seated. One panel bellied upward, black against the clouded sky. Every lift exposed six crowns. Every drop struck the mesh posts hard enough to split their ceramic collars.

Yara stood at the manual winch with Kem. "Drive chain jumped."

"Can we reset?" Lev asked.

"Not under tension."

"Cut the east corner."

"That loses half the bed."

"It saves the posts."

Yara shoved the cutting bar at him. "Your plan protects sunroot."

"Not by taking three workers under a falling panel."

"Then stop deciding between the crop and our bodies before you count available hands." She pointed uphill. Eight pear workers were running down with anchor ropes. "We can bleed tension across the open redleaf mesh."

Lev saw the path. The sacrificed orchard had empty panel rails and warmed soil loops no longer assigned to fruit. They could route the loose panel over those rails, sacrificing two more pear rows while preserving the sunroot bed.

"Ropes to west posts," he called. "No one under the center seam."

Workers moved without waiting for his restatement. Kem released the brake one tooth at a time. Yara fed rope through an orchard pulley. Lev climbed the lower frame to reach the iced corner lock.

His gloves had fallen somewhere between the housing and the beds.

Metal took heat from his fingers on contact. He wrapped one hand in his sleeve and struck the lock pin with the cutting bar. Ice had swollen inside the collar. The

pin moved a finger's width, stopped, then drove back when wind filled the panel.

"Come down," Yara shouted.

"Rope isn't seated."

"We can cut from here."

"It will whip through the crew."

Lev hooked his bare left hand through the safety ring. With his right, he cleared ice from the collar using the bar's point. Cold first hurt, then narrowed into pressure. He knew that change from orchard work. Pain leaving did not mean safety.

The anchor crews pulled. Black mesh slid across the abandoned redleaf rails. Two young trees bent until their trunks cracked. The corner pin loosened. Lev drove it out and let the panel run.

A rope snapped tight below him. The frame jumped. His wrist slipped through the ring, but two fingertips stayed pressed between metal and iced ceramic. He heard himself make a small sound. Yara climbed the first brace and struck the ring open with her chalk hammer.

"Hand," she said.

"I can move it."

"Hand."

He gave it to her. The two outer fingertips looked pale under orchard lamps. She pressed them into her armpit through her coat rather than rubbing them.

"You had a spare glove on the post," she said.

"I didn't see it."

"You saw a lock before you saw your hand."

Below them, the panel settled across empty pear rows. Sunroot leaves shuddered under their saved cover. No worker had gone beneath the seam.

At the warming shed, Yara put Lev's hand in cool water and increased the temperature by measured additions. Sensation returned as needles under the nails. The cooperative medic examined color, capillary fill, and skin firmness.

"Superficial frostbite in two fingertips," she said. "No field work with this hand for three days. Keep it dry. If blisters rise, you lose more time."

Lev looked toward the door. "Mesh inspection begins at light."

"Then inspect with your eyes."

Yara leaned against the medicine cabinet. Her twins slept on a bench under one shared blanket, brought over when the housing loop stuttered. "We do not need you injured to prove the cold exists."

"I understood that before tonight."

"You knew before tonight."

"Yes."

"Knowing did poor work."

He accepted that too.

Nera came on the first tram after receiving the emergency wafer. By then Lev's fingers carried white dressings and the sunroot cover held. She entered the warming shed with frost on her coat shoulders. The city had felt the same front.

"Show me," she said.

He held out the bandaged hand.

Her breath stopped, then resumed under control. "Can you feel pressure?"

"Yes."

"Heat?"

"Enough."

Yara answered from the bench. "The medic expects recovery. He is barred from wet work."

Nera looked at Lev. "Will you observe that?"

"If the crew does."

"The crew kept their gloves on," Yara said.

At daylight they walked the damage line. Two redleaf rows lay broken beneath the diverted mesh. Sunroot crowns remained insulated. Frost barley had taken a thin white edge but no stem break. Workers logged every loss before repair, because honest preparation required the city to see what happened before the planned test as well as during it.

Nera helped Lev fasten the final sensor with her uninjured hands while he read alignment marks. He resented needing the help for less than a minute. Then the wafer accepted both their seals, hers legal and his agricultural.

At sunset on Day 84, Bellweather crews took their stations. Yara held the lock board. Each team confirmed shelter heat, clothing, abort route, crop assignment, and who possessed authority to cut a panel without waiting for Lev.

Before the final lock, Kem ran a false alarm through the western circuit. A blue lamp flashed over the barley channel. The old response sheet assigned Lev sole authority to release its mesh, a delay of seven minutes if he stood at the sunroot beds. Yara crossed out his name and wrote three station leaders beneath it.

"One release can ruin the protected crop," Lev said.

"One missing supervisor can ruin the crew," she answered.

They tested again. This time Kem called a broken housing loop while the orchard alarm still flashed. Two workers moved toward the field before Yara stopped them. Their station card put crop rescue above shelter inspection because Lev had copied priorities from an ordinary frost plan. He took the card with his bandaged

hand and reversed the order. Housing heat, crew count, mesh control, crop salvage. The sequence cost barley under a compound failure. It also kept workers from preserving grain while their children lost floor heat.

On the third run, each station reported through separate speaking tubes. Lev heard people use the language they had built together: not save the orchard, but hold the barley channel; not protect Bellweather, but verify the south rooms; not acceptable loss, but two redleaf rows cut and entered. The trial ended with no body under tensioned mesh and every manual release held by more than one person.

He stood beside Nera at the master mesh wheel. His dressings prevented a full grip, so she placed her hand over the outer spoke while he held the inner one.

"This does not reconcile anything," he said.

"No."

"It does not make the lost pears payment in full."

"No."

Yara called from the board, "If you two finish explaining what it does not do, the temperature will decide for you."

Lev met Nera's eyes. Shared risk required no prettier name.

They turned the wheel together. Locks traveled outward through the orchard with a sequence of hard metallic knocks. Redleaf panels stayed open. Barley channels closed. The repaired cover settled over the sunroot beds. At the western boundary, the last mesh tooth engaged and Yara marked the time.

Bellweather stood locked for the thirty-one-percent test. Wind moved through the sacrificed pear rows without obstruction, carrying the smell of cut green fruit toward the city.

Interlude III — The House with Two Roofs

Ove Cas removed the sleeping-room door and laid it across two stools.

His daughter objected because the door carried seven years of height marks in blue pencil. Ove showed her how to lift from the hinge side without scraping the frame. Together they packed the hollow core with dried canal reed, then sealed both faces under tar cloth.

"You said doors have to breathe," Mila said.

"Not tonight."

"Will we put it back?"

"If it still fits after the cloth."

"That means you do not know."

Ove pulled the last binding knot tight. "Correct."

Their Low Glass rooms occupied the top floor of a radiator keeper's house. Hot water passed beneath the kitchen, but the roof kept only what its cracked tiles allowed. During ordinary moderated winters, Ove patched leaks with metal slips and deducted the work from his rent. For the controlled retained-load day, every gap had become an instrument.

His younger child, Pas, sat at the table dividing barley cakes into tomorrow's meals. One cake for breakfast before the temperature drop. Half each at midday. The remaining two stayed uncut in case the canal crew could not leave its station.

"Who chose our weather?" Pas asked.

Ove checked the kitchen gauge rather than answering too quickly. "People did."

"Which people?"

"Many who are dead. Some in Assembly Hall. Some who have not been born here yet."

Pas frowned at the cakes. "That does not fit in one room."

"No."

On the wall above the kettle, their latest heat receipt listed service, maintenance levy, and moderated-season credit. Ove took it down. Beren's new burden schedule had added a fourth line in red ink: retained load, measured locally. The amount remained blank until the test began.

Mila held a lamp while he climbed into the roof crawlspace. Dust stuck to his damp sleeves. He pressed strips of frost chalk along six rafters, then wired a temperature wafer beneath the northern tiles. The city wanted exterior loss, interior retention, wind direction, and time of first ice. Ove added a seventh mark over the children's room because official sampling points preferred square plans, and homes rarely obeyed them.

They altered three roof tiles. One received a narrow copper throat that whistled at a known pressure. Another held the wafer lead. The third stayed loose enough for Ove to lift from below and expose a measured patch of attic air if the sensor drifted. Data from these changes would enter Canal Nine's outgoing phase load. Someone in Year 360 could compare the house against inherited cold.

"Will they hear the whistle?" Mila asked.

"Not as sound."

"Then why make it?"

"So we know the wind before we send the number."

She considered this. "The future gets a receipt?"

"Part of one."

Ove tied the roof hatch open with cord so a swollen frame could not trap him. He had spent his working life receiving pressure from decisions made uphill. Tonight his roof would state what arrived.

At Holdfast Kest, Year 360, Siv Ren warmed a roof bolt inside her glove before turning it.

The bolt held a double skin of stonecloth above the family room. Wind-packed soil covered the lower skin. A scavenged cloud-comb panel formed the upper slope, polished enough to shed black snow when struck from beneath. Between them lay an air gap narrow enough for Orra to crawl through and broad enough to lose dangerous heat if the seals failed.

Orra waited inside that gap with a blue chalk nub. "Mark?" she called.

"Not yet."

Siv gave the bolt another quarter turn. The ceramic indicator beside it changed from white to gray. "Now."

Chalk moved overhead. Orra marked the bolt position on a calendar slip fixed to the rafter. Their shelter normally counted roof work in heat lost during access. This repair used a different account. They needed exact roof response during the inherited storm, including the phase correction that had arrived from Lume's retained-load preparation.

The households never stood beneath one sky together. Their roofs carried storms 137 years apart, linked by a delay that permitted numbers to cross but allowed no hand to pass a tool. Ove's copper throat had rusted away long before Siv's birth. Its pressure ratio survived inside a transmission record. Siv had used that ratio to set the angle of Kest's upper panel.

"Did they choose this storm?" Orra asked.

"They chose part of its shape."

"Did you choose theirs?"

"We sent what our roof had already endured. They decided what to keep."

Orra backed out of the gap, feet first. She carried three ceramic calendar slips between her teeth and kept both hands on the rope. Siv took the slips before helping her down.

Their meal waited in the wall recess: root mash, one cup of fermented grain broth, and a strip of preserved fungus for each of them. Siv had moved cooking two hours earlier so the room could recover its heat before the storm peak. Across the corridor, neighbors had combined meals in the larger shelter. Orra had refused to go. She wanted to watch the return needle mounted beside their household gauge.

"People in the past ate before cold too," she said.

"People usually do."

"Did they have root mash?"

"Some had barley cakes."

"How many?"

"The record did not say."

Orra disliked any record that omitted portions. She divided the fungus strips so their lengths matched, then pushed the wider cup of mash toward Siv. "Calendar keepers use more heat."

"That rule has no entry."

"It does now."

Siv let the cup remain.

After eating, they packed reed felt around the room's lowest vent. Kest could not close it completely without trapping breath moisture that later froze inside the wall. Siv left a gap equal to Orra's smallest finger. A phase wafer rested there inside a clay collar. It would record incoming temperature and the room heat sacrificed to maintain air.

Near the roof ladder, Orra had hung a copy of Lume's household sample. The old heat receipt looked extravagant even in facsimile: enough continuous canal service to assign a separate credit line, enough paper to preserve a single month's charge. Yet the red addition mattered more to her. Someone had left a blank for locally measured retained load.

"Will we fill their blank?" Orra asked.

"They filled it before either of us read the copy."

"Then why can we answer?"

"Because our answer helped them decide how."

Orra looked at the return needle. "That is a bad circle."

"It stays one history."

"Still bad."

Siv checked the roof wafer. Pressure rose over the northern terrace. She copied the first figure onto a ceramic slip, including uncertainty from a damaged wind cup. Past officials had hidden uncertainty when it threatened comfort. Kest had once hidden casualty figures when uncertainty threatened leverage. Siv no longer permitted either habit in her household record.

The upper roof shifted under a gust. Its cloud-comb panel flexed, transferred force through the warmed bolt, and held. Orra marked the time.

Far earlier in the same history, Ove replaced the insulated sleeping-room door. Tar cloth made it too wide. He shaved the latch edge with a valve knife while Pas carried the meal cakes into the warmest cabinet. Mila read the roof wafer through the hatch and entered every number beside the blank red line.

Wind came down Canal Nine and found the copper throat. A low whistle crossed the attic. Frost chalk on the northern rafter changed color from blue to white.

At Kest, Siv heard no whistle. Black snow began against the upper roof with a dry, granular hiss. The phase wafer brightened inside its clay collar, ready to

measure what the house kept and what the storm demanded.

Ove pressed his palm against one roof brace. Orra set both hands on another, 137 years beyond him. Their separate storms began at different roofs, under different measurements, with both household records open.

Act IV

Chapter 21 — Repayment Day

Sumi Quill checked the abort rail with the two fingers she no longer had.

Habit sent a phantom grip toward the red ceramic bar. The remaining fingers closed on empty air, then found metal. She pulled once. Every warning lamp in Canal Nine's phase room lit white. Workers at six stations raised their hands without leaving their controls.

"Manual abort available," Sumi said.

"Mechanical separation available," called Iren from the drum brake.

"Cloud-comb ground path available," said the tower relay.

"Low Glass canal relief available," Beren answered through a speaking tube.

The Warm Ward line clicked twice and produced no voice.

Sumi marked it incomplete. "Again."

Iren frowned toward the clock. "We have four minutes."

"We have four minutes to learn whether they can answer a tube."

He reset the brake. Around them, three black ceramic phase drums rotated below safe coupling speed. Frost chalk striped their housings. Each stripe had been signed by a worker after inspection, not by a supervisor on the worker's behalf. The room smelled of hot oil, canal mineral, and wool drying on bodies that would soon have less heat.

Nera stood behind the legal rail with Edrin and the retained-load schedule. The page carried Assembly authority, Bellweather crop conditions, Low Glass thresholds, ward benefit ratios, future claimant notice, and fourteen abort clauses. At its center, one figure had been written by hand to prevent the typesetter from softening it into a model range.

31%.

Sumi tapped the figure with a wrench handle. "Say what it means."

Edrin glanced at her. "We retain thirty-one percent of today's scheduled transfer in Year 223 for one operational day."

"Say what it does not mean."

"We are not repaying thirty-one percent of the historical account. We are not settling 4.7 billion degree-hours. We are not authorizing any further retention after the test window."

"Good."

Nera watched the Warm Ward lamp. "Their store controller passed inspection last night."

"Machines pass inspection when nobody asks them to betray their favorite instruction."

A voice finally entered through the tube. Nial Vosper, the communal-kitchen superintendent, spoke over a room full of other voices. "Ward station present. Garden vault station present. Civic store controller shows conditional release."

"Condition?" Sumi asked.

"Local temperature below ward minimum or steward key."

"That conflicts with the benefit schedule. Release begins on distribution draw, before local minimum."

"We entered the new schedule."

"Did you test the actuator under a simulated external draw?"

Silence lasted one second too long.

Nial said, "The council prohibited live draw before authorization."

"Then you tested ink."

Nera reached for the speaking tube. Sumi put her wrench across it. "No time to negotiate a council backward. Put a keyholder at the actuator. Keep the line open."

"Halden holds the civic tunnel key," Nera said.

"Then find him before Low Glass needs it."

The clock reached zero.

Outside the phase room, a bell struck across Canal Nine. Factory whistles answered. Roof watchers raised blue flags in Low Glass. Far west, Bellweather locked its final mesh circuits. The public boards had named the day Repayment Day, a phrase Sumi disliked for its implied completion. Engineers rarely got to name civic hope after it left their workrooms.

Nera pressed her seal into the schedule. Edrin witnessed. Sumi turned the coupling wheel three teeth.

The drums took load.

Sound altered first. Their ordinary rotation carried a low rub through the floor. Under retained weather, each drum found a separate pitch. The front housing hummed against Sumi's sternum. The center drum produced a thin note at the edge of hearing. The rear coupling struck once, paused, then settled.

"Phase separation?" she asked.

"Within nine marks," Iren said.

"Do not call that within."

"Eight and falling."

"Better."

Temperature needles across the wall moved left. Not quickly. Sumi trusted that

more than drama. A violent drop would have meant a valve error or a schedule released too much at once. This front entered the current atmosphere because the lattice no longer shifted its full scheduled share downstream. Cold gathered in roof tile, canal water, orchard soil, and every breath leaving a warm body.

At four minutes, the return needle lifted.

It had once required six hours to accumulate 180 characters. Now it wrote phase marks continuously, pale scratches across the receiving strip. Sumi bent close enough to feel static lift the hair along her cheek.

> KEST RECEIPT WINDOW OPEN. NORTHERN ROOF ARRAY READY. OUTER HOLDFAST DELAYS RANGE 11 TO 43 MINUTES.

Siv's identifier followed. Not a greeting. Sumi approved.

"Future timing live," she called. "Kest gives eleven to forty-three minutes across the winter line. Do not synchronize to the first receipt."

A junior technician reached toward the distribution dial. Sumi struck the console beside his hand with her wrench.

"I was matching Kest," he said.

"You would overrun every settlement behind it. Use the range."

"We cannot hold one phase for thirty-two minutes."

"We can step three plateaus. Calculate them."

The technician, Ari Seln, cleared the forecast display and built the steps from receipt delay rather than settlement size. Kest received first because its array stood closest to an intact northern phase brace. Two quarry shelters followed on damaged conductors. Outer terrace holds appeared last, spread across a delay longer than the entire opening sequence.

Ari proposed equal plateaus.

Sumi put her wrench across the first two figures. "That sends the highest correction to the easiest receiver."

"It gives us clean confirmation before the noisy sites."

"It also spends current load where Kest already sees relief."

"We need confirmation that the coupling works."

"We have it." She pointed to Siv's line. "Now we need coverage."

He rebuilt the sequence with a short first step, a longer middle hold, and a shallow final plateau that preserved phase stability while outer sites received. The arithmetic lowered Kest's signal clarity. It improved the chance that a damaged settlement would register enough change to distinguish present retention from its own storm noise.

"Read the hardware motion," Sumi said.

Ari traced it. "Front drum advances two marks. Center holds. Rear drum takes

differential through the slip collar. At second plateau, center advances one while front releases half. Final plateau grounds through Combs Two and Four."

"Comb Four has an old stability routine."

"Inspection cleared it."

"Mark it as a risk anyway."

He added amber chalk beside the final path. Sumi assigned one worker to the slip collar and another to each ground brush. Iren protested that the brake station could watch the collar.

"Watching two instruments means choosing which one you miss," she said.

"We have no spare phase worker."

"Use me."

"You are directing the room."

"Then use the apprentice at oil records."

The apprentice lifted her head. "I have never held a live collar."

"You will not hold it. You will read travel and call every half mark. If it crosses three, Iren releases pressure. Your hand stays behind the yellow line."

She moved to the collar with a chalk slate. Sumi watched her plant both boots before the drum speed increased.

They executed the first plateau. The front wheel resisted at one and a half marks. Sumi felt the drag through the floor before its gauge changed. A strip of sealing fiber had swollen under cold oil. Stopping the test to replace it would uncouple the return channel for nineteen minutes.

"Work it within one-quarter mark," she told Iren. "No full reversal."

He eased the wheel forward, back, then forward again. The fiber seated. Kest's signal narrowed but did not disappear. Quarry receipts began as broken vertical strokes on the return strip.

At the second plateau, the apprentice called collar travel in a steady voice. "One-half. One. One and one-half. Holding."

"Temperature?" Sumi asked.

"Rising two marks at the outer ring."

"Apply felt jacket."

Two workers wrapped the collar with mineral felt while it turned. They used hooked poles so neither sleeve crossed the yellow line. Oil flecked the jacket and darkened it. Heat stopped rising.

The quarry strokes resolved into a station code, followed by one word: RECEIVED.

No cheer passed through the phase room. Ari entered the time. The apprentice wrote collar travel. Iren added the swollen-fiber observation. Sumi shifted them

toward the final plateau because outer sites still waited.

He began the broader calculation. The future settlements had different roof mass, canal depth, and inherited load. Kest could relay timing but could not stand in for 2.1 million people. A wider return channel did not create a single future clock.

Before the last step, Sumi opened a direct station circuit to the public weather room. "Forecast, separate retained temperature from distribution temperature."

A model clerk answered, "They produce one observed curve."

"They produce one thermometer reading and two decisions. I need both."

"We can infer distribution after store release."

"Store release has not occurred."

"Then the fields overlap."

Sumi made him open a second board. One curve carried total local cooling. Another subtracted measured canal heat and marked every unmetered source as uncertainty instead of smoothing it away. Low Glass gained a wide gray band because household loops lacked complete meters. Warm Ward displayed a narrow band because protected stores counted each valve.

"Keep those widths," she said. "Do not make poorer instrumentation resemble lower harm."

The clerk locked the two-curve view to the public feed. Household watchers could now see where the city possessed exact knowledge and where neglect had produced ambiguity. Engineering could not repair that neglect during one test, but it could stop converting missing instruments into reassuring figures.

At Station Four, oil pressure fell.

The lamp changed from white to amber. Iren opened the reserve line. Pressure continued down.

"Filter," he said.

"Or aeration."

"Sight glass shows bubbles."

"Then the reserve intake pulled air. Close it. Hand pump."

Two workers took the pump levers. Their shoulders rose and fell out of rhythm until Sumi counted them into alternating strokes. Oil pressure stabilized one mark above caution. The rear drum's pitch deepened.

Nera leaned across the legal rail. "Does that trigger abort?"

"No."

"Will you tell me before it does?"

"I will tell the room."

"That includes me."

Sumi looked at her. "Then stand where the room can hear your answer."

Nera moved from the protected rail to the witness station beside the brake. Edrin followed with the schedule under his coat.

At twelve minutes, household data entered from Low Glass. Roof loss, room temperature, canal inlet, human count. Ove Cas's northern tile showed faster cooling than the city model because wind crossed a maintenance gap between two buildings. Sumi routed the discrepancy to Beren.

"Canal Nine to South Low Glass. Your roof sample runs point eight degrees below projection."

"We see it," Beren said. Machinery battered somewhere behind his voice. "Opening local bypass six."

"Confirm source."

"Factory return water."

"That source cools in nine minutes."

"It gets us nine minutes."

"Enter the limit."

"Entered."

Bellweather reported frost barley holding, sunroot stable, redleaf soil falling by design. Lev's voice sounded rough through distance. "Western mesh gives us another two degrees of radiator loss before tree-root threshold."

Sumi checked the crop board. "Do not spend root threshold without Yara's station."

A second voice cut in. "Yara here. We approved one degree. The other stays with next year's seed wood."

Lev did not contest it. "Correction entered."

The phase room absorbed each statement as work rather than confession. Sumi moved from station to station with her wrench tucked into her belt. Her shortened hand checked temperature by touch along housings. Her full hand wrote values. She refused help with either task.

Twenty-six minutes into the test, the center drum developed a bright line under its frost chalk.

Not a crack. Heat.

"Ground brush," Sumi said.

The tower relay answered, "Cloud-comb path drawing above schedule."

"How much?"

"Five percent."

"Source?"

"Comb Four refuses reduced storm stripping. Automatic stability routine."

Sumi swore once, with precision. "Disconnect Four."

"We lose western wind shaping."

"Bellweather already carries it."

Lev heard over the shared circuit. "Confirmed. We take natural west flow."

Yara added, "Within crew threshold."

"Comb Four separating," the relay said.

The ground brush retracted. Heat line faded. On the weather board, western wind moved outside the old comfort range and stayed inside the honest one. Sumi had expected the first failures to come from overstressed hardware. Instead, systems failed by preserving the purposes they had been built to serve.

At thirty-nine minutes, Kest sent its second timing block.

> CURRENT-SIDE RETENTION VISIBLE. COMPOUND FRONT LOAD FALLING AT KEST ARRAY. DO NOT LEVEL TO US. OUTER SITES RECEIVE LATER.

Sumi had already followed that instruction. Relief still moved through her, unwelcome and useful. She read the line to the room.

Nera said, "Can we answer?"

"Channel has capacity."

"Send measured conditions, not thanks."

Sumi looked at her with brief approval. She dictated current drum separation, Low Glass roof discrepancy, Bellweather crop state, Comb Four isolation, and the three future timing plateaus. The packet would reach Year 360 as part of the history that produced Siv's warning. The logic still made Sumi's teeth ache. Instruments did not owe her intuitive comfort.

The return strip accelerated. Characters arrived in clusters instead of solitary scratches. For a moment the signal carried more than text: a coarse amplitude rise with regular gaps.

"Voice structure?" the junior technician asked.

"Not yet. Too few harmonics."

"It repeats."

"So does a pump."

She saved the raw waveform anyway.

At forty-seven minutes, South Low Glass lost bypass six.

Beren's speaking tube filled with water noise. "Factory return gone cold. Main canal inlet dropping."

Sumi found his district on the temperature board. Three household points crossed caution together. Ove's roof sample followed ten seconds later.

"Ward distribution should begin now," Nera said.

"It should have begun at the first external draw."

Sumi opened the Warm Ward tube. "Canal Nine to protected stores. Release tier one under benefit schedule."

Nial answered, breathless. "Command accepted at kitchen manifold. Garden vault remains closed."

"Why?"

"Controller requests local minimum or steward key."

"You entered the distribution draw."

"It displays the draw and refuses actuation."

"Take manual control."

"Council seal covers the handwheel."

Nera's face changed at the witness station. "Who placed it?"

"Ward council, before the public vote. They classified the vault as long-duration summer reserve."

The phrase made Sumi understand the whole obstruction. Someone had changed schedules at the public layer while a deeper instruction still treated present comfort as the system's asset to preserve.

"Break the seal," she said.

Nial's reply came through several people arguing. "I lack custody authority."

"Temperature does not care."

"Marshals do."

Nera took the tube. "This is Senior Notary Voss. I authorize evidentiary breach to execute the Assembly schedule."

A clerk answered from the ward station. "Notarial authority authenticates transfer. It does not dispose protected civic stores."

"Then put the council chair on."

"Unavailable."

"Put any keyholder on."

No voice came.

Sumi checked Low Glass again. Ove's room had fallen one and four tenths degrees below projection. Two canal blocks crossed the first human exposure line. Beren opened emergency shelter loops, but their water entered colder than planned because the protected reserve had never joined the flow.

"Abort?" Nera asked.

"Not yet. Human minimum remains twenty-two minutes away under current slope."

"And if the slope steepens?"

"Twelve."

Edrin held the schedule against his chest. "The automatic abort observes temperature, not distribution breach."

"Correct," Sumi said.

"Can you add breach?"

"Not while coupled. We would blind three live sensors during rewrite."

Nera looked toward the civic tunnel door. "Halden has the key."

"Find him."

"If I leave the witness station, the test continues without senior authentication."

"If you stay, Low Glass continues without heat."

The choice required no speech. Nera handed her witness stylus to Edrin. "Record temporary custody under emergency necessity."

"I object to the form," he said.

"Record that too."

He took the stylus.

Nera ran for the tunnel.

At fifty-two minutes, a pressure sensor failed on the front drum. Its needle dropped to zero while the housing pitch held steady.

"False zero," Iren said.

"Prove it."

He laid a mechanical gauge across the port. The first connector leaked oil over his glove. He tightened it and read pressure within range.

"We keep coupling," Sumi said. "Mark electronic sensor unavailable."

The public board now carried three amber failures: oil reserve, cloud comb, pressure readout. Low Glass added four temperature cautions. Warm Ward stores showed green readiness and zero actual release.

A worker at the return station called, "Kest asks why current distribution changed."

Sumi read the incoming line.

> RELIEF CURVE FLATTENED BELOW YOUR RETENTION INPUT. DID CURRENT LOAD MOVE BETWEEN DISTRICTS?

Siv could infer injustice from weather 137 years later before the protected ward admitted it in its own canal.

"Answer yes," Sumi said. "Name the refusal."

Edrin looked up from his witness entry. "That becomes evidence against the ward council before provenance review."

"The controller log provides provenance."

"We do not have it."

"We have actuation refusal, local display, and Nial's authenticated station."

His objection lasted only long enough to test the chain. "Send."

Sumi dictated: present automatic stores withheld tier-one release despite authorized distribution draw; Low Glass temperature falling faster than model; manual access pending. The return needle took the packet at once.

Beren's voice entered over it. "South blocks at second caution. Shelter Four has twelve children and one working floor loop."

"Move them," Sumi said.

"Street exposure costs more than the loop if we get store heat in six minutes."

"You do not have six."

"I know my canal."

"You know its past behavior. Today's inlet has no precedent."

A pause. Then Beren ordered the move. Sumi heard doors, names, and body counts passing behind his tube.

At the Warm Ward station, Nial struck the sealed handwheel with something metal. "Glass cover cracked. Council seal intact beneath it. We need a key or a cutting tool."

"Do not cut the pressure stem," Sumi said.

"I run kitchens."

"Kitchen tools cut people too."

"Then send a vent apprentice."

The request entered the shared circuit. A young male voice answered from somewhere above the civic tunnel. "Tovan Voss, guild station. I have a collar saw and two apprentices."

Sumi pictured vent glass, stored soil heat, and a seventeen-year-old carrying a saw toward a mechanism whose controller had already refused the city. She almost ordered him to wait. That authority did not belong to her.

"Report to Nial," she said. "Touch no pressure member without a ward engineer."

"Understood."

The front drum pitch slipped half a note. Return characters smeared, recovered, then began arriving faster than the clerks could copy. More current cold created more backward bandwidth, but Sumi could not spend heat as communication without regard to where bodies lost it.

She placed her damaged hand on the abort rail. Scar tissue registered vibration before temperature. One pull would end further retention. It would not instantly

warm Low Glass, open the ward stores, or refund the cold already accepted. Aborting now could protect the drums while preserving the distribution failure as a completed fact.

“Sixty minutes,” Iren called.

The first hour of Repayment Day closed on the clock.

Across the public board, Warm Ward automatic stores still reported ready. Their valves remained shut. Low Glass continued falling, block by measured block, while Nera's custody mark moved through the civic tunnel toward a locked handwheel.

Chapter 22 — The City Gives Heat

The civic tunnel turned uphill beneath the Warm Ward, and Nera ran against its heat.

Pipes under the walkway carried stored water toward the protected quarter. Their iron sleeves remained warm while the Low Glass gauges clipped to her belt moved left. Every thirty steps, a public temperature dial showed comfort above. Beside it, Sumi's borrowed wafer showed the city below approaching danger.

Nera reached the first locked grille and found Halden Rusk on the other side.

He wore his steward's sash now. Two ward marshals stood behind him with a council clerk, none carrying tools. Halden held the civic-tunnel key but had not put it into the lock.

"Open it," Nera said.

"The ward council has invoked reserve preservation."

"The Assembly schedule supersedes their local seal."

"They contest that reading."

"Low Glass has crossed second caution."

"We have evacuation routes active."

"Evacuation does not distribute the authorized heat."

Halden's fingers closed around the key. "If we open a pressurized store through an untested controller, we may lose the tunnel manifold. Then neither district receives it."

"Your office certified the controller."

"My office certified its schedule interface. The mechanical release remains under ward custody."

Nera pulled Sumi's wafer from her belt. Four Low Glass blocks glowed amber. Shelter Four had begun moving children through streets already colder than planned. "This record no longer distinguishes your layers of authority from the temperature they produce."

The clerk said, "Senior Notary, you left the witness station."

"Yes."

"Your authentication may have lapsed."

"Deputy Sol holds it under emergency necessity."

"That form lacks precedent."

"So does a ward controller displaying readiness while refusing a lawful draw."

Halden looked through the grille at her. The tunnel lights made hollows beneath his eyes. He had spent decades preserving an institution by narrowing each danger

until procedure could contain it. Today every narrow choice opened into the same canal.

"Who has reached the actuator?" he asked.

"Nial Vosper. A ward engineer, if one answered. Three vent-glass apprentices."

"Tovan?"

"Yes."

"You let him go?"

Nera put both hands on the bars. "He answered a public work call. Do not turn that into my authorization so you can postpone yours."

Halden inserted the key.

The grille opened inward. Warm air struck Nera's face from the higher tunnel. She passed him without thanks. After two steps she heard his shoes follow.

The winter garden vault occupied the quarter's lowest terrace.

Stored soil heat entered through six ceramic tanks buried beneath citrus beds and seed rooms. Their combined manifold rose behind a wall of layered vent glass, visible to visitors but protected from hands. At the manual station, the glass cover had been cracked in a white star around Nial's kitchen mallet. A red council seal remained unbroken over the handwheel.

Tovan knelt beside the lower collar with a narrow saw. Two apprentices held lamps. He had tied his hair back with insulation string and removed his right glove for grip.

"Put that on," Nera said.

He looked up. "The teeth catch wool."

"Then use a leather guard."

"We do not have one."

A ward engineer crouched beyond him, tracing the pressure stem through a diagram. "His hand stays outside the cut plane."

"That was not an answer."

Tovan pulled the glove over his bare hand. "Now it is."

Nera wanted to take the saw from him. She knew vent glass only as windows, receipts, and the hospital corridor where her mother's body had come home. Tovan knew how its layers held stress. Removing him because she feared losing him would make his skill ceremonial at the moment it mattered.

Nial stood at a speaking tube connected to Canal Nine. "Kitchen manifold released tier one. It cannot feed the garden circuit backward."

"Low Glass slope?" Nera asked.

"Human minimum in nine minutes if it steepens. Seventeen if it holds."

The ward council arrived before Halden.

Five members entered in insulated coats, led by Chair Veyl, whose public vote had opposed the controlled day. Veyl took in the cracked cover and pointed to the red seal. "Any further damage voids ward custody."

Nial kept one hand on the kitchen tube. "Your custody has not opened a valve."

"The reserve protects hospitals, seed stock, communal food, and pressure stability through a shortened summer."

"It also protects a walking court," Nera said.

"We agreed to drain that circuit after engineering review."

"You had twenty-four days."

"We had twenty-four days to prepare for civic seizure under a formula the ward rejected."

Halden entered with the marshals. Veyl turned toward him. "First Steward, stop the breach."

Halden looked at the red seal, then at the store gauges. "What immediate release can the manifold sustain?"

The engineer answered. "Tier one for forty minutes. Tier two if we open the civic tunnel return and reduce soil circulation."

"Effect on seed rooms?"

"Six to nine degrees over the next day, depending on glass loss."

"Effect on hospitals?"

"Separate store. No direct draw."

Veyl said, "The pressure model has not been verified."

Nera held out Sumi's wafer. "Low Glass temperature has."

Council members formed a half circle around Halden. They spoke of fiduciary duty, ward residents who had not consented, glass failure, seed mortality, and the danger of surrendering emergency control during unrest. Each concern had substance. Together they created delay measured in other people's room heat.

Nera asked the engineer, "Can the manifold meter a 2.4 ratio?"

"By district draw, yes."

Veyl looked sharply at her. "The ward did not consent to 2.4 times Low Glass's per-capita burden."

"The Assembly adopted benefit-weighted distribution for this test."

"Under public pressure after an archival spectacle."

"Under a vote after disclosure."

"You ask residents to carry guilt for systems built before their births."

"I ask the best insulated district with the largest historical benefit to release stored heat before the least insulated district crosses danger."

"That phrasing still assigns moral rank to infrastructure."

Nera stepped closer. "Then give me a phrasing that opens the valve in six minutes."

Veyl had none.

Halden removed his steward's sash. For one impossible moment Nera thought he meant to resign in the vault. Instead, he wrapped the cloth around the key to insulate his hand and approached the manual station.

"The civic tunnel return requires steward access," he said. "I will open it. Ward custody remains on the store wheel until the council removes its seal or emergency authority overrides it."

"You hold that authority," Veyl said.

"I hold part of it."

"Then use it to protect the ward."

Halden met her eyes. "The ward includes the canal workers who maintain its supply."

He put the key into the tunnel return lock. The mechanism resisted. He leaned his full weight into the wrapped key until old tumblers struck inside the wall.

Tovan called, "Pressure path opening."

The engineer read both gauges. "Return seated. We can release when the cover clears."

Veyl placed her palm over the council seal. "I do not authorize its removal."

Nera took out her notarial blade. "Then I preserve your refusal and breach it under the Assembly schedule."

"You authenticate. You do not command stores."

"No. The Assembly commanded them. I am making the refusal admissible while workers execute the vote."

She cut the seal.

Tovan set the collar saw against the fractured cover. Its teeth scored a circle around the handwheel, each stroke producing white glass dust. One apprentice held a catch cloth below. The other watched stress lines with a polarized lamp.

"Upper layer moving," she said.

"Direction?" Tovan asked.

"Toward seven marks."

He changed the saw angle. "Nial, clear the left side."

Nera heard Sumi through the speaking tube. "Low Glass slope steepening. Five

minutes to first human minimum.”

The engineer reached for the saw. “Cut faster.”

Tovan did not surrender it. “Faster drives the crack into the pressure pane.”

“We only need hand access.”

“We need the cover to stay out of the stem.”

Halden held the tunnel key open with both hands. “Follow the apprentice.”

Tovan completed half the circle. The upper glass layer lifted by a fingernail's width. Air hissed into the gap between panes.

The polarized lamp flashed red.

“Stop,” the watching apprentice said.

Everyone stopped except the store.

A pressure pulse traveled from the newly opened tunnel return into the old cover frame. The handwheel seal, cut free, fluttered against the glass. One stress line crossed another beneath Tovan's saw.

Nera saw his shoulders tense.

“Back,” he said.

The outer pane shattered.

Glass did not fall like ordinary window glass. Tempered fragments burst outward in flat bright pieces. Tovan raised his left arm across his face. The pulse drove him backward over the low tool rail. His hip struck first. His left hand took the rest of the fall against stone.

Nera reached him before the last fragments stopped sliding.

Blood marked three narrow lines along his cheek. His eyes remained open. He drew one breath through clenched teeth and held his left wrist against his chest.

“Do not move it,” she said.

“I wasn't planning to.”

The joint had already begun to swell. No bone showed. His fingers moved when she asked, though the fourth brought pain across his face.

Nial pulled the apprentice first-aid case from the wall. Veyl called for the ward medic and, in the same breath, ordered the controlled day stopped.

“No,” Tovan said.

Nobody answered him. Halden still held the tunnel key. The engineer shut the pressure path at the stem, preserving the now-open return. Sumi's voice repeated the Low Glass countdown.

Veyl took the speaking tube. “Injury at protected store. Abort test.”

Tovan caught its cord with his right hand. “Not in my name.”

“You have been struck by ward equipment during the operation.”

"I was struck because the store stayed sealed until pressure moved through an old cover."

"You cannot know causation while injured."

"I know vent glass."

Nera supported his shoulders while Nial splinted the wrist with two tool slats. She wanted every drum stopped, every store closed, every future claimant kept waiting until her son no longer hurt. The wanting came complete and bodily. Authority demanded that she identify whose safety it would purchase.

"Tovan," she said, close to his ear. "Sumi still has manual abort. We can stop further retention."

"Will that warm Shelter Four?"

"Not immediately."

"Will it open this store?"

"No."

"Then stopping uses me to leave the wrong valve shut."

His face had gone gray. Nera hated the steadiness of his answer because it made room for no protective lie.

The medic arrived with a rigid splint and examined him on the vault floor. She cleaned the facial cuts, checked his pupils, and palpated the forearm while he swore into Nera's shoulder.

"Fractured left wrist," she said. "Closed injury. Circulation intact. He goes to the ward infirmary now."

Tovan looked toward the open glass cover. "After they turn it."

"Now," Nera said.

"Let me see it move."

The medic started to object. Nera stopped her with one raised hand, then faced the council. "You have used his injury to request an abort. He has refused that use. Open the store."

Veyl looked from the shattered cover to Low Glass temperatures on Sumi's wafer. Two points had crossed the human minimum. The automatic stores still displayed green readiness.

"I preserve the council's objection," Veyl said.

"Preserved," Nera answered.

Chair Veyl gripped the exposed handwheel with Nial. Together they turned it.

The first movement produced no water. A locking pawl beneath the controller held because its automatic instruction still awaited local cold. The ward engineer reached through the broken cover and lifted the pawl with a glass hook. Nial turned again.

Heat entered the civic tunnel.

Halden's wrapped key jerked under the new pressure. He held it. Pipes along the vault wall expanded with a sequence of knocks that traveled downhill toward Canal Nine. On Sumi's wafer, the Warm Ward draw rose.

"Tier one active," Nial called.

Sumi answered, "Distribution below schedule. Increase."

The engineer watched pressure. "We can give twelve percent more."

"Per-capita ratio remains under two," Nera said.

Veyl stared at the gauges. "Our soil beds will fall below citrus threshold."

"Close the citrus loop," Nial said.

"That kills fruit."

"Bellweather cut its redleaf harvest before today."

The comparison did not make the losses equal. It removed the council's claim that no current crop had already been spent.

Veyl shut the citrus circulation. Stored soil heat redirected into the manifold. The ward draw climbed past double the Low Glass rate. At 2.4 times, Sumi called hold.

"Ratio verified," she said through the tube. "Maintain until canal equalization or abort threshold."

Tovan watched the needle settle. "Now infirmary."

The medic and one apprentice lifted him onto a carry chair. Nera walked beside him as far as the vault door.

"You're coming?" he asked.

"I have to witness distribution."

"Good."

That permission hurt more than accusation.

She touched the uncut side of his face. "I will come when the store holds without us."

"Bring the receipt."

They carried him uphill.

Nera returned to the gauges before the chair reached the corridor. Low Glass had stopped falling, but stopping did not restore rooms already below minimum. Canal Nine needed another six degrees of inlet heat to reverse the district slope before night wind arrived.

"Tier two," Sumi said through the tube.

The ward engineer shook his head. "Not across the whole store. Tunnel pressure will lift the old joints."

"Then section it," Nial said.

Four store loops appeared on the control plate: citrus beds, seed rooms, walking court, and deep soil reserve. Citrus already stood closed. Seed rooms supplied strains the whole belt might need after the shortened summer. Deep soil protected pipes from freeze. The walking court held enough heat to change Low Glass but its curved panes would crack as the foundation cooled.

Veyl put one finger on the court loop. "That structure shelters residents during winter vent alarms."

"Can communal kitchens take them?" Nera asked.

Nial calculated occupancy from memory. "Standing, yes. Sleeping, not all."

"How many overflow?"

"One hundred twelve if every alarm district empties together."

Beren heard over the shared tube. "Low Glass meeting rooms can take forty after the test if ward trams run downhill."

Veyl looked at the speaker as if the offer had arrived in the wrong moral order. "Your district would shelter ward residents after we give you the court heat."

"Our district shelters people when rooms exist. Write the tram guarantee."

A council clerk opened the working folio. Nera authenticated the conditional exchange: court store released now, kitchens designated primary ward shelter, downhill trams guaranteed during vent alarms, forty overflow places registered in Low Glass. None of it reduced the benefit ratio. It prevented a present sacrifice from becoming a reason to neglect the next emergency.

The engineer isolated the court loop. Its valve opened halfway, stopped, and began knocking against a mineral ridge.

"Do not force it," Nial said.

"We need full travel," Sumi replied.

Nial fetched a kitchen descaling line. Ove answered from Canal Nine with the old court valve specification. He had serviced its return branch twelve years earlier and remembered that the stem turned against the modern direction after a repair crew reused a founder collar.

The engineer reversed the wrench. The valve moved.

Heat surged into the tunnel return. Somewhere above, a long glass report crossed the garden vault, one crack propagating through the walking court's empty floor. Veyl closed her eyes at the sound and kept her hand away from the wheel.

"Court loop active," the engineer said.

On Sumi's wafer, Low Glass inlet temperature climbed. Beren called each block as the change reached it. South Canal first. Foundry Steps after thirty seconds. Ove's row later because the maintenance gap between buildings still stripped heat from the branch. Shelter Four's floor loop coughed mineral sludge, cleared under

reverse pressure, then gained three degrees.

Nera made the ward meter and Low Glass meter share one display. Their curves differed in shape because insulation slowed loss above while cracked roofs spent warmth below. Per-capita burden could not be read from valve draw alone. Sumi's calculation included room retention, stored heat surrendered, and the current temperature each district would have held without release.

"Ratio drifting to two point two," Sumi said.

"Cause?" Nera asked.

"Low Glass recovery reduces its immediate burden faster than ward store loss rises."

Veyl stared at her. "You intend to take more because their rooms are warming?"

"We intend to hold the adopted distribution through the test," Nera said.

"That converts success into a larger demand."

"It converts changing conditions into a measured ratio. If Low Glass reaches equalized safety, the schedule steps down."

Beren interrupted. "Do not keep our blocks cold to make the arithmetic look stable."

"No one will," Sumi said. "I adjust the ward draw upward for twelve minutes, then reduce both when household minimum clears."

The engineer opened the court valve another mark. Ratio returned to 2.4 times. Seed rooms remained untouched, their preservation entered as a citywide food benefit rather than ward comfort. Deep soil reserve stayed above pipe threshold. Citrus beds and the walking court continued cooling.

By the time the infirmary door reported Tovan admitted, every Low Glass block had crossed back above human minimum. Beren did not call the district safe. He called its slope positive and named the households still requiring blankets. Nial sent two kitchen carts downhill with hot barley broth before any council motion could classify the food.

Veyl watched the carts enter the tunnel. "Those stores came from ward residents."

"They are going to city residents," Nial said.

Nera wrote both statements into the witness folio. One recorded loss of control. The other recorded an altered use. Neither needed erasure for the heat to move.

By the second hour of Day 86, heat from the garden vault had crossed the civic tunnel and entered Canal Nine.

Nera returned to the Office of Seasonal Accounts after the infirmary set Tovan's wrist in a full splint. His cuts had closed under three narrow strips. Pain medicine had made him sleepy but not agreeable. He had demanded his injury entry include

the controller refusal, the pressure pulse, and his own request not to abort in his name.

At her workroom, the legal observation slit had cleared of frost.

Hot canal water moved below with a heavier mineral smell. Beren's crews had opened district manifolds in benefit order. Ove's roof sample showed his children's room climbing above minimum. Shelter Four had moved its children back inside after the floor loop recovered. Warm Ward stores continued at the agreed ratio, losing citrus fruit, walking-court glass, and a measurable share of protected soil heat.

Nothing about the city had become painless. Its heat had changed direction.

Edrin met Nera beside the forecast glass with the witness folio. "Sumi requests authentication on a nontext receipt."

The return pane displayed pale bands instead of characters. Each carried amplitude, breath-length gaps, and a harmonic structure that had not appeared before the stores opened. Sumi had overlaid the timing against Canal Nine's distribution surge. The new bandwidth tracked city heat entering the coupled lattice.

Nera pressed her seal beside the waveform. "Source?"

"Kest household array, with relays from three outer sites."

"Content?"

"Unresolved. Sumi no longer thinks it comes from a pump."

Below the Office, warm water struck a canal gate and divided toward Low Glass. The glass under Nera's palm vibrated. A band on the return pane narrowed, rose, then broke into shapes that resembled a human mouth opening around breath.

Text had crossed the 137-year interval one character at a time. Now the signal held pitch.

Nera opened the observation slit. Canal heat entered the room in damp mineral air, carrying the city's released stores past the ledger. On the glass, the future waveform gathered enough harmonics for a voice.

Chapter 23 — The Returning Storm

The voice reached Nera through a tube meant to measure water pressure.

It had no words yet. A rough human vowel traveled along Canal Nine, broke against the iron bend below the Office, and rose through the gauge in her workroom. The glass needle shivered hard enough to shed frost chalk from its face. Outside the legal observation slit, heat released from the Warm Ward moved down the canal in pale muscular folds.

Nera held the gauge housing with both hands. Its bolts knocked against her palms in the rhythm Sumi had established for a clean carrier. Six long pulses. Six short. Then a third set arrived unevenly, stretched by load.

Edrin stood at the forecast glass with his sleeves rolled past his wrists. "The waveform is losing phase."

"Or gaining detail the instrument cannot hold."

"Those conditions look identical until one breaks something."

Below them, Low Glass crews opened the second civic gate. Released heat struck cold water retained by the repayment test. Steam filled the canal trench. The city had given warmth through a system built to send weather away, and every Office instrument now had to learn the opposite pressure.

Nera turned the gauge valve one mark left. The vowel narrowed. A consonant formed against it, vanished, then came again.

"Ne," the tube said.

Her name would have required another syllable. She did not supply it.

Sumi's runner climbed the outer stair without waiting for the public door. She carried a black ceramic strip in tongs and had blood on one cuff, none of it fresh enough to concern Nera before the message. "Phase room wants the senior notary. Drum Four has taken a return groove deep enough to read by touch."

"Tovan?" Nera asked.

"At the infirmary. Wrist set, cuts cleaned. He sent you back to work before I could tell him you were already here."

A hard relief moved through her, useful because it left no room for collapse. She sealed the gauge state, recorded the partial sound, then placed the Office under Edrin's witness. The floor pitched once beneath her shoes. Not an earthquake. Canal Nine had accepted another thermal surge.

On the stair, steam hid the lowest landings. Heatworkers in silver canvas hauled temporary pipes across the gantry while Warm Ward volunteers carried insulation tiles down from civic storage. Nobody moved with the grace of a planned operation. They moved with the competence that followed a public failure everyone could finally name.

Beren Olt stood knee-deep in vapor at the first gate. "Your canal has started talking."

"It belongs to your crews before it belongs to my seals."

"Tell that to the council who kept its keys." He pointed with a wrench toward the east rail. "We have six minutes before the next retained front reaches the intake. Sumi says the future side can see its shape."

"What can we send?"

"Nothing through here that isn't already happening."

That answer followed Nera into the phase room.

The black drums rotated inside their cages, each ceramic surface carrying a wet shine from emergency cooling. Sumi knelt beside Drum Four with her right hand braced on the floor. Her shortened left hand rested on the bearing frame, reading vibration through bone. Around her, technicians had tied every loose tool to a rail. The repayment load made metal travel.

A voice pattern crawled across the return plate. It resembled a storm seen from above, ridges and hollows pressed into blue wax. No civic instrument had vocabulary for it. Sumi had written timings directly on the floor in frost chalk.

"Say your name," she told Nera.

Nera bent toward the horn. "Nera Voss, Year Two Twenty-Three. Office seal open under public witness."

The nearest drum pulled at her breath. For three rotations she heard only water and the ceramic bearings. Then Siv Ren spoke from one hundred thirty-seven years ahead.

"Nera. Do not close west combs at the first pressure fall."

The words came thin, each edge scraped by distance, but the voice carried a person's choices. Siv paused in places the old text channel had not shown. A shallow breath preceded numbers. A quicker one followed the name of a settlement.

Nera looked at Sumi. "Can she hear me?"

"Not as voice. Text still rides backward on the same return. Give me the legal words; I will fit the packet."

A technician called the pressure fall from the north dial.

Sumi wrote without looking up.

> CONFIRM WEST COMBS REMAIN OPEN. REQUEST PHASE SEQUENCE, LOCAL EFFECT, CONFIDENCE.

The strip entered Drum Two. Heat vanished from the room so quickly that Nera's front teeth hurt. Immediate load bought the message a cleaner path. On the return plate, wax softened beneath a moving stylus.

Siv's answer came first as figures. Three pressure intervals, a wind shift, then a

warning that the second apparent lull would carry the worst ice. Sumi compared them with the city's forecast. The first interval differed by eleven minutes. The local model showed one front. Kest had lived through its delayed counterpart and recorded four.

"Storm-phase corrections," Sumi said. "Not a forecast. They are giving us observed structure from the burden we sent."

The distinction mattered to Nera. A forecast could be discounted, owned, or revised before injury. An observed storm in Year 360 had already crossed roofs and bodies. Its data returned with consequence attached.

She authenticated the first correction under contested emergency use. No charter line covered guidance from claimants outside the charter's present. The Founder Sum supplied standing but not procedure. Edrin, arriving with the red book under his coat, added his seal beside hers.

"This entry will be challenged," he said.

"If we survive, preserve the challenge."

"That sounded almost impatient."

"I have learned from the engineers."

Sumi heard and refused to smile. "West pressure in three minutes."

The city kept its cloud combs open.

Rain struck the phase-room roof with the sound of gravel poured across sheet metal. Within a minute it became ice. The western comb towers took the front broadside and bled charge into a sky that flashed green above the canal. Had their vanes closed on the official schedule, the pressure would have folded toward Bellweather.

Instead, the first ridge split over industrial roofs designed to shed it. Two vent stacks failed. A tram line came down near Low Glass. No orchard alarm sounded.

Sumi touched the next chalk mark. "First correction holds within forty seconds."

Nera entered the variance in the ledger. She did not write that Siv had saved the crop. The statement exceeded their evidence. She wrote that Kest's observed phase had altered a present action and that the resulting pressure crossed the city rather than the western fields.

The second packet arrived with crop strain codes appended.

Lev had sent them from Bellweather before the line went down: redleaf cambium temperature, sunroot bed saturation, frost-barley sheath tolerance. They had been meant for Sumi's local model. Kest asked for the full set.

"Why?" Edrin said.

Nera studied the return. "Their storm follows ours by history, not by identical ground conditions. They need to know which plants endured the first version."

"We cannot certify survival yet."

"We can certify measurements."

Sumi glanced toward the ceiling when a fastening snapped above them. "Send raw values. No conclusions. Their growers can decide what applies."

The packet exceeded the old one-hundred-eighty-character limit many times over. Every additional line demanded load now. Canal temperatures dropped as the phase drums held the channel open. Beren's crews called from the intake to say exposed pipes had begun to plate with ice.

Nera divided the data by legal character. Measurements carried Office authentication. Cultivar history carried Lev's cooperative mark. Predictions remained unsigned. The separation looked fussy under a falling roof, which made it more necessary.

Sumi fed the first packet. A drum cage twisted half a finger's width and stopped against its emergency pin.

"No more until the bearing cools," she said.

"How long?"

"Seven minutes if the canal keeps moving. Never if Gate Two freezes."

Nera took Edrin and two runners back into the storm.

Daylight had gone dark brown. Ice moved almost horizontally through the gantries. Workers crawled rather than trust the rails. At Gate Two, a skin had formed across the release channel, trapping Warm Ward heat behind a widening pressure seam. Beren had three people on the wheel and no leverage left.

"Office hands," he shouted. "You wanted physical procedure."

Nera wrapped both arms around the lower spoke. Edrin took the one above hers. The wheel resisted, jolted, then drove backward hard enough to bruise her shoulder. A heatworker jammed a frost-chalk box into the gear teeth. Chalk burst white over his gloves.

Beren moved the crew to the bypass. "When this opens, the steam can blind you. Turn your faces west."

They pulled together. Nera felt each increment through the spoke before she saw movement. The gate rose by the thickness of a receipt. Hot water forced beneath it and shattered the ice skin into plates. Steam struck her cheek. She kept her face turned and counted rotations because counting prevented panic from choosing a rhythm.

At six turns, Gate Two caught. At seven, it released.

Canal water ran east under the storm. A pressure horn sounded from the phase room. The crop packet had resumed.

Beren wiped chalk paste from his mouth. "Ask the future whether their councils make notaries turn wheels."

"If they do not, we can send instructions."

“Send them the injury rolls first.”

Nera carried that request back with the water.

By the fourth storm hour, three of Siv's corrections had proved exact enough to alter city action. Crews vented the northern tunnel before a trapped warm front could crack it. Bellweather lowered orchard mesh during the false lull instead of raising it. Low Glass sealed two school roofs with insulation the Warm Ward had released.

Reciprocity did not feel like trust. It felt like accepting an instruction whose author could not be inspected, then recording enough evidence to remain answerable if the instruction failed.

At the fifth hour, Kest stopped sending phase intervals.

The voice channel still carried pressure, but no words crossed. Sumi adjusted the bearing gap. Edrin checked every witness strip. Nera watched the future clock continue in exact step with theirs.

A text return surfaced after nine minutes.

> KEST STORE THREE DIVERTED. OUTER SHELTERS LOSING SOUTH WALLS. CHANNEL MAY FALL.

Nera read it twice. Holdfast Kest had withheld casualty figures from the outer holdfasts only weeks earlier in her own life. The deception still complicated every authenticated claim. Now Kest had moved one shelter store outward while its own delayed storm entered a severe phase.

“Can they spare it?” she asked.

Sumi's look carried no patience for the question. “Nobody spares shelter heat. They assign who goes without.”

Nera composed a return asking for the diversion terms, affected population, and whether the outer settlements had consented to Kest coordination. The legal questions consumed characters that could have carried comfort. She sent them anyway. Sentiment could not govern a store after the voice faded.

Siv answered in broken phrases. Kest's assembly had released Store Three for six hours. The receiving shelters had sent their wall temperatures and named their own priority rooms. Children, respiratory wards, water manifolds. Kest retained authority over its own two stores and accepted a lower residential target.

Then Siv's voice found the tube again.

“We sent what we said we would protect,” she said. “That includes people who distrusted us yesterday.”

Nera heard other sounds behind her: ceramic chimes, a child coughing, someone reading temperatures in a quick flat cadence. Kest had no quiet chamber for history. Its public channel ran through the room where shelter keepers worked.

“We received your crop measures,” Siv continued. “South Terrace has old

redleaf stock. The strain code survives here under another name. We are sending its blight tolerance."

Lev's data had traveled forward as burden knowledge and come back as cultivation memory. Nera took down every number the voice could hold. She asked Sumi to route the strain note toward Bellweather. No tram ran through the storm, but the buried civic line remained open.

Edrin turned a red-book page. "Do we certify future cultivar advice?"

"We certify source, transmission, and receipt. Lev certifies whether he uses it."

"And if the strain fails?"

"The record keeps all hands visible."

He entered the form without further objection.

Night brought the second lull Siv had warned them against. Above Canal Nine, cloud cover thinned enough to reveal a pale band of stars. Residents emerged onto protected landings. Some cheered when rain softened. Sumi sent runners to drive them back indoors.

Nera climbed to the Office roof with the final correction strip sealed against her chest. Western cloud combs stood black against the clearing. Their vanes had begun to turn toward rest.

"Hold them open," she told the tower crew.

The senior operator showed her the load gauge. "Another phase will strip the lower bearings."

"Kest observed ice after this lull."

"Kest has different towers."

"It has the storm we deferred."

The operator looked past her toward the Warm Ward, where layered vent glass reflected emergency lamps. Then he locked the combs in working angle and ordered his crew below.

The stars vanished.

The returning ridge hit from north rather than west. Ice wrapped the tower in a single roaring minute. Lower bearings screamed. One vane tore free and crossed the canal like a thrown blade, but the remaining combs held the pressure high. The storm broke over open water. Waves lifted against retaining walls and dropped black snow onto the gantries.

For forty minutes, Nera lay behind the roof curb with one hand clamped around the correction strip. The Office shook beneath her. She thought of Tovan's splinted wrist in the infirmary, Lev's exposed orchards, Kest rooms cooling while Store Three moved south. None supplied a reason to stop. Each supplied a claimant whose cost required an entry.

At dawn, the north ridge passed.

Lume had lost two tram roofs, twelve canal vents, one cloud-comb vane, and more glass than the ward crews could count. No district lost heat entirely. Bellweather reported damaged mesh and living root beds. The infirmary sent Nera a note in Tovan's uninjured hand: STOP ASKING WHETHER I AM RESTING.

She went there anyway, carrying Lev's strain report and a list of unopened shelter returns. The civic tunnel smelled of wet wool, boiled bandages, and canal mineral brought in on workers' boots. Every bench held someone with glass cuts or pressure bruises. Warm Ward residents lay beside Low Glass crews because the intake clerk had abandoned district columns when the roof alarms began.

Tovan occupied a cot under a vent patched with orchard mesh. Purple swelling showed above his splint. Fine cuts crossed his cheek, each sealed with clear resin. He had arranged witness strips across his blanket and enlisted the child in the next cot to turn them when his good hand cramped.

"You have made a branch office," Nera said.

"The branch has better tea."

"You were told to rest."

"I am resting the fractured part." He lifted Lev's report from her hand before she could stop him. "Did the future strain survive?"

"In another name. Kest sent blight tolerance figures."

"Then Bellweather needs the old glass recipes too. Redleaf grafts fail if the night cover sheds heat faster than the bark."

Nera looked at his bandaged face. "Your apprenticeship does not qualify you to advise an orchard."

"It qualifies me to know when glass makes weather."

A ward councillor on the next bench asked whether the phase channel had drawn enough immediate heat. He wore one shoe; his other foot had been wrapped in a silver work glove. Yesterday he had argued that opening stores beyond the ward would make its infirmary unsafe. Now he held a heat receipt marked with three districts.

"We do not know yet," Nera said. "Every draw remains under audit."

"Keep this one open." He tapped the receipt. "The intake crossed out residence. If the old rate code returns tomorrow, people here will go back to separate queues."

Tovan watched her rather than the councillor. Nera understood the request. Emergency conduct could vanish into ceremony once officials declared survival. She took the receipt, recorded the mixed intake as a storm action, and asked the councillor to witness it under his public office. His signature came slowly because his hand shook.

At the cot, she added Tovan's glass note to Lev's data packet with its source plainly named. Tovan objected to the word apprentice until she showed him the

line that gave Bellweather authority to test or refuse it.

“Good,” he said. “Do not make my wrist vote for them either.”

The child's mother laughed once, without humor, then covered her mouth. Nera sealed the packet on the infirmary table. A return had entered through crop strain, glass craft, a disputed queue, and one injured boy's refusal to become evidence for anyone else's convenience. The route looked untidy enough to be public life.

Kest restored its voice near midday. Store Three had reached the outer holdfasts before their south walls failed. Eleven rooms in Kest had dropped below shelter standard. No water manifold froze. Siv did not offer a casualty number until each roll could be witnessed.

Nera spent six days authenticating what both eras had done. She matched storm-phase corrections to tower logs, crop measures to future strain notes, store releases to room temperatures. Neither side had rescued the other. They had exchanged usable knowledge, accepted local cost, and left enough record for disagreement.

On Day Ninety-Two, she reopened the public channel. Her shoulder had purpled where the gate wheel struck it. Across the workroom, Tovan sat with his fractured left wrist held in a gray splint, sorting witness strips one-handed despite three written orders to remain home. Edrin placed the red book beside the voice gauge. Sumi waited below in the phase room. Lev listened from Bellweather through a line repaired before its casing had dried.

Nera entered her seal and addressed Siv without the shelter language of emergency. “We have governed one storm together without shared clocks, equal power, or complete trust. I propose that we stop treating this as an exception. Let both years negotiate claimant standing, future veto, present burden, reciprocal audit, and crop and shelter coordination under one public instrument. I propose a Treaty of Seasons.”

Chapter 24 — Treaty of Seasons

The first treaty draft could not fit through the channel that had made it necessary.

Nera spread its pages across the Assembly Hall floor because every table had warped during the storm. Forty-six clauses lay under brass clips in a long rectangle: claimant standing at the north end, crop exchange near the center, default and appeal beside the public doors. Meltwater moved between the stones in thin reflective seams. Above the dais, three workers replaced shattered vent glass while the negotiators argued beneath their falling dust.

Sumi had calculated the cost of sending the full instrument to Year 360. Even with the city retaining a measured share of weather load, transmission would take eleven days. Kest could answer only in compressed packets. A careless order of clauses would let Lume occupy the channel with definitions while future settlements waited to contest the powers hidden near the end.

“Send the veto first,” Siv had written.

Edrin objected before Nera finished reading the strip aloud. “Standing precedes remedy. A party cannot exercise a power before the instrument recognizes the party.”

“Recognition without power gives them a ceremonial chair,” Beren said. He sat on an overturned heat manifold with his boots leaving brown water on the floor. “We have those in Low Glass. They come with speeches and no valve keys.”

Lev leaned over the load schedule. Bandages covered two fingertips on his right hand, turning pages into slow work. “If you send the veto first, every grower hears a permanent stop before anyone sees the staged plan.”

“They are not voting on what growers hear,” Nera said.

“They will be when the frost-barley forecast reaches the public board.”

At the open hall doors, residents watched from behind a rope instead of privacy glass. Halden had permitted no closed sessions after publication of the ledger. He occupied the steward's chair without joining the drafting circle. His presence carried authority and its visible refusal to intervene.

Nera lifted the four pages dealing with sequence. “We send a table of powers before any full clause. Standing, veto, burden allocation, audit, appeal. One hundred eighty characters if Sumi removes our titles.”

“No titles,” Sumi said from the phase console near the dais. “No courtesy lines. One action per field.”

Edrin adjusted his damp cuffs. “The record must show that Siv speaks for Kest under Kest's own authority, not for all citizens in Years Three Sixty through Three Eighty-Eight.”

A return bell struck before Nera could answer. Sumi took the strip from the slot

with her shortened hand and held it to the hall light.

> KEST SPEAKS FOR KEST. OUTER ASSEMBLIES SIGN SEPARATELY. WINTER LINE MAY DELEGATE ONLY NAMED POWERS.

Beren gave Edrin a hard glance. "They have met a notary before."

"They have met the consequences of notaries," Edrin said. "That teaches precision."

Nera amended the heading. No single future voice would stand in for 2.1 million people. The channel could tempt the present to treat its first correspondent as a sovereign because one voice simplified reception. Siv had refused that convenience before Lume could convert respect into another form of erasure.

Each winter-line settlement would enter as a named claimant after authenticating population, weather burden, and decision authority. Kest could sponsor access but not absorb another settlement's claim. Outer assemblies could combine their bandwidth, appoint a temporary relay, or answer independently. No silence would count as consent.

Lev read the last sentence. "Then a broken shelter transmitter can stop a seasonal schedule indefinitely."

"Silence stops new borrowing," Beren said. "It does not stop weather already arriving."

"A farm cannot wait through three cycles to know whether cloud retention proceeds."

Nera moved two clips. "A future veto applies to new deferrals. Existing reciprocal schedules proceed under agreed tolerances unless a claimant sends a hazard notice. If the return fails, Lume retains the immediate load until contact or public emergency review."

"That assigns every channel fault to current crops," Lev said.

"It prevents us from treating inability to answer as permission."

His injured fingers rested against the page. The bandage had seeped pale yellow at one edge. "Write an emergency floor for food systems. Not automatic deferral. A floor below which the Assembly must choose in public among rationing, current heat release, or a named temporary burden submitted for later consent."

Beren shifted on the manifold. "Who receives that temporary burden while consent travels?"

"No one," Lev said. "Keep it local until they agree."

The answer cost him. Bellweather had already sacrificed redleaf fruit and exposed root beds to cold. Nera watched him use the remaining harvest as a limit on his own side's authority rather than a claim to exemption.

She added the food floor with public choice requirements. Edrin marked the clause provisional until claimant response. Sumi compressed it into a return packet

without changing “keep it local.”

That phrase survived the channel.

By Day Ninety-Five, the hall floor had acquired a geography. Claimant pages lay on dry stone near the dais. Burden schedules occupied a patch of boards over the worst leak. Reciprocal audits surrounded the phase console in narrow strips that technicians stepped over with care. Members of the public followed amendments by moving along the rope.

The first dispute to stop work concerned one verb.

The future settlements wanted authority to refuse any new deferral. Assembly counsel proposed review. Edrin offered withhold assent. Halden's legal office sent decline authorization.

Siv returned the same three words each time.

> BINDING VETO. NO SUBSTITUTE.

Beren read them to the gallery. A woman in a Warm Ward service coat asked whether one holdfast could prevent the city from moderating a lethal winter. Someone from Low Glass called back that Lume had prevented the holdfasts from moderating theirs for two centuries without asking.

Nera waited until the voices separated enough to hear questions again. “The veto governs borrowing, not survival measures that keep the burden in our year. We may open stores, move heat between wards, alter crops, change shelter targets, or ration public systems. We may not assign dissipation forward without assent from those receiving it.”

“What if they demand that children freeze here?” the service worker asked.

“Then they have demanded something they cannot order. The treaty gives future claimants power over what reaches them. Present distribution remains our obligation.”

That boundary quieted neither side, but it clarified what each could fight about.

Halden left his chair and crossed to the claimant pages. “We should state that no veto creates executive authority in Year Two Twenty-Three.”

Beren laughed without amusement. “You have discovered jurisdiction now that it limits someone else.”

Halden accepted the blow. “Jurisdiction often becomes visible at a boundary. We should write this one before fear writes it for us.”

Nera placed his sentence after the veto. Siv approved it with one addition: no current Assembly power created authority over future shelter distribution. Symmetry took sixty-three characters and two hours of retained load.

The burden schedule took longer.

Beren brought heat receipts from nine Low Glass blocks and stacked them on the boards. Each showed household moderation, service outages, maintenance

assessments, and injury contributions over thirty years. Warm Ward representatives arrived with insulation maps, seed-vault obligations, infirmity reserves, and the emergency store release from Repayment Day. Lev carried food-yield curves that made equal temperature shares look absurd before anyone spoke.

Equal cold would kill unevenly. Equal payment would preserve old benefit. The treaty needed a distribution that recognized both.

Beren's formula began with historical advantage. For the initial retained load, the Warm Ward would accept 2.4 times Low Glass's per-capita burden. Industrial systems could earn reductions only for verified public heat supplied elsewhere. Food systems received protection according to calorie yield and recovery time, never land value. No district could purchase exemption by promising later infrastructure.

A Warm Ward delegate named Sareth turned her receipt face down. "You are treating insulation as guilt."

"I am treating it as capacity," Beren said. "The guilt sits elsewhere, if you need it. I need the rooms that can lose three degrees without losing water."

"Our infirmity cannot lose three."

"Then mark the infirmity. Do not wrap the entire ward in its blankets."

Lev tapped the crop table with a bandaged finger, winced, then used his knuckle. "Bellweather can hold another four-hour frost if the following two nights stay above the root threshold. A blanket crop protection will overstate us. A daily auction will destroy planning."

Nera asked, "What interval can you certify?"

"Seven days with revision after any storm warning. Longer if the future strain data proves sound."

Beren wanted three-day revision for household heat. Sareth wanted fourteen. Sumi warned that different intervals would create phase noise if every district changed its draw independently.

They built a seven-day public schedule with hazard openings. Each district would publish capacity, protected systems, and actual loss. The Office could authenticate measurements but could not decide the burden formula alone. A distribution panel would include heat labor, growers, insulated wards, uninsulated districts, and future claimants through the return channel.

Edrin raised his pen. "A future claimant cannot attend every present adjustment."

"Then do not call it attendance," Beren said.

Siv's response arrived while they argued.

> WE NEED VETO ON NEW LOAD, NOT MANAGEMENT OF YOUR RADIATORS.
SEND RESULTS. WE AUDIT EFFECT AGAINST PROMISE.

Nera drew a line through “participating member” and wrote “receiving auditor.” The future side would not govern a valve it could not see. It would compare scheduled deferral with the burden that actually arrived, contest discrepancies, and suspend further assent when current records failed.

Lev studied the phrase. “They can suspend assent on error?”

“On unexplained variance beyond tolerance,” Nera said.

“Weather produces variance.”

“So do hidden transfers.”

“What tolerance?”

That question occupied Day Ninety-Seven.

Sumi built a test schedule on the phase console. She used the returning storm because both eras had dense measurements: comb angle, canal temperature, roof loss, wall temperature, crop strain, store diversion. Nera authenticated sources. Edrin separated direct observation from local calculation. Beren refused any metric that counted heatworker injuries as system noise. Lev required plant damage to appear after seven days rather than vanish at storm end.

They sent Kest a compact audit sheet. Kest returned its corresponding record with local names removed from public fields but preserved under claimant seal. The two sides disagreed on the storm's total burden by six percent.

Sumi found two points in current-side data where an automatic civic routine had opened a crust sink without entering the public schedule. The routine predated Nera's career. It had hidden minor loads by classifying them as stabilization.

“Disable it,” Beren said.

“We disable nothing during an audit without modeling the pressure,” Sumi replied.

“Then name it borrowing.”

“It may retain locally before it releases.”

“May?”

Sumi placed her shortened hand against the console. “That uncertainty belongs in the audit. Do not improve it with anger.”

Nera entered the routine as unresolved. Kest's figures exposed another discrepancy: Store Three's southward transfer had been measured at release, while outer settlements recorded less recoverable heat at receipt. Siv argued that transit loss should count against Kest's promise. The receiving assemblies argued that wall damage, not Kest, had consumed it.

Edrin proposed recording both gross action and delivered effect. “A promise concerns conduct. A claim concerns consequence. Neither replaces the other.”

For once, everyone accepted his wording before the return bell.

The audit clause grew from that dispute. Each side would publish scheduled

action, measured execution, observed arrival, local loss, and disputed variance. No single number could convert a broken pipe into compliance or a good-faith release into delivered shelter heat. Reciprocal audits would remain adversarial enough to matter.

A second difficulty emerged when they tried to write an appeal. Lume's Assembly could summon a technician, inspect a canal, or compel a ward to open its books. Kest could do none of those things across the fixed interval. Its only effective remedy involved withholding assent from later transfers, a power broad enough to punish districts unrelated to the disputed entry.

Edrin proposed a staged finding. First notice would identify the measurement, source, and claimed variance. Present operators would answer with raw records before argument. A joint panel could request one repeated observation, but it could not demand impossible reconstruction after weather had passed. If disagreement remained, both accounts would stay visible on the public ledger.

"That leaves lying without consequence," Beren said.

"It leaves contested evidence without forced agreement," Edrin replied. "Fraud has another clause."

Lev read the fraud provision. "Intent requires proof. A grower can hide a vent opening by calling it frost response."

"Then the schedule records heat movement, not motive," Nera said. "If the opening sends unapproved load, the future veto closes until correction. Nobody needs to prove what the grower wanted."

Sareth asked who would protect a district from a false accusation transmitted at public cost. Sumi answered before Nera could shape a legal response. "Checksums protect the record. Nothing protects you from being accused. Your measurements answer."

The future packet took the question outward. North Brace returned a demand that any suspended assent name the exact schedule affected. Two smaller holds required an expiry unless evidence renewed the claim. Kest objected to automatic expiry during transmitter failure.

They settled on a narrow suspension tied to the disputed transfer class, with no expiry while the claimant kept sending authenticated hazard marks. If contact failed entirely, Lume would retain the questioned load. The remedy could inconvenience a current district. It could no longer convert silence into weather delivered elsewhere.

On Day Ninety-Nine, they reached the repayment horizon.

The public ledger held 4.7 billion degree-hours across the audited belt. No honest schedule could erase that debt within one crop cycle or one administration. Sumi's survivability models ranged from twenty-two years under severe rationing to forty-one under conservative infrastructure change. Future settlements faced their hardest compound winters before the slow plan could finish.

Siv rejected forty-one years in the first packet.

Lev rejected twenty-two before Nera read the reasoning aloud. "That curve assumes we replant half the western belt as frost barley. It keeps calories on paper by removing orchards, soil crews, graft knowledge, and every food that takes years to return."

Beren pointed to the conservative line. "Forty-one assumes Warm Ward storage remains private capacity."

Sareth did not turn over her receipt. "It also assumes the ward's glass survives. The storm used seven percent of our repair stock."

Nera asked each side for a floor rather than a preference. What current annual load could their systems keep without concealed transfer? What reduction did the future settlements need before their most exposed shelter years? Which infrastructure work changed those limits instead of moving them?

The answers narrowed around 28 years.

Under the proposed schedule, present cities would accept a rising share during the first six, hold it through major radiator reconstruction, then increase again as crop conversion and public insulation matured. Future claimants could veto any new deferral and reopen annual shares when observed burdens exceeded modeled tolerance. The debt would remain visible after the initial term. Nobody could issue a satisfaction entry merely because the schedule ended.

Siv asked for the phrase "repayment horizon, not discharge."

Beren supported it. Lev added crop failure as a mandatory reopening condition. Edrin required a fixed date for review even if every measured target held. Halden suggested that each new Assembly inherit the agreement without power to shorten its protections through ordinary vote.

"That approaches entrenched law," Edrin said.

"It should," Halden replied.

"You sealed the annex that might authorize it."

"I inherited the seal. I continued its use."

The hall heard him. Heads turned along the public rope. Halden did not retreat to the chair.

Nera wrote the term and its limits into the instrument. Twenty-eight current years. Years Three Sixty through Three Eighty-Eight on the receiving side. Mandatory review every fourth year. No presumption that time paid what weather had not.

The full draft began crossing the channel on Day One Hundred.

Sumi sent powers first, then definitions, then burden schedules. Crop and shelter coordination followed. Audit and appeal traveled last only because Siv had approved that order after receiving the table. Each packet carried a sequence

number and a physical checksum cut into ceramic. Kest relayed copies outward under separate claimant seals.

Replies arrived with local amendments. North Brace wanted water manifolds listed before residential heat. Three agricultural holds demanded seed movement authority without Kest sponsorship. An outer assembly that had lost forty-two names from its shelter roll refused any clause calling the compound winter an emergency; it described the condition as an inherited schedule.

Nera preserved their wording.

At midday on Day One Hundred Three, the final assent strip emerged from the phase console. It carried Kest's seal, six outer assembly marks, and a winter-line relay register for those still reviewing. No absent settlement had been counted. No pending claimant lost the right to enter later.

Siv's personal line followed.

> TEXT AGREED FOR KEST. ORRA ASKS WHETHER A TREATY CAN HAVE WEATHER. I TOLD HER ONLY IF PEOPLE KEEP IT.

Nera did not include the child's question in the legal text. She copied it into the public transmission log where Tovan, waiting with his gray splint beside the gallery rope, could read it.

Edrin assembled the agreed pages on the driest table. Lev added Bellweather's crop seal with his bandaged fingers held clear of the wax. Beren pressed a canal-service mark beside the distribution schedule. Sumi authenticated the channel record. Sareth turned her heat receipt upward and signed for the ward's declared capacity, not for its innocence.

Nera read the instrument's name into both clocks: "Treaty of Seasons."

Before she could set the final season seal, Assembly counsel entered from the civic tunnel with a red defect notice held above the wet floor. The Founder annex, he announced, had never passed its second ratification reading. Its claimant clause might bind conscience and still fail to convey legal power. The text had crossed one hundred thirty-seven years and returned agreed. In Lume, it remained unsigned.

Chapter 25 — Foreclosure

Halden Rusk found the missing ratification in a box whose label promised complete minutes.

The box had occupied the steward's lower archive since before his appointment. Gray cord crossed its lid in the founder knot, two loops for public action and one for emergency consent. He cut the cord at dawn on Day One Hundred Four while the hearing clerks warmed Assembly Hall and citizens queued through sleet outside.

Inside, Year Eleven survived as paper compressed by old damp. The first reading of the claimant annex appeared under the Founding Crisis docket. Speakers had named future settlement, dissipation burden, and obligations no living Assembly wanted to price. Riot alarms interrupted the session. Three pages later, after a list of injuries and broken ration windows, the clerk recorded that the Founder Sum had been sealed pending a second reading.

No second reading followed.

Halden examined every sheet by the archive lamp. Page numbers continued without a gap. Witness threads showed no removed leaf. The emergency register contained a vote to preserve the annex from destruction, not to enact it. Later indexes called it a charter attachment because each index had copied the title from the prior one.

Assembly counsel had diagnosed the defect correctly.

A First Steward could certify the archive complete. Once he did, any court would treat the missing reading as evidence that enactment had never occurred. The treaty's claimant authority would lose its founder foundation before debate reached distribution. The current Assembly could still pass new law, but ordinary law allowed repeal by an ordinary majority. Future veto across 137 years would begin as a policy preference rather than a binding limit.

Halden placed the completeness form beside the box.

Its language asked for three findings. Custody intact. Sequence intact. No evidence omitted. All three fit the record he had inspected.

He had built a career by signing findings that fit records.

Outside the archive, the city prepared to discover whether summer had been legally defended or merely taken. Workers cleared storm glass from the hall roof. Low Glass delegates carried heat receipts in oilskin cases. Warm Ward residents arrived in winter coats designed for ornamental snow and found the cuffs too narrow for work gloves. From Canal Nine came the steady knock of a phase drum carrying treaty text toward people who would not hear today's hearing until history delivered it.

Halden dipped his pen.

The completeness certificate would not deny the deaths. It would not restore the old concealment or erase the published ledger. He could defend it as legal cleanliness. A defective annex should not acquire force because its secret had become morally useful. Institutions survived by refusing convenient irregularity.

He had said versions of that sentence across thirty years.

During the Founding Crisis, however, institutions had preserved the annex and hidden it. Every later steward had benefited from its ambiguity. When seasonal deferrals required authority, the archive called the Founder Sum technical foundation. When downstream claims threatened current comfort, officials called it unenacted study. The defect had never restrained borrowing. It had waited in the dark to restrain the creditor.

Halden set down the pen without touching the form.

His clerk knocked once and entered. Mira Pell had served four stewards and distrusted private pauses. "Counsel needs your archive certification before the first witness."

"I have not made one."

"Do you require another examiner?"

"No."

Her eyes moved from the cut cord to the unsigned sheet. "Then the record remains indeterminate."

"The pages remain exactly what they are."

"Counsel will ask whether the sequence is complete."

"Yes."

"What will you answer?"

Halden closed the box. "That the Office used this annex as authority for two centuries of transfer architecture. If the Assembly wants to argue it lacked effect, it may begin by accounting for every action taken under its title."

Mira did not show surprise. She lifted the completeness form by one corner. "Should this stay here?"

"Put it in the hearing file, unsigned."

He saw the consequence land in her posture. An unsigned certificate could be read by every member. Nobody could claim Halden missed the route out.

At the hall entrance, Nera Voss waited beside the Founder ledger in a dark coat still marked with storm chalk. Her shoulder made one side hang lower. Edrin Sol held the agreed treaty pages. Sumi Quill had brought a ceramic channel strip rather than dress for ceremony. Tovan watched from the public benches with his left wrist in a gray splint.

Nera looked at the archive box under Halden's arm. "Complete?"

"The physical sequence appears intact."

"Does that answer counsel?"

"It arms counsel."

She did not thank him for the warning. Gratitude would have made his act personal, perhaps even redemptive. Nera had become less willing to let a person borrow virtue from work not yet completed.

The hearing bell sounded.

Halden carried the box to the steward's table and placed the unsigned form on top.

Assembly counsel began with procedure. She established the Charter's two-reading requirement, the Year Eleven docket, the riot interruption, and the sealed status of the annex. Edrin authenticated the paper. Nera authenticated the connection between its language and the hidden cold accounts. Neither tried to invent a second vote.

"The Founder Sum contains an unratified proposal," counsel said. "Its moral relevance may inform this Assembly. It cannot create claimants, transfer authority, or bind successor bodies."

Public benches filled every pause with clothing and breath. No one shouted. The city had spent its louder fear during the ledger release.

Counsel turned to Halden. "First Steward, have you inspected the complete Year Eleven minutes?"

"I have inspected the box labeled complete minutes and its custody record."

"Did you find a second ratification reading?"

"No."

"Did you find evidence of removal?"

"No."

"Then will you certify the sequence as complete?"

Halden looked at the blank form. "Not without qualification."

"What qualification can alter intact numbering and witness thread?"

"The Assembly repeatedly invoked the annex as technical authority after Year Eleven. Phase expansion appropriations cite its debt calculations. Three continuity rulings cite its claimant model when permitting transfer into inhabited probability zones, then conclude that habitation remained too uncertain for notice. The archive treated the annex as operative whenever operation enlarged present power."

Counsel's face tightened. "Later error cannot ratify a defective act."

"No. Later reliance can prevent us from discovering the defect only when limitation becomes inconvenient."

"You are offering estoppel against the public."

"We are the public office that made both claims."

A member from the south canals asked whether Halden now considered the annex law.

He answered carefully. "I consider its legal status disputed. I refuse to certify it void through a defect the seasonal system ignored while borrowing."

The hearing clerk entered every word. Halden heard the pen move behind him.

Counsel lifted the unsigned certificate. "You could resolve this question."

"I could resolve it in favor of the authority I presently hold."

"Your office has a duty to preserve continuity."

"Continuity of what?"

The question left his mouth before institutional habit could soften it. Counsel did not answer. Beyond the high windows, a cloud comb turned under manual control because the automatic schedule remained limited after the storm.

Halden opened the archive box. He removed the annex and carried it to the witness rail. Its pages had once frightened founders who faced ration riots and failing climate machinery. They had sealed a claim rather than bear it. Later officers had inherited both the warning and the convenience.

"We called concealment stability," he said. "We called later people uncertainty. We allowed present institutions to rely on the Founder Sum as engineering foundation while denying its claimants legal existence. That practice cannot supply a neutral rule for this hearing."

Counsel objected to characterization.

The hearing chair sustained the objection and left his testimony in the record as steward policy. Procedure narrowed his language without erasing his choice.

Halden returned to his table. He signed one finding only: custody intact. Beside sequence he wrote DISPUTED BY SELECTIVE INSTITUTIONAL RELIANCE. Beside omission he wrote NO PHYSICAL OMISSION FOUND; LEGAL EFFECT UNRESOLVED.

The easy invalidation closed under his hand.

Nera watched Halden's signature dry while Assembly members divided the question into smaller powers.

They could recognize downstream claimants under current emergency legislation. They could adopt the negotiated burden schedule as public utility law. They could require reciprocal audits through the Office. Entrenching a future veto against successor repeal remained harder. The Seasonal Charter gave each Assembly broad control over weather in its present and no explicit power to surrender that control.

Counsel argued that binding future Assemblies violated the same principle Nera had used against deferral. No present body should govern people beyond its time.

Edrin asked for the speaking rail. "A veto does not govern the future Assembly's local weather. It limits delivery into another present. Every Assembly remains free to manage burdens within its own year."

A northern member frowned. "Year Three Sixty does not exist for us as a separate jurisdiction. It occupies our planet later."

"It exists for its citizens as now," Nera said.

"You cannot draft law around grammar."

"No. We draft around material effect. A transfer crosses a jurisdictional boundary because people on the receiving side can consent or refuse. The fixed interval does not make their rooms ours."

Sumi placed a return strip on the rail. Kest had sent wall temperatures during the hearing, not an argument. Store Three remained below reserve after helping the outer shelters. The figures showed where authority landed when doctrine finished speaking.

The member read them and asked, "What prevents a future holdfast from using veto to extract present resources?"

"Nothing prevents it from making demands," Nera said. "The treaty limits what the veto controls. They can refuse new burden. They cannot order our stores, crops, or labor. We can refuse unrelated demands without sending weather as pressure."

Beren testified next. He declined the witness chair and stood in his work coat, hands on the rail. Low Glass supported ratification only with benefit-weighted load fixed in law. Equal household shares would make treaty language cover another transfer from insulated rooms to exposed ones.

A member asked whether the 2.4 ratio punished Warm Ward residents who never knew the ledger existed.

"It assigns capacity and benefit," Beren said. "Punishment cares what a person deserves. Heat cares which room can cool without splitting pipes. Receipts show who received moderation. Insulation maps show who can hold cold. Use those facts. Leave innocence to private conversation."

Lev followed with his fingers freshly wrapped. He entered Bellweather's crop losses, the surviving sunroot beds, and the seven-day revision interval. Counsel asked if future veto could threaten food security.

"Yes," he said.

The hall shifted.

"So you oppose it?"

"No. Present borrowing already threatens food security. It moved crop failure to fields we declined to name. A veto forces us to see the choice before transfer. Seeing a danger does not remove it."

“Would you accept rationing rather than a deferral refused by Kest?”

Lev rested both hands on the rail. “I would require the Assembly to choose rationing, store release, crop change, or another local cost in public. If it asks consent for a limited transfer, the receiving people may answer. What I oppose is permission hidden inside machinery.”

Tovan testified without standing because his balance remained poor after pain medicine. He gave his name, age, apprenticeship, and injury in four plain lines. A member tried to ask whether Repayment Day had cost too much.

“My wrist cannot answer a public budget,” he said. “It can show that glass breaks when weather changes. You still have to decide whose consent counts.”

Nera kept her eyes on the treaty pages. Pride could turn a child into property as quickly as fear.

Before the witness list closed, she entered one final comparison. The Office had attached her mother's vent death to a Warm Ward winter scheduled twenty-one years later, yet no second Assembly reading had authorized that hidden classification either. Counsel objected that an unlawful practice could not validate an unlawful annex.

“I agree,” Nera said. “I offer it to show asymmetry. The system demanded no perfect ratification before taking. Now counsel demands perfect ratification before those burdened may refuse.”

A member asked whether she wanted the Assembly to ignore every procedural defect that favored repair.

“No. I want the defect exposed on both sides of the account. Ratify claimant standing today under present public authority. Preserve the Founder record with its incompleteness. Do not use old disorder as a private permission for transfer and a public barrier to consent.”

Edrin added a custody note to her exhibit. The page retained its missing vote, her mother's name, and the transfer credit in one authenticated record. Nothing in the treaty repaired Mara Voss's death. The comparison prevented the institution from making procedural purity another benefit distributed only to itself.

By afternoon, the Assembly had authority for every clause except permanence. Votes existed for claimant status, burden allocation, audit, channel funding, crop exchange, and shelter coordination. The future veto remained one vote short under the Charter amendment threshold.

Halden requested the rail again.

He carried no prepared statement. In his hand lay the brass steward seal used to authorize automatic seasonal schedules. Its face showed Orison under three softened weather bands. Nera had seen it on transfer warrants since her first year at the Office.

“The steward controls no Assembly vote,” he said. “The steward does control

delegated seasonal execution until law assigns it elsewhere. That delegation includes the automatic deferral table.”

Counsel rose. “First Steward, executive surrender cannot amend the Charter.”

“No. It can remove executive opposition from the question before you.”

Halden set the seal on the witness rail. “I transfer stewardship over intertemporal seasonal schedules to the public distribution panel established by this instrument, contingent on notarial authentication and claimant assent. I retain ordinary emergency administration within Year Two Twenty-Three until my term ends.”

The hall began to speak all at once. The chair struck the bell.

Nera understood the move. Halden could not bind successor Assemblies alone. He could deprive the current office of the machinery that made quiet deferral easy. Any later steward seeking to restore it would need a public grant rather than inherited habit.

Counsel called the transfer irregular.

Edrin answered from his table. “The original delegation came by Assembly schedule, not charter text. This body can accept its return by ordinary vote.”

A member asked Halden whether panic over shortened summer had changed his judgment.

“No,” he said. “Evidence changed what our stability measures contained. People may still reject the costs. They should reject or accept costs actually stated.”

Nera heard his old offer inside the words: managed disclosure, controlled language, comfort preserved through sequence. He had not become another person. He had carried his concern for continuity to a point where continuity could no longer mean unchecked institutional power.

The Assembly accepted return of the delegated authority by a clear majority. The veto amendment came again under a revised clause: no seasonal body could transfer burden beyond its current year without authenticated assent from materially burdened claimants. The clause governed the boundary, not the politics of a future Assembly.

The final vote passed by one.

No cheer followed immediately. Clerks had to verify marks. Edrin compared each ballot against the member register. Nera watched sleet gather at the high windows while the hall waited through the ordinary risk of counting.

At sunset, the chair declared ratification.

Nera carried the agreed pages to the phase console. Kest would receive the vote after the fixed interval already woven through its history. Siv's side had negotiated with a past that had not yet finished deciding. The instrument's sequence number showed that their assent and Lume's ratification belonged to one causal record, neither provisional because the other arrived later.

She pressed her notarial mark beside the claimant seals. Edrin followed. Halden did not sign as treaty author. He signed the transfer of his seasonal delegation and surrendered the brass seal to the clerk.

Sumi waited at Canal Nine with the automatic table exposed.

They descended through the civic tunnel after dark. Heat along the walls ran two degrees lower than before Repayment Day. Citizens had hung temporary curtains at cross-passages. Near the phase-room door, black snow melted from workers' boots into shallow trays.

The automatic table occupied a cabinet behind Drum One. Hundreds of pins encoded default weather movement: summer heat, winter cold, storm pressure, crust-sink release. Most entries had operated without a human hand once forecasts crossed preset margins. A small ivory lever enabled downstream assignment whenever local safety thresholds failed.

Sumi pointed to it. "That lever does not know about votes."

"Then let it learn through absence," Nera said.

Halden gave the cabinet key to Beren, whose panel now held physical custody under the treaty. Beren opened the lock but did not touch the lever. "Notary first. I will not inherit a secret action while everyone watches the ceremony."

Nera authenticated the current state. Edrin witnessed each pin count. Sumi recorded drum pressure and canal temperature. A return strip from Kest lay under glass beside the cabinet, carrying assent to the veto clause.

Halden read the disable order aloud. "This action forecloses unilateral borrowing of seasonal dissipation beyond the current year. It does not foreclose summer, emergency heat, crop protection, or any measure kept within present responsibility."

Nera amended one noun. "Use unilateral summer borrowing in the public index. People have heard your old phrase. They should see what has ended."

He accepted the correction.

Beren gripped the ivory handle. Sumi released its safety catch. Nera applied the first ratified treaty season seal to the cabinet record.

The lever moved with almost no resistance.

Across the automatic table, amber pins went dark one row at a time. Summer retention remained available. Ward exchange remained available. Cloud combs, radiator canals, emergency stores, and crop protections held their lawful controls. Only the silent route into Year Three Sixty lost power.

Drum One completed a rotation. No replacement schedule entered.

At eleven minutes past the night bell on Day One Hundred Eleven, the last automatic deferral disabled with a small mechanical click that every person in the room could witness.

Chapter 26 — The Unborrowed Sky

Snow entered Lume before the city had decided what to call its discomfort.

The public forecast offered exact terms. Twelve centimeters by midday. Canal skin likely on exposed branches. Warm Ward residential target reduced by four degrees. Low Glass reduction set lower under the benefit schedule, though its pipes faced greater risk. Bellweather frost-barley protection authorized. No downstream deferral requested.

People read the board under the Office awning while the first flakes crossed Canal Nine. Some called the schedule theft. Others called it repayment. A heatworker in a torn silver coat corrected both groups and called it Tuesday.

Nera stood beside the public seal desk with her gloves on. Snow had touched Lume many times during her life, but always after the Seasonal Continuity Lattice pared its depth, shifted its wind, or moved its hardest hours beyond the inhabited model. This fall arrived without such subtraction. Each flake carried only the weather of Year Two Twenty-Three.

At nine minutes past the morning bell on Day One Hundred Twelve, she authenticated the forecast as the first unshifted snow under claimant authority.

The brass cooled beneath her thumb.

No hidden column opened. No temporal-owner field refused her. The ledger displayed present load, local distribution, future effect, and a claimant-assent field marked NOT APPLICABLE: NO BURDEN TRANSFER. Its plainness unsettled her more than founder frost had on Day One. Honest records could look almost insufficient after concealment trained the eye to search for a chamber behind every line.

Edrin leaned over the witness copy. "Canal branch estimate changed by half a degree."

"Source?"

"Gate Three hand reading. Sumi disputes the public sensor."

Nera suspended that figure, not the schedule. "Send a worker with a reference bulb. Keep both values visible."

"You enjoy saying that now."

"I enjoy knowing what it costs."

Outside, the canal lost its usual steam. White accumulated on the gantry until workers swept paths with stiff reed brushes. Heat receipts began printing from the public slot every six minutes, each one naming district draw and protected services. Residents carried them home or pinned them to a board set beneath the awning.

A woman from the Warm Ward placed hers in front of Nera. "My building has infants."

The receipt showed a four-degree reduction in residential corridors and no reduction in the nursery loop. "The protected circuit remains at standard."

"Corridor pipes feed those rooms."

"Maintenance logged them as separate after the storm."

"And if maintenance lied?"

"Then we test."

Nera sent a public inspector rather than asking the woman to trust the schedule. The inspector returned two hours later with pipe temperatures, a resident witness, and a failed valve in the third-floor junction. The building had not lied. Its map had.

Under the old system, an automatic draw might have covered the fault and entered the extra cold downstream as stabilization. Under the treaty, the ward panel had to choose a present source. It lowered heat in an ornamental gallery, opened one soil store beneath the winter garden, and dispatched a valve crew.

The nursery kept its temperature. Frost formed inside the gallery glass.

Across Low Glass, households closed curtains over stairwells and moved evening meals into common kitchens. The district had fewer stored degrees to surrender, so its burden arrived through labor. Crews insulated exposed radiator joints. Neighbors checked heat receipts against pipe sound. Three blocks opened bathhouse boilers for residential warming under a fuel ration that would shorten hot-water hours for ten days.

Beren reported each exchange without praising sacrifice. "Block Seven requests more wool curtains. Block Twelve has enough hands and no dry sealant. The Warm Ward has sealant."

"Authority?" Nera asked.

"Distribution panel approved a swap for repair labor."

"Price?"

"No cash. Hours against material at public rates."

She authenticated the exchange and added injury reporting. One repair worker had already burned his wrist on a boiler collar. Keeping weather local did not keep harm abstract.

At Block Twelve, the common kitchen occupied a former tram depot whose doors could not close fully. Residents had hung orchard mesh under the roof and packed the gap with rolled work coats. Four stockpots stood over a boiler plate. Their heat kept the room above the household target, so families brought chairs, bedding, and their printed receipts rather than warm separate apartments.

An intake volunteer showed Nera three receipts with identical district estimates and different room readings. The top-floor household had lost twice the scheduled heat because wind reached a cracked vent collar. A basement family remained warmer than predicted beside the return pipe. District arithmetic had averaged

them into compliance.

"Do you want a hazard opening?" Nera asked.

"We want repair first," the volunteer said. "If you raise the block draw before closing that collar, half the heat leaves through the roof."

A phase mechanic at the food table offered two hours after her canal shift. Another resident supplied a ladder but refused to lend it without a second person on the feet. Beren assigned sealant from the ward swap. Nera recorded the sequence: inspect, repair, remeasure, then alter distribution if the rooms stayed below target.

The meal line moved while they worked. Nobody received extra portions for volunteering. An elderly man challenged the ration clerk over a half ladle and produced his heat receipt as if colder rooms should purchase more stew. The clerk returned the paper.

"Temperature is not food credit," she said. "If your household count changed, enter the count."

He cursed the treaty, the kitchen, and people who wrote rules in heated offices. Then he entered his sister's arrival from a damaged block. The clerk changed the portion because a person had joined the table, not because anger carried legal weight.

When the vent collar finally came free, black water poured from its insulation sleeve. The mechanic found an old automatic bypass beneath it, wired to open whenever indoor temperature crossed a founder threshold. Its downstream lead had gone dark with the deferral table, leaving the valve half-set and leaking heat into the roof cavity.

Beren wanted the part displayed in Assembly Hall. Nera wanted it preserved under technical custody. They did both. Sumi's crew sealed the valve in a clear evidence case after the local line had been repaired. Block Twelve kept the heat that its public schedule already assigned.

Near noon, Tovan arrived carrying a crate against his right hip. His left wrist remained in the gray splint. Snow clung to his hair and melted along the healed cuts on his cheek.

"You cannot carry with that hand," Nera said.

"I did not."

"You cannot balance a crate on a healing wrist either."

"I balanced it on my hip. The wrist came attached."

"What did the infirmary clear?"

"Walking and seated work after they checked the alignment this morning. Three more weeks in the splint. No fitting, no ladders, no lifting with the left hand."

Nera looked at the crate.

“Master Ilek weighed it before he let me leave.”

The crate held vent-glass patches from his apprenticeship shop. Each pane used a layered edge that could flex during cold contraction instead of shattering. Tovan had marked sizes in thick chalk because fine writing with his right hand exhausted him.

“Where are these assigned?” she asked.

“Low Glass clinic, Warm Ward nursery junction, Office slit.”

“The Office did not request one.”

“The crack between your panes widened during the returning storm.”

Nera looked toward the legal observation slit. A white line still crossed its sealed gap from the first cold account. “Does the patch close it?”

“It replaces the inner pane. The outer stays. The crack remains visible.”

“Then why patch it?”

“Because visible damage can still cut someone.”

He said it as craft, not lesson. Nera signed the material receipt and refused his attempt to join the installation crew. Tovan appealed to Beren, who assigned him to inventory at a table where his splint remained above the work.

At the table, one narrow pane slipped sideways as Tovan reached across the crate. His left hand moved by reflex. Pain stopped it before he touched the glass. He waited with his jaw set until the pulse eased, then asked an apprentice to square each pane while he called the dimensions. The count took longer. It did not lose a piece.

By afternoon, snow reached the lower rim of the canal. Automatic deferral never engaged. Instead, the public board changed as present decisions accumulated. Gallery heat moved to a nursery. Canal discharge protected two school loops. Bellweather accepted a lower night minimum in exchange for keeping its frost-barley pumps above freeze. No line claimed the city suffered equally.

At dusk, the winter garden opened its outer doors for the first time in Nera's memory.

Ward residents entered in coats to harvest herbs and late greens before soil temperature fell below the preservation target. The garden had once held warm rain through every winter, paid for partly by cold sent beyond the ledger's visible edge. Now its paths hardened under real frost. Children touched the white crust on leaves as if discovering a substance their lessons had described poorly.

The ward council kept one interior bed warm for seed stock. Beren challenged the allocation, inspected its inventory, then accepted it because the stock served the audited belt rather than decorative supply. Disagreement remained in the record beside consent.

Nera walked home through snow that squeaked under her boots. Their

apartment radiator ran cooler than any night Tovan could remember. He wore two shirts and complained while repairing a glove seam with his teeth and right hand.

"You told the inspector our bedroom loop could lose another degree," Nera said.

"It can."

"You also put a blanket around the gauge."

"The gauge does not need to feel brave."

She moved the blanket away from its intake. "It needs to measure the room."

Tovan surrendered it. "Orra's calendar says they sleep three to a mattress during compound fronts."

"We are not competing with their cold."

"I know." He looked at the radiator. "I am trying to learn what one degree means here."

They ate sunroot stew at the kitchen table. The root had grown at Bellweather before Repayment Day. Lev's cooperative mark appeared on the storage tag, along with a warning that later batches would run smaller. Tovan cut bread against the splint and made crumbs because he refused help.

At the evening bell, the apartment's public receiver printed a note from the distribution panel. Demand had exceeded the first forecast in three Warm Ward blocks and fallen below it in Low Glass common kitchens. The next day's schedule would alter corridor reductions while preserving the 2.4 ratio across the full week.

Nera read the calculations twice. "They used district averages."

Tovan frowned. "Is that wrong?"

"It can hide which buildings carried the cold."

She sent an objection with their household reading attached. Beren replied before midnight that the panel had already received nine similar notices. The morning board would publish building ranges as well as district means.

The treaty had lasted less than a day before its first current-side correction. Nera trusted it more for needing one.

On Day One Hundred Fifteen, the west tram resumed limited service. Nera rode to Bellweather with Sumi's crop packet sealed inside her coat. The trip took half a day because snowdrifts covered the outer line and crews stopped twice to clear switches by hand. Beyond Lume, cloud combs stood idle against a low white sky. Fields carried weather without the faint ribbed pattern deferral usually left in cloud bands.

Lev met her at the cooperative platform. Bandages still covered his damaged fingertips, now wrapped beneath fingerless mitts. He did not offer his hand. He took the packet under one arm and led her toward the redleaf rows.

Orchard mesh sagged under snow. Workers raised each section with poles, sending white slabs down between the trees. Many buds had blackened during the

split spring. Others held tight against the new cold, their bark marked with blue Kest strain codes.

"Old stock survived in South Terrace," Lev said. "Their blight record confirmed what our archive forgot. We grafted twelve trees before this front."

"Is that enough to matter?"

"In six years, perhaps. In six days, it matters because we did not discard them."

Yara Fen called from the sunroot house. Her twins followed at a careful distance, each dragging an empty harvest sling. The surviving crop lay in shallow bins under insulated cloth. Smaller roots had been left in soil because lifting them would expose the beds.

Yara pulled back one cover. "This row took the full night minimum. No shifted heat. We lost the wet corner and kept the center."

Nera crouched beside the bins. The roots showed cold scarring, pale rings under the skin that reduced storage life. "How much?"

"Thirty-eight percent by weight. Enough for seed and three district contracts. Not enough for the old market schedule."

Lev said, "We cut redleaf to protect it."

"You cut redleaf," Yara corrected. "The crews moved the covers, trimmed dead wood, and took the frost shifts."

He accepted the correction without the injured expression of a man offering humility for credit. "The cooperative cut redleaf. The crews kept these beds."

One twin lifted a root and asked whether future children would eat it.

Yara answered before any adult could enlarge the question. "Children here will eat this one. We sent growing notes forward. They sent other notes back."

"Did they send food?"

"No."

The child put the root in its bin. Material limits survived explanation.

Nera opened Sumi's packet at the worktable. It contained Kest's blight tolerances, South Terrace soil temperatures, and three shelter requests for cultivars suited to shorter summers. Lev compared them with Bellweather's surviving stock. Yara required that labor hours and seed ownership travel with each recommendation. A crop strain without its work could become another gift claimed by institutions.

They drafted the first reciprocal agricultural schedule. Bellweather would send sunroot storage results and glass-cover designs. South Terrace would send redleaf hardening records under its own seal. Neither side surrendered ownership by sharing measurements. Any seed movement in future years required separate local consent.

Nera authenticated sources, not outcomes.

When work ended, Lev walked her to the damaged redleaf block. Snow rested in branch forks. He moved stiffly around one tree because his bandaged fingers could not close on the support line.

"Let me," Nera said.

"You will tighten it like a document binding."

"That keeps pages in order."

"Trees grow."

He showed her how much slack to leave. Together they reset the line. Their shoulders touched once through winter coats. Neither used the contact to reopen what had ended between them or promise what might return.

Lev looked toward the sunroot house. "A shorter summer changes everything we planned."

"Yes."

"I still want one."

"So do I."

The admission required no argument. Summer remained desirable after borrowing ended. They had foreclosed a method, not the season or the human wish for warmth.

Snow began again before Nera reached the tram. Lev gave her a paper bag of scarred sunroot for Tovan and marked it against her household ration. Affection did not exempt the gift from the ledger.

Day One Hundred Eighteen brought the first complete future-side audit.

Sumi called Nera to Canal Nine before sunrise. Drum Four turned smoothly inside its repaired cage. The voice horn had lost its storm distortion, though words still arrived thin. Edrin waited with the treaty register. Beren attended for the distribution panel. Halden stood behind the public rail without his steward seal.

Siv spoke the measurements in shelter order. Kest's inherited load had fallen within the range predicted from Lume's retained snow. South Terrace recorded a shorter pressure crest. North Brace kept one water manifold above emergency threshold without opening its final heat store. Outer assemblies submitted separate variances rather than routing all figures through Kest.

"How much reduction?" Nera asked.

Sumi shook her head. "Do not make one number carry different rooms."

Siv seemed to hear the caution through history, though no present voice traveled that way. Her next packet listed settlement changes before the combined estimate. Only then came the audited total: the future winter line had received less deferred cold than the pre-treaty schedule assigned. The debt remained immense. This week had not repaid it. No new account had joined it.

Orra's voice entered for four words, coached and solemn. "We marked no day."

Nera waited.

Siv explained. A true day required weather free of inherited load. Year Three Sixty still carried old accounts from previous schedules. Orra would not improve the calendar because Lume had behaved differently once. The child had kept the category exact.

"Tell her that is correct," Nera said, then remembered the direction of voice. She wrote the return herself.

> AGREED. RECORD REDUCED LOAD, NOT TRUE DAY. DEBT OPEN. NO NEW DEFERRAL. SEND NEXT SHELTER WINDOW.

Sumi fed the strip into Drum Two. Present heat paid for the message. Beren entered the channel draw against the distribution schedule. Edrin witnessed both eras' seals on the received audit.

Nera assembled the reciprocal record at the phase-room table. Lume's forecast, building temperatures, canal draw, orchard loss, and public corrections occupied the left pages. Future shelter measurements, crop responses, store levels, and claimant variances occupied the right. No column described Year Three Sixty as empty forecast space.

She pressed the season seal through both page sets.

The brass met ceramic beneath the paper and left a joined impression: Orison under weather bands on one side, Kest's shelter mark on the other. The first reciprocal seal did not certify fairness. It certified actions, consent, measured consequence, and disagreements kept visible.

After the witnesses signed, Nera carried the register upstairs.

The Office workroom had cooled enough for binding paste to stiffen at its edges. Tovan's replacement inner pane leaned beside the legal observation slit, but he had followed her instruction and left installation to a licensed crew. They had fitted the safer glass around the old internal crack. The white line remained visible between layers.

Outside, Canal Nine moved dark beneath a narrowing rim of ice. Snow crossed the gantry in dry grains. A heatworker checked Gate Three with a sounding rod over one shoulder, following a schedule written for the weather under his boots.

Nera placed the reciprocal record where the cold account had first opened. The ledger accepted its season seal. Present load stayed in the present column. Future effect carried named claimant marks. No blank field waited for an invented owner.

Edrin reached for the observation-slit latch. "The room will lose heat."

"Record it."

"How far?"

"Until the thermometer settles."

He opened the red book.

Nera lifted the latch herself. The inner pane swung inward around its repaired edge. Cold entered without a pulse code, a hidden credit, or a future timestamp. It touched her face, crossed the desk, and softened the top layer of binding paste with clean snow.

She left the observation slit open while Edrin wrote the temperature down.

Epilogue — A Winter Kept

Orra Ren tested the snow with a cup before she marked the calendar.

She carried the sample through Holdfast Kest in both hands, careful not to warm it against her coat. The south passage remained narrow after last year's wall failure. New braces crossed the ceiling at head height for adults, though Orra could walk beneath them without bending. Shelter keepers had written temperature loss on every brace in blue chalk.

Outside, wind pressed snow against the agricultural terraces. It had buried two sensor posts and stripped the cover from a water manifold before dawn. Three families had moved into the schoolroom while repairs continued. Difficult weather had arrived without asking whether history had improved.

Inside the calendar room, Siv waited beside the phase plate. She did not touch the ceramic slips arranged across the table. Orra had kept them since she could write a stable number. White marks recorded ordinary inherited weather. Black marked account storms. Blue remained rare.

"South Terrace confirms local fall," Siv said.

"North Brace?"

"Same phase within tolerance. No old-load signature."

"Lume record?"

"Authenticated and open."

Orra set the cup on the test ring. Its melt line matched Kest's local pressure curve. No delayed crest appeared beneath it. No continuity margin thickened the storm after arrival. The snow had formed from Year Three Sixty-One's own sky, corresponding to weather kept in Year Two Twenty-Four.

She checked all three records again.

Kest still carried unpaid cold accounts. Two older fronts would reach the northern shelters before spring. The Treaty of Seasons had reduced their modeled force, not removed them. Store Three remained below ideal reserve. This day required rationed heat, roof labor, and a shortened meal cycle.

None of those facts disqualified it.

Orra took up a blue mineral chalk and wrote the date on a fresh slip. Beneath the date she entered local temperature, wall loss, manifold damage, and the verified absence of inherited dissipation. Her hand paused over the category line

only once.

TRUE WINTER DAY.

Siv breathed out. "Do you want me to witness?"

"You already checked the return."

"That is not the same work."

Orra considered this, then turned the slip around. Her mother added a calendar-keeper mark beside Orra's entry, not over it.

The schoolroom bell rang for the midday heat reduction. Pipes clicked inside the walls. Someone in the south passage called for another brace clamp. A true day made no promise of comfort. It named weather that had arrived without a concealed hand pushing more into it.

The phase plate tapped once.

A packet from Lume opened across its damp surface: sunroot storage loss, vent-glass edge tolerances, and a note from Tovan Voss. He had signed as a second-year glassworker rather than as Nera's son. The note included a revised collar that could flex around Kest's roof braces during deep contraction.

Orra read the last line aloud. "Please return ice thickness at first crack."

Siv looked toward the buried sensor posts. "He asks excellent favors."

"We can measure it."

"After the manifold cover."

They carried the blue slip into the schoolroom, where three families slept behind hanging blankets and children sorted brace bolts by size. Orra fixed the true-day record to the public calendar. People stopped long enough to read it. Then the shelter keeper sent two adults back to the roof and asked Orra to count clamps.

Blue chalk did not repair a wall.

It remained visible while they worked.

One hundred thirty-seven years earlier, Tovan Voss received the calendar mark through Canal Nine.

His left wrist had healed, though cold made it stiff when he cut thick vent glass. He rotated the hand before lifting Orra's packet from the return tray. Across the phase room, Nera authenticated South Terrace's snow data while Sumi compared ice thickness with the revised collar design.

Tovan read TRUE WINTER DAY twice. Last year Orra had refused the words when inherited load remained. This mark needed no praise from Lume. Its measurements carried the proof.

He added Kest's ice figures to his work sheet. The first crack had begun at the brace corner, exactly where his shop's test rig predicted trouble. He changed the

next collar radius by two millimeters and wrote the reason beside the drawing. No gift language. No claim that the present had saved anyone. Material moved forward only as knowledge; consent and labor waited where it arrived.

Bellweather's crop note lay beside him. Yara reported smaller sunroot yield and viable seed. Lev had sent redleaf graft temperatures with a warning against extrapolating from twelve trees. Kest replied with shelter humidity from rooms that used the old strain as food storage. The two eras had begun a shared agricultural ledger full of failures, revisions, and useful limits.

Nera brought the treaty register to the workbench. "The distribution panel wants the summer line entered before today's seal."

Tovan looked at the paired schedules. Lume's coming summer ran nineteen days shorter than the forecast published before the ledger opened. Kest's summer estimate had also shortened, though less compound cold meant its thaw no longer had to spend every warm day repairing inherited damage.

He entered both lengths without balancing them into equality.

Sumi set Orra's blue slip beneath the season seal. "Ready?"

Nera checked source, claimant mark, phase interval, and local measurements. "Ready."

The seal joined the calendar record to both ledgers. On Tovan's page, the Year Two Twenty-Four summer column ended at a black rule. Across the return plate, Orra drew a second rule on the Year Three Sixty-One ceramic schedule. Each line stopped earlier than old promises allowed.

Between them, the channel carried crop notes, shelter figures, and the measured record of one true winter day.